

Applied Educational Psychology

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Jonathan O'Neill



www.kaufmanpress.com



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Published by Kaufman Press,
1201 N. Orange Street, Suite 600,
Wilmington, DE 19801 U.S.A

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Includes bibliographical references and index.
ISBN 978-1-6668-8828-7

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Printed and bound in U.S.A

Library of Congress Cataloging-in-Publication Data
A catalog record for this book is available from the Library of Congress

For more information regarding Kaufman Press and its products, please visit the publisher's website
www.kaufmanpress.com

Table of Contents

Preface

xi

Chapter 1	Introduction	1
	Aims of Educational Psychology	1
	Scope of Educational Psychology	2
	Meaning of Psychology	3
	Meaning of Psychology	5
	Psychology – The Science of Soul	6
	Psychology – The Science of Mind	6
	Observational Learning	6
	Intrinsic Reinforcement	6
	The Modelling Process	7
	Relationship between Psychology and Education	7
	Psychology and Aims of Education	9
	Importance of Educational Psychology for Teacher	10
	Problem of Discipline	13
	School and Class Administration	13
	Concept of Growth and Development	20
Chapter 2	Importance of Educational Psychology in Child Development....	24
	Proper Atmosphere	24
	Proper Treatment	24
	Later Childhood-(from to Years)	26
	Educational Importance of Later Childhood	27
	Proper Physical Development	27

Adolescence Stage (Period From to / Years)	29
Physical Development	29
Height and Weight	29
Adulthood Stage (Period after Years of Age)	30
Old Stage (Period after Years of Age)	30
Chapter 3 Psychological Approaches to Learning for Educational Practice	42
Psychoanalysis	42
Behaviourism	43
Humanistic Psychology	44
The Neurobiological Approach	45
Cognitive Psychology	45
Transformative Learning	46
Other Perspectives	46
Transformative Learning in Practice	48
The Role of the Educator	49
Benefits of Transformative Learning	52
Challenges of Transformative Learning Environments	52
Acknowledging the Role of Self-Efficacy in Transformative Learning ..	52
Academic Self-efficacy	53
Social Learning Theory	53
The Role of the Rational and the Affective	54
Educational Neuroscience	54
A Brain-based Theory of Learning	55
Multiple Intelligences	55
Multimedia Learning	55
Other Learning Theories	55
Learning Style Theory	56
Informal and Post-modern Theories	56
Philosophical Anthropology	56
Criticism of Learning Theory	56
Hull's Reinforcement Theory	56
Transfer of Learning and its Theories	57
Reinforcement Theory	57
Classical Conditioning	61
Classical Conditioning Procedures	62
Phenomena Observed in Classical Conditioning	64
Models of the Learning Process	67
Gagné's Hierarchy of Learning	67
Learning and Motivation	69
Ethology	70
Conditioning	70

Individual Learning Variation	71
The Role of Needs	74
The Effect of Generalisation	74
Reading and Writing	75
Transfer of Learning	76
Transfer Taxonomies	76
The Effect-perspective: Positive vs. Negative Transfer	77
The Situation Perspective: Specific vs. General, Near vs. Far Transfer ..	78
The Process Perspective	79
Traditional Fields of Transfer Research	79
Education Science: Teaching for Transfer	80
Implications for School Psychology Practice	80
From Formal Discipline to Meta-cognition	81
The Generality Predicament: Return to the Specificity View	82
Nature of Learning/characteristics of Learning	82
Factors Affecting Learning	83
Factors Belonging to the Teacher	87
What is Learning Behaviour?	91
Environmental Factors Affecting Learning	91
Heredity	91
What Learning Involves	95
 Chapter 4 Trends in Educational Psychology and Classroom Management	96
Constructivism	96
Student Diversity in the Classroom	98
Out-of-School Influences	99
Classroom Management and the Learning Environment	100
Why Classroom Management Matters	100
Preventing Management Problems by Focusing Students on Learning	101
Instructional Strategies	102
Instructional Planning	106
Student Diversity	109
Individual Styles of Learning and Thinking	109
Student Development	113
Why Development Matters	113
Cognitive Development	118
Revolution in Education Induced by Psychology	119
Child-Centred Education	119
The Critical Role of Classroom Management	124
Deep and Surface Approaches to Learning	127
Deep and Surface Approaches	127
Designing for Deep Learning	128

Basic Principles and the Approaches to Learning	128
Putting Theory into Practice	129
Behaviourist Theories of Learning	131
Behaviourist Approach	131
Varieties of Behaviourism	131
Social Learning	132
Basic Concepts	132
Modeling Process	133
Social Constructivism	133
Psychology: Motivation and Learning	135
Research Fundamentals	135
Theories of Teaching	136
Implications for Teaching Practice of Some Key Ideas from Learning Theories	136
Chapter 5 Modern Theories of Intelligence	138
Nature of Intelligence	139
History of the Concept	146
Measurement of Intelligence in Educational Settings	154
Chapter 6 Critical Educational Psychology	160
Traditional Education Psychology	161
Critical Psychology	161
Origins	162
Abilities and Intelligence	165
Intelligence and Information Processing	167
Three Parts of Intelligence	167
Educational Psychology: The Concept of Intelligence is an Artificial Construct	168
Sternberg's Triarchic Theory of Intelligence	169
Critical Foundations for Educational Psychology	170
Learning	171
Educational Assessment	174
Forward Moves	177
Chapter 7 Educational Psychology: A Foundation for Teaching	179
Relevance of Educational Psychology for Secondary Teachers	209
The Practical Value of Psychology to the Teacher	212
The Teacher as Director	212
The Teacher as Student	214
The Teacher as Educator	214
The International Level	222

Chapter 8	Measurement and Evaluation in Psychology and Education	226
	Methodology	226
	Standardization of Measurement Units	227
	Standards	228
	Units and Systems	228
	Units of Measurement	228
	Systems of Units	230
	System of Measurement	231
	Imperial and US Customary Measurement Systems	236
	Intelligence Tests as Clinical Instruments	247
	Intelligence	247
	Determines	250
	Testing for Special Aptitudes	251
	Motivation	253
	Integrity Inventory	256
	Projective Test	257
	Theory	257
	Projective Hypothesis	257
	Common Variants	258
Chapter 9	Motivation in Educational Psychology	265
	Introduction	265
	The Hierarchal Model of Achievement Motivation	267
	Implicit and Self-Attributed Motives	268
	Motivational Theories	288
	<i>Bibliography</i>	293
	<i>Index</i>	297

Preface

Educational psychology is a discipline within the larger field of psychology which is focused on studying how people learn. People have been curious about the processes behind learning for thousands of years, with educational psychology as a distinct scientific discipline arising in the 1800s. Advancements in this field are occurring all the time, including periodic reversals of previously stated conclusions. Some examples of famous psychologists who have performed research in this field include Jean Piaget and B.F. Skinner. This field incorporates several disciplines from within psychology. Educational psychologists are very interested in the study of developmental psychology, which looks at the stages of human development and the processes which can impact development, and they are also interested in social and behavioural psychology, as well as abnormal psychology. As with other types of science concerned with human subjects, educational psychology is subject to a number of ethical restrictions which can make experiments challenging, with many research psychologists in this field using observation as a tool to increase knowledge.

Educational psychology is the branch of psychology concerned with the scientific study of human learning. The study of learning processes, from both cognitive and behavioral perspectives, allows researchers to understand individual differences in intelligence, cognitive development, affect, motivation, self-regulation, and self-concept, as well as their role in learning. The field of educational psychology relies heavily on quantitative methods, including testing and measurement, to enhance educational activities related to instructional design, classroom management, assessment, which serve to facilitate learning processes in various educational settings across the lifespan.

Educational psychology is a social science. Like any other social science it studies human beings relating to education. Like the economist, political scientist, sociologist or anthropologist, the educational psychologist conducts investigations and experiments, collects data and statistics, and makes interpretations and inferences. He does not work in an ivory tower nor makes any arm- chair speculations.

Educational psychology is psychology in relation to education. It deals with the behaviour of the individual in various educational environments. Psychology studies the behaviour of the individual in different conditions and situations. Therefore, scope of general psychology is broader than that of educational psychology, which is comparatively limited.

Educational psychology deals with the child as a whole-his physical, mental, emotional and social development at various stages. Such development is resulted by heredity and environment, different biological, social and cultural factors. Hence all these aspect and factors come under the purview of educational, psychology.

Learning is the key-concept in educational psychology. It is so important and broad-based that educational psychology is also called learning psychology. The learning process, learning and maturation, nature and conditions of learning, factors influencing, learning, motivation, attention and interest, various kinds of learning and laws of learning come under the scope of educational psychology.

The psychological principles underlying various methods of teaching, educational innovations and experiments, educational objectives, mental health and hygiene, special provisions for slow- learners, gifted, handicapped and deprived children are usefully applied to educational development. These are, therefore, part and parcel of educational psychology. The psychological tools and techniques, methods and approaches are profitably used by the educational psychologists for undertaking research studies and experiments in various fields. New methods and techniques are also developed by them for collection, interpretation and analysis of data. All these constitute the scope of educational psychology.

The uniqueness of this book hence lies in the editor's way of reconstructing the chapter under review by delving deep into the areas of the subject.

– *Author*

Introduction

AIMS OF EDUCATIONAL PSYCHOLOGY

Educational psychology aims at the harmonious growth and rightful conduct of the children in too. It aims at the development of ‘wholesome personality’ and ‘continuous growth’.

It aims at helping the teacher in providing facts and generalisation in his task of assisting the child to develop the harmonious personality.

A few of the specific aims and objectives are discussed here with:

- (1) Educational psychology aims at developing right attitudes in the teacher about the educational problems. An effective management of learning is a key problem in educational psychology. From educational psychology the teacher knows the proper method of imparting instruction. He also knows the amount of learning which can be acquired by the child.
- (2) Educational psychology aims at assisting the teacher to organise the material which to be taught to the child. The teacher studies the whole child and his mental make-up and chooses and organises the subject-matter properly.
- (3) Educational psychology aims at studying heredity growth and maturation, environmental influences, language, thinking, the development of language and the process of socialisation in relation to their effect on the child as a learner. It assists the teacher to set up appropriate educational situations in order to bring about desirable change.

- (4) It aims at assisting the teacher in treating their pupils with sympathy and understanding. It also aims at creating positive attitude towards learning.
- (5) It aims at helping the teacher to understand his own task. The teacher faces many problems in the classroom as teaching situation. Educational psychology develops in the teacher a scientific attitude to solve different problems of education faced by him.
- (6) Educational psychology aims at teaching, teachers how best to help their pupils to learn more effectively both in and out of class.
- (7) It aims at providing the teacher with the proper method of teaching. The various teaching procedure are put in practice in teaching the child. Those which are psychologically sound are recommended for the teacher's use.
- (8) Conducting research is another important objective of educational psychology. Knowledge in this field keeps on developing from this practice of research methods.
- (9) Educational psychology aims at the application of research findings in the learning situation itself.
- (10) It also aims at developing sound methods of measuring and evaluating the achievements of the pupils objectively with pure objectivity.
- (11) Another significant aim of educational psychology is to guide the administrators in the organisation and administration of the educational instruction and to provide a scientific basis for the supervision of instruction.
- (12) Educational psychology also aims at helping the teacher to provide proper guidance programme in the school having thorough knowledge of individual difference.

SCOPE OF EDUCATIONAL PSYCHOLOGY

In this age of science and technology, psychology has been considered as one of the youngest, yet one of the most influential sciences. It has influenced education in many different ways and has give a new turn; a psychological turn to the human mind. For a skilful teacher in this day and age, a great deal of knowledge of educational psychology is highly indispensable.

The subject psychology has two aspects pure and applied. Pure psychology formulates techniques for the study of human behaviour, which finds the practical shape in its applied aspects, *i.e.* branches of applied psychology like clinical psychology, crime psychology, industrial psychology, occupational psychology and educational psychology. Educational Psychology as a Branch of Applied Psychology.

As discussed above educational psychology is nothing but one of the branches of applied psychology. It is an attempt to apply knowledge of pure psychology to the field of education. It consists of application of psychological principles and techniques to human behaviour in educational situations. In other words, Educational Psychology is

a study of the experience and behaviour of the learner in relation to educational environment. In order to develop a clear understanding of the term educational psychology it is necessary to understand the meaning of psychology and education separately.

MEANING OF PSYCHOLOGY

Curiosity in man has led to know his surroundings which mainly conclude nature and other fellow men. There is always a desire to 'know' what 'one' is, what is his background, what is it made of, what are the associated factors and in what way can it be useful to one self? Each question will lead to more questions, Psychology forces to answer many of the questions we have about ourselves, other people and the nature of human life; why do we feel lonely? Why do we forget? How people learn? What makes someone creative? Why do we take drugs? What makes some one help others? and so on. Psychology had its formal beginning when Wilhelm Wundt established his psychological laboratory in Leipzig Germany in 1879. But in real sense, interest in psychology as a discipline dates back to the work of Plato, Aristotle and other philosophers.

Psychology – The Science of Soul

The Greek Philosophers conceived psychology as a science of soul, as early as 400 B.C. In fact, the term psychology literally means the science of soul. Etymologically, it is composed of two Greek words "Psyche" and Logos means soul and science respectively. Goeckel named it as psychologia.

Soul is a being which dwells on the body and with the end of life it leaves the body. Soul is a metaphysical idea. It can neither be perceived on imagined nor its nature and function can be studied by scientific methods of observation, experiment etc. Therefore, definition of psychology as the science of soul has been discarded by the modern psychologists.

Psychology – The Science of Mind

Some regard psychology as the science of mind. Historically the French philosophers like Descartes (1596-1650) and the Britisher philosophers like Locke considered psychology as the science of mind. Descartes tried to understand body mind relationship in terms of their interaction, the study of nervous system, and interest of references as innate actions, etc. The definition of psychology as the science of mind is not acceptable at present. Mind is an ambiguous a concept as the soul. It is not at all possible to carry on scientific observation and experimentation on mind. This definition also does not include the overt behaviour of man and animal which are also important subject matter of psychology. Therefore, the definition of psychology as the science of mind has been discarded.

Psychology – The Science of Consciousness

Psychology has also been defined as the science of consciousness. Historically such a definition has been propounded by the Leipzig school of psychologists led by Wilhelm Wundt (1832-1920). Wundt defined psychology as the science of immediate experience with consciousness being the main subject matter.

He postulated that conscious experience can be reduced in to elements and the primary aim psychology is the analysis of conscious experience in to its elements. But the definition of psychology as the science of consciousness is not acceptable. That is because mental life does not consist only of consciousness. There are unconscious and subconscious mental process which influence our behaviour in various ways without our knowledge.

Psychology – The Science of Experience

Titchner (1867 – 1927), the leader of the structuralists defines psychology as the science of conscious experience which is dependent upon the experiencing person. To give an example the physicist and the psychologist may be investigating about sound. But while the former investigates the phenomena as such, the latter is interested as to how it is perceived by the observer.

The mind is nothing but the sum total of the conscious experiences as perceived by a person. The subject matter of psychology is the study of such conscious experience which constitutes mind. The method of study of conscious experience is through the introspection of a trained observer.

Psychology – Study of Behaviour

Watson (1878-1958), an American brought about a revolution in psychology called behaviourism. He argued that psychology is to be regarded as a science and as a science it is to limit itself to the study and analysis of publicly observable events such as the behaviour of the subject rather than subjective matters like his private mental states. He defined psychology as “the science of behaviour.”

Meaning of Science and Behaviour

Science has been defined as “Systematic study of knowledge” concerning the relationship between the cause and effect of a particular phenomenon. In order to collect the scientific data and systematised material, science employs various kinds of methods of enquiry such as observation classification formulation of hypothesis, analysis of evidence etc.

It also organises and develops our knowledge of the world we live on, Psychology too aims at same thing. It uses scientific methods to study human behaviour. It also helps us to understand control and predict human behaviour.

What is Social Learning Theory?

The social learning theory proposed by Albert Bandura has become perhaps the most influential theory of learning and development. While rooted in many of the basic concepts of traditional learning theory, Bandura believed that direct reinforcement could not account for all types of learning. His theory added a social element, arguing that people can learn new information and behaviours by watching other people. Known as observational learning (or modelling), this type of learning can be used to explain a wide variety of behaviours.

Nature and Scope of Educational Psychology

In this age of science and technology, psychology has been considered as one of the youngest, yet one of the most influential sciences. It has influenced education in many different ways and has given a new turn; a psychological turn to the human mind.

For a skilful teacher in this day and age, a great deal of knowledge of educational psychology is highly indispensable. The subject psychology has two aspects pure and applied. Pure psychology formulates techniques for the study of human behaviour, which finds the practical shape in its applied aspects, i.e. branches of applied psychology like clinical psychology, crime psychology, industrial psychology, occupational psychology and educational psychology. Educational Psychology as a Branch of Applied Psychology.

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OBSERVATIONAL LEARNING

In his famous “Bobo doll” studies, Bandura demonstrated that children learn and imitate behaviours they have observed in other people. The children in Bandura’s studies observed an adult acting violently toward a Bobo doll. When the children were later allowed to play in a room with the Bobo doll, they began to imitate the aggressive actions they had previously observed.

Bandura identified three basic models of observational learning:

1. A live model, which involves an actual individual demonstrating or acting out a behaviour.
2. A verbal instructional model, which involves descriptions and explanations of a behaviour.
3. A symbolic model, which involves real or fictional characters displaying behaviours in books, films, television programs, or online media.
2. *Mental states are important to learning.*

INTRINSIC REINFORCEMENT

Bandura noted that external, environmental reinforcement was not the only factor to influence learning and behaviour. He described intrinsic reinforcement as a form of

internal reward, such as pride, satisfaction, and a sense of accomplishment. This emphasis on internal thoughts and cognitions helps connect learning theories to cognitive developmental theories. While many textbooks place social learning theory with Behavioural theories, Bandura himself describes his approach as a 'social cognitive theory.'

3. *Learning does not necessarily lead to a change in behaviour.*

While behaviourists believed that learning led to a permanent change in behaviour, observational learning demonstrates that people can learn new information without demonstrating new behaviours.

THE MODELLING PROCESS

Not all observed behaviours are effectively learned. Factors involving both the model and the learner can play a role in whether social learning is successful. Certain requirements and steps must also be followed.

The following steps are involved in the observational learning and Modelling process:

- **Attention:** In order to learn, you need to be paying attention. Anything that detracts your attention is going to have a negative effect on observational learning. If the model is interesting or there is a novel aspect to the situation, you are far more likely to dedicate your full attention to learning.
- **Retention:** The ability to store information is also an important part of the learning process. Retention can be affected by a number of factors, but the ability to pull up information later and act on it is vital to observational learning.
- **Reproduction:** Once you have paid attention to the model and retained the information, it is time to actually perform the behaviour you observed. Further practice of the learned behaviour leads to improvement and skill advancement.
- **Motivation:** Finally, in order for observational learning to be successful, you have to be motivated to imitate the behaviour that has been modelled. Reinforcement and punishment play an important role in motivation. While experiencing these motivators can be highly effective, so can observing other experience some type of reinforcement or punishment. For example, if you see another student rewarded with extra credit for being to class on time, you might start to show up a few minutes early each day.

RELATIONSHIP BETWEEN PSYCHOLOGY AND EDUCATION

- According to Aristotle, "Education is the creation of a sound mind in a sound body."
- According to Pestalozzi, "Education is the natural, harmonious and progressive development of man's innate powers."

- According to Gandhi ji, “By education I mean around drawing out of the best child and man, body, mind and spirit.”

The relation between psychology and education is very intimate. Psychology has been defined as the science of behaviour. It seeks to understand and explain behaviour in terms of mental and bodily activities. Its chief problem is how and why we behave, how we Think, know, feel and act and why we think, know, feel and act in the way in which we do. It tries to understand the conditions from which acts of behaviour arise and to arrive at general principles which govern behaviour so as to interpret, control and predict it. Education, as we have seen above, is an attempt to mould and shape behaviour. It tries to help young people to grow and develop along certain lines, to acquire knowledge and skill and to learn certain ways of thought and feeling so that they may be absorbed in adult social life.

The science of psychology must be basic to such an attempt, for any influence on behaviour, to be effective, must be planned and worked according to the principles of psychology. Education, therefore, must be based on psychology and from the very first step which he takes to educate the child, the educator must depend upon psychological knowledge. Education deals with young people and the conditions that promote or retard growth and development; it selects and strengthens those influences which promote healthy growth and tries to eliminate and weaken those which retard it.

As a result of this study it formulates certain principles on which organisation and administration in schools should be based; it has to study the needs and interests of children and provide for their healthy satisfaction and expression; it has to devise effective methods of teaching so that children may learn more quickly and better.

All this is not possible without a knowledge of psychology which explains how young people grow, what dominant interests mark the several stages of their growth, how they differ from one another and grow at different rates, how they learn new skills or acquire new knowledge, how they react to the influence of teachers and classmates.

Psychology is expanding rapidly and our growing knowledge of the minds and behaviour of young people promises to be an effective guide in the solution of our educational problems. The Education and Psychology are complementary to each other. Psychology is an essential element to education. Without its help problems of education cannot be solved. Both education and psychology are concerned with behaviour.

Modern education is based and founded on psychology. The child is imparted education only after making a thorough study of his interests, aptitudes, intelligence and personality both is two distinct branches of knowledge but they are closely related.

- According to Jha, “The process of education is entirely at the mercy of psychology.”
- According to Davis, “Psychology has made a distinct contribution to education through its analysis of pupils’ potentialities and differences.”
- According to Skinner, “The entire range of behaviour and personality is related to educational psychology.”

Psychology is the science of behaviour and education in its narrower sense is the modification of behaviour. The modification in behaviour cannot be easily brought unless we know the science of behaviour. The teacher must know about the developmental stages, personality development and emotions of the students in order to be successful teacher.

Unless the teacher is fully aware of psychological characteristics of the child, he may not be successful in bringing the desirable changes in the behaviour of students. This leads us to believe that education and psychology are intimately related to each other.

Below are some points which show how far education and psychology are related to each other:

- (1) Education is concerned with aims, ideals and standards of life and psychology determines whether these aims are attainable or not.
- (2) Education demands the teacher to know the child as well as the subject matter of instruction, where the psychology helps to know about the child.
- (3) Psychology also helps the teacher to teach effectively undertaking effective and appropriate teaching techniques.

On the basis of the following points the relationship between Psychology and education can easily be seen:

PSYCHOLOGY AND AIMS OF EDUCATION

Psychology helps the educator in the realisation of educational aims by helping him to bring out improvement in the quality of instruction by providing him ability and insight into the child's attitudes, ideas, aptitudes, interests and emotions, etc.

Psychology and Teacher

Psychology helps the teacher to understand the learner, learning process and the learning situations. Psychology states that teacher should have sympathetic and affectionate attitude towards the learner. They should have genuine interest in the teaching profession.

Psychology and Curriculum

Psychology suggests that the curriculum should be integrated, flexible, co-related and child-centred. There should be different co-curricular activities in the school. Co-curricular activities are considered as an important part of education because they are important media for sublimation of instincts and for the development of personality.

Psychology and Methods of Teaching

Various methods of teaching like Project method, Heuristic method, Montessori Method, Play-way method are based on sound psychological principles.

Psychology and Text Books

Psychology tells the teachers and the students that text books should be attractive, well illustrated and according to the mental level of the pupils. These may act as good aids to the learners.

Psychology and Innovations

Psychology has made significant contribution by introducing innovative ideas for improving the process of teaching and learning such as-Activity-centred teaching, Micro-teaching, Programmed instruction, Interaction analysis.

Psychology and Audio-visual Aids

Psychology states that to develop interest among students, teacher should properly use audio-visual aids. Use of audio-visual aids makes the learning easy, interesting and effective.

Psychology and Time Table

Time table is prepared according to the psychological principles. While preparing it, the teacher should keep in mind the relative importance of different subjects and their toughness and the fatigue of students.

Psychology and School Administration

Psychology helps in solving problems of administration by mutual discussion among various agencies of school. It provides a scientific basis for the supervision of instruction.

Psychology and Discipline

Psychology tells us the ways of dealing with problems of delinquent, backward, handicapped and gifted children and helps in maintaining discipline. It states that discipline should be self-discipline, dynamic and constructive through participation in purposeful activity.

Psychology and Evaluation

Psychological tools help the teacher to evaluate the achievement of the pupils *and suggests improvements in examination. Teacher can control, direct and predict the behaviour of students on the basis of research studies in classroom teaching. Thus, education and psychology are closely and intimately related.

**IMPORTANCE OF EDUCATIONAL PSYCHOLOGY
FOR TEACHER**

The contribution of educational psychology to the theory and practice of education is rich and varied. The knowledge of educational psychology is important as it provides

teachers with some basic skills and guidelines to solve the problems of teaching- learning process. According to John Adams, “Teacher should know John as well as Latin”. It means teacher should know child and subject- matter. A teacher should know the nature, capacities, likings and aptitudes and attitudes of the child. Child is like a book, teacher should know each and every page of it.”

- Skinner’s View, “The teacher needs psychology to bridge the lives of the young and the aims of education in our democratic society.”
- Kuppuswamy’s View, “Psychology contributes to the development of the teacher by providing him with a set of concepts and principles.”

Teaching is an art. Knowledge of educational psychology is very useful and indispensable for the teacher because it gives knowledge to the teacher.

Knowledge of Innate Nature

The child has got natural urges instincts, potentialities and propensities. These innate qualities are the “Prime movers” of his behaviour. The teacher who knows psychology can make his teaching very successful while keeping in view innate nature of the child.

Knowledge of Behaviour

Educational psychology assists the teacher in knowing the behaviour of the child at different stages of development. It also helps the teacher in understanding the physiological and psychological basis of behaviour, *i.e.*, nervous system, glands, instincts, emotions, sentiments, motives, play, intelligence, heredity and environment, etc.

Knowledge of Guidance

It helps the teacher in giving guidance to the pupils by having an understanding of interests, abilities, aptitudes, achievements, problems, educational and vocational plans of the pupils.

Knowledge of Unconscious Mind

It helps the teacher in knowing the unconscious mind of the students and plays very important role in the development of the personality of the individual.

Knowledge About Himself

It helps the teacher to know about himself. He learns the psychology of being a teacher and acquaints himself with the traits of a successful teacher.

Understand Development Characteristics

Children pass through different stages of development as infancy, childhood and adolescence. These developmental stages have their own characteristics. If the

prospective teacher knows the characteristics emerging at different stages of development, he can utilise these characteristics in imparting instruction and moulding their behaviour according to the specified goals of education.

Understand The Individual Differences

No two individuals are alike. The teacher with the knowledge of the kind of individual differences may adjust his teaching to the needs and requirements of the class and thus may be helpful in creating conducive environment in the schools where the students can develop their inherent potentialities to the maximum. Understand the Nature of Classroom Learning. The knowledge of educational psychology provides a teacher the knowledge of learning process in general and problems of classroom learning in particular. The teacher by the knowledge of educational psychology can understand the principles of learning and various approaches to learning process, problems of learning and their remedial measures and also about factors affecting and guidance for effective learning.

Understand Effective Teaching Methods

Educational psychology gives us the knowledge of appropriate teaching methods. It helps in developing new strategies of teaching. It also provides us with the knowledge of different approaches evolved to tackle the problems of teaching at different age levels.

Understand the Problems of Children

By studying educational psychology a teacher may understand the causes of the Problems of the children which occur at different age levels and can successfully solve them.

Knowledge of Mental Health

By studying educational Psychology teacher can know various factors which are responsible for mental ill health and maladjustment and can successfully help in Central hygiene.

Measurement of Learning Outcome

Psychological tools help the teacher to assess the learning outcome of the students and also to evaluate his teaching methods for required modification.

Curriculum Construction

Psychological principles are also used in formulating curriculum for different stages. Needs of the students, their developmental characteristics, learning patterns and needs of the society all are to be included in curriculum construction.

Research

Educational psychology helps in developing tools and devices for the measurement of various variables which influence the behaviour and performance of students.

Helps to Develop Positive Attitude

The teacher training programme aims to develop positive attitude towards teaching profession and provides the prospective teachers with the necessary competencies to meet the classroom challenges. Training colleges provide the knowledge of organising the subject matter in a sequential order to suit the needs of the class. The trainees are also acquainted with the techniques of motivating children for learning.

Understanding of Group Dynamics

Educational psychology helps the teacher to recognise the importance of social behaviour and group dynamics in classroom teaching learning.

PROBLEM OF DISCIPLINE

With the knowledge of educational psychology teacher utilises the importance of indirect discipline rather than corporal punishment. It tells the teacher that discipline should be self-discipline, dynamic, positive and constructive through participation in purposeful activity. Pleasure and pain, reward and punishment, praise, etc., should be judiciously used. If the teacher is unaware of the principles of educational psychology he may be unable to solve the problems of his students and thereby fail to induce order and discipline among them.

SCHOOL AND CLASS ADMINISTRATION

Former autocratic method of administration in school and class has been changed by democratic way of life wherein the teachers and administrators are more democratic, co-operative and sympathetic and problems of administration solved by mutual discussion.

Use of Audio-Visual Aids

Educational psychology has helped the teachers to make use of various types as audio-visual aids in classroom teaching so as to make the concept more clear, definite and learning to last longer.

Time-Table

The knowledge of psychology is helpful to the teacher in preparing time-table. He should keep in main the relative importance and toughness of different subjects and level and index of fatigue of the students.

Use of Innovations

Activity-centred teaching, discussion method, micro-teaching, etc., are some innovative ideas adopted to improve the teaching learning process.

Co-curricular Activities

Activities like debate, drama, games are given due importance along with theoretical subjects for the harmonious development of the personality of children.

Production of Text Books

Educational psychology has helped in planning of text books according to the intellectual development of children, their needs and interests at different age levels. Undoubtedly the study of educational psychology may be very helpful to equip our prospective teachers with necessary skills to deal with classroom teaching learning problems.

Difference between Psychology and Educational Psychology

Psychology, as a field of study, is very comprehensive, in terms of the various concepts which come under the ambit of the subject. It is, essentially, the study of behaviour and the way in the Mind works. This could include Humans and other life forms, under the sun. Educational Psychology is a specialized field of study which aims to understand the psyche of people in the ecosystem of an educational institution.

To give you an example, Educational Psychology would essentially entail understanding the different levels of intellect of the various students and integrating that in the learning plans of the various courses and subjects taught at the institute to make the subject matter as relatable as possible.

Psychology vs Educational Psychology

The key to the difference between psychology and educational psychology is that educational psychology is one sub-discipline of psychology. Psychology can simply be defined as the scientific study of the human mind and behaviour.

This is a discipline that encompasses a wide range of sub-disciplines such as abnormal psychology, social psychology, developmental psychology so on and so forth. Educational psychology is also one such sub-discipline which falls under the main discipline of psychology. Educational psychology pays specific attention to the study of learning throughout the human life span.

So the main difference between psychology and educational psychology stems from the fact that while psychology in general possesses a broader view which entails all aspects of human life, educational psychology pays specific attention to the learning process. The objective of this article is to provide an understanding of the two terms,

while emphasizing the difference that lies between the two terms, psychology and educational psychology.

What is Psychology?

For many years, humans have been fascinated by the capabilities of the human mind that led to the establishment of a discipline known as psychology. In this sense, it can be defined as the scientific study of the mental processes and behavioural patterns of human beings. The ability to examine the human mind and behaviour in experimental conditions began with the setting up of the first laboratory in Germany in 1879 by Wilhelm Wundt, who was later considered as the father of Psychology. Psychology is a discipline that has a very large scope. Even though at the initial phase medicine (biology) and philosophy provided the roots for psychology to grow as a field, now it has become a discipline which has expanded much more not only influencing other disciplines but also being influenced by them, which highlights that it is continuously changing and moving forward in the academic as well as scientific sphere. It studies the human development, personality, abnormalities, education, social interactions and almost all aspects of human life. When we learn about psychology, we also hear about schools of psychology. These refer to the different approaches that have been used when analyzing and examining human life over the years. Structuralism, Functionalism, Behaviourism, Psychoanalysis, Gestalt and Humanistic psychology are some of these schools of psychology.

What is Educational Psychology?

Educational psychology is a sub-discipline of psychology that specifically studies on human learning. It explores a variety of themes such as motivation, conditioning, memory, intelligence, cognition, etc.

Educational psychologists are interested in studying the learning processes of individuals in different settings, under different circumstances through their life span. They adopt cognitive as well as behavioural approaches in this field.

The theories which fall under the educational psychology discipline come from different schools such as behaviourism, Gestalt psychology, humanistic psychology and functionalism. Especially the behaviourist theories of classical conditioning brought by Ivan Pavlov and Operant conditioning by B.F Skinner are popular in educational psychology for their applicability to real life and education related processes. Not only Behaviourists, under different schools of psychology psychologists have presented various theories that denote an analysis and understanding of the human learning process.

What is the difference between Psychology and Educational Psychology?

- To sum up, psychology is mainly the study of the mental processes and behavioural patterns of human beings whereas educational psychology is the study of human learning process.

- This highlights that while educational psychology explores only the learning aspect of human life, psychology in general explores all human activities throughout the life span which goes beyond the learning process.

Human Growth and Development

The Human Growth and Development exam (infancy, childhood, adolescence, adulthood, and aging) covers material that is generally taught in a one-semester introductory course in developmental psychology or human development. An understanding of the major theories and research related to the broad categories of physical development, cognitive development, and social development is required, as is the ability to apply this knowledge.

The questions on the CLEP Human Growth and Development exam adhere to the terminology, criteria, and classifications referred to in the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders (DSM-5)*.

The exam contains approximately 90 questions to be answered in 90 minutes. Some of them are pretest questions that will not be scored. Any time test takers spend on tutorials and providing personal information is in addition to the actual testing time.

Knowledge and Skills Required

Questions on the Human Growth and Development exam require test takers to demonstrate one or more of the following abilities.

- Knowledge of basic facts and terminology
- Understanding of generally accepted concepts and principles
- Understanding of theories and recurrent developmental issues
- Applications of knowledge to particular problems or situations

The subject matter of the Human Growth and Development exam is drawn from the following categories. For each category, several key words and phrases identify topics with which candidates should be familiar. The percentages next to the main categories indicate the approximate percentage of exam questions on that topic.

Theoretical Perspectives (10%)

- Cognitive developmental
- Evolutionary
- Learning
- Psychodynamic
- Social cognitive
- Sociocultural
- Biological
- Ecological.

Research Strategies and Methodology (6%)

- Case study
- Correlational
- Cross-sectional
- Cross sequential
- Experimental
- Longitudinal
- Observational.

Biological Development Throughout the Life Span (12%)

- Development of the brain and nervous system
- Genetic disorders
- Heredity, genetics, and genetic testing
- Hormonal influences
- Influences of drugs
- Motor development
- Nutritional influences
- Perinatal influences
- Physical growth and maturation, aging
- Prenatal influences
- Sexual maturation
- Teratogens.

Perceptual Development Throughout the Life Span (6%)

- Habituation
- Sensitive periods
- Sensorimotor activities
- Sensory acuity
- Sensory deprivation.

Cognitive Development Throughout the Life Span (12%)

- Attention
- Environmental influences
- Executive function
- Expertise
- Information processing
- Memory
- Jean Piaget's cognitive development theory
- Play

- Problem solving and planning
- Thinking
- Lev Vygotsky's sociocultural theory
- Wisdom.

Language Development (8%)

- Bilingualism
- Development of syntax
- Environmental, cultural, and genetic influences
- Language and thought
- Pragmatics
- Semantic development
- Vocalization and sound.

Intelligence Throughout the Life Span (6%)

- Concepts of intelligence and creativity
- Developmental stability and change
- Giftedness
- Heredity and environment
- Intelligence tests
- Reaction range.

Social Development Throughout the Life Span (12%)

- Aggression
- Attachment
- Gender
- Interpersonal relationships
- Moral development
- Prosocial behaviour
- Risk and resilience
- Self
- Social cognition
- Social learning and modeling
- Wellness.

Family, Home, and Society Throughout the Life Span (8%)

- Abuse and neglect
- Bronfenbrenner, Urie
- Death and dying

- Family relationships
- Family structures
- Media and technology
- Multicultural perspectives
- Parenting styles
- Social and class influences.

Personality and Emotion (8%)

- Attribution styles
- Development of emotions
- Emotional expression and regulation
- Emotional intelligence
- Erikson, Erik
- Freud, Sigmund
- Psychosocial theory
- Stability and change
- Temperament.

Schooling, Work, and Interventions (6%)

- Applications of developmental principles
- Facilitation of role transitions
- Intervention programs and services
- Learning styles
- Occupational development
- Operant conditioning
- Preschool care, day care, and elder care
- Retirement.

Developmental Psychopathology (6%)

- Antisocial behaviour
- Anxiety and mood disorders
- Asocial behaviour, fears, phobias, and obsessions
- Attention-deficit/hyperactivity disorder
- Autism spectrum disorders
- Chronic illnesses and physical disabilities
- Cognitive disorders, including dementia
- Learning disabilities
- Intellectual Disability
- Trauma-based syndromes.

CONCEPT OF GROWTH AND DEVELOPMENT

Definition of Growth and Development

Growth is the progressive increase in the size of a child or parts of a child. Development is progressive acquisition of various skills (abilities) such as head support, speaking, learning, expressing the feelings and relating with other people. Growth and development go together but at different rates.

Importance of Assessing Growth and Development

The assessment of growth and development is very helpful in finding out the state of health and nutrition of a child. Continuous normal growth and development indicate a good state of health and nutrition of a child. Abnormal growth or growth failure is a symptom of disease. Hence, measurement of growth is an essential component of the physical examination.

Stages of Growth and Development in Children

Growth and development goes side by side, and both have profound effects upon each other. When a child is born, passes through various stages and ultimately a whole picture of a person emerges. The following are the stages of growth and development through which every children passes:

Infancy: (From Birth to 2 Years)

Common features of infancy are

1. A child jumps, walk and learn to talk.
2. A child starts learning about simple, social concepts.
3. The body growth is accelerated.

Early Childhood (2-6 Years)

The common features of this stage are:

1. Infancy features are strengthened.
2. Physical growth occurs like, expansion of muscles, speed in body actions, changes occur in respiration, blood pressure develops etc.
3. Language skill is developed and learns new words.
4. New concepts about social relations develop.
5. Memory increases.
6. They ask questions about the environment.
7. Socially he/she develops new friends, and wants social approval of his/her actions.
8. Emotional development starts to develop.

Adolescence (12-19 Years)

This is the period of adulthood and to some extent of maturity, physically and socially. This is called the spring, romantic, and the period of “storm & stress”. The development and growth vary from culture to culture and environment factors that influence a person at this stage. Some basic characteristics of this stage are as follows:

Physical Growth

Here children are active, and sharp. They are keen observers. They give more attention to body – beauty and strength. Boys love strength and play the games, where power is shown.

Mental or Intellectual Development

1. They think in abstraction.
2. They can anticipate the future needs and plan for that.
3. The children ability to solve complex problems increase. They try to understand complicated issues, and to solve the complex problems
4. They develop the communicative abilities and talk for hours.
5. Confidence is developed which make them to make decisions about future.
6. Moral concepts are developed and know that what is morally and socially good and wrong.

Emotional Development

Here the adolescent period is not rational but emotional. That's why it is called the period of stress and storm. They are entering a new stage, which therefore, want emotional adjustment. Emotional disturbance is produced when they don't adjust to new situation or role for which they are suppose to be.

Characteristics of Emotions in Adolescence

1. Complexity: They experience complexity in various issues and start adjustment. When that adjustment doesn't develop, storm and stress develop.
2. Development of Abstract Emotions: They develop emotions with those things which do not exist in real situation. Wants and desires are generated by abstract emotions.
3. Widening of Emotional Feelings: Here they select a hero/heroine. Idealization and imagination start. And get emotional attachment with that idealization.
4. Bearing Tension: They start to learn that how to bear a difficult emotional situations.
5. Sharing of Emotions: They share emotions, especially with peer groups. That develops loyalty and emotional confidence among peer groups.

6. Hopes & aspirations develop about the future.
7. This stage gives increase to compassion in them.

Common Emotional Patterns

The following are some emotional patterns which can be seen in students:

1. Worries/Anxieties: These can be imaginative or real. These can be the products of school work, exam, school problems or home problems.
2. Phobias: Phobias or fears can be from material objects, meeting with people, talking to strangers etc.
3. Anger, Love and Hate: These are some emotional patterns, which have its own causes, and effects.

As for emotions are concerned, the fear of failure make you to work hard. Emotions can be realized through good and positive activities. Now it is the responsibility of home, school, teachers & parents to understand the emotional state of the children and to avoid the bad effects of emotional disturbance. “Become emotions can make or mar one’s life”.

Difference between Growth and Development

Growth and development is one of the important studies for the teachers and parents. Growth is different from development. But both are correlated and one is dependant on other. We can say that growth is a part of development, which is limited in physical changes.

Growth is physical changes whereas development is overall development of the organism. The main differences between both are-

- (1) Growth is change of physical aspects of the organism. Development is overall changes and progressive changes of the organism.
- (2) Growth is cellular but development is organizational.
- (3) Growth is the change in shape, form, structure, size of the body. Development is structural change and functional progress of the body.
- (4) Growth stops at maturation but development continues till death of the organism.
- (5) Development also includes growth. Growth is a part of development.
- (6) Growth and development go side by side.
- (7) Growth and development is the joint product of heredity and environment.
- (8) Growth is quantitative and development is qualitative in nature.
- (9) Growth can be measured accurately but development is subjective interpretation of one’s change.

Both growth and development are interrelated aspects of psychology. There are some basic differences as per their structure but it is difficult to separate them. They have some basic similarities also. In study of educational psychology study of their differences and similarities have equal importance for a teacher.

Growth vs Development

The terms Growth and Development are used with every aspect of life. There might be some confusion when using the terms as they are often used interchangeably. Growth is just 'getting bigger', whereas development is improvement.

Growth can be explained as becoming bigger or larger or having more importance. Growth is termed as a physical change, where as development is said to be physical as well as social or psychological change. Development also means transformation or improvement. While growth is related to quantitative improvement, development is related to quantitative as well as qualitative improvement.

When the term growth is related to living beings, it can mean the increase in size. Here, the growth that comes over a living being is a physical change and it indicates the increase in weight, height and bone size.

On the other hand, development is the process of developing skills and capacities. It deals with the behavioural aspect of a living being. For example, we only say that the tumour has grown and not that the tumour has developed. On the other hand, we generally say that 'he has developed into a better man' or 'he developed into a better citizen' and so on.

In terms of economy, we say there was a steady growth in the country's economy over the last few years. We also say that there was a tremendous growth in the case of hospitals in the region. We also refer to countries as developed and developing, which means that the country has made great strides in all spheres.

Summary

1. Growth is just 'getting bigger', whereas development is improvement.
2. Growth can be explained as becoming bigger or larger or having more importance.
3. Growth is termed as a physical change, where as development is said to be physical as well as social or psychological change.
4. While growth is related to quantitative improvement, development is related to quantitative as well as qualitative improvement
5. When the term growth is related to living beings, it can mean the increase in weight, height and bone size. On the other hand, development is the process of developing skills and capacities. It deals with the behavioural aspect of a living being.
6. In terms of economy, we say there was a steady growth in the country's economy over the last few years. We also refer to countries as developed and developing, which means that the country has made great strides in all spheres.

Importance of Educational Psychology in Child Development

Early childhood is considered to be “an ideal period for learning.” Watson thinks, ‘the scope and intensity of learning during this period exceeds that of any other period of development.’ The following points should be kept in view by the educator while planning the education of the child at this stage.

PROPER ATMOSPHERE

A healthy, peaceful and secure atmosphere should be provided for the child. This is necessary both at home and the school.

PROPER TREATMENT

The child has to depend on others for the satisfaction of his requirements. Hence he should always receive an affectionate, sympathetic, and courteous treatment from others. An atmosphere of fear and repression will not allow the child to develop properly.

Developing the Endowments of the Child

A child takes birth with certain natural endowments. The educator should try to locate them and plan for their maximum development.

Satisfaction of Curiosity

Every child is curious to know the details of his social environment. The parents and the teachers should try to pacify this need of the child by answering the questions asked by him.

Developing Self-Sufficiency

The infant depends on others for the satisfaction of his needs. The child has to do everything himself. The educator should try to develop a spirit of self-sufficiency in the child and should provide him opportunities so that the child is able to do most of the things himself.

Developing a Social Sense

Efforts should be made by the teacher to develop a rational social séances in the child.

The Child starts developing a social sense by the end of this stage.

Encouraging the Instinctive Development of the Child

The child at this stage is guided by his instincts. The teacher should not try to suppress the instinctive development of the child.

It should be sublimated in the best interests of the personality development of the child.

Formation of Good Habits

This is considered to be the most flexible age of the child. Steps should be taken towards the formation of good habits by the child.

Learning by Doing

The child is active by nature. This ought to be satisfied through the use of activity method which advocates learning by doing.

Use of the Play-Way

Play is natural to the child. Let him play and learn. This will make learning easier and acceptable to the child.

Sensory Training

There are five senses of the child. These should be properly developed at this stage. This sensory training was also emphasized by Madam Montessori.

Use of Stories and Pictures

Children at this stage are interested in stories and multi-coloured pictures. They should be made to learn by using story telling method and with the help of multi-coloured pictures and charts.

LATER CHILDHOOD-(FROM 7 TO 12 YEARS)

Physical Development

The development is slow and uniform. The child at the age of 12 is about 55 inches in height, which means an annual increase of 2 to 3 inches. Boys are slightly taller than girls. Body proportions change considerably and the overall appearance of the child also alters. The nose becomes larger and the lower jaw also increases in size. Arms, legs and trunk increase in length and look thinner. This rather gives 'an ugly look' to the child. He also builds up fine muscular co-ordinations and improves many muscular and motor skills. Boys show superiority to girls and special interest in games and skills requiring physical strength, while girls are superior to boys in skills involving the use of finer muscles, such as drawing, painting, sewing, etc.

Emotional Development

The child now comes to have a control over his emotions. His emotional behaviour is guided by a rational expression. He is still jealous, may be of his younger brothers or his class-fellows. He may tease and make fun of those children who are the objects of his jealousy. He may be afraid of things but tries to look brave. His expression of joy is expressed by a smile. He satisfies his curiosity by asking questions from the teacher and by studying books.

Social Development

At this stage the child is engaged in social give and take. Sometimes he is teased and bullied by others and he also teases and bullies others. He attaches great value to friendship and takes part in games and excursions. Now he is fairly independent in his behaviour. He moves with his friends and comes to have a 'group' spirit. Sometimes he may assume the leadership of the group as well. Boys and girls make their separate groups. The children also come to have knowledge about their caste, religion bias and prejudices.

Mental Development

Children develop their mental power by studying in the schools, reading books, and by visiting places. They develop their reasoning power, memory, attention and sensory

discrimination during this period. Children also judge their own actions and criticize the actions of others and in this way they come to have their own opinions. In religious and racial matters they take up the opinions of their elders. They also take up the aesthetic standards of their elders. Their vocabulary increases by leaps and bounds. Now they are able to speak and write sentences of more than average length.

Moral Development

To him the moral code is now determined by the group to which she belongs. He considers that stealing is generally bad. He judges the actions of others accordingly. He has a strong sense of fair-play and justice. Sometimes conditions in the family play a very important role in determining these attitudes.

EDUCATIONAL IMPORTANCE OF LATER CHILDHOOD

This period of childhood is the time when the basic outlook, values and ideals of the child are finally shaped. Hence it is duty of the parents, teachers and the society to plan for their proper development. The School, however, has a very important role to play. The following points should be kept in view by the teacher.

PROPER PHYSICAL DEVELOPMENT

Well-organised evening games and sports should be provided for children so that they are able to have a proper physical development. The teacher should see that every child takes part in these games.

Morning physical training may also be organised for resident students. In the new educational pattern of 10+2+3, physical development is being highly emphasized by educators.

Provision for Excursions and Scouting

Children start developing surplus energy at this stage. They are interested in adventure and roaming about here and there. It is, therefore, necessary that suitable provision should be made for excursions and outings for them. The scouting should be organised in the school to satisfy this need of the child.

Proper Provision of Extra-Mural Activities

Hidden qualities have to be found out and properly developed. This is possible only if a large number of extra-mural activities are provided in the school this will enable the child to choose those activities in which he is interested. In the organisation of these activities democratic ways should be adopted.

Planning for Proper Social Development

Children develop the 'group spirit' at this stage. In order to direct it in the right channels it is necessary that group games on a group competition basis should be arranged. It is a period of competitive socialisation. It is through these group competitions that the child will develop various qualities like discipline, self-control, sympathy and co-operation.

Developing the Creative Instinct

Children want to create something new. This creative trend should be properly utilised for educational ends. The teacher should provide opportunities where children are in a position to satisfy their creative instinct.

Proper Emotional Outlet

This stage, however, is considered to be a unique stage in the emotional development of the child. It is possible only if proper outlets for the emotional expressions of children are provided. The teacher should not try to suppress these outlets. He should try to redirect them in the right channels. He should place examples of great men of the world before them.

Satisfying the Acquisitive Instinct

Children want to acquire those things which they like. If it is not properly directed they start stealing they should, therefore, be encouraged to collect those things which have an educational value.

Moral Training

The child at the age of eight years starts accepting the moral values of the society in which he lives. Therefore, it is necessary that some moral training should be provided. Instructions may be provided which have a moral value for the child. A democratic approach should be followed as morality is something which springs from the heart of the child.

Learning Through Self-Activity

Children like activity. It comes to occupy a more important place in the educational system when it originates from the children themselves. Self-activity leads to self-expression, thus leading to self-realisation.

Teacher's Role in Academic Development

The teacher has a very important role to play in the mental development of the child. *His role may be in the following directions:*

- (a) Children are interested in language learning. They may learn two or three languages at this stage.

- (b) Children should be encouraged to choose those subjects which have a functional value. Preference should be given to those subjects who are useful in life.
- (c) Interests of children differ and change. Hence, it is necessary that their text-books should have different types of lessons. They may include adventurous stories, biographies of great men, drama, dialogue, etc.
- (d) It is the duty of the teacher to make necessary changes in the curriculum and methods of teaching.

ADOLESCENCE STAGE (PERIOD FROM TO / YEARS)

The term 'Adolescence' is derived from the Latin word 'Adolescence' which means 'to grow to maturity?' This is a period lying between childhood and maturity. Thus, it is a period of change, when the individual is neither a child nor an adult. The changes which take place are physical, sexual, emotional, social, mental and moral. A harmonious development of adolescence means full adult height, weight, stature and strength, and complete sexual maturity; the proper adjustment of the adolescent in the society; he also takes up his duties and responsibilities; the complete development of his mental powers and emotional stability and economic independence and professional competence.

PHYSICAL DEVELOPMENT

HEIGHT AND WEIGHT

Growth is rapid just before and during the early years of adolescence. Both boys and girls develop their height and weight.

Bodily Proportion

Different parts of the body grow at different rates and reach their final size. Arms and legs grow longer and hands and feet grow much bigger.

Sexual Development

This stage is the development of sex maturity. The pituitary glands secrete 'gonadotropic' hormone which finally stops bodily growth and leads to sex maturity. The sex glands of male produce spermatozoa which lead to the development of sexual growth. In the same way, female sex glands are called the ovaries which produce 'Ova' sex hormones.

Emotional Development

This is the period of increased and charged emotions as the instincts of the child attain maturity. His feeling also undergoes a great change. He is sensitive and easily

offended. He feels he is all alone, unloved, and in a world which is hostile. He is required himself to adjust to the new situations and behave as the new adult. He may face emotional difficulties. He is not able to control his emotions and this leads to temper outbursts. Sometimes he is happy and sometimes he is sad, sometimes he also suffers from emotional moods. He may also indulge in day-dreaming. Love is also predominant feeling in this period. This is due to the development of sex instinct. This love may be of most selfless kind. Adolescents are prepared to sacrifice everything simply to gain the love of another child. In the later period of adolescence this is directed to the opposite sex. The adolescents also direct love towards great men, which is known as hero worship.

Social Development

At this stage the adolescent is self conscious and lacks confidence and is very modest. He wants seclusion. All these things disappear as he develops. He starts choosing his own friends and chooses them on the qualities they possess. Gradually he forms small group. He takes part in various social groups as a member of these groups. Thus he gains experiences which are valuable to him in adult life. In this period adolescents also come in conflict with parental demands. They demand independence of action whereas their parents treat them as children and hence they come in conflict with them.

Moral Development

At this stage the child develops his critical ability and judgement. He also comes across various individuals and thus he does not accept religious principles without criticism. He becomes conscious of the idea of right and wrong.

Mental Development

At this stage the span of attention increases. Memory, reasoning and judgement also increase. He understands and appreciates poetry, music, art, literature and mathematics.

ADULTHOOD STAGE (PERIOD AFTER YEARS OF AGE)

Adulthood period is the longest of all the periods or stages of life. It is characterised by an all round maturity physical, intellectual, emotional and social. At this stage the individual develops into a functioning member of the society and acts according to social and moral standards. He is self-reliant and accepts responsibility for his decisions and actions. He develops ability to think rationally and intelligently, ability to understand the world and to make maximum adjustments.

OLD STAGE (PERIOD AFTER YEARS OF AGE)

This stage starts from 58 years till death. At this stage one spends one's most of time in worshipping social service and excursions.

Piagets Theory of Cognitive Development

The **Theory of Cognitive Development**, one of the most historically influential theories was developed by Jean Piaget, a Swiss psychologist (1896–1980). His theory provided many central concepts in the field of developmental psychology and concerned the growth of intelligence, which for Piaget, meant the ability to more accurately represent the world and perform logical operations on representations of concepts grounded in the world. The theory concerns the emergence and acquisition of schemata—schemes of how one perceives the world—in “developmental stages”, times when children are acquiring new ways of mentally representing information. The theory is considered “constructivist”, meaning that, unlike nativist theories (which describe cognitive development as the unfolding of innate knowledge and abilities) or empiricist theories (which describe cognitive development as the gradual acquisition of knowledge through experience), it asserts that we construct our cognitive abilities through self-motivated action in the world. For his development of the theory, Piaget was awarded the Erasmus Prize. Piaget divided schemes that children use to understand the world through four main periods, roughly correlated with and becoming increasingly sophisticated with age:

- Sensorimotor period (years 0–2)
- Preoperational period (years 2–7)
- Concrete operational period (years 7–11)
- Formal operational period (years 11–adulthood).

Sensorimotor Period

According to Piaget, this child is in the *sensor-motor period* and primarily explores the world with senses rather than through mental operations. Infants are born with a set of congenital reflexes, according to Piaget, in addition to a drive to explore their world. Their initial schemes are formed through differentiation of the congenital reflexes.

The ***sensorimotor period*** is the first of the four periods. According to Piaget, this stage marks the development of essential spatial abilities and understanding of the world in six sub-stages:

1. The first substage, known as *the reflex schema* stage, occurs from birth to six weeks and is associated primarily with the development of reflexes.
2. The second substage, *primary circular reaction phase*, occurs from six weeks to four months and is associated primarily with the development of habits.
3. The third substage, the *secondary circular reactions phase*, occurs from four to nine months and is associated primarily with the development of coordination between vision and prehension.
4. The fourth substage, called *the coordination of secondary circular reactions stage*, which occurs from nine to twelve months, is when Piaget (1954) thought that object permanence developed.

5. The fifth substage, *the tertiary circular reactions phase*, occurs from twelve to eighteen months and is associated primarily with the discovery of new means to meet goals.
6. The sixth substage, considered “beginnings of symbolic representation”, is associated primarily with the beginnings of insight, or true creativity.

Preoperational Stage

The Preoperational stage is the second of four stages of cognitive development. By observing sequences of play, Piaget was able to demonstrate that towards the end of the second year a qualitatively new kind of psychological functioning occurs. (Pre) Operatory Thought in Piagetian theory is any procedure for mentally acting on objects. The hallmark of the preoperational stage is sparse and logically inadequate mental operations. During this stage the child learns to use and to represent objects by images and words, in other words they learn to use symbolic thinking. Thinking is still egocentric: The child has difficulty taking the viewpoint of others.

The child can classify objects by a single feature: e.g. groups together all the red blocks regardless of shape or all the square blocks regardless of colour. According to Piaget, the Pre-Operational stage of development follows the Sensorimotor stage and occurs between 2–7 years of age. In this stage, children develop their language skills. They begin representing things with words and images. However, they still use intuitive rather than logical reasoning. At the beginning of this stage, they tend to be egocentric, that is, they are not aware that other people do not think, know and perceive the same as them. Children have highly imaginative minds at this time and actually assign emotions to inanimate objects. The theory of mind is also critical to this stage.

The Preoperational Stage can be further broken down into the Preconceptual Stage and the Intuitive. The Preconceptual stage (2-4 years) is marked by egocentric thinking and animistic thought. A child who displays animistic thought tends to assign living attributes to inanimate objects, for example that a glass would feel pain if it were broken.

The Intuitive (4-7 years) stage is when children start employing mental activities to solve problems and obtain goals but they are unaware of how they came to their conclusions. For example a child is shown 7 dogs and 3 cats and asked if there are more dogs than cats. The child would respond positively. However when asked if there are more dogs than animals the child would once again respond positively. Such fundamental errors in logic show the transition between intuitiveness in solving problems and true logical reasoning acquired in later years. (Child Psychology, 1st Canadian Edition, Hetherington, Parke & Schmuckler).

Concrete Operational Stage

The Concrete operational stage is the third of four stages of cognitive development in Piaget's theory. This stage, which follows the Preoperational stage, occurs between

the ages of 6 and 12 years and is characterized by the appropriate use of logic. Important processes during this stage are:

Seriation—the ability to sort objects in an order according to size, shape, or any other characteristic. For example, if given different-shaded objects they may make a colour gradient.

Classification—the ability to name and identify sets of objects according to appearance, size or other characteristic, including the idea that one set of objects can include another. A child is no longer subject to the illogical limitations of animism (the belief that all objects are alive and therefore have feelings).

Decentering—where the child takes into account multiple aspects of a problem to solve it. For example, the child will no longer perceive an exceptionally wide but short cup to contain less than a normally-wide, taller cup.

Reversibility—where the child understands that numbers or objects can be changed, then returned to their original state. For this reason, a child will be able to rapidly determine that if $4+4$ equals 8, $8-4$ will equal 4, the original quantity.

Conservation—understanding that quantity, length or number of items is unrelated to the arrangement or appearance of the object or items. For instance, when a child is presented with two equally-sized, full cups they will be able to discern that if water is transferred to a pitcher it will conserve the quantity and be equal to the other filled cup.

Elimination of Egocentrism—the ability to view things from another's perspective (even if they think incorrectly). For instance, show a child a comic in which Jane puts a doll under a box, leaves the room, and then Melissa moves the doll to a drawer, and Jane comes back. A child in the concrete operations stage will say that Jane will still think it's under the box even though the child knows it is in the drawer.

Formal Operational Stage

The formal operational period is the fourth and final of the periods of cognitive development in Piaget's theory. This stage, which follows the Concrete Operational stage, commences at around 11 years of age (puberty) and continues into adulthood. It is characterized by acquisition of the ability to think abstractly, reason logically and draw conclusions from the information available.

During this stage the young adult is able to understand such things as love, "shades of gray", logical proofs, and values. Lucidly, biological factors may be traced to this stage as it occurs during puberty (the time at which another period of neural pruning occurs), marking the entry to adulthood in Physiology, cognition, moral judgment (Kohlberg), Psychosexual development (Freud), and social development (Erikson). Some two-thirds of people do not develop this form of reasoning fully enough that it becomes their normal mode for cognition, and so they remain, even as adults, concrete operational thinkers.

General Information Regarding the Stages

These four stages have been found to have the following characteristics:

- Although the timing may vary, the sequence of the stages does not.
- Universal (not culturally specific)
- Generalizable: the representational and logical operations available to the child should extend to all kinds of concepts and content knowledge
- Stages are logically organized wholes
- Hierarchical nature of stage sequences (each successive stage incorporates elements of previous stages, but is more differentiated and integrated)
- Stages represent qualitative differences in modes of thinking, not merely quantitative differences

Challenges to Piagetian Stage Theory

Piagetians accounts of development have been challenged on several grounds. First, as Piaget himself noted, development does not always progress in the smooth manner his theory seems to predict. 'Decalage', or unpredicted gaps in the developmental progression, suggest that the stage model is at best a useful approximation.

More broadly, Piaget's theory is 'domain general', predicting that cognitive maturation occurs concurrently across different domains of knowledge (such as mathematics, logic, understanding of physics, of language, etc.). However, more recent cognitive developmentalists have been much influenced by trends in cognitive science away from domain generality and towards domain specificity or modularity of mind, under which different cognitive faculties may be largely independent of one another and thus develop according to quite different time-tables.

In this vein, many current cognitive developmentalists argue that rather than being domain general learners, children come equipped with domain specific theories, sometimes referred to as 'core knowledge', which allows them to break into learning within that domain.

For example, even young infants appear to understand some basic principles of physics (e.g. that one object cannot pass through another) and human intention (e.g. that a hand repeatedly reaching for an object has that object, not just a particular path of motion, as its goal). These basic assumptions may be the building block out of which more elaborate knowledge is constructed.

Post Piagetian and Neo-Piagetian Stages

There are three major changes to the number of stages and their definitions. First and foremost, the half stages are now shown to be stages. Pascual Leone discovered this. Almost all Post Piagetians accept this. Second, postformal stages have been shown to exist. Kurt Fischer suggested two, Michael Commons presents evidence for four postformal stages: the systematic, metasytematic, paradigmatic and cross paradigmatic.

Fischer has considered a stage suggested by Biggs and Biggs. It is a stage before the early preoperational. Commons and Richards call this stage the sentential because organisms can sequence representations of concepts.

Learning Theory as Cognitive Development

Traditionally, learning theory has focused on forms of human change that were by definition somewhat peripheral to human development. Learning theory was about relatively rapid associations between stimuli or between responses and reinforcements. When stimuli, responses and consequences were all simple enough and when conditions were properly controlled, the fast-paced (by developmental standards) stimuli and responses displayed lawful properties and allowed for testable predictions about (short-term) behaviour change. The resulting changes were called “learning.” Connections between stimuli were named associative learning or “classical” conditioning, and those between a response and its consequence were named operant conditioning. Those between a response and consequence that were merely observed by an individual were called Modelling, observational learning or “vicarious conditioning.”

Assessed in terms of developmental theory and of SBTL, the problem with behaviourally oriented learning theory had three blind spots: it did not say enough about long-term change, about the experience of learning, or about how learning and behaviour became organized in the minds and lives of learners. Observational learning theory was an early effort to remedy the last two of the three problems (Bandura, 1977), and represented an initial turn toward what SBTL theorists eventually called meaning-making. In observational learning, a person somehow “sees” relationships between a relatively complex behaviour and its consequences. A student does not merely see a classmate elevate an arm and vocalize, for example, but sees the classmate “answer a teacher’s question.” If the teacher then tightens facial muscles, the student sees not merely muscular action, but sees the teacher “smile in response to the student.” In observational learning theory, observed concrete behaviours become expressions of meanings perceived as happening in others (Bandura, 1995). This approach is still far from the thorny issues of interpretation discussed by social constructivists, but it is nonetheless a nod toward students’ lived experience and personally constructed meanings.

Recent versions of learning theory have moved even more toward meaning-making than did Bandura and his colleagues. In one relatively radical formulation of learning theory, for example, conditioning has been interpreted as not about behaviour change per se, but about making meaning (DeGrandpre, 2000). In this view, linkages among overt stimulus cues, responses and consequences are merely cases of this broader process. Operant conditioning is not merely about reinforcing or strengthening specific responses as a result of specific consequences, but about creating perceived relationships among responses, the stimuli prior to responses, and the consequences that follow. “Reinforcement” of a behaviour is therefore not simply the consequences of a behaviour,

as in earlier versions of learning theory, but the relationship among all elements of a learning trial-stimuli, responses, and consequences. If a teacher smiles pleasantly at a young child's interesting comment during a circle-time discussion, for example, what is reinforced is not just the child's tendency to make verbal comments in the future. What is reinforced is actually the child's tendency to associate or connect the teacher's smile, the child's own verbal contributions, and the setting or conditions when doing so is appropriate. To say that "the teacher reinforces the child with her smile," in fact is misleading because it implies more control on the part of the teacher than the teacher really has. She does contribute to the reinforcement-relationship by smiling, but so do the classroom circumstances and the child's own behaviour. All must happen together, and together they constitute reinforcement-and the creation of meaning.

Learning theory has also become more constructivist by developing more complex models of how children learn, and therefore more able to explain concepts and skills that truly matter in the lives of children and caregivers. An example is the "overlapping waves theory" proposed recently by Robert Siegler (1996). This theory makes three assumptions: 1) that children normally use a variety of strategies for solving problems, 2) that strategies can co-exist for long periods even when some are suboptimal or incompatible, and 3) that experience changes the relative salience of particular strategies, rather than eliminating any outright.

Changes in salience alter the likelihood of particular strategies actually appearing as behaviour. When graphed across time, relative salience rises and falls like a series of overlapping waves; hence the name of the model. Learning theory-or at least Siegler's overlapping waves version of learning theory-is about studying these changes in salience and their eventual expression as altered behaviour. The result, compared to older models of learning, are predictions about learning that are more realistic. In actual classroom practice, a child is indeed likely to try a variety of strategies in ways that are often somewhat trial-and-error. A child may work two-digit subtraction problems, for example, by using "borrowing" algorithms officially recommended by the teacher, but also persist in counting on fingers some of the time, and at other times also try personalized (often bug-ridden) one-digit procedures that work only with particular subsets of problems.

Even when a child gains skill with a new strategy-in this case the "borrowing" algorithm-he or she still often regresses to earlier methods now and then, just as adults do. The child is sometimes also likely to spend time perfecting existing, suboptimal approaches rather than trying new ones-learning to count on fingers faster or more surreptitiously, for example-since doing so usually has short-term payoff. New, more optimal strategies appear both suddenly and intermittently when and if they finally do.

The relative importance of a child's alternative subtraction strategies will gradually change with practice, but suboptimal ones will not totally disappear for quite a while, if ever. The results are patterns of behaviour that resemble Siegler's "overlapping waves" model of learning (Siegler, 2000). The educational value of a more meaning-oriented version of learning theory can be seen in the recent professional writing of early educators.

Discussions of behaviour problems in young children, for example, still draw heavily on concepts of applied behaviour analysis. Teachers are often still urged to frame behaviour problems in terms of operant conditioning (or as antecedents, behaviour, and consequences, which amount to the same idea), in order to influence a child's unwanted behaviour. But though this literature is quite behaviourist, it often is also concerned with the meaning of antecedents, behaviour, and consequences for the child: with the relationships she perceives among these elements, and with how a teacher might influence the perceptions in desirable directions. One result is that teachers are urged to alter not just consequences, but also the connections (or lack thereof) among antecedents, behaviour, and consequences.

A temper tantrum may be highly disruptive, for example, but the solution may not be simply to avoid paying attention to it (withholding reinforcement in the narrow sense). It may also be necessary to avoid connecting antecedents to the tantrum behaviours—avoid the triggers for tantrums in the first place. Not only is the broader strategy more likely to alter behaviour, it is also more likely to help a child “see” his tantrum in new, more acceptable ways—to alter, that is, its personal meaning. Much of the literature on behaviour management also implies a learning theory of “overlapping waves.” In discussing how to deal with a tantrum, for example, teachers are reminded that relapses are likely to occur, and that teachers should persevere with altering relationships among antecedents, responses, and consequences in spite of them. Expecting relapses implies that a child is guided by competing responses that are gradually shifting in relative importance or salience. Helping a child respond to frustration in consistently desirable ways is really a matter of developing a high ratio of desirable responses to undesirable ones, and thereby also making the desirable responses highly “meaningful” to the child in the sense of strengthening their connections to antecedents and consequences.

In addition to becoming more compatible with early childhood practice, such revisions to learning theory have had significant influences both on learning theory itself and on traditional meanings of human development, including cognitive development. The ideas about meaningful reinforcement described earlier bring learning theory closer to SBTLs in that it becomes more concerned than before about how learning is experienced by the child and how the child can take responsibility for “constructing” new, learned behaviours. The shifting, overlapping strategies described by Siegler, on the other hand, blur the conventional distinction between short-and long-term change which has been central to the idea of cognitive development. At the heart of the model is a process that is continuous, yet never complete, and in this sense it is neither short-nor long-term, but always both and always lifelong.

Mingling the short-and long-term is consistent with the professional needs of teachers and others with direct responsibilities for children. As professionals, we are fundamentally concerned with connecting the past and the future to a child's present: with helping children to relate previous experience and future intentions to current skills and plans.

Kohlberg's Stages of Moral Development

Lawrence Kohlberg's stages of moral development constitute an adaptation of a psychological theory originally conceived of by the Swiss psychologist Jean Piaget. Kohlberg, while a psychology postgraduate student at the University of Chicago expanded and developed throughout the course of his life. The theory holds that moral reasoning, the basis for ethical behaviour, has six identifiable developmental stages, each more adequate at responding to moral dilemmas than its predecessor. Kohlberg followed the development of moral judgment far beyond the ages studied earlier by Piaget, who also claimed that logic and morality develop through constructive stages.

Expanding on Piaget's work, Kohlberg determined that the process of moral development was principally concerned with justice, and that it continued throughout the individual's lifetime, a notion that spawned dialogue on the philosophical implications of such research. Kohlberg relied for his studies on stories such as the Heinz dilemma, and was interested in how individuals would justify their actions if placed in similar moral dilemmas. He then analyzed the form of moral reasoning displayed, rather than its conclusion, and classified it as belonging to one of six distinct stages.

There have been critiques of the theory from several perspectives. Some argue that it emphasizes justice to the exclusion of other moral values, such as caring; or that there is such an overlap between stages that they should more properly be regarded as separate domains; or that evaluations of the reasons for moral choices are mostly *post hoc* rationalizations (by both decision makers and psychologists studying them) of essentially intuitive decisions. Nevertheless, an entirely new field within psychology was created as a result of Kohlberg's theory, and according to Haggbloom et al.'s study of the most eminent psychologists of the 20th century, Kohlberg was the 16th most frequently cited psychologist in introductory psychology textbooks throughout the century, as well as the 30th most eminent overall.

Stages

Kohlberg's six stages can be more generally grouped into three levels of two stages each: pre-conventional, conventional and post-conventional. Following Piaget's constructivist requirements for a stage model, as described in his theory of cognitive development, it is extremely rare to regress backward in stages—to lose the use of higher stage abilities.

Stages cannot be skipped; each provides a new and necessary perspective, more comprehensive and differentiated than its predecessors but integrated with them.

Level 1 (Pre-Conventional)

1. Obedience and punishment orientation
(*How can I avoid punishment?*)

2. Self-interest orientation
(*What's in it for me?*)
Level 2 (*Conventional*)
3. Interpersonal accord and conformity
(*Social norms*)
(*The good boy/good girl attitude*)
4. Authority and social-order maintaining orientation
(*Law and order morality*)
Level 3 (*Post-Conventional*)
5. Social contract orientation
6. Universal ethical principles
(*Principled conscience*).

Pre-Conventional

The pre-conventional level of moral reasoning is especially common in children, although adults can also exhibit this level of reasoning. Reasoners at this level judge the morality of an action by its direct consequences. The pre-conventional level consists of the first and second stages of moral development, and is solely concerned with the self in an egocentric manner. In Stage one (obedience and punishment driven), individuals focus on the direct consequences of their actions on themselves. For example, an action is perceived as morally wrong if the perpetrator is punished. "The last time I did that I got spanked so I will not do it again." The worse the punishment for the act is, the more "bad" the act is perceived to be. This can give rise to an inference that even innocent victims are guilty in proportion to their suffering. It is "egocentric", lacking recognition that others' points of view are different from one's own. There is "deference to superior power or prestige".

Stage two (self-interest driven) espouses the "what's in it for me" position, in which right behaviour is defined by whatever is in the individual's best interest.

Stage two reasoning shows a limited interest in the needs of others, but only to a point where it might further the individual's own interests. As a result, concern for others is not based on loyalty or intrinsic respect, but rather a "you scratch my back, and I'll scratch yours" mentality. The lack of a societal perspective in the pre-conventional level is quite different from the social contract (stage five), as all actions have the purpose of serving the individual's own needs or interests. For the stage two theorist, the world's perspective is often seen as morally relative.

Conventional

The conventional level of moral reasoning is typical of adolescents and adults. Those who reason in a conventional way judge the morality of actions by comparing them to society's views and expectations. The conventional level consists of the third and fourth

stages of moral development. In Stage three (interpersonal accord and conformity driven), the self enters society by filling social roles. Individuals are receptive to approval or disapproval from others as it reflects society's accordance with the perceived role.

They try to be a "good boy" or "good girl" to live up to these expectations, having learned that there is inherent value in doing so. Stage three reasoning may judge the morality of an action by evaluating its consequences in terms of a person's relationships, which now begin to include things like respect, gratitude and the "golden rule".

"I want to be liked and thought well of; apparently, not being naughty makes people like me." Desire to maintain rules and authority exists only to further support these social roles. The intentions of actions play a more significant role in reasoning at this stage; "they mean well ...".

In Stage four (authority and social order obedience driven), it is important to obey laws, dictums and social conventions because of their importance in maintaining a functioning society. Moral reasoning in stage four is thus beyond the need for individual approval exhibited in stage three; society must learn to transcend individual needs.

A central ideal or ideals often prescribe what is right and wrong, such as in the case of fundamentalism. If one person violates a law, perhaps everyone would—thus there is an obligation and a duty to uphold laws and rules. When someone does violate a law, it is morally wrong; culpability is thus a significant factor in this stage as it separates the bad domains from the good ones. Most active members of society remain at stage four, where morality is still predominantly dictated by an outside force.

Post-Conventional

The post-conventional level, also known as the principled level, consists of stages five and six of moral development. There is a growing realization that individuals are separate entities from society, and that the individual's own perspective may take precedence over society's view. Because of this level's "nature of self before others", the behaviour of post-conventional individuals, especially those at stage 6, can be confused with that of those at the pre-conventional level.

In Stage five (social contract driven), individuals are viewed as holding different opinions and values. Similarly, laws are regarded as social contracts rather than rigid dictums. Those which do not promote the general welfare should be changed when necessary to meet "the greatest good for the greatest number of people". This is achieved through majority decision, and inevitable compromise. Thus democratic government is ostensibly based on stage five reasoning.

In Stage six (universal ethical principles driven), moral reasoning is based on abstract reasoning using universal ethical principles. Laws are valid only insofar as they are grounded in justice, and a commitment to justice carries with it an obligation to disobey unjust laws. Rights are unnecessary, as social contracts are not essential for deontic moral action. Decisions are not reached hypothetically in a conditional way but rather categorically in an absolute way, as in the philosophy of Immanuel Kant. This involves

an individual imagining what they would do in another's shoes, if they believed what that other person imagines to be true.

The resulting consensus is the action taken. In this way action is never a means but always an end in itself; the individual acts because it is right, and not because it is instrumental, expected, legal, or previously agreed upon. Although Kohlberg insisted that stage six exists, he found it difficult to identify individuals who consistently operated at that level.

Psychological Approaches to Learning for Educational Practice

Psychology is the branch of science that is concerned with the study of behaviour and experience. It is one of the largest and most ubiquitous of all the sciences, and, given its concern with human behaviour and methods of changing same, has had an enormous influence on educational thinking. Although psychology is undoubtedly a well-established and universally-recognised scientific discipline, it is still in many ways an ‘immature science’, in the sense of the term used by the historian and philosopher of science Thomas S. Kuhn. In a seminal book published in 1962, Kuhn presented the thesis that a branch of science only reaches true maturity when it acquires a paradigm - a generally-accepted underlying model, set of beliefs and methodology. By Kuhn’s criterion, psychology has certainly not yet attained such a status, since it is still characterised by several different approaches, or schools, none of which has gained universal acceptance among practitioners. The most important of these are psychoanalysis, behaviourism, humanistic psychology, the neurobiological approach and cognitive psychology. Let us now take a look at each of these approaches, and see how they have influenced our ideas about the nature of the learning process.

PSYCHOANALYSIS

Psychoanalysis was one of the first schools of psychological thought to develop, and is still what many people think of when the word ‘psychology’ is mentioned, largely due to

the influence of its founder, Sigmund Freud. Freud worked in Vienna at the turn of the century, and developed the theory that the individual human being lives in a continuous state of internal conflict due to demands made by different parts of the personality, particularly at a subconscious level. He identified three basic components of personality, the ego, superego and id. Of these, the ego can be thought of as the rational, conscious part of the mind, whereas the superego and id are conflicting aspects of the subconscious mind. According to Freud, the superego is a primeval, instinctive force which represents a drive towards the individual's ideal self. The id is a similar primeval force which expresses itself either through the love instinct (*eros*) and the internal drive known as the *libido*, or through the destructive or death instinct (*thanatos*). According to Freud, the well-adjusted individual uses the ego to maintain a reasonable balance between the conflicting demands of the superego and id, only encountering problems when the two drives become seriously out of balance. Psychoanalysis involves trying to resolve such internal conflicts by looking for clues as to what is happening within the individual's subconscious mind.

The ideas of Freud were developed by a number of 'post-Freudians' such as Carl Jung, Erik Erikson and Alfred Adler, who modified and extended his ideas and founded their own 'schools' of psychoanalysis. Although many modern psychologists disagree profoundly with several of the ideas put forward by Freud and his successors, the psychoanalytic approach has had an enormous influence both on psychology and on society at large. It has not, however, had very much influence on mainstream educational thinking or the theory of learning, and is not an approach that teachers and tutors are advised to have anything to do with. Without proper training in psychoanalysis, any attempt to 'play the therapist' with one's students could be extremely dangerous, and could cause permanent psychological damage. In other words, psychoanalysis should be left to the experts. If a student does start to become introspective during a tutorial or counselling session, suggest a break or use the appropriate referral system.

BEHAVIOURISM

Origins of behaviourism lie in the Russian psychologist, Pavlov's, work with dogs. At about the same time as Freud was developing his basic ideas of psychoanalysis in Vienna, an American psychologist, John Watson, was laying the foundations for the development of Pavlov's and Freud's work into what was to become a whole new - and highly influential - school of psychology. Watson argued that the Freudian approach based on self-observation (introspection) had yielded poor results, and that if psychology was to be considered to be a true science, its data would have to be both observable and measurable, like Pavlov's. His approach, which was later to become known as 'behaviourism', thus adopted the systematic study of observable behaviour as its focus. Behaviourists argue that nearly all behaviour is learned, and that the main function of psychology should be to seek to discover what the basic laws of learning are. Largely because of the work of the later American behavioural psychologist B.F. Skinner during the 1950's, behaviourism has had a tremendous influence on educational thinking.

Behavioural psychological theory is based on what is commonly referred to as stimulus/response or S-R learning. It assumes that learning has occurred if a specific response is elicited from a learner when he or she is placed in a particular situation and is given a particular stimulus. Learning of relatively complex behaviour can (it is claimed) be achieved through an appropriate series of stimulus-response situations. At each stage, the learner must actively participate by performing a set task, after which he or she is then supplied with immediate feedback in the form of the correct answer. This is known as successive reinforcement. Skinner also argued that each successive stimulus-response step should be small enough to ensure that the learner is almost always correct in their response. Use of these small steps, plus successive reinforcement, led to what behavioural psychologists believed was an efficient way of 'shaping behaviour'. Skinner's original work was with animals, mostly with pigeons. His later work, which evolved from this, was with humans, and was largely responsible for triggering the bandwagon programmed learning movement that so dominated progressive educational thinking during the 1960's and early 1970's. This, in turn, led to more recent developments such as open learning, distance learning, computer-based learning and multimedia. Although Skinner's original behavioural model of learning has since been rejected or at least greatly modified by many educational psychologists, he has probably had a greater influence on educational thinking than any other psychologist. Indeed, every teacher who makes use of individualised learning methods today owes him a very real debt.

HUMANISTIC PSYCHOLOGY

Humanistic psychology emerged during the 1950's and 1960's out of a reaction against the two schools that had dominated psychology up till then - psychoanalysis and behaviourism. Humanistic psychologists such as Abraham Maslow and Carl Rogers rejected the psychoanalytical approach as regarding human beings as being little better than animals, driven and controlled by a bundle of unconscious, instinctive forces. They similarly rejected the behaviourist approach as regarding people as nothing more than unthinking products of their environment, shaped and programmed by the patterns of rewards and punishments that they receive in the course of their lives. Humanistic psychologists rejected these two models of man in favour of one which emphasized the uniqueness and essential 'humanness' of every individual person, and concerned themselves mainly with concepts such as self-fulfilment and actualisation, the importance of subjective experience, and the development of human values. As an ex-Freudian, Rogers developed a new, 'client-centred therapy' in which he aimed to give his patients the self-knowledge and skills needed to find their own solutions to their problems rather than simply telling them what to do.

Humanistic psychology has had a considerable influence on progressive educational thinking since the early 1970's, when the early research findings of people like Rogers started to feed back into educational development. Just as the earlier work of Skinner led to the programmed learning movement, so the work of Rogers led to the modern

student-centred learning approach. In this, the teacher is no longer seen as an expert who hands down knowledge and understanding to the student. Rather, the teacher facilitates learning, first providing the student with guidance on how to learn and then providing a variety of learning opportunities and experiences through which such learning can occur. One of the most important manifestations of the student-centred learning approach has been the dramatic increase in the use of group learning since the mid-1970's, and the more recent appearance of flexible learning and competence-based learning. This last attempts to link student-centred learning with behaviourist reinforcement.

THE NEUROBIOLOGICAL APPROACH

This essentially physiological approach to psychology attempts to relate human behaviour to electrical and chemical activities taking place in the brain and central nervous system. Much of what has been discovered via this approach has proved extremely useful in improving the effectiveness and efficiency of human learning. Increased knowledge about how the eyes, ears and other sensory organs work and pass information to the brain, for example, has led to a greater understanding of how to attract and maintain a learner's attention. Similarly, increased knowledge about how sensory information and other types of information is actually processed in the brain is leading to a greater understanding of such things as the nature of memory.

Ultimately, it may be possible to explain complex human behaviour purely in terms of neurobiological activities, but, at present, such a prospect seems highly remote. In the meantime, the neurobiological approach is proving extremely useful in providing basic data for the fifth and last major branch of psychology - cognitive psychology.

COGNITIVE PSYCHOLOGY

The early behavioural psychologists treated the human mind as a 'black box', being concerned only with the relationship between the input to the system (the stimulus) and the output from the system (the response). None of the processes that occurred between the stimulus and the response were considered to be the legitimate concern of psychology, since they could not be directly observed. Dissatisfaction with this strict view, together with an increasing realisation that internal higher-order mental processes are an important part of psychology, has since led to the development of cognitive psychology. This is concerned with what actually happens within the mind/brain system when we think, reason, remember, develop language skills, etc., as well as with how we process information received from the outside world. Cognitive psychology is currently having an increasing influence on educational thinking, especially through the burgeoning field of artificial intelligence and the design of expert systems.

Intelligence. A concept which does not fit neatly into any of the above perspectives of psychology but which has profound implications for educational practice is the idea

of intelligence, whereby verbal, numerical, reasoning and spatial skills can be measured to give a quotient of a person's ability. Intelligence Quotient (IQ) is supposedly normally distributed amongst the population, and is the concept that ostensibly justifies selection at both secondary school and university level.

TRANSFORMATIVE LEARNING

At the core of Transformative Learning theory, is the process of “perspective transformation”, with three dimensions: psychological (changes in understanding of the self), convictional (revision of belief systems), and behavioural (changes in lifestyle).

“Transformative learning is the expansion of consciousness through the transformation of basic worldview and specific capacities of the self; transformative learning is facilitated through consciously directed processes such as appreciatively accessing and receiving the symbolic contents of the unconscious and critically analysing underlying premises.”

Perspective transformation leading to transformative learning occurs infrequently. Mezirow believes that it usually results from a *disorienting dilemma*, which is triggered by a life crisis or major life transition, although it may also result from an accumulation of transformations in meaning schemes over a period of time. Less dramatic predicaments, such as those created by a teacher, also promote transformation.

An important part of transformative learning is for individuals to change their frames of reference by critically reflecting on their assumptions and beliefs and consciously making and implementing plans that bring about new ways of defining their worlds. This process is fundamentally rational and analytical.

OTHER PERSPECTIVES

The theory and practice of “Transformative Education” was very effectively used by Russia, China, fascist Italy and Nazi Germany in the last century. The goal was to remove and dismantle all obstructionist elements and beliefs, like history, traditions, religions and opposing viewpoints. The purpose is always the same. It is to ridicule, remove and silence those who might oppose expanded government control of the population.

There are a number of critical responses to Mezirow's theory of transformative learning have emerged over the years. One criticism of Mezirow's theory is its emphasis upon rationality. Some studies support Mezirow. Others conclude that Mezirow grants rational critical reflection too much importance. Taylor has since suggested neurobiological research as a promising area that may offer some explanation about the role emotions play, closing the gap between rationality and emotion in the transformative learning process. Taylor implies that with available modern technology such as magnetic resonance imaging (MRI) and positron emission topography (PET), these once obscure factors can now be examined through determining which neurological brain systems

are at work during disorienting dilemmas and the journey of recovery that follows. This research also stresses the importance of the role of implicit memory, from which emerges habits, attitudes and preferences that are related to unconscious thoughts and actions.

While this learning process is certainly rational on some levels, it is also a profound experience that can be described as a spiritual or emotional transformation as well. The experience of undoing racist, sexist, and other oppressive attitudes can be painful and emotional, as these attitudes have often been developed as ways to cope with and make sense of the world. This type of learning requires taking risks, and a willingness to be vulnerable and have one's attitudes and assumptions challenged.

Other theorists have proposed a view of transformative learning as an intuitive and emotional process. Dirkx, Boyd, Myers and Ruether link Mezirow's rational, cognitive and analytical approach to a more intuitive, creative and holistic view of transformative learning. This view of transformative learning is based primarily on the work of Robert Boyd, who has developed a theory of transformative education based on analytical (or depth) psychology. For Boyd, transformation is a "fundamental change in one's personality involving [together] the resolution of a personal dilemma and the expansion of consciousness resulting in greater personality integration." This calls upon extrarational sources such as symbols, images, and archetypes to assist in creating a personal vision or meaning of what it means to be human.

First, an individual must be receptive or open to receiving "alternative expressions of meaning," and then recognise that the message is authentic. Grieving, considered by Boyd to be the most critical phase of the discernment process, takes place when an individual realises that old patterns or ways of perceiving are no longer relevant, moves to adopt or establish new ways, and finally, integrates old and new patterns. More recent research has specifically explored the process of transformative learning as it occurs in bereaved elders, maintaining that the "disorienting dilemma" deemed necessary by Mezirow is present in the loss of a loved one, with an additional devastating factor being the isolation that the elderly in particular are likely to face.

In another study, transformative learning in the context of suicide bereavement; in these cases the dilemma is compounded by the questioning of conceptions or misconceptions that were held about the relationship with the deceased and resolving the meaning of that relationship during the grieving process.

Unlike Mezirow, who sees the ego as playing a central role in the process of perspective transformation, Boyd and Myers use a framework that moves beyond the ego and the emphasis on reason and logic to a definition of transformative learning that is more psychosocial in nature.

Another definition of transformative learning was put forward by O'Sullivan:

Transformative learning involves experiencing a deep, structural shift in the basic premises of thought, feelings, and actions. It is a shift of consciousness that dramatically and irreversibly alters our way of being in the world. Such a shift involves our understanding of ourselves and our self-locations; our relationships with other humans

and with the natural world; our understanding of relations of power in interlocking structures of class, race and gender; our body awarenesses, our visions of alternative approaches to living; and our sense of possibilities for social justice and peace and personal joy. Positing that understanding transformative learning may have been hindered by perspectives of rational thought and Western traditions, King provides an alternate model grounded in a meta-analysis of research, the *Transformative Learning Opportunities Model*. Recent considerations of these varying perspectives seem to indicate that one perspective does not need to exclude the other. For example, Mezirow and Dirkx discussed their views on transformative learning at a 2005 International Transformative Learning Conference. This dialogue, facilitated by Cranton, continued via e-mail after the conference and the overview was published in *The Journal of Transformative Education*. Dirkx focuses on subjectivity, in the power of the inner world in one's shift in view of the outer world. Mezirow emphasises critical assessment of assumptions. Although their approaches are different, they agree that their perspectives are similar in several aspects. This includes transforming frames of reference that have lost meaning or have become dysfunctional and fostering enhanced awareness and consciousness of one's being in the world. Both perspectives are required to deepen understanding and to incorporate these ways of learning into transformative education. One of the difficulties in defining transformative learning is that it bleeds into the boundaries of concepts such as "meaning making" or "critical thinking". The term "meaning making" (*i.e.*, constructing meaning) is found most frequently in constructivist approaches to education, based on the work of educators such as John Dewey, Maria Montessori, Jean Piaget, Jerome Bruner, and Lev Vygotsky. In the constructivist view, meaning is constructed from knowledge.

Mezirow posits that all learning is change but not all change is transformation. There is a difference between transmissional, transactional and transformational education. In the first, knowledge is transmitted from teacher to student. In transactional education, it is recognised that the student has valuable experiences, and learns best through experience, enquiry, critical thinking and interaction with other learners. It could be argued that some of the research regarding transformative learning has been in the realm of transactional education, and that what is seen as transformative by some authors is in fact still within the realm of transactional learning. According to Brookfield, learning can only be considered transformative if it involves a fundamental questioning or reordering of how one thinks or acts; a challenge to hegemonic implications. In other words, reflection alone does not result in transformative learning unless the process involves a critical reflection, a recognition and analysis of taken for granted assumptions.

TRANSFORMATIVE LEARNING IN PRACTICE

On the surface, the two views of transformative learning presented here are contradictory. One advocates a rational approach that depends primarily on critical

reflection whereas the other relies more on intuition and emotion. However, the differences in the two views may best be seen as a matter of emphasis. Both utilise rational processes and incorporate imagination as a part of a creative process. The two different views of transformative learning described here as well as examples of how it occurs in practice suggest that no single model of transformative learning exists.

When transformative learning is the goal of adult education, fostering a learning environment in which it can occur should consider the following:

THE ROLE OF THE EDUCATOR

In order to foster transformative learning, the educator's role is to assist learners in becoming aware and critical of assumptions. This includes their own assumptions that lead to their interpretations, beliefs, habits of mind or points of view as well as the assumptions of others. Educators must provide learners practice in recognising frames of reference. By doing so, educators encourage practice in redefining problems from different perspectives. The goal is to create a community of learners who are "united in a shared experience of trying to make meaning of their life experience".

Educators need to provide learners with opportunities to effectively participate in discourse. Discourse involves assessing beliefs, feelings, and values. This dialogue has the goal of assessing reasons behind competing interpretations through critical examination of evidence, arguments, and alternate points of view. Learners are able to validate how and what they understand as well as develop well-informed judgements regarding a belief. Educators can encourage critical reflection and experience with discourse through the implementation of methods including metaphor analysis, concept mapping, consciousness raising, life histories, repertory grids, and participation in social action. The educator must encourage equal participation among students in discourse. One strategy is to encourage procedures that require group members to take on the roles of monitoring the direction of dialogue and ensuring equal participation. Educators can also encourage dialogue from different perspectives through controversial statements or readings from opposing points of view. It is necessary that the educator avoids shaping the discussion.

The role of educators is also to set objectives that include autonomous thinking. By fostering learners' critical reflection and experience in discourse, autonomous thinking is possible. The foundations to thinking autonomously begin in childhood and continue in adulthood. The educator assists adult learners in becoming more critical in assessing assumptions, better at recognising frames of references and alternate perspectives, as well as effective at collaborating with others to assess and arrive at judgements in regards to beliefs. It is the role of the educator to promote discovery learning through the implementation of classroom methods such as learning contracts, group projects, role play, case studies, and simulations. These methods facilitate transformative learning by helping learners examine concepts in the context of their lives and analyse the justification of new knowledge.

The educator's role in establishing an environment that builds trust and care and facilitates the development of sensitive relationships among learners is a fundamental principle of fostering transformative learning. The educator also serves as a role model by demonstrating a willingness to learn and change. As a result, professional development is important to assist educators in becoming authentic and critically reflective.

The Role of the Learner

The educator becomes a facilitator when the goal of learning is for learners to construct knowledge about themselves, others, and social norms. As a result, learners play an important role in the learning environment and process. Learners must create norms within the classroom that include civility, respect, and responsibility for helping one another learn. Learners must welcome diversity within the learning environment and aim for peer collaboration.

Learners must become critical of their own assumptions in order to transform their unquestioned frame of reference. Through communicative learning, learners must work towards critically reflecting assumptions that underlie intentions, values, beliefs, and feelings. Learners are involved in objective reframing of their frames of reference when they critically reflect on the assumptions of others. In contrast, subjective reframing occurs when learners critically assess their own assumptions.

The role of the learner involves actively participating in discourse. Through discourse, learners are able to validate what is being communicated to them. This dialogue provides the opportunity to critically examine evidence, arguments, and alternate points of view which fosters collaborative learning.

The Role of Professional Development for the Educator

Transformative learning about teaching occurs when educators critically examine their practice and develop alternative perspectives of understanding their practice. It is essential that this becomes the role of professional development. With this taken into consideration, the role of professional development is to assist educators in bringing awareness to their habit of minds regarding teaching. When this occurs, educators critically examine the assumptions that underlie their practice, the consequences to their assumptions, and develop alternative perspectives on their practice.

Teachers need education and professional development that will help them to question, challenge and experience critical discussions on school improvement. Transforming teachers so they see themselves as agents of social change can be a challenge within education. Strategies for transformative professional development include action plans, reflective activities, case studies, curriculum development and critical theory discussions. Action plans and reflective activities provide practice and modelling of critical reflection on the profession of education and provide guidance for the teaching and learning experience. Through the use of real-life examples, case studies provide opportunity to

analyse assumptions as well as the consequences of choices and actions. This provides focus for reflection on practice and the opportunity to be involved in discourse to analyse philosophical and practical aspects of educators' practice. Curriculum development creates the opportunity to connect theory and practice. In addition to introducing new teaching techniques, educators can test and compare these new concepts and practices with previous techniques. This moves away from uncritically accepting new teaching methods. Critical theory discussions can be implemented to guide educators to question the purpose and meaning of information. This then fosters opportunity for educators to question the selection of information they provide for their students.

Examples of Transformative Learning in Educator Professional Development

Mentoring is another strategy for transformative professional, personal and organisational development. By creating a supportive culture, mentoring can provide the environment for transformative learning to occur. Through this experience mentoring becomes a transformative relationship in which individuals reconstruct possible selves. As a two-way process, mentoring is a learning tool for both the mentor as well as the mentee. In a recent study, Swanson applied theories of critical reflection, critical incident model and a learning partnership to design a programme for practising teachers with the purpose of transformative teacher development. Experiences were created to support teachers in reflecting on their assumptions, ask them to consider alternative perspectives and develop a language for making connections between theory and practice. Over this two year period, teachers were able to develop ownership and transfer this knowledge into their practice. To be effective, transformative teacher development must value what teachers bring from their personal and professional experience and acknowledge that learning is both an individual and social experience. The process involved: transparency through negotiation of curriculum and finding a common language through discussion of individual assumptions, continuous feedback through critical incident questions, and action research through teacher application of the programme within their classrooms. Autobiography and journaling were additional techniques used to within this learning partnership. Both teachers and faculty participating in the programme were transformed as beliefs were challenged and knowledge was co-constructed through the experience.

New academics often find expectations ambiguous and lack the self-awareness and understanding to navigate the educational environment. A transformative learning framework was used in a Foundations course for participants to build on individual and collective analysis of teaching experiences and help them to reframe their practice. The author investigated 'transformation narratives' which emerged from written reflection of participants teaching practice. Reflections were based on individual reflection and group discussion. By using Mezirow's work along with Kegan's constructive developmental theory Constructivism (learning theory), the author found the following themes: a move from non-reflective habitual action to a more conscious practice, a

change in perspective to a more sophisticated view of teaching, an increased sense of agency including the concept that academic practice is an object which can be controlled and shaped rather than something externally imposed, increased confidence to take risks and experiment, and a more multifaceted idea of what it means to be an academic. The conducive environment allowed for transformative experiences to occur. Through creation of a safe social context where ‘disorienting dilemmas’ can be examined, questioned and explored participants were able to develop a new ‘frame of reference’ and reintegrate learning into practice. The author cautions that there are limitations to the framework such as the possibility that participants will conform to expectations in their reflections. The author notes that the ‘transformation narratives’ examined are not a single, final narrative of the self but a snapshot for further exploration. Similar to Fletcher’s findings, transformative learning helps to make sense of a complex and often ambiguous work environment which requires multiple selves.

BENEFITS OF TRANSFORMATIVE LEARNING

When learning is contextualised in a community, connected to one’s social identities, and when it asks a group of learners to be better as a people, it is not generally forgotten, because it holds tremendous importance for the learner. As a result, transformative learning develops autonomous thinking that is transferable to other learning contexts.

CHALLENGES OF TRANSFORMATIVE LEARNING ENVIRONMENTS

While supporting engagement of the entire class, often the most self-efficacious students will voice their suggestions, ideas and concerns. Those with lower self-efficacy can become spectators if not given equal encouragement to participate. Transformative learning inherently requires taking risks and a willingness to be vulnerable and have one’s attitude and assumptions challenged.

ACKNOWLEDGING THE ROLE OF SELF-EFFICACY IN TRANSFORMATIVE LEARNING

Psychologist Albert Bandura has defined self-efficacy as one’s belief in one’s ability to succeed in specific situations. One’s sense of self-efficacy can play a major role in how one approaches goals, tasks, and challenges. The theory of self-efficacy lies at the centre of Bandura’s social cognitive theory, which emphasizes the role of observational learning and social experience in the development of personality. The main concept in social cognitive theory is that an individual’s actions and reactions, including social behaviours and cognitive processes, in almost every situation are influenced by the actions that individual has observed in others. Because self-efficacy is developed from external experiences and self-perception and is influential in determining the outcome of many events, it is an important aspect of social cognitive theory. Self-efficacy

represents the personal perception of external social factors. According to Bandura's theory, people with high self-efficacy—that is, those who believe they can perform well—are more likely to view difficult tasks as something to be mastered rather than something to be avoided. In a transformative learning environment, consideration of one's current level of self-efficacy can assist both the Educator and student in preparing for engagement in discourse. As frames of reference play a critical role in shaping one's past and present assumptions or beliefs, so to do they influence the self-efficacy of an individual participating in adult learning activities.

ACADEMIC SELF-EFFICACY

In a study, the majority of a group of students questioned felt they had a difficulty with listening in class situations. Instructors then helped strengthen their listening skills by making them aware about how the use of different strategies could produce better outcomes. This way, their levels of self-efficacy were improved as they continued to figure out what strategies worked for them. Understanding how to foster the development of self-efficacy is important for educators, and others in leadership positions, and to anyone seeking to build happier, more productive, and highly efficacious students.

Through transformative learning, the Educator plays a similar role in facilitating the development of trust, care and sensitivity, so that students feel empowered to discuss their assumptions and beliefs in a way that works best for them, while incorporating strategies that best accommodate their current level of self-efficacy.

Bandura also showed that difference in self-efficacy correlates to fundamentally different world views. People with high self-efficacy generally believe that they are in control of their own lives, that their own actions and decisions shape their lives, while people with low self-efficacy may see their lives as outside their control.

Such evidence speaks directly to the outcomes of transformative learning, in that the assumptions, beliefs and values of individuals (as a result of their level of self-efficacy), can potentially shape the direction of the critical discourse taking place in the classroom, based on their feelings of either being in control or not in control of their destiny.

SOCIAL LEARNING THEORY

Social learning theory describes the acquisition of skills that are developed exclusively or primarily within a social group. Social learning depends on how individuals either succeed or fail at dynamic interactions within groups, and promotes the development of individual emotional and practical skills as well as accurate perception of self and acceptance of others. According to this theory, people learn from one another through observation, imitation, and modelling. Self-efficacy reflects an individual's understanding of what skills he/she can offer in a group setting.

Transformative learning ideally promotes equal participation among members of a group, as they work towards establishing a consensus in exploring the depths of subject

matter. Should an individual feel they can either offer a lot, in comparison with one who feels they can offer little, the equal participation in transformative learning may become more difficult to attain.

THE ROLE OF THE RATIONAL AND THE AFFECTIVE

Transformative learning has two components that at times seem to be in conflict: the cognitive, rational, and objective and the intuitive, imaginative, and subjective. Both the rational and the affective play a role in transformative learning. Although the emphasis has been on transformative learning as a rational process, teachers need to consider how they can help students use feelings and emotions both in critical reflection and as a means of reflection.

There are a number of educational and research institutions that are based on the principles of transformative learning. Some examples include the Transformative Studies Institute and The Transformative Learning Centre at the Ontario Institute for Studies on Education (OISE) of the University of Toronto. Evidences are emerging on the applicability of transformative theory in new educational programmes. Some examples include in business and industry education, health professional education, community education such as courses offered by YMCA, and informal settings such as self-help groups. A study in the UK describes how students in a service-learning section experienced perspective transformation and shifts of their world-view when their training was derived in part by the principles of transformative learning theory.

EDUCATIONAL NEUROSCIENCE

American Universities such as Harvard, Johns Hopkins, University of Southern California and others, in the first decade of the twenty-first century, began offering majors and degrees dedicated to educational neuroscience or neuroeducation. Such studies seek to link an understanding of brain processes with classroom instruction and experiences. Neuroeducation seeks to analyse the biological changes that take place in the brain as new information is processed. It looks at what environmental, emotional and social situations are best in order for new information to be retained and stored in the brain via the linking of neurons, rather than allowing the dendrites to be reabsorbed and the information lost. The 1990s were designated “The Decade of the Brain,” and advances took place in neuroscience at an especially rapid pace. The three dominant methods for measuring brain activities are: Event-related potential, Functional magnetic resonance imaging and MEG.

The integration and application to education of what we know about the brain was strengthened in 2000 when the American Federation of Teachers stated: “It is vital that we identify what science tells us about how people learn in order to improve the education curriculum.” What is exciting about this new field in education is that modern brain imaging techniques now make it possible, in some sense, to watch the brain as it learns,

and the question then arises: can the results of neuro-scientific studies of brains as they are learning usefully inform practice in this area? Although the field of neuroscience is young, it is expected that with new technologies and ways of observing learning, the paradigms of what students need and how students learn best will be further refined with actual scientific evidence. In particular, students who may have learning disabilities will be taught with strategies that are more informed.

A BRAIN-BASED THEORY OF LEARNING

The differences of opinion and theory in psychology indicate that the learning process is not yet understood. Neuroscience shows that the brain can be modelled not with a central processor where “intelligence” lies, but in having perhaps 70 functional areas. Mental activity requires several areas to work together. What appear as different types of intelligence result from different combinations of well-developed functional areas. Learning is a process by which neurons join by developing the synapses between them. Knowledge is arranged hierarchically, with new knowledge being linked to existing neural networks.

MULTIPLE INTELLIGENCES

The existence of multiple intelligences is proposed by psychologist Howard Gardner, who suggests that different kinds of intelligence exists in human beings. It is a theory that has been fashionable in continuous professional development (CPD) training courses for teachers.

MULTIMEDIA LEARNING

Multimedia learning refers to the use of visual and auditory teaching materials that may include video, computer and other information technology. Multimedia learning theory focuses on the principles that determine the effective use of multimedia in learning, with emphasis on using both the visual and auditory channels for information processing.

The auditory channel deals with information that is heard, and the visual channel processes information that is seen. The visual channel holds less information than the auditory channel. If both the visual and auditory channels are presented with information, more knowledge is retained. However, if too much information is delivered it is inadequately processed, and long term memory is not acquired. Multimedia learning seeks to give instructors the ability to stimulate both the visual and auditory channels of the learner, resulting in better progress.

OTHER LEARNING THEORIES

Other learning theories have also been developed for more specific purposes. For example, andragogy is the art and science to help adults learn. Connectivism is a recent theory of networked learning which focuses on learning as making connections.

LEARNING STYLE THEORY

Learning style theory proposes that individuals learn in different ways, that there are four distinct learning styles – feeling, watching, thinking and doing – and that knowledge of a learner's preferred learning style will lead to faster and more satisfactory improvement.

INFORMAL AND POST-MODERN THEORIES

In theories that make use of cognitive restructuring, an informal curriculum promotes the use of prior knowledge to help students gain a broad understanding of concepts. New knowledge cannot be told to students, it believes, but rather the students' current knowledge must be challenged. In this way, students will adjust their ideas to more closely resemble actual theories or concepts. By using this method students gain the broad understanding they're taught and later are more willing to learn and keep the specifics of the concept or theory. This theory further aligns with the idea that teaching the concepts and the language of a subject should be split into multiple steps. Other informal learning theories look at the sources of motivation for learning. Intrinsic motivation may create a more self-regulated learner, yet schools undermine intrinsic motivation. Critics argue that the average student learning in isolation performs significantly less well than those learning with collaboration and mediation. Students learn through talk, discussion, and argumentation.

PHILOSOPHICAL ANTHROPOLOGY

Every well-constructed theory of education has at its center a philosophical anthropology.

CRITICISM OF LEARNING THEORY

Critics of learning theories that seek to displace traditional educational practices claim that there is no need for such theories; that the attempt to comprehend the process of learning through the construction of theories creates problems and inhibits personal freedom.

HULL'S REINFORCEMENT THEORY

Hull explained learning on the basis of reinforcement. According to this theory, any stimulus occurring 'in close temporal contiguity' with a reinforced response becomes connected with that response making it more likely to recur. It is the reinforcing stimulus that determines the relative strength of a possible alternative response in a learning situation.

Hull's theory has been much useful in the field of education. Since, it stresses on motivation and motivation is the very heart of learning. As such Hull's theory has been

of much help in facilitating learning in children, especially. Also, this theory helps the students to attain knowledge of different subjects. It develops interest among the students for the subject-matter.

This theory helps in creative works. It also helps in the development of intellect, logic and thinking. It enables the students to solve complex problems. It aids in the teaching of literature, art and music. It helps in habit formation. It helps in learning the skills. This theory encourages carrying on researches in various fields.

TRANSFER OF LEARNING AND ITS THEORIES

An important issue in optimising learning is the extent to which the learning of one thing facilitates the learning of something else. If everything we learned was specific to the situation in which, it was learned, the amount of learning that would have to be crammed into a lifetime would be phenomenal. Fortunately, most learning is readily transferable with some phenomenal. Fortunately, most learning is readily transferable with some modification to a number of different situations.

The influence that learning one task may have on the subsequent learning of another is called transfer of learning. The term positive transfer is used when learning one task does facilitate learning another. If one is a good tennis player, it is easier to learn to play squash; this is positive transfer. But, transfer is not always positive; when interference occurs, we have negative transfer.

There are numerous examples of negative transfer in everyday life. When driving a car with automatic transmission after having been accustomed to one with a stick shift, we may find ourselves depressing a non-existent clutch pedal. When changing from a pedal-brake to a hand-brake bicycle, we may still try to press back on the pedal when we have to stop quickly. And the transition from driving on the right-hand side of the street to the British procedure of driving on the left is difficult for many American visitors to Great Britain. The original habit is so over-learned that even after driving successfully on the left for some time, an individual may revert to right-side driving when required to act quickly in an emergency.

REINFORCEMENT THEORY

Reinforcement is an event that increases behaviour. In the classroom, reinforcement occurs as teachers manage the environmental events that follow students' desired ways of behaving so to increase the strength and future likelihood of that behaviour.

Reinforcement comes in two types—positive and negative. Positive reinforcement occurs when desired behaviour is strengthened by the presentation of a contingent stimulus. The attractive, behaviour-increasing, contingent stimulus used during positive reinforcement is referred to as a positive reinforcer. A positive reinforcer is defined as any environmental event that, when given in response to the behaviour, increases the strength and frequency of that behaviour. Some commonly used positive reinforcers in

the classroom are praise, attention, tokens, and stickers. Negative reinforcement occurs when desired behaviour is strengthened by the removal of a contingent stimulus. The aversive, behaviour-increasing, contingent stimulus that is removed during negative reinforcement is referred to as a negative reinforcer. A negative reinforcer is defined as any environmental event that, when taken away in response to the behaviour, increases the strength and frequency of that behaviour. Some commonly used negative reinforcers in the classroom are taking away an aversive assignment (*e.g.*, homework), withdrawing an intrusive stare, or cancelling a chore. Table provides a sample of reinforcers commonly used in K-12 classrooms. The left side of the table lists a variety of positive reinforcers that teachers give to strengthen students' behaviour, while the right side of the table lists a variety of negative reinforcers that teachers take away to strengthen students' behaviour. What is common to all positive reinforcers in the list is the idea that after the student engages in a particular behaviour, he or she receives an attractive consequence for doing so. That is, when the student turns in her homework, the teacher then places a sticker on the paper. The sticker is given after the homework has been handed in and with the intention of strengthening the likelihood that future homework assignments will be forthcoming. What is common to all negative reinforcers in the list is the idea that when the student engages in a particular behaviour, he or she gets a break from an unattractive or aversive consequence for doing so. That is, when the student turns in her homework, the teacher then exempts the student from an arduous assignment of long division problems. The arduous assignment is removed after the homework has been handed in and with the intention of strengthening the likelihood that future homework assignments will be forthcoming.

Table. Illustration by GGS Information Services. Cengage Learning, Gale.

Example of Positive and Negative Reinforcers Commonly-Used in K—12 Classrooms	
Positive Reinforcers	Negative Reinforcers
Giving Praise	Taking Away a Deadline
Giving Stickers	Taking Away Surveillance
Giving Privileges	Taking Away an Unattractive Assignment
Giving Tokens	Taking Away a Threat
Giving Attention	Removing a Negative Evaluation
Giving Good Grades	Removing Bad Grades
Giving Scholarships	Taking Away Criticism
Giving an Honor	Stopping One's Ridicule
Giving a Trophy	Stopping One's Teasing
Giving a Prize	Stopping One's Yelling or Screaming
Giving an Award	Stopping One's Crying
Giving Food	Stopping one's Whining
Giving Money	Stopping One's Staring
Giving a Smile	Stopping One's Pleading or Pouting
Giving Positive Feedback	Stopping Laughing (at Someone)

Reinforcers and Rewards

Reinforcement is defined by its effect on behaviour. Only environmental events that actually increase behaviour are reinforcers. If an event such as a smile, candy bar, or break from chores does not increase the student's behaviour, then the event is not actually a reinforcer. This qualification of which environmental events are reinforcers and which are not is an important point because teachers often implement consequences haphazardly (non-contingently) or inconsistently. Praise represents a good example of commonly used attractive environmental event that is used for a dozen different reasons, only one of which is to reinforce (increase) students' behaviour. Further, teachers do not reinforce positive behaviour as often as they could (Wehby, Symons, Canale, and Go, 1998). Instead of delivering contingent and strategic consequences for desired behaviour (*i.e.*, reinforcers), what teachers more typically offer to students are promises of reward.

A reward is the offering of an environmental event in exchange for the student's participation, service, or achievement. When a teacher promises an award if the student will complete an assignment or when a teacher promises a prize to acknowledge a successful performance, she introduces a reward into the learning environment. Adding a discussion of rewards to a discussion of reinforcement highlights the instructional practice of soliciting students' behaviour or acknowledging their achievement, regardless of whether those rewards actually reinforce behaviour. The discrepancy between reinforcers and rewards is that a reward functions as a reinforcer only when the student values it enough to affect a change in his or her behaviour. Sometimes teachers use rewards that students do not value (*e.g.*, public recognition), and unappealing rewards do not increase the frequency of the desired behaviour. If the student is embarrassed, rather than gratified by, the public recognition then it will not reinforce (*i.e.*, strengthen) the behaviour, and it might even act as a punisher to some students. The important point is that teachers offer rewards in hopes of soliciting behaviour, while they deliver contingent reinforcers to increase desired behaviour which has already occurred.

Assumptions that Relate Reinforcement to Learning

Reinforcement is an educational concept rooted in behavioural learning theory. Behavioural learning theory does not focus on mental knowledge, such as learning information. Neither does it focus on cognitive and sociocultural concepts, such as creating meaning, understanding concepts, using memory, and the experience of conceptual change. Instead, behavioural learning theory focuses on behaviour. Specifically, it focuses on voluntary, intentional, and situationally appropriate behaviour. So, the learning highlighted by behavioural learning theory is learning how to adapt successfully to one's environment (*e.g.*, raise one's hand before speaking, wait one's turn in the lunch line). Reinforcers play the important role they do in helping students learn how to adapt to the classroom (and school) environment by signalling which behaviours are desirable (those that are reinforced) and which are not (those that are

not reinforced). Students learn which behaviours are desirable and adaptive by learning which behaviours are associated with reinforcers. The assumption that relates reinforcement to learning is that the presence of a reinforcer signals that a particular behaviour is desirable, and this signalling process, therefore, helps students learn how to adapt more successfully to the classroom environment.

Assumptions that Relate Reinforcement to Motivation

Reinforcing students' desirable behaviours with positive and negative contingencies is an extrinsic motivational strategy. Motivation researchers outside behavioural learning theory find that extrinsic rewards often undermine students' intrinsic motivation and capacity for autonomous self-regulation. That is, reinforcers and rewards produce complicated, and often undesirable, motivational side-effects. The undermining effect that extrinsic rewards often have on intrinsic motivation and autonomous self-regulation is important for two reasons. First, many teachers incorrectly assume that adding extrinsic rewards on top of students' intrinsic motivation will create a supermotivation in which the two types of motivation (extrinsic and intrinsic) will combine and complement one another. Instead, extrinsic rewards typically interfere with and undermine intrinsic motivation and autonomous self-regulation. Second, teachers often offer students extrinsic rewards when introducing uninteresting activities in the hope that the reward will be able to generate the motivation to engage in the task that the task itself cannot generate (because it is uninteresting). From a motivational point of view, teachers therefore use positive reinforcers, negative reinforcers, and extrinsic rewards to bolster students' otherwise low motivation. In doing so, teachers are essentially trying to set up (manage) conditions that make desired target behaviours, such as engaging in uninteresting (but educationally important) lessons, more likely.

Classroom Management to Promote Educational Outcomes

Many positive educational outcomes can be framed in terms of desirable behaviour. Teachers, administrators, and parents all consider behaviours such as attending school regularly and promptly, being actively engaged during classroom lessons, showing respect for others, making good grades, and graduating from school as desirable ways to behave. Because educators value these ways of behaving and because they would like to see students engage in these ways of behaving more frequently, they find merit in classroom management practices, the most popular and the most effective of which involves the strategic use of positive reinforcers, negative reinforcers, and extrinsic rewards (Landrum and Kauffman, 2006). The fundamental task of classroom management is to create a caring, supportive, inclusive, and engaging community in which students frequently engage in desirable, constructive, and prosocial behaviour. When practised effectively, classroom management is a proactive strategy that creates a classroom environment in which desirable behaviour is expected, supported, and reinforced. From this point of view, reinforcement

plays only one part in the teacher's larger effect to manage the classrooms. In addition to reinforcing desirable behaviour after it occurs and in addition to offering extrinsic rewards to solicit students' desired behaviour, teachers can further support students' desirable ways of behaving by adding additional classroom management strategies such as modelling and observational learning, scaffolding and tutoring, behavioural supports, and promoting students' effective self-management.

CLASSICAL CONDITIONING

Classical conditioning (also Pavlovian conditioning or respondent conditioning) is a form of learning in which the *conditioned stimulus* or CS, comes to signal the occurrence of a second stimulus, the *unconditioned stimulus* or US. (A stimulus is a factor that causes a response in an organism.) The conditioned response is the learned response to the previously neutral stimulus. The US is usually a biologically significant stimulus such as food or pain that elicits a response from the start; this is called the *unconditioned response* or UR. The CS usually produces no particular response at first, but after conditioning it elicits the *conditioned response* or CR. Classical conditioning differs from *operant* or *instrumental* conditioning, in which behaviour emitted by the subject is strengthened or weakened by its consequences (*i.e.*, reward or punishment).

There are many examples of these innate (unconditioned) reflexes (corneal, coughing, swallowing, withdrawal reflexes, etc.) or learned (conditioned) reflexes.

"In any animal, regardless of its prior history, painful stimulation of the foot causes the leg to be withdrawn by bending at all its joints. This *flexor reflex* is an example of an *unconditioned reflex*, an innate response based on fixed connections in the chain of neurons from the receptor (sensor) to the effector. Of still more interest in everyday life are the *acquired* or *conditioned reflexes*, in which the functional connections between the excited sensors and the patterns of activity in effector organs become established by learning process." Conditioning is usually done by pairing the two stimuli, as in Pavlov's classic experiments. Pavlov presented dogs with a ringing bell followed by food. The food elicited salivation (UR), and after repeated bell-food pairings the bell also caused the dogs to salivate. In this experiment, the unconditioned stimulus is the dog food as it produces an unconditioned response, saliva. The conditioned stimulus is the ringing bell and it produces a conditioned response of the dogs producing saliva.

It was originally thought that the process underlying classical conditioning was one where the conditioned stimulus becomes associated with, and eventually elicits, the unconditioned response. But many observations do not support this hypothesis. For example, the conditioned response is often quite different from the unconditioned response. Learning theorists now more commonly suggest that the CS comes to signal or predict the US. In the case of the salivating dogs in Pavlov's experiment, the bell tone signalled and predicted the arrival of the dog food, thus resulting in the dog salivating. Robert A. Rescorla provided a clear summary of this change in thinking, and its consequences, in his 1988 article "Pavlovian conditioning: It's not what you think it is."

Origins

I. P. Pavlov provided the most famous example of classical conditioning, though E. B. Twitmyer published his findings a year earlier (a case of simultaneous discovery). During his research on the physiology of digestion in dogs, Pavlov noticed that, rather than simply salivating in the presence of food, the dogs began to salivate in the presence of the lab technician who normally fed them. Pavlov called this anticipatory salivation *psychic secretion*. From this observation he predicted that, if a particular stimulus in the dog's surroundings was present when the dog was given food, then this stimulus would become associated with food and cause salivation on its own. In his initial experiment, Pavlov used a bell to call the dogs to their food and, after a few repetitions, the dogs started to salivate in response to the bell. Pavlov called the bell the *conditioned* (or *conditional*) *stimulus* (CS) because its effect depended on its association with food. He called the food the *unconditioned stimulus* (US) because its effect did not depend on previous experience. Likewise, the response to the CS was the *conditioned response* (CR) and that to the US was the *unconditioned response* (UR). The timing between the presentation of the CS and US is integral to facilitating the conditioned response. Pavlov found that the shorter the interval between the bell's ring and the appearance of the food, the more quickly the dog learned the conditioned response and the stronger it was. As noted earlier, it is often thought that the conditioned response is a replica of the unconditioned response, but even Pavlov noted that saliva produced by the CS differs in composition from that produced by the US. In fact, the CR may be any new response to the previously neutral CS that can be clearly linked to experience with the conditional relationship of CS and US. It was also thought that repeated pairings are necessary for conditioning to emerge, however many CRs can be learned with a single trial as in fear conditioning and taste aversion learning.

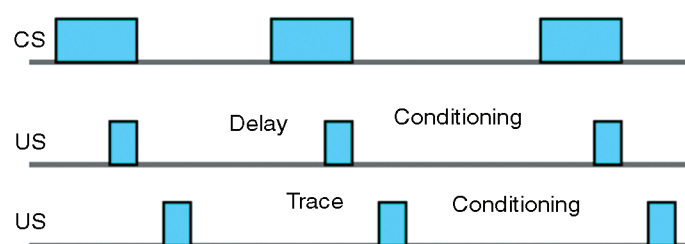


Fig. Diagram Representing Forward Conditioning. The Time Interval Increases from Left to Right.

CLASSICAL CONDITIONING PROCEDURES

Forward Conditioning

Learning is fastest in forward conditioning. During forward conditioning the onset of the CS precedes the onset of the US in order to signal that the US will follow.

Two common forms of forward conditioning are delay and trace conditioning.

- *Delay conditioning:* In delay conditioning the CS is presented and is overlapped by the presentation of the US.
- *Trace conditioning:* During trace conditioning the CS and US do not overlap. Instead, the CS is presented, a period of time is allowed to elapse during which no stimuli are presented, and then the US is presented. The stimulus-free period is called the *trace interval*. It may also be called the *conditioning interval*.

Simultaneous Conditioning

During simultaneous conditioning, the CS and US are presented and terminated at the same time.

Second-order Conditioning

This form of conditioning follows a two-step procedure. First a neutral stimulus ("CS1") comes to signal a US through forward conditioning. Then a second neutral stimulus ("CS2") is paired with the first (CS2-CS1) and comes to yield its own conditioned response. For example, a bell (CS1) might be paired with food (US) until the bell elicits salivation. If a light is then paired with the bell the light also may come to elicit salivation.

Backward Conditioning

Backward conditioning occurs when a CS immediately follows a US. Unlike the usual conditioning procedure, in which the CS precedes the US, the conditioned response given to the CS tends to be inhibitory. This presumably happens because the CS serves as a signal that the US has ended, rather than as a signal that the US is about to appear.

Temporal Conditioning

When a US is presented at a number of regularly timed intervals a CR often develops shortly before the US is scheduled to appear. This suggests that animals have a biological clock that can serve as a CS. This method has also been used to study timing ability in animals.

Zero Contingency Procedure

In this procedure, the CS is paired with the US as usual, but the US also occurs at other times, so that the US is just as likely to happen in that absence of the CS as it is following the CS. Here, in other words, the CS does not "predict" the US. In this case, conditioning fails: the CS does not come to elicit a CR. This finding - that *prediction* rather than CS-US pairing is the key to conditioning - greatly influenced subsequent conditioning research and theory.

Extinction

In the extinction procedure, the CS is presented repeatedly in the absence of a US. This is done after a CS has been conditioned by one of the methods above. When this is done the CR frequency eventually returns to pre-training levels. However, spontaneous recovery shows that extinction does not completely eliminate the effects of the prior conditioning.

PHENOMENA OBSERVED IN CLASSICAL CONDITIONING

Acquisition

During acquisition the CS and US are paired in one of the ways described above. The extent of conditioning may be tracked by test trials in which the CS is presented alone and the CR is measured.

A single CS-US pairing may suffice to yield a CR on test, but usually a number of pairings are necessary, during which the strength and/or frequency of the CR gradually increases. The speed of conditioning depends on a number of factors, including the nature and strength of both the CS and the US, the animal's motivational state, and previous experience. During acquisition, which is a phase of classical conditioning, there is a gradual increase in learning. It starts off slow, rapidly rises and then slowly fades away.

Extinction

The process of presenting a CS alone, without the US, that causes the learned behaviour to disappear eventually is called *extinction*. When the presentation of the CS with the US absent is done often enough, the CS ordinarily stops eliciting a CR; that is, it has been *extinguished*.

External Inhibition

External inhibition may be observed if a strong or unfamiliar stimulus is presented just before, or at the same time as, the CS. This causes a reduction in the conditioned response to the CS.

Recovery from Extinction

Several procedures can cause an extinguished CR to recover. The outcomes of these procedures suggest that extinction does not entirely erase the effects of conditioning. Each of the following descriptions assumes that a CS has been conditioned and then the CR extinguished through the procedures described above.

- Reacquisition:

If the CS is again paired with the US, a CR is again acquired, but this second acquisition usually happens much faster than the first one.

- Spontaneous recovery:

If the CS is tested at a later time (for example an hour or a day) after conditioning it will again elicit a CR, a phenomenon called *spontaneous recovery*. This renewed CR is usually much weaker than the CR observed prior to extinction

- External disinhibition:

If the CS is tested just after intense but associatively neutral stimulus has occurred, there may be a temporary recovery of the conditioned response to the CS

- Reinstatement:

If the US used in conditioning is presented to a subject in the same place where conditioning and extinction occurred, but without the CS being present, the CS often elicits a response when it is tested later.

- Renewal:

If conditioning and extinction are conducted in different contexts (for example, conditioning occurs in one experimental chamber and extinction occurs in a chamber with a different colour and shape), a return to the conditioning context may cause the CS to again elicit a CR.

Stimulus Generalisation

Stimulus generalisation is said to occur if, after a particular CS has come to elicit a CR, another test stimulus elicits the same CR. Usually the more similar are the CS and the test stimulus the stronger is the CR to the test stimulus. The more the test stimulus differs from the CS the more the conditioned response will differ from that previously observed.

Stimulus Discrimination

One observes *stimulus discrimination* when one stimulus (“CS1”) elicits one CR and another stimulus (“CS2”) elicits either another CR or no CR at all. This can be brought about by, for example, pairing CS1 with an effective US and presenting CS2 in extinction, that is, with no US.

Latent Inhibition

In this procedure, a CS is presented several times before paired CS–US training commences. This slows the rate of CR acquisition relative to that observed without such pre-exposure.

Conditioned Suppression

This is one of the most common ways to measure the strength of learning in classical conditioning. In a typical example of this procedure, a rat first learns to press a lever

through operant conditioning. Classical conditioning follows: in a series of trials the rat is exposed to a CS, often a light or a noise. This CS is followed by the US, a mild electric shock. As an association between the CS and US develops, the rat slows or stops its lever pressing when the CS comes on. The rate of pressing during the CS measures the strength of classical conditioning; that is, the slower the rat presses, the stronger is the association of the CS and the US.

Conditioned Inhibition

Three phases of conditioning are typically used:

Phase 1

A CS (CS+) is paired with a US until asymptotic CR levels are reached.

Phase 2

CS+/US trials are continued, but interspersed with trials on which the CS+ is paired with a second CS, (the CS-) but not with the US (*i.e.*, CS+/CS- trials). Typically, organisms show CRs on CS+/US trials, but stop responding on CS+/CS- trials.

Phase 3

- *Summation test for conditioned inhibition:* The CS- from Phase 2 is presented together with a new CS+ that was conditioned as in Phase 1. Conditioned inhibition is found if the response is less to the CS+/CS- pair than it is to the CS+ alone.
- *Retardation test for conditioned inhibition:* The CS- from Phase 2 is paired with the US. If conditioned inhibition has occurred, the rate of acquisition to the previous CS- should be less than the rate of acquisition that would be found without the Phase 2 treatment.

Blocking

This form of classical conditioning involves two phases.

Phase 1

A CS (CS1) is paired with a US.

Phase 2

A compound CS (CS1+CS2) is paired with a US.

Test

A separate test for each CS (CS1 and CS2) is performed. The blocking effect is observed in a lack of conditional response to CS2, suggesting that the first phase of training blocked the acquisition of the second CS.

Conditioned and Unconditioned Reflexes

In their textbook on human physiology, Nikolai Agajanyan and V. Tsyarkin list five criteria for demarcation between unconditioned and conditioned reflexes. In contrast

to the former, the latter are learned and unstable reactions which can be inhibited and extinguished under definite conditions. A leading role in the performance of unconditioned reflexes, in contrast to the conditioned ones, is played by the lower divisions of the higher nervous system, the subcortical nuclei, brain stem and spinal cord. Conditioned reflexes, on the contrary, are a function of the cerebral cortex and can be developed under the action of the most varied stimuli applied to different receptive fields.

MODELS OF THE LEARNING PROCESS

Let us now turn our attention to some of the models of learning that educational psychologists and educationalists have developed over the years. We will start by examining the three that have probably proved most influential - Gagné's 1956 hierarchy of learning, Piaget's 1969 model of cognitive development and Kolb's 1984 experiential cycle - before turning to one of the most recent - Race's 1993 'ripples' model of learning.

GAGNÉ'S HIERARCHY OF LEARNING

In 1956, the American educational psychologist Robert M. Gagné proposed a system of classifying different types of learning in terms of the degree of complexity of the mental processes involved. He identified eight basic types, and arranged these in the hierarchy shown in Figure.

According to Gagné, the higher orders of learning in this hierarchy build upon the lower levels, requiring progressively greater amounts of previous learning for their success. The lowest four orders tend to focus on the more behavioural aspects of learning, while the highest four focus on the more cognitive aspects.

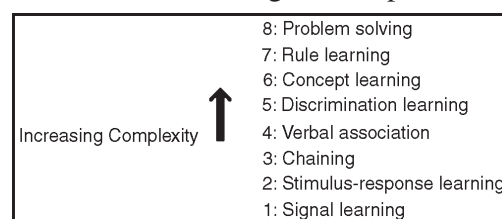


Fig. Gagné's Hierarchy of Learning

Let us now take a closer look at Gagné's eight categories of learning.

Signal Learning

This is the simplest form of learning, and consists essentially of the classical conditioning first described by the behavioural psychologist Pavlov. In this, the subject is 'conditioned' to emit a desired response as a result of a stimulus that would not normally produce that response.

This is done by first exposing the subject to the chosen stimulus (known as the conditioned stimulus) along with another stimulus (known as the unconditioned stimulus) which produces the desired response naturally; after a certain number of repetitions of the double stimulus, it is found that the subject emits the desired response when exposed to the conditioned stimulus on its own. The applications of classical conditioning in facilitating human learning are, however, very limited.

Stimulus-response Learning

This somewhat more sophisticated form of learning, which is also known as operant conditioning, was originally developed by Skinner. It involves developing desired stimulus-response bonds in the subject through a carefully-planned reinforcement schedule based on the use of 'rewards' and 'punishments'. Operant conditioning differs from classical conditioning in that the reinforcing agent (the 'reward' or 'punishment') is presented after the response.

It is this type of conditioning that forms the basis of programmed learning in all its various manifestations.

Chaining

This is a more advanced form of learning in which the subject develops the ability to connect two or more previously-learned stimulus-response bonds into a linked sequence. It is the process whereby most complex psychomotor skills (*e.g.*, riding a bicycle or playing the piano) are learned.

Verbal Association

This is a form of chaining in which the links between the items being connected are verbal in nature. Verbal association is one of the key processes in the development of language skills.

Discrimination Learning

This involves developing the ability to make appropriate (different) responses to a series of similar stimuli that differ in a systematic way. The process is made more complex (and hence more difficult) by the phenomenon of interference, whereby one piece of learning inhibits another. Interference is thought to be one of the main causes of forgetting.

Concept Learning

This involves developing the ability to make a consistent response to different stimuli that form a common class or category of some sort. It forms the basis of the ability to generalise, classify, etc.

Rule learning. This is a very-high-level cognitive process that involves being able to learn relationships between concepts and apply these relationships in different situations, including situations not previously encountered. It forms the basis of the learning of general rules, procedures, etc.

Problem solving. This is the highest level of cognitive process according to Gagné. It involves developing the ability to invent a complex rule, algorithm or procedure for the purpose of solving one particular problem, and then using the method to solve other problems of a similar nature.

LEARNING AND MOTIVATION

Recognition of the meaning of a moment is a perception evolved from one's neurological equipment, past experience shaping the perception, a mixture of hormones in the blood stream, and the extent of knowledge to which one is aware. Most people consider learning as a level playing field that is available to everyone as just something we do. We are faced with a stimulus, something happens, and behaviour comes out the other side. "Learning" is the process that happens between the stimulus and response. Since the process of learning has been found among humans and most animals, we can assume that it is part of an evolutionary agenda.

Philosophy tends to examine cultural concepts as rational constructs; but, there's something deeper than that. For instance, the theist and the atheist can look at the same thing, as a friend and I were both enjoying a sunny afternoon in the backyard, and both drew support from this same experience for our very different concepts. We were both in awe, but one found no doubt that this glorious experience had to be created, while the other saw no indication of a creator.

When the temporal lobes are excited, as in temporal lobe epilepsy, the result is an intense heightening of the patient's sensory appreciation of the world and intense empathy for all beings to the extent of seeing no barriers between himself and the cosmos. This might be a clue as to the neurological basis of religious and mystical experiences: a neurological effect not experienced by the atheist. Einstein said "the most beautiful and most profound experience is the sensation of the mystical. It is the source of all true science. He to whom this emotion is a stranger, who can no longer wonder and stand rapt in awe, is as good as dead. To know that what is impenetrable to us really exists, manifesting itself as the highest wisdom and the most radiant beauty which our dull faculties can comprehend only in their primitive forms - this knowledge, this feeling is at the center of true religiousness."

We can see in this example: it can be the perception, or the experience, that is causal to the concept. Due to the philosophical tendency to categorise, and deliberate concepts through reason, we can ignore, or do not understand, the role of perceptual variations. Understanding the variation of perceptual processing is necessary for understanding how and what we learn.

ETHOLOGY

Ethology is the study of “fixed action patterns” of behaviour among animals in their natural setting. Ethology introduced the concept of inherent perceptual equipment. Examples include: suckling by infant mammals, the dance of honey bees returning to their nest to communicating the location of nectar, and the imprinting of ducklings on the first moving object that might be a mother duck among many other examples. The Ethologists carefully observed animals to describe what these fixed action patterns of behaviour were; their adaptive value to the species; the trigger or “releasing stimuli” that precedes the behaviour; the embodied physiological mechanisms that produce that behaviour; and what supportive learning is necessary to shape the behaviour. The Ethologists found there is virtually always some amount of experience and resulting reinforcement to shape the fixed action pattern beyond the initial inclination.

The importance of Ethology is that neurological “hard wiring” is an additional consideration to the Behaviourist’s various strategies of reinforcement in the development of behaviour. This study of the capabilities of numerous species and their species specific behaviour underscores the importance of having the necessary equipment for certain behaviours: flying requires an aerodynamic physique and an appropriate nervous system to use it; just as literacy (a human capability) requires the ability for symbolic perception and the appropriate nervous system to interpret, and use it. This means that each species, even each individual, including mankind, is limited in capabilities by their nature.

CONDITIONING

Pavlov’s Experiments

Ivan Petrovich Pavlov (1849-1936) in 1927 performed an experiment that demonstrated the conditioned reflex. In this experiment a dog was prepared with a cannula in a saliva gland. The dogs were not fed and were hungry. In the first part of the experiment a light was lit just before a drawer of food was opened in front of the dog. Naturally, the saliva gland reacted to the sight of food. At the same time the dog associated the light with the appearance of the food. After a short training period the dog would salivate at the sight of the light without the appearance of the food. The lessons to be learned are:

1. Glandular secretion can be due to stimulation through association in the cortex. An associated stimulus can lead to the secretion of glands, including those associated with emotions. Later understanding has lead us to believe that this is the case with virtually every stimulus. Even our thinking can produce a variation in hormonal levels in the blood. The hormones in our blood are one method our internal cells communicate what we think is happening in our environment. There are hormonal combinations for stress, freight, sex, success, and failure.

2. The dog's responses were inappropriate to the associated signal (the light). The dogs responded to the signal (light) with ptyalin in the saliva, pepsin in the stomach, and trypsin in the pancreas. This association, or sequential memory, is a very primitive capability. In this case, the dog was shown to have sequential memory in that it quickly learned that the appearance of a light was followed by food. Even a flat worm can learn some significant sequences. An electric shock preceded by the appearance of a light will eventually cause the animal to curl up at the appearance of the light without the electric shock. The flat worm is avoiding the shock.
3. The dog has specialised cells that detect the need for food and signal that message to the dog's mind. Since the time it was a puppy, the dog has learned from an inclination (fixed action pattern) to identify food with his nose, eyes, and motor skills. Pavlov's dogs had their food appreciation shaped to value the Russian dog food and started preparing for eating by exciting their digestive glands and attempting movement against the harness towards the food tray. The dog's cortex associated the light with the food as sequential events, and eventually prepared for the food with the appearance of the light.

Pavlov demonstrated this in several digestive glands. In other situations, danger causes the secretion of epinephrine, the hormone of fear. A threatening situation causes the secretion of norepinephrine, the hormone of aggression and anger. An attractive potential mate causes the secretion of ethylphenylamine from the hypothalamus and causes feelings of love. It should be noted that hormones both affect behaviour and are produced by behaviour. For instance, exercise increases testosterone in females as well as males. Then the testosterone sensitises aggressive parts of the brain and builds muscle tissue. Of course, these were not the conclusions that Pavlov drew. But there is no reason that in the light of subsequent findings regarding hormones and learning that we should not apply more sophisticated interpretations than he possibly could.

INDIVIDUAL LEARNING VARIATION

Every individual human falls within a variety of capabilities that makeup their learning performance characteristics. The secret to a good education is to expose that individual to as many subjects as possible so they might discover where their talents lie. A great education allows a person to reach their ultimate potential capabilities. Some of these capabilities, such as the acquisition of language, or the establishment of attachments and bonding, emerge during a window of time during the maturational development of the individual while others are able to be developed at anytime during the life of the individual. Even though there have been many misdirected biases and prejudices established over time, such as the idea that males are better at math than females, each of us, even the most gifted, have limitations. We are limited by our established collection of knowledge, by what we are inherently able to comprehend, by what we believe is our ability to comprehend, and by our fears.

When I was in graduate school, and behaviourism was the answer, I was required to virtually memorise a book of hundreds of animal experiments trying to decipher what learning was all about. One experiment jumped off the page, even though I have not seen it since. This seemingly insignificant experiment explained the difference between positive (reward) and negative (fear based) reinforcement. It is a simple experiment with two white rats in two shoe box sized containers. The idea was to train them to jump through a sliding door from one side of the box to the other. In both cases there was a light turned on just prior to the delivery of the reinforcement. The only difference was the motivation for learning this trick. For rat “A” they placed food on the other side of the door. This encouraged him to make the jump. For rat “B” they electrified a grid under his feet. He jumped around till he realised that he could escape the shock by jumping to the other side. This encouraged him to make the jump. It took both rats about the same number of trials to learn the trick. The rat that learned to jump for the food was experiencing “positive reinforcement” while the rat that jumped to avoid the shock was experiencing “negative reinforcement”.



The important lesson in this experiment was shown during what they call the “extinction trials”. Why they chose that name, I do not know. During the extinction trials, the light was turned on, but no food or shock was delivered. Soon the rat jumping for food no longer associated food with the light and began sniffing around for another solution. However, the rat jumping to avoid the shock continued to jump and jump for many trials after the shock was turned off. The animal’s behaviour was rigid and did

not change even though the shock was no longer present. The rat which learned with positive reinforcement was adaptive and flexible while the rat which learned with negative reinforcement became rigid and fixed. Similarly to Pavlov's dogs, there is an inappropriate response to an associated signal (the bell or light) when the original stimulus is gone. One of the characteristics of neurotic behaviour is that it takes place inappropriately. Neurotic behaviour and neurotic emotional reactions can often be traced back to a negatively reinforced experience.

Another important aspect of learning was introduced by B. F. Skinner who illustrated general learning principles with pigeons. Skinner probably was too enthusiastic about the application of his concepts. The Behaviourist movement overlooked the fixed action patterns of behaviour and attributed everything to reinforcement. As you are introduced to these few experiments of the many he produced, you will easily be able to identify with the experience of the pigeon.

In all his experiments Skinner deprived the pigeons of food for twenty-four hours to insure the need for food would be a good motivation. The animal was then placed in a cage with a small target disc attached to a lever in such a way that when the pigeon pecks at the disc, a pellet will drop into a nearby tray. The number of pecks on the disc per minute is recorded. As would be expected, the hungry pigeon discovers the disc pecking behaviour, and the number of pecks per minute increases over time to create an upward inclined graph. This is called the "learning curve".

There are three factors that he found to affect the learning curve.

1. One to one reinforcement. In this experiment the pigeon received one pellet per peck. The learning curve was fast, but the animal was soon satiated and the curve crashed.
2. Fixed interval reinforcement. In this experiment the pellets were released at regular intervals, such as one every fourth peck. The learning curve is not as steep, but the pigeon had to peck more times before it was satiated. The animal hurried the process along by pecking faster. Do you want faster work? Pay people by commission.
3. Random-interval reinforcement. The intervals were selected from a list of random numbers; so, there was no way to predict how many pecks will be necessary to get the next pellet. In this study the learning curve was the least rapid, but the pecking behaviour is eventually the most vigorous by far. This is the principle of the addiction to the unpredictable fruits of gambling. There are other factors involved, of course, but this is one.

All these aspects of learning are motivated by the fulfilment of needs. Each need is initially detected by some neuro-sensitivity that triggers a fixed action pattern. In time, this pattern is shaped by neurologically successful need satisfaction and the resulting repetition. This process can often be a long, subtle evolution of development, beginning in maturation and continuing over a lifetime. Neurobiology tells us that the continued use of a neural pathway strengthens that pathway with each use. This is why we learn

and remember things in school: repetition. Meeting needs on a continual basis is the evolution of establishing pathways. Sometimes need fulfilment is not so immediate. Such as in school. Our frontal cortex has something to do with our ability to keep our eye on a distant goal.

THE ROLE OF NEEDS

The fulfilment of needs is our motivation for learning. The struggle to fulfil our needs is the driving force of our life as we have evolved. The fulfilment of these needs holds the fabric of human society together by motivating human communication and community work. Needs are the essence of what we pursue, what we strive for, what reinforces our efforts. It stands to reason, in a fruitful society, that each one of us should feel the desire and the possibility of successfully fulfilling our needs. Karl Marx postulated that it is the frustration of need fulfilment that leads to aggression and revolt. The association of stress and aggression has been biologically linked. Some examples of our needs are: food, shelter, waste removal, activity, sex, safety, belongingness, power, achievement, knowledge, encouragement, beauty, and creativity. The order is not important. Everyone has these needs, but to different degrees depending on environment, culture, and individual variation. All these roads to need fulfilment are shaped from some original fixed action pattern or heritable inclination that emerges during the maturational development of the individual. The achievement of need fulfilment is the reinforcement necessary for learning.

THE EFFECT OF GENERALISATION

Generalisation is a vague concept describing a combination of neurological events. We are generalising when we see an object in different positions and we can recognise it as the same object. This is known as “gestalt”. We generalise when we have an emotional response to a specific subject through experience and then ascribe that emotion to all other similar subjects. As we have shown, this would be more lasting with a negatively reinforced experience than with a positively reinforced experience. An example of generalisation is people who say, “all women are the same!” as a person ascribes a characteristic to every member of a class. On the other hand, generalisation could be the prejudicial application of a characteristic of a class to a specific member.

Here we have examples of two natural neurological events, both propensities are evolved through time by nature because they are beneficial to survival. Once again we see problems of rigidity when negative reinforcement is involved. This generalisation characteristic is the basis for racism and prejudice. It has been a primary ingredient in the dehumanisation of a class of people as a preliminary preparation for their eradication in warfare. On the other hand it is the basic requirement of a positive philosophy, or religion, that includes all people.

READING AND WRITING

This is an interesting subject because of its correlation with general intelligence and education. I say “correlation” because the two do not always go hand in hand. For instance Charlemagne, an important leader in the eighth century Europe, was credited with many letters but was noted by his biographer as not being able to write his name. On the other hand, I have known a few families with children who learned to read, with little encouragement, a couple of years before they entered school.

Reading and writing are, without a doubt, tied to the maturational development of specific parts of the brain. This is part of the post birth recapitulation of humanity’s evolutionary development through the threshold of literacy. Most children begin to learn to read, with instruction, by age six. About five percent of the children learn to read on their own by age four. Twenty-five percent will not learn to read till age ten; but, will have caught up to the others by age twelve. A small percentage of children will never learn to read. The lack of acceptance of this simple principle of the maturing brain has cost untold thousands of academically shattered lives. Since the pressure has been on the schools to show results by second grade, teachers have been trying every method they can think of to cram reading down the throats of students who are slow to mature. As a result, when they finally do mature, they don’t want anything to do with reading.

Reading readiness tests are very accurate and well developed. Children who are determined to be unready to read should be deferred from reading instruction. By fourth grade virtually all children are ready for reading instruction and by sixth grade those children are performing as well as the other students. Recognising this phenomenon, a talented teacher, who introduced me to these realisations, started a private learning facility specialising in reading instruction to only fourth, fifth and sixth graders. He was careful to use teaching methods that were different from their previous schools. The teaching methods were actually no more effective, but were able to introduce the student to reading in a way that the student had not been previously negatively conditioned.

With people who cannot draw, cannot spell, cannot sing, cannot learn foreign languages, it is amazing to me that it is difficult for some people to understand there are people who cannot read, or are poor readers. Low reading ability is a perfectly natural and normal occurrence for a percentage of the population just as is advanced reading ability. For the poor readers, the written word makes as much sense as a blackboard full of advanced calculus does to most of us. The percentage of non-readers and poor readers is growing; but, it is not the fault of the schools or the parents. Reading is not a heritably guaranteed ability. There are powerful political forces which are unfairly punishing teachers and schools who happen to be populated by children of low reading ability.

Some would like to think that anyone can be anything they want to be; that every human is capable of becoming a college graduate, a physician, a classical musician, a fighter pilot, or an author. Unfortunately, this quixotic idea comes under the heading of

“unrealistic idealism” for it is opposed to all evidence. Every attempt to measure the performance of any human behaviour (when displaying performance measure against numbers of people) always produces some sort of Gaussian curve (a bell shaped curve) as would be predicted by the requirement of variation in biological evolution. The majority of the performers fall in a range around the average while the high and low performers fall off in number.

TRANSFER OF LEARNING

Transfer of learning is the study of the dependency of human conduct, learning, or performance on prior experience. The notion was originally introduced as *transfer of practice* by Edward Thorndike and Robert S. Woodworth. They explored how individuals would transfer learning in one context to another context that shared similar characteristics – or more formally how “improvement in one mental function” could influence another related one. Their theory implied that transfer of learning depends on the proportion to which the learning task and the transfer task are similar, or where “identical elements are concerned in the influencing and influenced function”, now known as *identical element theory*. Transfer research has since attracted much attention in numerous domains, producing a wealth of empirical findings and theoretical interpretations. However, there remains considerable controversy about how transfer of learning should be conceptualised and explained, what its probability occurrence is, what its relation is to learning in general, or whether it may be said to exist at all.

Most discussions of transfer to date can be developed from a common operational definition, describing it as the process and the effective extent to which past experiences (also referred to as the transfer source) affect learning and performance in a current novel situation (the transfer target) (Ellis, 1965; Woodworth, 1938). This, however, is usually where the general consensus between various research approaches ends.

There are a wide variety of viewpoints and theoretical frameworks apparent in the literature. For review purposes, these are categorised as follows:

- A taxonomical approach to transfer research that usually intends to categorise transfer into different types;
- An application domain-driven approach by focusing on developments and contributions of different disciplines that have traditionally been interested in transfer;
- The examination of the psychological scope of transfer models with respect to the psychological functions or faculties that are being regarded; and
- A concept-driven evaluation, which reveals underlying relationships and differences between theoretical and empirical traditions.

TRANSFER TAXONOMIES

Of the various attempts to delineate transfer, typological and taxonomic approaches belong to the more common ones. Taxonomies are concerned with distinguishing

different types of transfer, and therefore less involved with labelling the actual vehicle of transfer, *i.e.*, what is the explanatory mental unit of transfer that is carried over. Hence, a key problem with many transfer taxonomies is that they offer an excessive number of labels for different types of transfer without engaging in a discussion of the underlying concepts that would justify their distinction; *i.e.*, similarity and the nature of transferred information. This makes it very difficult to appreciate the internal validity of the models.

The following table presents different types of transfer, as adapted from Schunk (2004, p. 220).

Type	Characteristics
Near	Overlap between situations, original and transfer contexts are similar.
Far	Little overlap between situations, original and transfer settings are dissimilar.
Positive	What is learned in one context enhances learning in a different setting.
Negative	What is learned in one context hinders or delays learning in a different setting.
Vertical	Knowledge of a previous topic is essential to acquire new knowledge.
Horizontal	Knowledge of a previous topic is not essential but helpful to learn a new topic.
Literal	Intact knowledge transfers to new task.
Figural	Use some aspect of general knowledge to think or learn about a problem.
Low Road	Transfer of well-established skills in almost automatic fashion.
High Road	Transfer involves abstraction so conscious formulations of connections between contexts.
High Road/ Forward	Abstracting situations from a learning context to a potential transfer context.
Reaching	
High Road/ Backward	Abstracting in the transfer context features of a previous situation where new skills and knowledge were learned.
Reaching	

Apart from the effect-based distinction between negative and positive transfer, taxonomies have largely been constructed along two, mostly tacit, dimensions. One concerns the predicted relationship between the primary and secondary learning situation in terms of the categorical overlap of features and knowledge specificity constraints. The other concerns general assumptions about how transfer relationships are established, in terms of mental effort and cognitive process.

THE EFFECT-PERSPECTIVE: POSITIVE VS NEGATIVE TRANSFER

Starting by looking at the effect side of transfer – in terms of the common performance criteria, speed and accuracy – transfer theories distinguish between two broad classes that underlie all other classifications: *negative* and *positive* transfer. Negative transfer refers to the impairment of current learning and performance due to the application of non-adaptive or inappropriate information or behaviour. Therefore, negative transfer is a type of interference effect of prior experience causing a slow-down in learning,

completion or solving of a new task when compared to the performance of a hypothetical control group with no respective prior experience. Positive transfer, in contrast, emphasizes the beneficial effects of prior experience on current thinking and action. It is important to understand that the positive and negative effects of transfer are not mutually exclusive, and therefore real-life transfer effects are probably mostly a mixture of both. Positive transfer: transfer of learning or training is said to be positive when the learning or training carried out in one situation proves helpful to learning in another situation. Examples of such transfer are:

- The knowledge and skills related to school mathematics help in the learning of statistical computation;
- The knowledge and skills acquired in terms of addition and subtraction in mathematics in school may help a child in the acquisition of knowledge and skills regarding multiplication and division;
- Learning to play badminton may help an individual to play ping pong (table tennis) and lawn tennis.

THE SITUATION PERSPECTIVE: SPECIFIC VS GENERAL, NEAR VS FAR TRANSFER

The situation-driven perspective on transfer taxonomies is concerned with describing the relation between transfer source (*i.e.*, the prior experience) and transfer target (*i.e.*, the novel situation). In other words, the notion of novelty of the target situation *per se* is worthless without specifying the degree of novelty in relation to something that existed before. Butterfield and Nelson (1991), for example, distinguish between *within-task*, *across-task*, and *inventive* transfer. A similar classification approach reappears in many situation-driven transfer taxonomies (*e.g.*, *similar* vs. *different* situations, *example-to-principle* and vice versa, *simple-to-complex* and vice versa) and can be noted as distinctions made along the *specific* vs. *general* dimension. Mayer and Wittrock (1996, pp. 49ff.) discuss transfer under the labels of general “transfer of general skill” (*e.g.*, “Formal Discipline”, Binet, 1899), “specific transfer of specific skill” (*e.g.*, Thorndike’s, 1924a, b, “identical elements” theory), “specific transfer of general skill”, and “meta-cognitive control of general and specific skills” as a sort of combination of the previous three views.

Haskell’s (2001) taxonomy proposes a more gradual scheme of similarity between tasks and situations. It distinguishes between non-specific transfer (*i.e.*, the constructivist idea that all learning builds on present knowledge), application transfer (*i.e.*, the retrieval and use of knowledge on a previously learned task), context transfer (actually meaning context-free transfer between similar tasks), near vs. far transfer, and finally displacement or creative transfer (*i.e.*, an inventive or analytic type of transfer that refers to the creation of a new solution during problem solving as a result of a synthesis of past and current learning experiences). Both near and far transfer are widely used terms in the literature. The former refers to transfer of learning when task and/or context change slightly but

remain largely similar, the latter to the application of learning experiences to related but largely dissimilar problems.

THE PROCESS PERSPECTIVE

The *specific vs. general* dimension applies not just to the focus on the relation between source and target, *i.e.*, from where to where is transferred, but also to the question about the transfer process itself, *i.e.*, what is transferred and how. *Reproductive vs. productive* transfer are good examples of this type of distinction, whereas reproductive transfer refers to the simple application of knowledge to a novel task, productive transfer implies adaptation; *i.e.*, mutation and enhancement of retained information.

A similar dichotomous distinction is the one between *knowledge* transfer and *problem-solving* transfer (Mayer and Wittrock, 1996). Knowledge transfer takes place when knowing something after learning task *A* facilitates or interferes with the learning process or performance in task *B*. Knowledge used is referred to by many different terms, such as declarative or procedural types (Anderson, 1976), but it means that there are representational elements that suit *A* and *B*. Problem solving transfer, on the other hand, is described as somewhat more “fluid knowledge” transfer, so that experience in solving a problem *A* helps finding a solution to problem *B*. This can mean that the two problems share little in terms of specific declarative knowledge entities or procedures, but call for a similar approach, or solution search strategies (*e.g.*, heuristics and problem solving methods).

The issues discussed in problem-solving transfer literature are also closely related to the concepts of *strategic* and *theoretic* transfer (Haskell, 2001, p. 31), and cognitive research on analogical reasoning, rule-based thinking and meta-cognition. Indeed, far transfer can be considered as the prototypical type of transfer, and it is closely related to the study of analogical reasoning. Within the problem-solving literature the distinction between specific and general methods is made mostly with reference to Newell and Simon’s (1972) strong vs. weak problem solving methods.

Another concern that is frequently addressed in transfer taxonomies is the question of conscious effort. *High-road vs. low-road* transfer (Mayer and Wittrock, 1996; Salomon and Perkins, 1989) expresses a distinction between such instances of transfer where active retrieval, mapping, and inference processes take place, as opposed to those instances that occur rather spontaneously or automatically. Hence, low-road transfer concerns frequently employed mental representations and automated, proceduralised knowledge, and occurs preferably in near transfer settings. In contrast, high-road transfer is more conception-driven, and requires cognitive and meta-cognitive effort.

TRADITIONAL FIELDS OF TRANSFER RESEARCH

There are a nearly unlimited number of research fields that share some applied interest into the study of transfer, as it pertains to learning in general. Three fields that contributed

in most substantial ways to the progress of transfer research, both from a conception and empirical point of view, are the fields of education science, linguistics, and human-computer interaction (HCI). In fact, most transfer research has been conducted in reference to one of these applied settings, rather than in basic cognitive psychological laboratory conditions.

EDUCATION SCIENCE: TEACHING FOR TRANSFER

Due to their core concern with learning, educational science and practice are the classic fields of interest regarding transfer research, and probably the prime target for the application of theories. Transfer of learning represents much of the very basis of the educational purpose itself. What is learned inside one classroom about a certain subject should aid in the attainment of related goals in other classroom settings, and beyond that it should be applicable to the student's developmental tasks outside the school; the need for transfer becomes more accentuated. This is because the world educators teach in today is different from the world they themselves experienced as students, and differs equally from the one their students will have to cope with in the future. This is why the promotion of instruction designed to educate students on the process of enquiry so that they can apply it readily for themselves is recommended.

By nature of their applied interest, educationalists' main concern has been less with the question of how transfer takes place, and much more with under what conditions, or, that it happens at all. The basic conviction that student's learning and achievement levels depend primarily on learning and achievement prerequisites, has constituted a central part in educational learning theories for quite some time. The major focus in educational transfer studies has, therefore, been on what kind of initial learning enables subsequent transfer: *teaching for transfer*. Research on learning and transfer has identified key characteristics with implications for educational practice.

IMPLICATIONS FOR SCHOOL PSYCHOLOGY PRACTICE

It is important for teachers to facilitate the transfer of learning to their students and not merely regurgitate the things they learned during their university experience. However, many instructors have been out of the classroom for a decade or more and the study of transfer was not a part of their curriculum. Therefore, it is important that these instructors understand the process of transfer so that they can design their lesson plans with it as a goal. School psychologists have a general understanding of how pedagogical applications of psychology in the classroom affect learning. Therefore, it is important when considering systems of support using the Response to Intervention (RTI) model that school psychologists first assess through observation if a teacher's current pedagogical practices promote transfer. If current practices do not, this could prevent the transfer of classroom knowledge which can affect a child's behaviour and their perceived intellectual ability.

FROM FORMAL DISCIPLINE TO META-COGNITION

Educational transfer paradigms have been changing quite radically over the last one hundred years. According to the doctrinaire beliefs of the Formal Discipline (Binet, 1899) transfer was initially viewed as a kind of global spread of capabilities accomplished by training basic mental faculties (*e.g.*, logic, attention, memory) in the exercise of suitable subjects, such as Latin or geometry. With the turn of the 20th century, learning, and therefore transfer of learning, was increasingly captured in behavioural and empiricist terms, as in the Connectionist and Associationist theories of Thorndike (*e.g.*, 1932), Guthrie (*e.g.*, 1935), Hull (*e.g.*, 1943), and Skinner (*e.g.*, 1938). Thorndike (1923, 1924a and b) attacked the Formal Discipline empirically and theoretically and introduced the theory of “identical elements”, which is probably still today the most influential conception about transfer (Thorndike, 1906; Thorndike and Woodworth, 1901a, b and c). Thorndike’s belief that transfer of learning occurs when learning source and learning target share common stimulus-response elements prompted calls for a hierarchical curricular structure in education. “Lower” and specific skills should be learned before more complex skills, which were presumed to consist largely of configuration of basic skills. This small-to-large learning, also referred to as part-to-whole or vertical transfer, has been popular with theories of learning hierarchies (Gagné, 1968).

It has later been challenged from conceptualistic point of views, which argue that learning is not just an accumulation of pieces of knowledge (*i.e.*, rote memorisation), but rather a process and product of active construction of cognitive knowledge structures. Knowledge, from a constructivist perspective, was no more believed to be a simple transfer by generalisation to all kinds of situations and tasks that contain similar components.

The critical issue was the identification of similarities in general principles and concepts behind the facades of two dissimilar problems; *i.e.*, transfer by insight. This idea became popular in the Gestaltists’ view on transfer (*e.g.*, Katona, 1940), and, in combination with growing interest in learners as self activated problem-solvers (Bruner, 1986), encouraged the search for abstract problem-solving methods and mental schemata, which serve as analogy-enhancing transfer-bridges between different task situations.

Emerging from these developments, a new theme started to dominate educationalists’ research in transfer: meta-cognition (Brown, 1978; Brown and Campione, 1981; Campione and Brown, 1987; Flavell, 1976). In contrast to classical knowledge forms like declarative and procedural knowledge, different types of meta-knowledge and meta-cognitive skills such as strategic knowledge, heuristics, self-monitoring skills, and self-regulation quickly became the road to learning and transfer. Characterised as self-conscious management and organisation of acquired knowledge (Brown, 1987) it is evident that meta-cognitive awareness of task features, problem structures, and solution methods makes relations between different situations cognitively salient: only an individual who learns from learning, learns for future learning. Soini (1999) developed on the same core ideas an examination of the preconditions for active transfer. Her

emphasis is on the active and self-reflected management of knowledge to increase its accessibility. To some researchers, meta-cognition and transfer have become so entangled that the argument was generated that only the measurement of positive transfer effects truly supports inferences that meta-cognitive learning has taken place.

THE GENERALITY PREDICAMENT: RETURN TO THE SPECIFICITY VIEW

Ever since the introduction of the meta-knowledge theme in education science, transfer discussions have been oscillating between the position taken by those representing the meta-cognitive view and those who stress that generic knowledge forms alone do not allow an effective transfer of learning. When knowledge stays “on the tip of the tongue”, just knowing that one knows a solution to a problem, without being able to transfer specific declarative knowledge (*i.e.*, know-what) or automated procedural knowledge (*i.e.*, know-how), does not suffice. Specific teaching of the cognitive and behavioural requisites for transfer marked in principle a return to the identical element view, and can be summarised with Dettermann’s (1993) conclusion that transfer does not substantially go beyond the restricted boundaries of what has been specifically taught and learned. The basic transfer paradigms in educational psychology keep replicating themselves, and fundamental promotion of transfer itself is seen to be achievable through sensibilisation of students by creating a general culture and “a spirit of transfer” inside the classroom on the one hand, and by allowing concrete learning from transfer models on the other (Haskell, 2001).

NATURE OF LEARNING/CHARACTERISTICS OF LEARNING

Firstly, All Living is Learning, *i.e.*, the individual is in active relation with his environment.

Secondly, It results in Change in Behaviour. We note a change in the planarian’s response to the light. It is a change of behaviour influenced by previous behaviour. It is any activity that leaves a more or less permanent effect on later activity.

Thirdly, Learning is an Adjustment. Most learning in children consists in modifying, adapting, and developing their original nature. In later life the individuals acquire new forms of behaviour.

Fourthly, It comes about as a result of practice. This characteristic eliminates sources of change such as illness and maturation. Potent effects of motivation on behaviour are worth considerable here.

Fifthly, Learning is Universal. Every creature that lives learns. Man learns most. Human children have the longest period of immaturity and helplessness and hence the longer period for opportunity for learning. The human nervous system is very complex, so are human reactions and so are human acquisition.

Sixthly, Learning is a relatively Permanent Change. After a rat wake up from his nap he still remembers the path to the food. Even if you have been on a bicycle for years, in just a few minutes practice you can be quite proficient again. Seventhly, Learning is Growth. It is never ending growth. At reach stage the learner acquires new visions of his future growth and news ideals of achievement in the direction of his effort. Each achievement forms are basis of a fresh endeavour and thus the constant urge of his soul to newer and higher ideals of work and achievement is progressively fulfilled.

Lastly, Learning is not directly observable. The only way to study learning is through some observable behaviour. Actually, we cannot observe learning; we see only what precedes performance, the performance itself, and the consequences of performance.

FACTORS AFFECTING LEARNING

Learning is a comprehensive process. The success of this process depends not only upon the effective teaching but also upon so many group factors. The factors related with the learner, the teacher, and the environment are seen responsible as a determinant of quantity, nature and speed of learning. The main factors that affect learning may be mentioned as below-

- A) Factors belonging to the learner.
- B) Factors belonging to the teacher.
- C) Environmental Factors.

Factors Belonging to the Learner

The factors affecting learning related with the learner may be specified as under:-

The Child Himself

Child is the pivot of any learning activity. All activities rotate around him. As the aim of education is all round development of child's personality, from this point of view, the activities based and carried on account of child's needs, interests, attitudes, aptitudes, potentialities, capabilities, individuality, intelligence, etc., may be effective and of immense importance. Child is the basis of teaching- learning process, hence, ignorance of child at any stage will make all the learning process futile and more imagination, nothing else. Hence, it is suggested that each and every activity related with the learning should be carried out in accordance with child's interests, needs and his physical and mental capacity. This all is possible only when the teacher has a deep insight to understand his pupil thoroughly.

His Intelligence

Intelligence has been found to be one of the main factors that influence learning. The proof for this factor is very commonly found in the class-rooms. The schools which

are meant for children of average ability usually are not able to help a retarded child. The teacher does not have the time to spend on this child who needs more help than an average child. Similarly, these schools are of no help to a child who is more intelligent than the average one. Cases have been found of a child indulging in delinquent acts simply because they are far above average in intelligence but the school that they attend is poor in stimulation. As a result these children satisfy their intellectual curiosity by indulging in delinquent and thrilling tricks. So we may say that holding the conditions of learning constant, the level of intelligence of a child makes a difference to the amount of their learning output on the classroom.

His Age

The test of learning given to subjects of a wide range of ages have shown that learning efficiency increases with age to a certain extent after which it stays stationary for some time and ultimately tends to decrease. This phenomenon can be understood easily if we keep in mind the developmental curves. Maturation of individual, experience also accounts for a difference both of which are linked up with age. So we find that children are speedier and more efficient at learning tasks as they grow older. As we get into adulthood and old age we find the output of the subjects going down because of the fact that though their experience has increased they cannot keep up the speed. So that when the tasks have a time limit on them, the older subjects cannot do so well as the younger subjects can. However, the experiment on learning is a continuous process and it spreads over the whole life with a desire in the individual to learn more and more. The only difference that exists is caused by the type of tasks given for learning.

His Will to Learn

The will to learn is always labelled as a factor determining the amount of learning. It is believed that in order to be able to grasp and retain a certain material the individual must have an inner urge to learn, a drive that can motivate him into learning. This motivation is determined by the interests, desires and the purpose of a particular individual. All these three are more or less developed with the help of a teacher. An intelligent and effective teacher, teacher helps his pupils in developing wide interests and habits, desire to pursue certain activities and at large a purpose in life. If a teacher is successful in helping his pupils to develop worthwhile interests and habits and a desire to understand things, he has helped them in becoming good learners.

Guidance

Usually trial and error is considered to be the method of learning. When faced with a new task an individual attempts it, in case it is a failure, he tries again, perhaps using a different technique this time. After a few trials and errors he evolves the correct method of tackling the task. In practice we find that it allows the hit and miss method,

a pupil wastes a great deal of his time and while facing failure, he goes through a certain amount of tension and frustration. A teacher, through his guidance, can save his pupils a lot of frustration. He can help the child in making the right trials and avoiding the ones which would not yield any fruitful results. In the initial stages of learning, a limited amount of guidance can help the student avoid unnecessary errors. The word “limited” is mentioned here because very often teachers are found to indulge in too much of guidance as far as the learning is concerned. The usual result of excessive guidance is failure of the purpose. Proper guidance should aim at developing initiative in the learner and discourage the tendency to seek ready made solution.

Educational Background

Educational background of the learner is an important factor that affects his learning. Student may be backward from educational point of view as a general case or specific one. Some of the students are weak in most of the subjects while some students are weak in one or two particular subjects.

The first category of the students is termed as general backwardness. While second category of students is known as specifically backward. Educational background of generally backward students is more challenging than the educational background of specifically backward children. It is very common that if a particular student is weak in any subject, he will feel difficulty in learning new knowledge of that subject. On the contrary if one possesses superior ability, beyond normally, it will be easier for him to gain new knowledge in a very comfortable and effective manner. Thus, educational background contributes a lot in further learning.

Health Status of Child

The health of the student should be very sound. A mal-nourished, ill or otherwise physically handicapped, cannot realise his potentialities as a learner. A pupil who is unhappy, discouraged or otherwise does not possess the balanced emotional tone is sure to be handicapped in his learning attempts. It is rightly said that “a sound mind rests in a sound body” from this point of view a student should be physically and mentally fit so that he may learn easily and comfortably.

Child’s attention, interests, concentration, etc., have direct link with the physical and mental health level of the students. Any sort of mental illness or deficiency or abnormality, physical illness or mental tension and frustration will affect directly students learning. Students who are physically and mentally feel tired very easily. They fail to concentrate upon their studies for a long and get easily disturbed and depressed. They feel boredom and make very few efforts to get success in any activity. They feel complex and always complain of headache and other poor bodily symptoms. Thus, it is the first and foremost duty of the parents and the teachers to take note of their ward’s physical and mental health seriously.

Motivation

Motivation is one of the basic principles for effective teaching, because no purposeful learning can take place apart from it. The human mind cannot absorb knowledge like a sponge. In order to learn, a person must do some activity.

The urge to activity shows itself along different instinctive tendencies. Successful motivation depends on the successful use of these natural powers and tendencies to action. The teacher who identifies and intensifies motives in learners, is able to arouse interest in them.

He succeeds in setting up an excellent learning situation. It is said that “as is the motivation, so is the learning”, or “learning will proceed best, if motivated”. These sentences clearly depict the importance of motivation in the process of learning.

Thus, it is the first and foremost duty of the teacher to create interest in the child before starting anything new. Because, in the absence of motivation all the efforts made by the teacher will go waste. Proper motivation helps in making the desire for learning strong.

Attitude of the Learner

Learning has ever been the chief activity of man. Each generation after something new to cultural heritage hands it down to the succeeding generation. For several centuries the heritage has been so large and so important that society has given primarily to the school the task of selecting, organising and presenting to the children the most valuable situation possible.

Life pattern of an individual in adulthood, his attitudes, habits, mental, moral and physical well being is largely determined by the forces which were prevalent in childhood.

Pope's couplet says——

Tie education forms the common mind;
Just as the twig is bent the tree's inclined.

Favourable or positive attitude is must to get success in any field of endeavour. Favourable attitude towards the job or work makes one more active and enthusiastic. The same is related with the students' class- room activities. If the student has positive attitude towards any subject he will grasp the knowledge imparted by the teacher interestingly and whole heartedly and on the contrary, if he is not involved in the subject he will hate the subject, as well as the teacher of that subject. Liking for anything is must in learning.

The same view is supported by Thorndike's law of learning namely, 'law of use or dis-use'. If the student has negative attitude towards the subject all efforts made by the teacher will go in vain.

Thus, it is the first and foremost duty of the teacher to make positive attitude of the student towards his entire academic activities.

FACTORS BELONGING TO THE TEACHER**Knowledge of Subject**

Teacher's knowledge related with his subject, his experience and ability has a direct bearing on the learning of the students. If a teacher does not possess deep knowledge of his subject he can't give much enough to his students. On the contrary if a teacher has full command over his subject and has a mastery over the subject matter, he will be capable in giving new knowledge to his student with full confidence and his teaching will be effective undoubtedly. Generally student like those teachers on the other hand, face very few disciplinary problems. On the other side, teachers who lack competency on their parts deal students in a negative manner and indulge themselves in non-academic activities or cheap politics in the college campus. Good teachers are praised by their students like anything.

Teacher's Behaviour

Teachers behaviour influence the learning of students directly. A teacher should inherit all essential qualities of a good teacher. Sympathy, co-operation, objectivity, sweet tempered, polite, etc., are all such traits that should reflect in teacher's behaviour always. These traits will help in making the environment of the school congenial and praise worthy. Student will feel relaxed in the company of such teachers and face with them difficulty in discussing their personal problems. Apart from it, if a teacher is rough and tuff in his behaviour, the students will not like his subject, his company and will be compelled to leave the class and moreover they may turn to be truants. Also, the teacher should be objective in his dealing with his students. He must be impartial, no subjective attitude. Because, if it all happens, the student will start hating their teaching and may develop a hatred attitude for him permanently in the near future. Thus, the teacher should not be biased while dealing so many activities related with school and classroom teaching.

Knowledge of Psychology

Every teacher must have extensive knowledge of psychology without which he can neither know the student nor set the stage for learning. The elementary and secondary school teachers are most concerned with the psychology of childhood, psychology of adolescence and educational psychology. The latter deals with the application of psychological facts in such a way that the educational growth of the individual is efficiently directed and controlled. So it is rightly said that knowledge of educational psychology is must for a teacher. In this context it is quiet apparent that until a teacher does not acquire proper understanding of the principles of psychology, it is very difficult for him to make use of these principles in the field of education. Hence it is quiet

justified that he should have a deep knowledge of most common concepts of psychology such as process of child development, heredity, individual differences, motivation, theories of learning, etc. Making use of all these a teacher can make his teaching effective. Further, in the classroom situation he has to deal a variety of students at the same time. To gain mastery over such situations and to handle problems of individuals it is unavoidable for a teacher to have a detailed knowledge of educational psychology otherwise he will be utmost failure with regard to so many classroom complex situations. It is only knowledge of psychology that makes teacher fully competent and enthusiastic in dealing with his students.

Methods of Teaching

The traditional methods of teaching were more formalised, conservative, teacher dominated as opposed to modern, more flexible pupil involved, learner centred methods. In modern era it is now recognised that the subjects should be taught as activities. This activity principle was sponsored by Dewey. Teaching methods have a direct link with learning process. Consequently each teacher has a unique method of teaching. Similarly, all the students cannot be taught by a single method. If the method of teaching is scientific in nature, it will help in making the teaching effective. Also the learning process will be easier and purposeful. Realising this very need of the students educationists have developed modern methods and techniques of teaching. Methods are Playway method, Learning by doing, Learning by observation, Project method, Heuristic method, Discovery method, etc. Apart from above methods a teacher has to adopt his unique teaching style. It is commonly seen that a teacher is academically and professionally sound but his teaching is not effective in the classroom. There may be so many reasons behind the same but the most sensed factor is teacher's personal problem orientation technique, in the absence of which a student does not keep pace with the teacher and hesitates in sharing his personal problem with the teacher. Hence, personal rapport which the teacher is also needed for the success of teaching.

Knowledge of Individual Differences

Knowledge of individual difference is must for a teacher. Each student has his own interests, attitudes, aptitudes, needs, potentialities, capabilities, values, etc., and on account of these some students grasp the knowledge imparted by their teachers easily while some grasp with some difficulty and some of the students totally fail to grasp anything given by the teacher. Generally a teacher has to face three types of students in his class- backward, normal, and gifted children. To plan teaching accordingly the needs of these students or to satisfy the needs of all these students at the same time is really a difficult and challenging task for the teacher. To keep pace with this problem the teacher plans his teaching strategies keeping in mind the average children. This helps a bit to backward as well as to the gifted children. Though both of these categories seem to be

dissatisfied with this approach of the teacher. But, apart from it there is no proper channel for the teacher that he may try. To go through such channels only knowledge of individual differences can help him to some extent.

Personality

Good and appealing personality is the basis of successful and effective teaching. From this point of view the personality of the teacher must be very attractive and influencing. He must create an impression or put a stamp on his students keeping appropriate balance between his deeds and actions. Students learn so many things indirectly from the teacher. It is thus that teacher is said to be the best motivator for his students. He forms an idol for his students. Students imitate his each and every part of his behaviour gladly. So it is on the part of the teacher how he creates an example for others. For this he has to be very much cautious in his behaviour and has to exhibit all those traits through his personality that are welcomed most by most of the societal members. Personality should be viewed as the entire qualitiveness of a person or as an integrated pattern of traits. Teacher should keep in his mind that personality is shaped by and inter –woven with the social environment and culture is the ground from which personality emerges. To shape students' personality he should always borne in his mind that personality which is just a bundle of ideas, attitudes and intelligence depends a good deal on the people with whom the individual constantly associates.

No Care to Foster Individuality

The aim of education should be to develop to the full of potentialities of every child in school in accord always with the general good of society. It is also to develop each individual into a happy well coordinated personality with socially desirable qualities. It is to produce good citizens who will have sense enough to judge of public affairs, discernment enough to choose the right officers, self control, enough to accept the decision of majority, honesty, enough to seek the general welfare rather than his own at the expense of the community, public spirit, enough to face trouble or even danger for the good of community. It would be no exaggeration if we call traditional school a prison, the pupils prisoners and the teacher superintendent of the prison where every thing was teacher centred and subject centred, where no attention was paid towards the psychology of child. The children, in the schools were just like parrots in the cage who were just like parrots in the cage who were made to cram certain bits of dis-organised, dis-connected facts of knowledge which had no value in their later life.

Lack of Personal Contact between Teacher and Taught

The size of classes in our schools is too large to permit close pupil and teacher contacts. Therefore teacher cannot look towards the individuals habits, attitudes aptitudes personal traits needs potentialities, etc. The increase in the size of classes has considerably

reduced personal contacts between teacher and students. Thus the training of character, inculcation of proper discipline have been seriously undetermined. Teaching in our schools had an aim of imparting only factual knowledge through certain exercises in which the teacher was an expounder, drill master and a disciplinarian and the child a passive recipient of verbal and visual impressions. It was assumed that the natural inclination of the child were against such subject matter. The intermediary factor which was used to bridge the gap between the teacher and taught was discipline, an external force that made the child to memories that subject matter. There was nothing that could cater the intellectual, emotional and physical growth of the child. The child could not develop into anything but a suppressed personally in the absence of proper rapport between the teacher and the taught.

No Emphasis on Co-curricular Activities

The purpose of traditional school was to import certain fragments of isolated knowledge through different subjects of the curriculum. All other activities were regarded as something for which the school was not legitimately responsible hence, were called extra curricular activities. But the modern school aims at all round development of the child. The school is responsible for the education of the whole individual, his physical, mental, emotional, moral and social self. To realise this aim the importance of introducing certain activities having relation to different subjects of curriculum become clear and hence named as co-curricular activities. Secondary Education Commission observes that, “The school has to formulate a scheme of hobbies, occupations and projects that will appeal to, and draw out, the powers of children of varying temperaments and aptitudes.” The co-curricular activities satisfy the needs of young people, promote meaningful learning and develop them into good citizens.

Narrow Curriculum Approach

Previously education in India was under foreign patronage. So it was divorced from our cultural, social and economic life. Secondary Education Commission writes – “The curriculum as formulated and as presented through the traditional methods of teaching does not give the student insight into everyday world in which they are living. When they pass out of school they feel ill-adjusted and cannot take their place confidently and competently in the community.”

The education given in our schools is purely of academic nature. It could contribute towards the development of cognitive aspect of personality only whereas the non-cognitive aspect was altogether ignored. The secondary Education commission observes that, “Education given in our schools is narrow and one sided and fails to train the whole personality of the student. For many decades it has provided only academic instructions. The non-cognitive aspects of his personality – his practical aptitudes, his emotions, his tastes were largely ignored.”

WHAT IS LEARNING BEHAVIOUR?

Learning Behaviour emphasises the crucial link between the way in which children and young people learn and their social knowledge and behaviour. In doing this the focus is upon establishing positive relationships across three elements of self, others and curriculum. The principles of learning behaviour have wide ranging implications for pupils, teachers, parents and other professionals. The principles can be applied to all children at any age and not just those perceived as being “difficult to manage”. They apply as much to teachers and their relationship with children as much as they apply to the children themselves. All NCfLB programmes and our partners are based on this relationship framework, a concept that has emerged from a review of theories of effective behaviour management. Tutors and trainees should recognise that a learning behaviour approach is fundamentally linked to a view that ‘behaviour’ in classrooms and whole schools/settings does not occur in isolation – it is the product of a variety of influences and not simply the product of a pupil’s unwillingness to behave or learn as required by the teacher (an approach which has frequently been referred to as an ‘ecosystemic approach’). In summary, the three sets of relationships which contribute to a culture/ethos of ‘learning behaviour’ are:

- Relationship with Self: a pupil who does not feel confident as a learner and who has ‘internalised’ a view that s/he is unable to succeed as a learner will be more likely to engage in the challenge of learning and (in consequence) may be more inclined to present ‘unwanted behaviours’
- Relationship with Others: all ‘behaviour’ needs to be understood as ‘behaviour in context’. Behaviour by pupils is triggered as much by their interactions with others (pupils, teachers or other adults in schools/settings) as it is by factors internal to the child.
- Relationship with the Curriculum: pupil behaviour and curriculum progress are inextricably linked. Teachers who promote a sense of meaningful curriculum progress in learning for each pupil will be more likely to create a positive behavioural environment.

In order to maximise the potential for learning schools should proactively facilitate Access, Participation and Engagement in learning through enabling teaching and supporting pupils to develop relationships with the curriculum, others and themselves supported by their School, Education Services, Family and Community.

ENVIRONMENTAL FACTORS AFFECTING LEARNING

HEREDITY

Children of bright parents tend to be bright and the children of very inferior parents are often inferior. Though, heredity follows the principle of regression that says that

traits when passing from parents to children tend to move towards the average. The genius son of a genius father is a very rare thing. Again, there is a tendency among the off-spring to be higher in the traits in which their parents are low and lower in which their parents are high. They move towards the average that is they regress. The reason for not showing consistency in the attainment is due to the complex nature of heredity, as heredity of a child is not solely determined by his immediate parents but from all his forefathers and ancestors. Apart from this heredity plays an important role in motivating and deciding the future course of learning according to the status of the family and heredity characteristics.

Example: Well known cricketers Yuvraj sing and Mohammad Kaif got full support from their family. Kaif's Father had been a good cricketer and he made tremendous efforts to get his son place in Indian squad. Yuvraj got coaching in the early days from his father itself. His father prepared pitch at his home for this purpose.

Environment

Our environment is our habitation in the fullest sense. Not only our physical surrounding but also the people around us, social customs and traditions, culture, education and training all constitute our environment. A number of studies conducted show that environment has remarkable influence on the learning of the individuals.

Example: I.Q. of the fathers corresponds to their occupational status and that the intelligence of the children would correspond to the occupation of the fathers.

Freedom and Self-discipline

Until recently, school discipline was regarded narrowly as merely a means of restraining pupils from creating disorder. So that learning activities may be carried effectively. Restrain commonly took the form of stern teacher imposed authority involving the extensive use of punishments. It was based on the assumption that the child is a bundle of evil qualities and that he should be corrected and moulded into shape by the iron rod of authority. Thus no child could escape beating. But now gone are the days of such beliefs and no truly progressive education or school can tolerate this authoritarian method of life. The education in every child centred school is carried on in an atmosphere of freedom. Freedom means that the child is given opportunity to think for him, to feel for himself and to get for himself. Opportunities are provided for using and developing all his powers. The child is given the chance to develop his own personality, to learn to discipline himself, build up his own character.

Recognition of Teachers' Authority

One should not accrue that the teacher has lost its importance in the modern time. Rather, modern school situations demand a more carefully selected and well prepared teacher than was necessary for the traditional schools. Infact education on an atmosphere

of freedom demands on teacher than does education guided by the old ideas of authority imposed. It demands a new atmosphere in the school and a new attitude in the teacher. It also demands new methods which advocate free activity of body, mind and soul of the pupils. The teacher instead of being a slave driver forcing the unwilling pupils to learn becomes a guide, makes himself one of the group, brings to the help of the group his larger experiences and thus helps the group in the search for truth, encourages them in every way to think, feel and work for themselves. Hence the teacher should be given a key place in the modern times. His authority of course is not that of an autocrat but of a leader, friend and a guide.

Nature Neglect

A teacher should have a complete knowledge of the biological laws. He should know how the whole organism works. He should realise that the whole child comes to school and not only his mind. The mind and the body are an entity. They cannot exist without each other. Latin proverb says that “Mens Sano in Corpore Sano” means “A sound mind in a sound body”. Mind lived in the body as in a clay cottage but it would be more true to say that the mind grows in the body as a plant grows in its soil. Poorer the soil, the more arrested the growth; richer the soil better the growth. Thus the mind and the body are independent, one influences the other. The body responds to various states of mind. Fear, joy, anger and all other emotions are reflected in the body. On the other hand mental activity uses up nervous energy which has to be replenished by the body. These considerations have natural bearing upon education and the teacher being conversant with these laws aims at the development of both, the body and the mind. In brief one should not neglect the laws of nature and should work accordingly.

Physical Environment of School

In general, the school environments present an educative atmosphere. The modern school site, building and equipments are adequate, safe, sanitary comfortable and attractive. It is only in the best environmental setting that the most effective learning takes place.

The improvement of the environment no doubt, lies in the hands of school authorities rather than teachers. Yet, the requisitions, request and suggestions by the teacher can correct the environment to a considerable extent. The teacher can suggest about lighting arrangements in the class, as due to defective lighting and ventilation arrangements the crowded classrooms lead to overheated conditions which reduce working capacity and encourage the spread of respiratory infections. Further, it will amount to restlessness, inattention and unsatisfactory work. It is a well know fact that individuals work much better in a room with an attractive atmosphere. Although a teacher cannot completely recondition a dull and dilapidated room yet a better planning and efforts can work wonders to improve its condition. Actually it all depends upon the upkeep of the room.

Careless disorder in the room, waste papers on the floor, and the untidy state of storage space, book shelves, teacher's table and pupils desks definitely give a shabby look to a room.

Thus, the rooms should be attractively decorated. Neatness, cleanliness, orderliness and suitable decoration in a room will induce pupils to take a pride in their room and should largely eliminate scribbling on the wall and desks. This all will prove to be a boosting environment for learning.

Gap between School and Community

The traditional schools are directed from social life and social activities. It has confined itself to book learning which has no utility for the child in his later life. But, the modern school represents the society in miniature. It has an intimate and organic relationship with the society.

It incorporates such experiences which are intimately connected with actual living and enable the individual to lead a successful life in society. Secondary Education Commission points out that there exists a "two way traffic" between the school and the larger community outside and the problems that arise in the home and community life and the realistic experiences gained there should be brought into school so that education may be based on them and be intimately connected with real life and on the other hand the new knowledge, skills and values acquired in the school should be carried into home life, solve its problems, to raise its standards and link up the teachers, parents and children in one compact and naturally helpful group.

Wholesome Experiences

Real education is the acquirement of experiences which will help the pupil to adjust himself to his environments. Modern school, therefore, is an experience giving institution where pupils learn things through purposeful experiences. The old conception of the pupils mind as tabula rasa or clean slate, upon which the teacher is to write are gone. Now the school is an institution where learning takes place through experience. The importance of an experience will be recognised by the child only when it responds to his needs and once the child feels its importance he will really be interested in it. Also the experience provided must have a clear objective. Actually we forget when we do not see clearly what we are supposed to be learning, because in that case we do not know what we are trying to be. We do not know our goal. Apart from this experience should have its value and meaning for use in daily life. We forget when we do not make use of what we have been asked to learn because we cannot make use of it in our daily living or in any other meaningful way.

Thus, wholesome experiences provide a basis for better learning as most of the things common to our daily life situations need some sort of experience this way or that way significant to find a solution of the problems.

WHAT LEARNING INVOLVES

Learning can be defined as ‘a relatively permanent change in behaviour which occurs as a result of practice or experience’. Here, the term ‘relatively permanent’ is usually taken to mean that the learned behaviour can be demonstrated at least a few hours after the learning process occurs. To put it another way, learning can be thought of as a process whereby a person acquires the ability to carry out a task or procedure of some sort in a natural and relatively effortless way, as if it were an innate part of his or her behaviour pattern.

The way in which such learning comes about will, of course, depend on the exact nature of the task or procedure that is being learned. In the case of a relatively simple cognitive task such as learning the names of the various parts of an object or system, for example, it may simply involve scanning the material sufficiently often for it to pass through the short-term memory (where material is stored on a purely temporary basis) into the long-term memory (where material is stored on a long-term or permanent basis). In the case of a more complicated learning task involving higher-level cognitive activities (such as analysis or synthesis) or a mixture of cognitive and non-cognitive activities, on the other hand, it will probably involve actually carrying out the task sufficiently often for the behaviour patterns needed to perform it without difficulty to become firmly established. This major key to effective learning is called repetition. The other major key relates to the concept of reinforcement, which refers to the consequences

of actions. Behaviour which is rewarded is likely to be repeated. Behaviour which is ignored is likely to fade. Behaviour which is punished tends to be suppressed in the short-term rather than changed in the long-term.

Another feature of the learning process is that both learned material and behaviour tend to disappear with time if the material or behaviour is not revisited periodically. The ability to recall simple factual material, for example, generally falls to something like 25 per cent of its original value roughly a week after the material was learned, and continues to fall thereafter until it eventually disappears completely unless the material is subsequently revisited. If the material is reviewed at suitable intervals, on the other hand (say, after a day, a week, a month, and so on) it will become progressively more firmly lodged in the long-term memory until it eventually becomes permanently fixed. Exactly the same is true of other learned skills, most of which require to be practised or used regularly if they are to become a permanent part of a learner’s repertoire.

Trends in Educational Psychology and Classroom Management

CONSTRUCTIVISM

A quiet, yet dynamic change in both instructional theory and practice has been emerging in many classrooms of America, a change called constructivism. Simply put, constructivism means that students construct their own understanding of the world. We're not talking about some simple change in a teaching technique but, rather, the way we think about knowledge acquisition and the assessment of that knowledge. Think about it for a moment. If students construct their own understanding, what does this imply for the teacher's role? One immediate conclusion is that anyone who thinks "teaching is telling" is sadly mistaken. Shuell (1996, p. 743) has neatly summarised the meaning of constructivism:

The learner does not merely record or remember the material to be learned. Rather, he or she constructs a unique mental representation of the material to be learned and the task to be performed, selects information perceived to be relevant, and interprets that information on the basis of his or her existing knowledge and existing needs. In the process, the learner adds information not explicitly provided by the teacher whenever such information is needed to make sense of the material being studied. This process is an active one in which the learner must carry out various operations on the new materials

in order for it to be acquired in a meaningful manner. The two key words here are active and meaning. Learners don't just sit there and copy what's put on the board or told to them. They take their own knowledge-that enormous reservoir of personal experiences they have accumulated in their lives-and interpret this new material according to what's in their reservoir.

Have you ever wondered why, when you're in a class with many others like you, and you're all subjected to the same lesson, lecture, or reading material, your answers to the same questions can differ greatly? Let's admit at the start that differences in attitude, motivation, and attention are all at work, but a major reason for the differences lies in the way that you, with your special knowledge, interpreted the material. You took in the material through your personal filter of experiences and constructed your understanding of it in light of your exclusive network of knowledge. You stamped your own meaning on the material.

As you can imagine, different interpretations of constructivism have arisen. The basic distinction to keep in mind is between those who believe that the individual alone-each student-constructs meaning (often referred to as individual or psychological constructivism). Others argue that individuals in a social situation-each student in the class, influenced by peers, home, and so on-construct meaning (often referred to as social constructivism). To enhance your understanding of constructivism, let's examine two classes who are studying the opening of the American West. One teacher has the class read the chapter that includes the Battle of Big Horn. The teacher then summarises the chapter and indicates the important points to be remembered. He then gives them time to write an essay about the battle, telling them to "get their facts straight." Finally, he tells his students that they will be tested tomorrow.

The second teacher has also prepared carefully and decided that this era in our history is too exciting to be restricted to text reading. She comes to class and poses a problem for her students. "I want you to assume that Custer, although critically wounded, survives the battle. He then has to stand trial for his leadership and the decisions he made leading up to and during the battle." The students are told to use all the sources they want, and to discuss the feeling of the country and the political climate of the times.

We trust you can see how the teacher who understands that students construct their own meanings will adopt different techniques in the classroom.

Teachers who follow a constructivist pathway often do the following:

- Wrap their teaching in a cloak of problems for their students, problems that are real, meaningful, and age-appropriate.
- Use their students' perspectives to interpret their responses and solutions, that is, take into account such important variables as cognitive level, home experiences, and motivation.
- Know that their students' responses reflect their current level of understanding.
- Accept the conflicts and confusion that initially accompany the search for meaning.

More about constructivistic approaches to instruction and learning will be discussed later, but let's now turn to the issue of student diversity.

STUDENT DIVERSITY IN THE CLASSROOM

Another topic that more and more educational psychologists are addressing in their research and practice is the diversity of students in schools today. Psychologists, for more than a century, have been interested in individual differences, but today more attention is being given to how schools can accommodate differences in ability, race, ethnicity, regional origin, family makeup, gender, and sexual orientation so that all students have opportunities to learn.

Developmental contextualism, popularised by Richard Lerner (1991), provides a rationale for recognising and capitalising on the richness and diversity of students' backgrounds. It also incorporates recent research relating to cultural constructivism, which means that students use the particular environment around them to construct their own worldview. Developmental contextualism attempts to analyse and understand development in the light of the multiple levels of interactions between individuals and their environments. That is, all students' characteristics, psychological as well as biological, interact with the environment (the context in this theory).

Context is an inclusive term that attempts to portray the complexity of students' backgrounds by identifying four major forces of development:

1. The physical settings through which your students move, such as the home, classroom, and workplace.
2. Social influences, such as students' families, peers, and significant others.
3. The personal characteristics of students, such as physical appearance, temperament, and language fluency.
4. The influence of time, that is, change brought about by the sheer chronology of living; to put it simply, the longer we're able to survive, the more changes we experience.

Consequently, the crucial element in learning and development is the changing relationship between the complexity students bring to the classroom and a multilayered context (school, home, peers, etc.). If you think about this deceptively simple statement, you can appreciate the need to study teaching, learning, and development from many different perspectives. For example, consider what's going on with students. The genes provide a blueprint that is passed on to the cells, tissues, and organs of the body, influencing the growth of such widely divergent growth features as brain development and temperament, to name only two. On the other hand, the intricate and involved layers of the context, ranging from family to peers to schools and to the wider social sphere, simultaneously weave their networks of influence. Simple explanations? Hardly. What is needed is a perspective equally as intricate as the behaviour it attempts to clarify. As Lerner (1991, p. 31) noted: The revised understanding of what constitutes the basic process of human development brings to the fore the cutting-edge importance

of continued empirical focus on individual differences, on contextual variations, and on changing person-context relations. Nothing short of these emphases can be regarded as involving scientifically adequate developmental analysis of human life.

OUT-OF-SCHOOL INFLUENCES

A natural outgrowth of developmental contextualism is the concern that educational psychologists have expressed about out-of-school influences that act decisively on students' learning. Educational psychologists have traditionally focused on those forces that are directly related to the classroom and school (instructional techniques, time-on-task, etc.), but now the time has come to adopt a broader perspective. Understanding the wellsprings of students' achievement demands that we know more about their lives beyond the classroom. What do we know about their families, their peers, their socioeconomic circumstances, and their cultural setting? It's becoming increasingly clear that all of these influences can play a significant role in motivating students to study and achieve in school.

In a recent survey of nine high schools involving about 20,000 students, Steinberg (1996) concluded that school is only one influence that affects what students learn and how well they do on tests of that learning. For example, the existence of differences in ethnic groups was the most important finding: Asian students outperform whites, blacks, and Latinos. Initially thinking that Asian students might believe that academic success correlated closely with out-of-school success, that is, there is a "payoff" for academic success, Steinberg was surprised this wasn't the case.

All students believed that doing well in school would have a payoff. The students actually differed, however, in their belief that failing in school would have negative consequences. The Asian students clearly felt that poor academic performance would definitely and negatively affect their future. Non-Asian students didn't share this belief, with black and Latino students not really believing that doing poorly in school would hurt their chances for future success.

Another of the findings related to the students' home: Parents exert a profound and lasting effect on their children's achievement in school by three things they do:

1. Deliberately or casually, they communicate specific messages to their children about teachers, schools, and learning. Their children quickly learn whether school is or isn't important, and whether they should expend much effort there.
2. Parental behaviour sends clear and unmistakable signals about the importance the parents place on schooling. Disregarding notices from the school, not attending parental functions, refusing to volunteer in school activities, all paint a stark picture for children- "School just isn't that important, no matter what I may say."
3. Their parenting style encourages, or discourages, engagement in school. Interestingly, Steinberg found that such parental activities as checking

homework or encouraging children to do better in school were not the most significant forms of parental engagement. What seemed to make a real difference was the actual physical presence of the parents at school: attending school programmes, participating in teacher conferences, joining in extracurricular events, and so on.

Quite obviously, not all students will be fortunate enough to have cooperative parents, and you'll work with students from different types of families. Families change, and as they do, they exercise different effects on a child's development—some significant, others not so (Scarr, 1992). For example, children who remain in an intact family, or who experience the death of a parent, or who go through a parental divorce—even though the experiences are quite different—all undergo changes that must affect development. You can see the need to carefully consider the out-of-school influences of students to better understand their classroom behaviour.

CLASSROOM MANAGEMENT AND THE LEARNING ENVIRONMENT

WHY CLASSROOM MANAGEMENT MATTERS

Managing the learning environment is both a major responsibility and an on-going concern for every teacher, even those with years of experience. There are several reasons. In the first place, a lot goes on in classrooms simultaneously, even when students seem to be doing only “one” task together. Twenty-five students may all be working on a sheet of math problems. But look more closely: several may be stuck on a particular problem, but each for different reasons. A few others have worked only the first problem or two and are now chatting quietly with each other instead of continuing. Still others have finished and are wondering what to do next. At any one moment each student needs something different—different information, different hints, different kinds of encouragement. The diversity increases even more if the teacher deliberately assigns multiple activities to different groups or individuals (for example, if some are doing a reading assignment while others do the math problems).

Another reason that managing the environment is challenging is because a teacher can never predict everything that will happen in a class. A well-planned lesson may fall flat on its face, or take less time than you expect, and you find yourself improvising to fill class time. On the other hand an unplanned moment may become a wonderful, sustained exchange among students; so you have to drop previous plans and “go with the flow” of their discussion. Interruptions happen continually: a fire drill, a quick drop-in visit from another teacher or from the principal, a call on the intercom from the office. An activity may turn out well, but also end up rather differently than you intended; you therefore have to decide how, if at all, to adjust the next day to allow for this

surprise. A third reason for the importance of management is that students form opinions and perceptions about your teaching that may coincide neither with your own nor with other students'. What seems to you like encouragement of a shy student may seem to the student herself like "forced participation." A more eager, outgoing classmate watching your special effort to encourage the shy student, however, may not see you as either encouraging or coercing, but as overlooking or ignoring other students who are already more willing to participate. The variety of perceptions can lead to surprises in students' responses to you—most often small ones, but occasionally more major.

At the broadest, society-wide level, management challenges teachers because public schooling is not voluntary, and students' presence in a classroom is therefore not a sign, in and of itself, that they wish to learn. Students' presence is instead just a sign that an opportunity exists for teachers to motivate students to learn. Many students, of course, do enjoy learning and being in school—but not all. Others do enjoy school, but primarily because teachers have worked hard to make classroom life pleasant and interesting. They become motivated because you have successfully created a positive learning environment and have sustained it through skilful management. Fortunately it is possible to earn this sort of commitment from students. We begin with some ways of preventing management problems in the first place by increasing students' focus on learning. The methods include the arrangement of classroom space, the establishment of procedures, routines, and rules, and communicating the importance of learning both to students and to parents. After these prevention oriented discussions, we look at ways of refocusing students when and if their minds or actions do stray from the tasks at hand. As you probably know from your own experience as a student, bringing students back on task can happen in many ways, ways that vary widely in the energy and persistence required of the teacher. We try to indicate some of this diversity, but because of space limitations and because of the richness of classroom life, we cannot describe them all.

PREVENTING MANAGEMENT PROBLEMS BY FOCUSING STUDENTS ON LEARNING

The easiest management problems to solve are ones that do not happen in the first place! You can help to prevent problems even before the first day of school by...

Pacing and Structuring Lessons and Activities

One of the best ways to prevent management problems is by pacing and structuring lessons or activities as smoothly and continuously as possible. Reaching this goal depends on...

Responding to Student Misbehaviour

So far we have focused on preventing behaviours that are off-task, or inappropriate or annoying. Our advice has all been pro-active or forward-looking: plan the classroom

space thoughtfully, create reasonable procedures and rules, pace lessons and activities appropriately, and communicate the importance of learning clearly. Although we consider these ideas to be important, it would be naive to imply they are enough to prevent all behaviour problems...

Keeping Management Issues in Perspective

There are two messages. One is that management issues are important, complex, and deserve any teacher's serious attention. The other is that management strategies exist and can reduce, if not eliminate, management problems when and if they occur. We have explained what some of those strategies are—including some intended to prevent problems from happening and others intended to remedy problems if they do occur.

But there is a third message: that good classroom management is not an end in itself, but a means for creating a climate where learning happens as fully as possible. During the stress of handling problem behaviours, there is sometimes a risk of losing sight of this idea. Quiet listening is never a goal in itself, for example; it is desirable only because (or when) it allows students to hear the teacher's instructions or classmates' spoken comments, or because it allows students to concentrate on their work or assignments better. There may therefore actually be moments when quiet listening is not important to achieve, such as during a "free choice" time in an elementary classroom or during a period of group work in a middle school classroom. As teachers, we need to keep this perspective firmly in mind. Classroom management should serve students' learning, and not the other way around.

INSTRUCTIONAL STRATEGIES

Forms of Thinking Associated with Classroom Learning

Although instructional strategies differ in their details, they all function to encourage certain major forms of learning and thinking, each with distinctive educational purposes. The forms sometimes overlap, in the sense that one form of thinking may contribute to a student's success with another form. To see what we mean, look first at three somewhat complex forms of thinking that are common goals of classroom learning:

- Critical thinking,
- Creative thinking, and
- Problem-solving.

Critical Thinking

Critical thinking is the mental skill for analysing the reliability and validity of information, as well as an attitude or disposition to do so. The skill and attitude may be

expressed or displayed with regard to a particular subject matter or topic, but in principle it can occur in any realm of knowledge or living. A critical thinker does not necessarily have a negative attitude in the everyday sense of being critical of someone or something.

Instead he or she can simply be thought of as astute: the critical thinker asks key questions, evaluates the evidence for ideas accurately, reasons about problems logically and objectively, and expresses ideas and conclusions clearly and precisely. Last (but not least), the critical thinker can apply these habits of mind in more than one realm of life or knowledge, though he or she may not always do so in fact. With such a broad definition, it is not surprising that educators have nominated a wide variety of specific cognitive skills as contributors to critical thinking. In one study, for example, the researcher found that critical thinking about a published article was stimulated by annotation—writing questions and comments in the margins of the article (Liu, 2006). In this study students who were initially instructed in ways of annotating reading materials. Later, when the students completed additional readings for assignments, it was found that some students in fact used their annotation skills much more than others—some simply underlined passages, for example, with a highlighting pen. When essays written about the readings were later analysed, the ones written by the annotators were found to be more well-reasoned—more critically astute—than the essays written by the other students.

But the skills comprising critical thinking are not just written ones. In another study, for example, a researcher found that critical thinking can also involve oral discussion with classmates of personal issues or dilemmas (Hawkins, 2006). In this study, students were asked to describe to classmates a recent personal incident that disturbed them. Classmates then discussed the incident together in order to identify the precise reasons why the incident was disturbing to the individual, as well as the assumptions that the student had made in thinking about the incident. The original student—the one who had first told the story—then used the results of the group discussion to frame a topic for a research essay.

In one story of a troubling incident, for example, a student told of a time when a store clerk has snubbed or rejected the student during a recent shopping errand. Through discussion, classmates decided that an assumption underlying the student's disturbance was her suspicion that she had been a victim of racial profiling based on her skin colour. The student then used this idea as the basis for a research essay on the topic of "racial profiling in retail stores." The group discussion thus stimulated critical thinking in the student and the classmates, but it also relied on their prior critical thinking skills at the same time.

Notice that in both of these research studies, as in others like them, what made the thinking "critical" was students' use of metacognition—strategies for thinking about thinking and for monitoring the success and quality of one's own thinking. There we pointed out that when students acquire experience in building their own knowledge, they also become skilled both at knowing how they learn, and at knowing whether they

have learned something well. These two defining qualities of metacognition are part of critical thinking as well. In fostering critical thinking, then, a teacher is really fostering a student's ability to construct or control his or her own thinking and to avoid being controlled by ideas unreflectively.

How best to teach the skills of critical thinking, however, remains a matter of debate. One issue is whether to infuse critical skills into existing courses or to teach them through separate, free-standing units or courses. The first approach has the potential advantage of demonstrating how critical thinking relates to students' entire educations. But it does so at the risk of diluting students' understanding and use of critical thinking simply because critical thinking takes on so in many different forms—its details and appearance varying among courses and teachers. The free-standing approach has the opposite qualities: it stands a better chance of being understood clearly and coherently, but by the same token its connections to other courses, tasks, and activities may not be as clear to students.

Unfortunately, research to compare the infusion versus free-standing strategies for teaching critical teaching does not settle the matter; it suggests that either approach can work as long as it is implemented thoroughly and the teachers are committed to the value of critical thinking.

A related issue about teaching critical thinking is about who needs or should learn critical thinking skills the most. Should it in fact be all students? This goal seems the most democratic and therefore appropriate for educators. Surveys of teachers have found, however, that teachers sometimes favour teaching of critical thinking to high-advantage students—the ones who already achieve well, who come from relatively high-income families, or (for high school students) who take courses intended for university entrance. Presumably the rationale for this bias is that high-advantage students can benefit and/or understand and use critical thinking better than other students. There is little evidence to support this idea, however, even if it were not ethically questionable. The study by Hawkins (2006) described above, for example, achieved good success teaching critical thinking even with students usually considered low-advantage.

Creative Thinking

Creativity is the ability to make something new that is also useful or valued by others. The “something” can be an object (like an essay or painting), a skill (like playing an instrument), or an action (like using a familiar tool in a new way).

Problem-solving

Somewhere between open-ended, creative thinking and the focused learning of content lies problem solving, the analysis and solution of tasks and situations that are somewhat complex or ambiguous and that pose difficulties, inconsistencies, or obstacles of some kind.

Major Instructional Strategies and Their Relationships

Because the forms of thinking just described—critical thinking, creativity, and problem solving—are broad and educationally important, it is not surprising that educators have identified a lot of strategies to encourage their development. There are so many possibilities, in fact, that just keeping them all in mind—let alone choosing among them—can be difficult.

Lectures and Readings

Lectures and readings are traditional staples of educators, particularly when teaching older students (including university students!). At their best, they are the good examples of pre-organised information, so that the student only has to remember what was said in the lecture or written in the text in order to begin understanding it.

Mastery Learning

This term refers to an instructional approach in which all students learn material to an identical, high level, even if some students require more time than others to do so (Gentile, 2004). In mastery learning the teacher directs learning, though sometimes only in the indirect sense of finding, writing, and orchestrating..

Direct Instruction

Sometimes this term serves as a synonym for teacher-directed instruction, but more often direct instruction refers to a relatively scripted version of mastery learning, meaning that it not only organises the curriculum into small modules or units, but it also dictates how teachers should teach, including.

Madeline Hunter's Effective Teaching Model

Many teacher-directed strategies have been combined by Madeline Hunter into a single, relatively comprehensive approach that she calls mastery teaching (not to be confused with the related term mastery learning) or the effective teaching model.

Student-centred Models of Learning

Student-centred models of learning shift some of the responsibility for directing and organising learning from the teacher to the student. Being student-centred does not mean, however, that a teacher gives up organisational and leadership responsibilities completely. It only means.

Instructional Strategies: An Abundance of Choices

Looking broadly, you can see that choices among instructional strategies are numerous indeed, and that deciding among them depends on the forms of thinking that you want

to encourage, the extent to which ideas or skills need to be organised by you to be understood by students, and the extent to which students need to take responsibility for directing their own learning. Although you may have personal preferences among possible instructional strategies, the choice will also be guided by the uniqueness of each situation of teaching—with its particular students, grade-level, content, and purposes.

If you need to develop students' problem solving skills, for example, there are strategies that are especially well suited for this purpose. If you want the students to take as much initiative as possible in organising their own learning, this too can be done. Yet having this knowledge is still not enough to teach well. What is still needed are ideas or principles for deciding what to teach, we have still not addressed an obvious question: How do I find or devise goals for my teaching and for my students' learning? And assuming that I can determine the goals, where can I find resources that help students to meet them?

INSTRUCTIONAL PLANNING

Selecting General Learning Goals

At the most general or abstract level, the goals of education include important philosophical ideas like “developing individuals to their fullest potential” and “preparing students to be productive members of society.” Few teachers would disagree with these ideas in principle, though they might disagree about their wording or about their relative importance. As a practical matter, however, teachers might have trouble translating such generalities into specific lesson plans or activities for the next day's class. What does it mean, concretely, to “develop an individual to his or her fullest potential”? Does it mean, for example, that a language arts teacher should ask students to write an essay about their personal interests, or does it mean that the teacher should help students learn to write as well as possible on any topic, even ones that are not of immediate interest? And what exactly should a teacher do, from day to day, to “prepare students to be productive members of society” as well? Answers to questions like these are needed to plan instruction effectively. But the answers are not obvious simply by examining statements of general educational goals.

National and State Learning Standards

Some (but not all) of the work of transforming such general purposes into more precise teaching goals and even more precise objectives has been performed by broad national organisations that represent educators and other experts about particular subjects or types of teaching (Riley, 2002). The groups have proposed national standards, which are summaries of what students can reasonably be expected to learn at particular grade levels and in particular subjects areas. In the United States, in addition, all state

governments create state standards that serve much the same purpose: they express what students in the state should (and hopefully can) learn at all grade levels and in all subjects.

Because they focus on grade levels and subject areas, general statements of educational standards tend to be a bit more specific than the broader philosophical goals we discussed above. As a rule of thumb, too, state standards tend to be more comprehensive than national standards, both in coverage of grade levels and of subjects. The difference reflects the broad responsibility of states in the United States for all aspects of public education; national organisations, in contrast, usually assume responsible only for a particular subject area or particular group of students. Either type of standards provides a first step, however, towards transforming the grandest purposes of schooling (like developing the individual or preparing for society) into practical classroom activities. But they provide a first step only. Most statements of standards do not make numerous or detailed suggestions of actual activities or tasks for students, though some might include brief classroom examples—enough to clarify the meaning of a standard, but not enough to plan an actual classroom programme for extended periods of time. For these latter purposes, teachers rely on more the detailed documents, the ones often called curriculum frameworks and curriculum guides.

Curriculum Frameworks and Curriculum Guides

The terms curriculum framework and curriculum guide sometimes are used almost interchangeably, but for convenience we will use them to refer to two distinct kinds of documents. The more general of the two is curriculum framework, which is a document that explains how content standards can or should be organised for a particular subject and at various grade levels. Sometimes this information is referred to as the scope and sequence for a curriculum. A curriculum framework document is like a standards statement in that it does not usually provide a lot of detailed suggestions for daily teaching. It differs from a standards statement, though, in that it analyses each general standard in a curriculum into more specific skills that students need to learn, often a dozen or more per standard. The language or terminology of a framework statement also tends to be somewhat more concrete than a standards statement, in the sense that it is more likely to name behaviours of students—things that a teacher might see them do or hear them say. Sometimes, but not always, it may suggest ways for assessing whether students have in fact acquired each skill listed in the document.

Teachers' need for detailed activity suggestions is more likely to be met by a curriculum guide, a document devoted to graphic descriptions of activities that foster or encourage the specific skills explained in a curriculum framework document. The descriptions may mention or list curriculum goals served by an activity, but they are also likely to specify materials that a teacher need, time requirements, requirements for grouping students, drawings or diagrams of key equipment or materials, and sometimes even suggestions for what to say to students at different points during the activity. In

these ways the descriptions may resemble lesson plans. Since classroom activities often support more than one specific skill, activities in a curriculum guide may be organised differently than they might be in a framework document. Instead of highlighting only one standard at a time, as the framework document might, activities may be grouped more loosely—for example, according to the dominant purpose or goal of an activity (“Activities that encourage the practice of math facts”) or according to a dominant piece of equipment or material (“Ten activities with tin cans”). Note that the activity meets several educational objectives at once—tracing shapes, knowledge of leaves and of colours, descriptive language skill. Each of these skills may reflect a different curriculum standard.

Formulating Learning objectives

Given curriculum frameworks and guides like the ones just described, how do you choose and formulate actual learning objectives? Basically there are two approaches: either start by selecting content or topics that what you want students to know (the cognitive approach) or start with what you want students to do (the behavioural approach).

Taxonomies of Educational objectives

When educators have proposed taxonomies of educational objectives, they have tended to focus on one of three areas or domains of psychological functioning: either students’ cognition (thought), students’ feelings and emotions (affect), or students’ physical skills (psychomotor abilities). Of these three areas, they have tended to focus the most attention on cognition. The taxonomy originated by Benjamin Bloom, for example, deals entirely with cognitive outcomes of instruction.

Students as a Source of Instructional Goals

So far our discussion of instructional planning has described goals and objectives as if they are selected primarily by educators and teachers, and not by students themselves. The assumption may be correct in many cases, but there are problems with it. One problem is that choosing goals and objectives for students, rather than by students, places a major burden on everyone involved in education—curriculum writers, teachers, and.

Enhancing Student Learning through a variety of Resources

Whether instructional goals originate from curriculum documents, students’ expressed interests, or a mixture of both, students are more likely to achieve the goals if teachers draw on a wide variety of resources. As a practical matter, this means looking for materials and experiences that supplement—or occasionally even replace—the most traditional forms of information, such as textbooks.

STUDENT DIVERSITY

INDIVIDUAL STYLES OF LEARNING AND THINKING

All of us, including our students, have preferred ways of learning. Teachers often call these differences learning styles, though this term may imply more consistency across situations than is really the case. One student may like to make diagrams to help remember a reading assignment, whereas another student may prefer to write a sketchy outline instead. Yet in many cases the students could in principle reverse the strategies and still learn the material: if coaxed (or perhaps required), the diagram-maker could take notes for a change and the note-taker could draw diagrams. Both might still learn using the alternate style, even if neither felt as comfortable as with the strategies they prefer.

This reality suggests that a balanced, middle-of-the-road approach may be a teacher's best response to students' learning styles. Or to put it another way: it is good to support students' preferred learning strategies where possible and appropriate, but neither necessary nor desirable to do so all of the time. Most of all, it is neither necessary nor possible to classify or label students according to seemingly fixed learning styles and then allow them to learn only according to those styles.

A student may prefer to hear new material rather than see it; he may prefer for you to explain something orally, for example, rather than to see it demonstrated in a video. But he may nonetheless tolerate or sometimes even prefer to see it demonstrated. In the long run, in fact, he may learn it best by encountering the material in both ways, regardless of his habitual preferences.

That said, there is evidence that individuals, including students, do differ in how they habitually think. These differences are more specific than learning styles or preferences, and psychologists sometimes call them cognitive styles, meaning typical ways of perceiving and remembering information, and typical ways of solving problems and making decisions. In a style of thinking called field dependence, for example, individuals perceive patterns as a whole rather than focus on the parts of the pattern separately. In a complementary tendency, called field independence, individuals are more inclined to analyse overall patterns into their parts. Cognitive research from the 1940s to the present has found field dependence/independence differences to be somewhat stable for any given person across situations, though not completely so. Someone who is field dependent (perceives globally or "wholistically") in one situation, that is, tends to a modest extent to perceive things globally or wholistically in other situations. Field dependence and independence can be important understanding students because the styles affect students' behaviours and preferences in school and classrooms. Field dependent persons tend to work better in groups, it seems, and to prefer "open-ended" fields of study like literature and history.

Field independent persons, on the other hand, tend to work better alone and to prefer highly analytic studies like math and science. The differences are only a tendency, however, and there are a lot of students who contradict the trends. As with the broader notion of learning styles, the cognitive styles of field dependence and independence are useful for tailoring instruction to particular students, but their guidance is only approximate. They neither can nor should be used to “lock” students to particular modes of learning or to replace students’ own expressed preferences and choices about curriculum.

Another cognitive style is impulsivity as compared to reflectivity. As the names imply, an impulsive cognitive style is one in which a person reacts quickly, but as a result makes comparatively more errors. A reflective style is the opposite: the person reacts more slowly and therefore makes fewer errors. As you might expect, the reflective style would seem better suited to many academic demands of school. Research has found that this is indeed the case for academic skills that clearly benefit from reflection, such as mathematical problem solving or certain reading tasks.

Some classroom or school-related skills, however, may actually develop better if a student is relatively impulsive. Being a good partner in a cooperative learning group, for example, may depend partly on responding spontaneously (*i.e.*, just a bit “impulsively”) to others’ suggestions; and being an effective member of an athletic team may depend on not taking time to reflect carefully on every move that you or your team mates make.

There are two major ways to use knowledge of students’ cognitive styles. The first and the more obvious is to build on students’ existing style strengths and preferences. A student who is field independent and reflective, for example, can be encouraged to explore tasks and activities that are relatively analytic and that require relatively independent work. One who is field dependent and impulsive, on the other hand, can be encouraged and supported to try tasks and activities that are more social or spontaneous. But a second, less obvious way to use knowledge of cognitive styles is to encourage more balance in cognitive styles for students who need it. A student who lacks field independence, for example, may need explicit help in organising and analysing key academic tasks (like organising a lab report in a science class). One who is already highly reflective may need encouragement to try ideas spontaneously, as in a creative writing lesson.

Multiple Intelligences

For nearly a century, educators and psychologists have debated the nature of intelligence, and more specifically whether intelligence is just one broad ability or can take more than one form. Many classical definitions of the concept have tended to define intelligence as a single broad ability that allows a person to solve or complete many sorts of tasks, or at least many academic tasks like reading, knowledge of vocabulary, and the solving of logical problems. There is research evidence of such a

global ability, and the idea of general intelligence often fits with society's everyday beliefs about intelligence. Partly for these reasons, an entire mini-industry has grown up around publishing tests of intelligence, academic ability, and academic achievement. Since these tests affect the work of teachers. But there are also problems with defining intelligence as one general ability. One way of summing up the problems is to say that conceiving of intelligence as something general tends to put it beyond teachers' influence. When viewed as a single, all-purpose ability, students either have a lot of intelligence or they do not, and strengthening their intelligence becomes a major challenge, or perhaps even an impossible one. This conclusion is troubling to some educators, especially in recent years as testing school achievement has become more common and as students have become more diverse.

But alternate views of intelligence also exist that portray intelligence as having multiple forms, whether the forms are subparts of a single broader ability or are multiple "intelligences" in their own right. For various reasons such this perspective has gained in popularity among teachers in recent years, probably because it reflects many teachers' beliefs that students cannot simply be rated along a single scale of ability, but are fundamentally diverse.

One of the most prominent of these models is Howard Gardner's theory of multiple intelligences. Gardner proposes that there are eight different forms of intelligence, each of which functions independently of the others. Each person has a mix of all eight abilities—more of one but less of another—that helps to constitute that person's individual cognitive profile. Since most tasks—including most tasks in classrooms—require several forms of intelligence and can be completed in more than one way, it is possible for people with various profiles of talents to succeed on a task equally well.

In writing an essay, for example, a student with high interpersonal intelligence but rather average verbal intelligence might use his or her interpersonal strength to get a lot of help and advice from classmates and the teacher. A student with the opposite profile might work well alone, but without the benefit of help from others. Both students might end up with essays that are good, but good for different reasons.

As evidence for the possibility of multiple intelligences, Gardner cites descriptions of individuals with exceptional talent in one form of intelligence (for example, in playing the piano) but who are neither above nor below average in other areas. He also cites descriptions of individuals with brain damage, some of whom lose one particular form of intelligence (like the ability to talk) but retain other forms. In the opinion of many psychologists, however, the evidence for multiple intelligences is not strong enough to give up the "classical" view of general intelligence. Part of the problem is that the evidence for multiple intelligences relies primarily on anecdotes—examples or descriptions of particular individuals who illustrate the model—rather than on more widespread information or data.

Nonetheless, whatever the status of the research evidence, the model itself can be useful as a way for teachers to think about their work. Multiple intelligences suggest

the importance of diversifying instruction in order to honour and to respond to diversity in students' talents and abilities. Viewed like this, whether Gardner's classification scheme is actually accurate is probably less important than the fact there is (or may be) more than one way to be "smart." In the end, as with cognitive and learning styles, it may not be important to label students' talents or intellectual strengths.

It may be more important simply to provide important learning and knowledge in several modes or styles ways that draw on more than one possible form of intelligence or skill. A good example of this principle is your own development in learning to teach. It is well and good to read books about teaching (like this one, perhaps), but it is even better to read books and talk with classmates and educators about teaching and get actual experience in classrooms. The combination both invites and requires a wide range of your talents and usually proves more effective than any single type of activity, whatever your profile of cognitive styles or intellectual abilities happens to be.

Gender Differences in the Classroom

Gender roles are the patterns of behaviours, attitudes, and expectations associated with a particular sex—with being either male or female. For clarity, psychologists sometimes distinguish gender differences, which are related to social roles, from sex differences, which are related only to physiology and anatomy.

Differences in Cultural Expectations and Learning Styles

A culture is the system of attitudes, beliefs, and behaviours that constitute the distinctive way of life of a people. Although sometimes the term is also used to refer specifically to the artistic, intellectual and other "high-brow" aspects of life, I use it here more broadly to refer to everything that characterises a way of life—baseball games as well as symphony concerts, and McDonald's as well as expensive restaurants.

Accommodating Diversity in Practice

Briefly, then, Students differ in a multitude of ways, both individually and as groups. Individually, for example, students have *preferred learning styles* as well as preferred *cognitive or thinking styles*. They also have unique *cognitive profiles*, intelligence or competence that affect how and what they learn most successfully.

In addition to individual diversity, students tend to differ according to *behaviours associated with gender*, although there are many individual exceptions. *Motor abilities* as well as *motivation* and experience with athletics gradually differentiate boys and girls, especially when they reach high school and begin high school. *Socially*, boys tend to adopt relationships that are more active and wide-ranging than do girls. *Academically*, girls tend to be a bit more motivated and to receive slightly higher marks in school. *Teachers sometimes contribute to gender role differences*—perhaps without intending—by paying attention to boys more frequently and more publicly in class, and by distributing

praise and criticism in ways differentiated by sex. Students also differ according to *cultures, language, and ethnic groups* of their families. Many students are *bilingual*, with educational consequences that depend on their fluency in each of their two languages. If they have more difficulty with English, then programmes that add their first language together with English have proved to be helpful. If they have more difficulty with their first language, they are at risk for language loss, and the consequences are also negative even if more hidden from teachers' views.

In addition to language differences as such, students differ according to *how they use language*—in taking turns at speaking, in eye contact, social distance, wait time, and the use of questions. Some of these differences stem from cultural differences about self-identity, with non-Anglo cultures sometimes (but not always) supporting a more interdependent view of the self than Anglo culture or the schools. Differences in attitudes and in use of language have several consequences for teachers. In particular—where appropriate—they should consider using cooperative activities, avoid highlighting individuals' accomplishments or failures, and be patient about students' learning to be punctual.

Hopefully, therefore, we have persuaded you—if you ever really needed persuading—that students are indeed diverse. The question that follows from this point is *what to do* about the diversity. We partially answered that question by making a number of teaching-related suggestions. But there is obviously more to be said about accommodating diversity—about actually working with students' differences and making them a resource rather than a burden or challenge. In the rest of this book therefore we offer more suggestions, not only about knowing how different one student can be from another, but also about diversifying teaching to acknowledge this fact. Differences among students are inevitably a challenge during all phases of teaching, from planning instruction, to implementing lessons and activities, to assessing students' learning after lessons or activities are all finished.

STUDENT DEVELOPMENT

WHY DEVELOPMENT MATTERS

Students' development matters for teachers, but the way it matters depends on the kind of teaching. If you teach a single, "self-contained" grade-level, then the benefits of knowing about development will be less explicit, but just as real, as if you teach many grade levels. Working exclusively with one grade (like, say, a third-grade classroom) highlights differences among students that happen in spite of their similar ages, and obscures similarities that happen because of their having similar ages.

Under these conditions it is easy to notice students' diversity, but somewhat harder to know how much of it comes from differences in their long-term development, as opposed to differences in shorter-term learning or immediate experiences. Yet this

knowledge is useful in planning appropriate activities and in holding appropriate expectations about students. What changes can you expect to happen relatively soon simply from your current programme of activities, and which ones might not show up this year at all? These are questions that developmental psychology can help to answer.

If you teach multiple grade levels, as often happens to specialists or to teachers in middle school or high school, then your need for developmental knowledge will be more obvious because you will confront wide age differences on a daily basis. As a physical education teacher, for example, you may teach kindergarten children as well as sixth-graders, or seventh-graders as well as twelfth-graders, all in the course of a single day, or even of a single hour.

Diversity among your students will therefore be a more explicit mix of the developmental and non-developmental—of complex differences due to age as well as differences due to “other” factors, such as skills or knowledge learned only recently. Nonetheless, your instructional challenge will be the same as the one faced by teachers of single-grade classes: you will want to know what activities and expectations are appropriate for your students. To answer this question, you will need to know something not only about your students, but also about general developmental trends during childhood and adolescence.

Note that developmental trends vary in two important ways. The first, as I indicated already, is in their generality. Some theories or models of development boldly assert that certain changes happen universally—happen, that is, to virtually every person on the planet. For example, a theory might assert that virtually every toddler acquires a spoken language, or that every teenager forms a sense of personal identity.

Individuals who do not experience these developments would be very rare as well as significantly disabled, no matter what family, community, or society they belong to. Other theories, however, propose developmental changes that are more limited in scope, claiming only that the changes happen to some people and only under certain conditions. Developing a female gender role, for example, does not happen to everyone, but only to the female part of the population, and the process differs according to the family, community, or society in which a child lives.

The second way that developmental trends vary is in how strictly they are sequenced and hierarchical. In some views about development, changes are thought to happen in a specific order and to build on each other—sort of a “staircase” model of development. For example, a developmental psychologist (and many of the rest of us, for that matter) might argue that young people must have tangible, hands-on experience with new materials before they can reason about the materials in the abstract.

The order cannot be reversed. In other viewpoints, developmental change happens, but not in a uniform sequence or with a common end point; change is more of a “kaleidoscope” than a staircase. A person who becomes permanently disabled, for example, may experience complex long-term changes in personal values and priorities. But the changes can in principle happen at any time during life, including childhood,

and can take a variety of forms. Except in the realm of physical development, educational psychologists have tended to emphasize explanations of development that are relatively general, universal and sequential, rather than specific to particular cultures or “merely” kaleidoscopic. Such models (sometimes called “grand theories”) have the advantages of concisely integrating many features of development, while also describing where “typical” children or adolescents end up in life. The preference for integrative perspectives makes sense given educators’ need to work with and teach large numbers of diverse students both efficiently and effectively. But the approach also risks overgeneralising or oversimplifying what happens to particular children and youth. It can also confuse what does happen as certain children (like the middle-class ones) develop with what should happen to all children. To understand these risks, imagine any two children of about the same age who have clearly had very different experiences—for example, one child who grows up in poverty and another who grows up financially well-off. In what sense can we say that these two children experience the same underlying developmental changes as they grow up?

And how much should they even be expected to? Developmental psychology, and especially the broad theories of it emphasized in educational psychology, highlights the “sameness” or common ground between these two children as they mature, no matter how different they seem on the surface. Developmental knowledge, then, serves as counterpoint or balance to knowledge of the obvious complementary reality of teaching, diversity among students.

Physical Development During the School Years

Although it may be tempting to think that physical development is the concern of physical education specialists, many aspects of it are actually quite relevant to other teachers as well. In first-grade, for example, it is important to know whether children can successfully manipulate a pencil. In later grades, it is important to know how long students can be expected to sit still without discomfort—actually a physical challenge itself. In all grades, it is important to have a sense of students’ health needs related to their age or maturity, if only to know who is likely to become ill with what illness, and what physical activities are reasonable and needed.

Trends in Height and Weight

Typical growth in students’ height and weight are easy to summarise. The figure shows not only the average for each age from preschool through the end of high school, but also the range within which 90 per cent of children fall for each age. At age 6, for example, about when most children begin school, the average boy or girl is about 45 ½ inches tall, but 90 per cent of six year olds are between about 43 and 49 inches tall. Average weight at the same age is about 39 pounds, but ranges between about 35 and 53 pounds.

There are two important points to keep in mind. The first is that boys and girls are quite similar in height and weight during childhood, but diverge in the early teen-age years, when they reach puberty. For a time (approximately age 10-14), the average girl is taller, but not much heavier, than the average boy. After that the average boy becomes both taller and heavier than the average girl—though there remain individual exceptions. The preteen difference can be especially awkward for some children and youth, at least among those who aspire to looking like older teenagers or young adults. For young teens less concerned with “image,” though, the fact that girls are taller may not be especially important, or even noticed.

The second point is that as children get older, individual differences in weight diverge more radically than differences in height. Among 18-year-olds, the heaviest youngsters weigh almost twice as much as the lightest, but the tallest ones are only about 10 per cent taller than the shortest. Nonetheless, both height and weight can be sensitive issues for some teenagers. Society (and therefore teenage peers) tends to favour relatively short women and tall men, as well as a thin body build, especially for girls and women.

Yet neither “socially correct” height nor thinness is the destiny for many individuals. Being overweight, in particular, has become a common, serious problem in modern society, due to the prevalence of diets high in sugar and trans fats and lifestyles low in activity. The educational system has unfortunately contributed to the problem as well, by gradually restricting the number of physical education courses and classes in the past two decades.

Puberty and Its Effects on Students

The most universal physical development in students is puberty, which is the set of changes in early adolescence that bring about sexual maturity. Along with internal changes in reproductive organs are outward changes, such as growth of breasts in girls and of the penis in boys, as well as relatively sudden increases in height and weight. By about age 10 or 11, most children experience increased sexual attraction to others (usually heterosexual, though not always) that can affect social life both in school and elsewhere. By the end of high school, more than half of boys and girls report having experienced sexual intercourse at least once—though it is hard to be certain of the proportion because of the personal sensitivity of the information. In addition to creating obvious physical differences, puberty accentuates gender role differences for at least some teenagers. Girls who excelled at math or science in elementary school may curb some of their enthusiasm for and displays of success at these subjects. Boys who previously were not especially interested in sports may feel a new need to dedicate themselves to athletics in order to affirm their masculinity in the eyes of others.

Individual boys and girls who might have once worked together successfully on class projects may no longer feel comfortable doing so—or alternatively may now want to work together for non-academic rather than academic reasons. Such changes do not affect all youngsters equally, nor affect any one youngster equally on all occasions. An

individual student may act like a young adult on one day, but more like a child on the next. In teaching children who are in the midst of puberty (especially early in the process), you will need to respond flexibly and supportively.

Development of Motor Skills

Students' fundamental motor skills are already developing when they begin kindergarten, but are still not yet perfectly coordinated. Five-year-olds generally can walk satisfactorily for most school-related purposes (if they could not, the schools would have to be organised very differently!). For some fives, running still looks a bit like a hurried walk, but usually it becomes more coordinated within a year or two. Similarly with jumping, throwing and catching: the large majority of children can do these things, though often a bit clumsily, by the time they start school, and most improve their skills noticeably during the early elementary years.

Whether or not physical skills become a special focus of teachers' concern, however, they can be quite important to students themselves. Whatever their grade level, students who are clumsy are aware of that fact and of its potentially negative effect on peer relationships. In the long term, self-consciousness and poor self-esteem can result for a child who is clumsy, especially if peers (or teachers and parents) place high value on participation in athletics.

One research study confirmed, for example, what teachers and coaches sometimes suspect: that losers in athletic competitions tend to become less sociable and are more apt to miss athletic practices than winners.

Health and Illness

By world standards, children and youth in the United States and other developed societies tend to be remarkably healthy, though much depends on how well-off their families are and on how much health care is available to them. Children from middle- and higher-income families experience far fewer serious or life-threatening illnesses than children whose families lack resources. All across the economic spectrum, however, parents and teachers often rightly note that children—especially the youngest ones—get far more minor illnesses than do adults. In 2004, for example, one government survey estimated that children get an average of 6-10 colds per year, but adults get only about 2-4 per year (National Institute of Allergies and Infectious Diseases, 2005). The difference probably exists because children's immune systems are not as fully formed as adults', while at the same time children are continually exposed to others at school, many of whom may be contagious themselves.

An indirect result of children's frequent illnesses is that teachers (along with airline flight attendants, incidentally) also report more frequent minor illnesses than do adults in general—about five colds per year, for example, instead of just 2-4. These "simple" illnesses are responsible for many lost days of school, both for students and for teachers,

as well as many days when a student is present physically, but functions below par and simultaneously infects classmates. Learning, and often also teaching, therefore suffer because health is suffering. The problem is not only the prevalence of illness as such (in the United States in winter, approximately one person gets infected with a minor illness every few seconds), but the fact that illnesses are not distributed uniformly among students, schools, or communities. Whether it is a simple cold or something more serious, illness is particularly common where living conditions are crowded, where health care is scarce or unaffordable, and where individuals live with frequent stresses of any kind.

As students get older and become teenagers, illnesses become less frequent, but other health risks emerge. The most widespread is the consumption of alcohol and the smoking of cigarettes. As of 2004, about 75 per cent of teenagers reported drinking an alcoholic beverage at least occasionally, and 22 per cent reported smoking cigarettes. The good news about these figures is that they are part of a small, but steady decline in the frequencies over the past ten years or so.

The bad news is that teenagers also show increases in the abuse of some prescription drugs, such as inhalants, that act as stimulants. As with the prevalence of illnesses, the prevalence of drug use is not uniform, with a relatively small fraction of individuals accounting for a disproportionate proportion of usage. One survey, for example, found that a teenager was 3-5 times more likely to smoke or to use alcohol, smoke marijuana, or use drugs if he or she has a sibling who has also indulged these habits. Siblings, according to the research, seem to be more influential as role models than parents.

COGNITIVE DEVELOPMENT

Cognition refers to thinking and memory processes, and cognitive development refers to long-term changes in these processes. While cognition has been studied from several different perspectives and in the light of several theories, the one that is both the most widely known among educators and the most frankly “developmental” is the cognitive stage theory of a Swiss psychologist named Jean Piaget.

Social Development: Relationships and Personal Motives

Social development refers to the long-term changes in relationships and interactions involving self, peers, and family. It includes both positive changes, such as how friendships develop, and negative changes, such as aggression or bullying.

Moral Development: Forming a Sense of Rights and Responsibilities

Morality is a system of beliefs about ethics, about what is right and good compared to what is wrong or bad. Moral development refers to the changes in moral beliefs as a person grows older and gains maturity.

Understanding “The Typical Student” versus Understanding Students

As every teacher knows, however, development is not that simple. A class of 25 or 30 students will contain 25 or 30 individuals who will be learning and developing along 25 or 30 distinct pathways. Why then study developmental patterns at all? Because underlying their obvious diversity, students do indeed show important similarities. My references to “the” student should not be understood, therefore, as referring to simple-minded stereotypes, but to tendencies around which real, live students often group themselves. Pointing to developmental changes is like pointing to a flock of birds in flight: the flock has a general location, but individual birds also have their own locations and individual flight paths. As a teacher, I believe that the most helpful approach is to understand development and diversity not separately, but jointly. There are indeed similarities woven among the differences in students, but also differences woven among students’ commonalities.

**REVOLUTION IN EDUCATION
INDUCED BY PSYCHOLOGY**

The extent to which psychology has produced into the field of teaching and learning education is so impressive that only the term revolution can be applied to it with any accuracy. In pointing to the changes in education caused by psychology Ryan has written, “In many schools of today one finds an atmosphere of friendliness and happy activity. Much of the traditional formality, the forced silence, the tension, the marching is gone. Children’s voices are heard in the halls and class-rooms.

The younger children come gaily down the stairways, naturally and relatively unstrained, the older boys and girls through the corridor or outside walk making their way to school rooms, shops, studios, libraries, laboratories and playing fields to tasks that mean something to them, that make demands upon their energies and their imagination, that often involve hard, difficult work, but work that they recognise as creative. Beauty of surroundings is considered a first, requirement in these schools—there are flowers about, brightly coloured murals painted by the children, attractive informal work rooms for the various groups. Art and music has begun to play the role that being to them as fundamental in education and life.”

CHILD-CENTRED EDUCATION

In early times the teachers were mainly engaged in imparting various kinds of information to the children. In those times children were less the centre to attention, the school claimed more of their time. Today, it is the child who is the centre of attention in education.

Every child is individually considered and treated. Efforts are always made to adapt the curriculum of study to the needs of the child with the result that the brilliant and

dull witted children do not study the same subjects. Applied psychology has directed attention to the differing abilities and capacities of people. Such an attitude has made it possible for the teacher to gauge the special abilities of his wards and to make the best arrangements for the guidance of individual children.

The teacher of today not only must be acquainted with teaching but also with his students since his profession is now centred about the child and not about the subject or teacher. The consequence of this kind of reasoning is that what matters isn't the beauty, knowledge or ability of the teacher but the extent to which he succeeds in developing the personality of his students.

Process of Education

Previously, education was believed to be a comprehensive process that trained the individual, provided him with a moral character and made him more cultured. In psychology as it is understood today, the conception of mind has completely changed.

Analysis has led to the discovery that many kinds of forces and mental activities take place within the human being so that for the development of the child it is necessary to provide him with different kinds of programmes and subjects so that he may succeed in properly developing all his mental faculties and abilities. At present, variety is believed to be essential not only in education itself but it is equally held to be true that the education of one individual subject includes many different activities and functions. The teacher should be aware of all these functions. It is only then that he can understand the difficulties of the children in understanding the subject. If a child fails to memorise some one lesson, it can hardly be taken to indicate only a lack of effort on his part and no other cause. It is equally probable that he has a weak memory, or is not intelligent, or has no interest in the subject or that he is suffering from ill health. Learning is a complex activity and it is only through a complete knowledge of its various parts that one can diagnose the difficulty of a particular child in memorising or learning a lesson. The teacher has no alternative but to attend to the activity of learning.

Reform in Curriculum

Psychology has also managed to bring about important variations and reforms in educational curricula. Now it is the mental age and not the chronological age of the child that helps to determine the course of study to which he is to be subjected. Intelligence tests reveal the mental age of the child. Different courses are devised for brilliant and deficient children.

After the eighth or delta class, the curriculum is sub-divided into various parts such as literary, scientific, agricultural or art class. The child then has the opportunity of selecting his subject according to his interest and aptitude.

Various extra-curricular activities are organised for the emotional and sentimental development of the child. These include picnics, dramatic societies, etc. Other

programmes are organised in order to develop the qualities of leadership, public speaking, etc., in the child. But no course of study or programme is foisted upon the child against his inclination.

Inspiration is awakened in him according to his interests and his tastes. Education has now been made more interesting and appealing with the use of audio-visual aids such as television, film projection, etc. Many schools are also equipped with radio sets that provide both entertainment and information to the child.

The use of analogies to penetrate the defences of the child is an important method in education. Children who choose to study history are also taken to visit historical places and even a glance at the curriculum of basic education makes clear the extent to which psychology has revolutionised education. And special curricula are also devised to meet the needs of adult education. They suit the psychology of adult individuals.

Discipline

Discipline has an important place in education. One of the aims of education is to instill respect for discipline in the child. But the recent discoveries of psychology have changed the ways and means in which discipline is taught. Previously physical punishment was one of the chief methods of teaching and imparting discipline in the children. The main motive behind the discipline was fear of physical violence and injury. But instead of injuring the body of the child it injured his personality by making it distorted.

Psychologists attracted the attention of enlightened people to the injurious effects of physical punishment and also suggested alternative means of making children disciplined. One suggestive example is here described. The inmates of a hostel were given to indiscipline during lunch time. They threw morsels of food at each other and wasted much more. The warden made every effort to stop this nuisance used every means of arousing fear in the children but without success.

He never discovered the name of the mischief makers and neither did the boys give up their behaviour. Sometime later he was replaced by a new warden who resorted to psychological methods for solving this menace. He made an announcement to the effect that the best behaved table with the most seemly boys would win a flag that would entitle them to twice the normal ration of the most succulent food. The strategy took immediate effect and the indiscipline promptly ceased.

The children seated at various tables were now busily engaged in trying to look more disciplined and seemly than all the others so that they now even prevented the more hardened and incorrigible boys among them from mischief. In this way, the warden took advantage of the psychology of children to effortlessly perform a task that had completely foiled all the efforts of the earlier warden with his shock and fear tactics.

Nowadays it has become customary in schools to allow the children to adopt discipline rather than foist it upon them. At the college level it is from among the students that the prefects are chosen to create and maintain discipline among their fellow mates and to

help the proctor in maintaining order within the college. Very recently there was a high level debate on the spreading indiscipline among university students in India and almost all speakers put forward psychological solutions because the problem of indiscipline is fundamentally and basically a psychological one. It need hardly be pointed out that the more a teacher knows of child psychology, the easier will it be for him to create discipline in the motley group with which he is faced in the class.

Education on Personality

The modern methods of education place great emphasis upon the education of personality. Education is now understood to mean something more than mere reading or writing, it is also understood to involve the development of personality. Many schools have resident psychologists who solve problems related to personality of the students and give advice to teachers and parents in this regard. At places there are also child guidance clinics that make special efforts towards the correction and normal rehabilitation of problem children or juvenile delinquents.

A study of psychology is an essential part of the curriculum of teacher's training. In addition to these individual efforts most states and districts have a bureau of psychology whose functions are to solve all problems of students relating to their personality, to give personal guidance and to give advice to their parents and teachers regarding them. The truth of the matter is that the psychologist has become a link for the child's adjustment between the home and the school since such an adjustment helps to improve his personality. In the role of specialist his contribution to the school has become indispensable.

Teaching Method

In the manner already described psychology has changed teaching methods lock stock and barrel. All new research aims at evolving teaching methods that induce the child to learn for himself and thus himself achieve his development. The means of a changed teaching method are radio, cinema, cultural programmes, debates and other competitions, picnics, visits to historical places, games, election contests, students' unions, etc. It is psychology that has contributed the Kinder garden and Montessori methods of teaching. Special learning and teaching methods have been evolved for the use of blind, dumb and deaf children and mentally backward or handicapped children.

Individual Differences

In days gone by, the same curriculum was prescribed for all students in the class. The development of applied psychology led to the fact that different individuals differ from each other in respect of their interests, intelligence, ability, capabilities, etc. This knowledge of individual differences proved a variable source of revolution for education. Now-a-days the essential pre-requisite to guidance of an educational nature is knowledge

of the subject's ability and interests. Choice of his curriculum and other extra-curricular activities are definitely governed by these individual differences.

Process of Learning

The experiments that psychology has carried on in connection with the process of learning have led to the discovery of many laws that economies of time and yet produce good results in terms of material learned. One example would be Thorndike's laws. It is the psychologist who has stressed the importance of reward and punishment in learning. Research in psychology has aimed at discovering the efficacy and inefficacy of the various methods of learning. The employment of the various methods of learning is improved and made more scientific by knowledge of their qualities and drawbacks.

Mental Testing and Guidance

Mental testing and guidance are important factors in applied psychology. Psychologists have evolved scientific tests to measure intelligence and other abilities. Students can be guided in educational and vocational matters with the help of these tests. Now, education aims both at the complete development of the individual as well as his best possible adjustment to his vocation and of his vocation to his natural abilities. Teachers need the help of psychologists in either of these two matters.

Psychologists solve the problems of students through personal guidance and inform them of the job they are best suited to through vocational guidance. In this way, the co-operation and co-ordination between the teacher and the psychologist in the field of education is on the increase.

Reform of Problem Children

One of the most notable contributions of psychology to education is the improvement and reform of juvenile delinquents, absconding children, morons, backward children, unsocial and problem children, and advice to their parents regarding them. To this end the state and district psychological bureau assist the educational institutions.

Extra-Curricular Activities

The aim of education is the complete development of the child so that mere book knowledge is insufficient to gain this end. In this connection many other programmes are included in the curriculum on the suggestion of psychologists apart from the normal theoretical knowledge.

In this manner, educational psychology provides the teacher with the means to attaining knowledge of the student's nature, capacity, characteristics, needs and motive. This knowledge is the equipment for him to modify his teaching methods sufficiently to attain the goal for education.

THE CRITICAL ROLE OF CLASSROOM MANAGEMENT

Teachers play various roles in a typical classroom, but surely one of the most important is that of classroom manager. Effective teaching and learning cannot take place in a poorly managed classroom. If students are disorderly and disrespectful, and no apparent rules and procedures guide behaviour, chaos becomes the norm. In these situations, both teachers and students suffer. Teachers struggle to teach, and students most likely learn much less than they should. In contrast, well-managed classrooms provide an environment in which teaching and learning can flourish. But a well-managed classroom doesn't just appear out of nowhere. It takes a good deal of effort to create—and the person who is most responsible for creating it is the teacher.

We live in an era when research tells us that the teacher is probably the single most important factor affecting student achievement—at least the single most important factor that we can do much about. To illustrate, as a result of their study involving some 60,000 students, S. Paul Wright, Sandra Horn, and William Sanders (1997) note the following:

The results of this study will document that the most important factor affecting student learning is the teacher. In addition, the results show wide variation in effectiveness among teachers. The immediate and clear implication of this finding is that seemingly more can be done to improve education by improving the effectiveness of teachers than by any other single factor. *Effective teachers appear to be effective with students of all achievement levels regardless of the levels of heterogeneity in their classes.* If the teacher is ineffective, students under that teacher's tutelage will achieve inadequate progress academically, regardless of how similar or different they are regarding their academic achievement.

Researcher Kati Haycock (1998) uses the findings of this study and others conducted by William Sanders and his colleagues to paint a dramatic picture of the profound impact an individual teacher can have on student achievement. The differences in achievement between students who spend a year in class with a highly effective teacher as opposed to a highly ineffective teacher.

The classes of teachers classified as the most effective can be expected to gain about 52 percentile points in their achievement over a year's time. Students in the classes of teachers classified as least effective can be expected to gain only about 14 percentile points over a year's time. This comparison is even more dramatic when one realizes that some researchers have estimated that students will exhibit a gain in learning of about 6 percentile points simply from maturation—from growing one year older and gleaned new knowledge and information through everyday life. The least effective teachers, then, add little to the development of students' knowledge and skill beyond what would be expected from simply growing one year older in our complex, information-rich society.

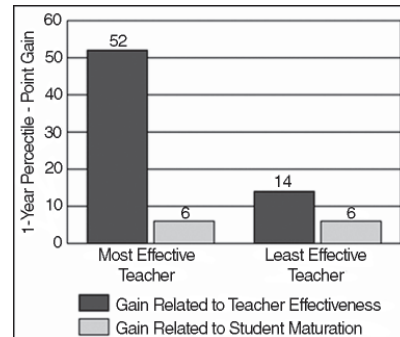


Fig. Impact of Teacher Effectiveness on Student Achievement.

Sanders and his colleagues, who gathered their data from elementary school students in Tennessee, are not the only ones to document dramatic differences in achievement between students in classes taught by highly ineffective versus highly effective teachers. Haycock (1998) reports similar findings from studies conducted in Dallas and Boston.

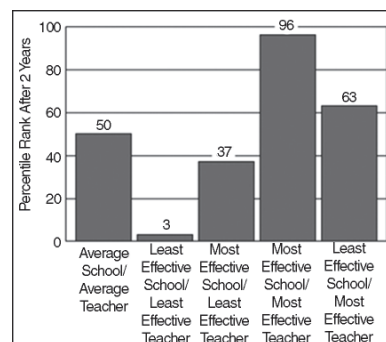


Fig. Effects of a School vs. a Teacher on Student Entering at 50th Percentile.

To understand the impact that a teacher can make, let's consider each of the five scenarios. A student begins at the 50th percentile in mathematics, let's say, and attends an average school and has an average teacher, her achievement will still be at the 50th percentile at the end of about two years. The student has learned enough to keep pace with her peers. But what happens to that student if she attends a school that is considered one of the least effective and is unfortunate enough to have a teacher who is classified as one of the least effective? After two years she has dropped from the 50th percentile to the 3rd percentile.

She may have learned something about mathematics, but that learning is so sporadic and unorganized that she has lost considerable ground in a short time. In the third scenario, the same student is in a school classified as most effective, but she has a teacher classified as least effective. Although the student entered the class at the 50th percentile, two years later she leaves the class at the 37th percentile. In contrast to the two previous scenarios, the fourth presents a very optimistic picture. The student is not only in a school classified as most effective, but also is in the class of a teacher classified as most effective. She enters the class at the 50th percentile, but she leaves at the 96th

percentile. The fifth scenario most dramatically depicts the impact of an individual teacher. Again, the student is in a school that is considered least effective, but she is with a teacher classified as most effective. The student now leaves the class at the 63rd percentile—13 percentile points higher than the point at which she entered. It is this last scenario that truly depicts the importance of individual teachers. Even if the school they work in is highly ineffective, individual teachers can produce powerful gains in student learning.

Although the effect the classroom teacher can have on student achievement is clear, the dynamics of how a teacher produces such an effect are not simple. Rather, the effective teacher performs many functions. These functions can be organized into three major roles: (1) making wise choices about the most effective instructional strategies to employ, (2) designing classroom curriculum to facilitate student learning, and (3) making effective use of classroom management techniques. The first role deals with instructional strategies and their use. Effective teachers have a wide array of instructional strategies at their disposal. They are skilled in the use of cooperative learning and graphic organizers; they know how best to use homework and how to use questions and advance organizers, and so on. Additionally, they know when these strategies should be used with specific students and specific content. Although cooperative learning might be highly effective in one lesson, a different approach might be better in another lesson. Some general strategies that have a good research “track record” in terms of enhancing student achievement have been detailed in *Classroom Instruction That Works: Research-Based Strategies for Increasing Student Achievement*.

The second role associated with effective teaching is classroom curriculum design. This means that effective teachers are skilled at identifying and articulating the proper sequence and pacing of their content. Rather than relying totally on the scope and sequence provided by the district or the textbook, they consider the needs of their students collectively and individually and then determine the content that requires emphasis and the most appropriate sequencing and presentation of that content. They are also highly skilled at constructing and arranging learning activities that present new knowledge in different formats (*e.g.*, stories, explanations, demonstrations) and different media (*e.g.*, oral presentations, written presentations, video presentations, Web-based presentations, simulations, hands-on activities).

The third role involved in effective teaching is classroom management. This, of course, is the subject of this book.

Before delving into classroom management, however, it is important to note that each of these three roles is a necessary but not sufficient component of effective teaching. That is, no single role by itself is sufficient to guarantee student learning, but take one out of the mix and you probably guarantee that students will have difficulty learning. Nevertheless, a strong case can be made that effective instructional strategies and good classroom curriculum design are built on the foundation of effective classroom management. As Long and Frye (1985) note in their book, *Making It Till Friday: A*

Guide to Successful Classroom Management, it is a myth to believe that... effective teachers can prevent all discipline problems by keeping students interested in learning through the use of exciting classroom materials and activities. The potential for problems exists beyond academics. Students experience difficulties at home which spill over into the classroom; students experience problems with peers during class breaks and in the classroom which often involve the teacher; and students experience mood changes which can generate problems, to name just a few.

Similarly, in their synthesis of the research, Edmund Emmer, Julie Sanford, Barbara Clements, and Jeanne Martin (1982) note that

At all public school grade levels, effective classroom management has been recognized as a crucial element in effective teaching. If a teacher cannot obtain students' cooperation and involve them in instructional activities, it is unlikely that effective teaching will take place... In addition, poor management wastes class time, reduces students' time on task and detracts from the quality of the learning environment.

DEEP AND SURFACE APPROACHES TO LEARNING

Approaches to learning describe what students do when they go about learning and why they do it. The basic distinction is between a deep approach to learning, where students are aiming towards understanding, and a surface approach to learning, where they are aiming to reproduce material in a test or exam rather than actually understand it.

The concept of preferences to different individual learning styles was introduced in an accompanying document. In this document we look at the associated concept of approaches to learning. The original work on approaches to learning was carried out by Marton and Saljo (1976). Their study explored students' approaches to learning a particular task. Students were given an academic text to read, and were told that they would subsequently be asked questions on that text. The students adopted two differing approaches to learning. The first group adopted an approach where they tried to understand the whole picture and tried to comprehend and understand the academic work. These students were identified with adopting a deep approach to learning. The second group tried to remember facts contained within the text, identifying and focusing on what they thought they would be asked later. They demonstrated an approach that we would recognise as rote learning, or a superficial, surface approach.

DEEP AND SURFACE APPROACHES

Deep and surface approaches to learning are words that most academics will have heard. In fact the idea that students can and do take a deep or surface approach to their learning is probably one of the most used bits of educational research in higher education. It is a very powerful and useful principle that we should apply most of the time to the way we teach. It is particularly applicable in engineering, and failure to apply it and

apply it properly explains how an awful lot goes wrong with the learning processes. Simply stated, deep learning involves the critical analysis of new ideas, linking them to already known concepts and principles, and leads to understanding and long-term retention of concepts so that they can be used for problem solving in unfamiliar contexts. Deep learning promotes understanding and application for life. In contrast, surface learning is the tacit acceptance of information and memorization as isolated and unlinked facts. It leads to superficial retention of material for examinations and does not promote understanding or long-term retention of knowledge and information.

Critical to our understanding of this principle is that we should not identify the student with a fixed approach to learning, but it is the design of learning opportunity that encourages students to adopt a particular approach.

DESIGNING FOR DEEP LEARNING

Very crudely: deep is good, surface is bad, and we should teach in a way that encourages students to adopt a deep approach; although achieving this is not so easy.

Perhaps the major influence on the students' approach to learning is the assessment methods. It is often argued that the explicit setting of "straightforward" assessments involving short questions testing separate ideas will encourage surface learning. However, again this is not necessarily the case as even the most apparently simple assessment questions can require students to demonstrate that their knowledge can be applied. For example, students can be asked to apply the laws of Ohm, Kirchhoff etc. albeit in simple cases rather than merely to quote them.

BASIC PRINCIPLES AND THE APPROACHES TO LEARNING

The evaluation of process is very valuable in determining the depth of learning, but if we concentrate on process alone we risk losing sight of the structure of the material being learnt. Engineering, like mathematics and science, is a hierarchical subject. As argued above, there is little point in trying to comprehend Kirchhoff's 2nd Law without first developing at least a working comprehension of potential, potential difference, emf., current, etc. and the ability to apply Ohm's Law reliably. This is not to say that understanding of the subject proceeds in a simple linear fashion (the naive bricks in the wall model of learning).

Working with the laws of Kirchhoff, Thévenin, Norton etc. will undoubtedly lead to a deeper understanding of earlier principles, but learning cannot start there. Attempting to work with more complex principles without a good grasp of the more basic principles from which they are built can only lead to frustration and a surface learning approach in which students attempt to memorise solutions to complex problems they cannot understand. Encouraging students to practice the application of basic principles will not force them adopt a deep approach to learning, but it at least makes it possible.

PUTTING THEORY INTO PRACTICE

The following table compiled from the work of Biggs (1999), Entwistle (1988) and Ramsden (1992) provides some very valuable characteristics of the approaches and illustrates the importance of how we manage the curriculum impacts on the learning process. For example, clearly stated academic aims, opportunities to exercise some choice and well aligned assessment strategies that help students to build confidence can be found among the factors identified as encouraging a deep approach.

Table. Compares the characteristics and factors that encourage Deep and Surface Approaches to learning.

	Deep Learning	Surface Learning
Definition:	Examining new facts and ideas critically, and tying them into existing cognitive structures and making numerous links between ideas.	Accepting new facts and ideas uncritically and attempting to store them as isolated, unconnected, items.
Looking for meaning.	Focussing on the central argument or concepts needed to solve a problem.	Relying on rote learning. Focussing on outwards signs and the formulae needed to solve a problem.
Characteristics	Interacting actively.	Receiving information
Distinguishing between	passively. argument and evidence. Making connections between different modules. Relating new and previous knowledge. Linking course content to real life. Having an intrinsic simply as material to be well and mentally engaging when doing academic work.	Failing to distinguish principles from examples. Treating parts of modules and programmes as separate. Not recognising new material as building on previous work. Seeing course content curiosity in the subject. Being determined to do learnt for the exam.
Encouraged by Students'	Having the appropriate background knowledge for a sound foundation. Having time to pursue interests, through good time management. Positive experience of education leading to	Studying a degree for the qualification and not being interested in the subject. Not focussing on academic areas, but emphasising others (<i>e.g.</i> social, sport). Lacking background knowledge and understanding necessary to understand material. Not enough time / too

Showing personal interest	confidence in ability to understand and succeed.	high a workload. Cynical view of education,
High	believing that factual recall	in the subject. Bringing is what is required.
Encouraged by	subject. Concentrating on	out the structure of the anxiety.
Teachers'	and ensuring plenty of time for key concepts.	Conveying disinterest or even a negative attitude to the material. Presenting material so that it can be perceived as a series of unrelated facts and ideas. Allowing students to be passive. Assessing for independent facts (short answer questions).
	Confronting students' misconceptions. Engaging students in active learning.	Rushing to cover too much material. Emphasizing coverage at the expense of depth. Creating undue anxiety or low expectations of success by discouraging statements or excessive workload. Having a short assessment cycle.
	Using assessments that require thought, and requires ideas to be used together.	
	Relating new material to what students already know and understand.	
	Allowing students to make mistakes without penalty and rewarding effort.	
	Being consistent and fair in assessing declared intended learning outcomes,	
	and hence establishing trust.	

The last row of the table provides us with some simple guidelines as the “do’s” and “don’ts” in teaching. A particular example is to use problem based learning. Rather than producing assessments that require rote application of Kirchhoff’s 2nd Law, such as working out the current in an abstract network, we need to provide assessments where students need to link multiple ideas and concepts together, such as using Kirchhoff’s Laws, Ohm’s Law and their understanding of electrical principals, to design an amplifier for a particular purpose.

Therefore, in order to encourage active learning we need to be positive about the study of engineering. We need to concentrate on the key concepts, not just in isolation, but also by demonstrating the way that the components link together. We can also see that over reliance on traditional lectures, where students are passively taking notes and not being required to engage actively with material, will not encourage a deep approach. Similarly, over assessment, through repeated testing, while seen to regularly focus the learners on the material, is likely to have the opposite effect to that desired by just encouraging memorising of facts. Fewer assessments in general, and assessments that encourage and require students to engage with problems, will also encourage the students to use and apply their learning, facilitating the deep approaches that we require.

We need to think carefully about the assessment and assessment processes, as it is this part of the curriculum that affects the students' approaches to learning most. We need to construct assessment that gives students opportunity to receive feedback, but also must make the assessment relevant to the real world of engineering.

BEHAVIOURIST THEORIES OF LEARNING

BEHAVIOURIST APPROACH

Behaviourism (also called the Behaviourist approach) was the primary paradigm in psychology between 1920s to 1950 and is based on a number of underlying assumptions regarding methodology and Behavioural analysis:

- Psychology should be seen as a science. Theories need to be supported by empirical data obtained through careful and controlled observation and measurement of behaviour. Watson (1913) stated that “*psychology as a Behaviourist views it is a purely objective experimental branch of natural science. Its theoretical goal is ... prediction and control*”.
- Behaviourism is primarily concerned with observable behaviour, as opposed to internal events like thinking and emotion. Observable (*i.e.* external) behaviour can be objectively and scientifically measured. Internal events, such as thinking should be explained through Behavioural terms (or eliminated altogether).
- People have no free will – a person's environment determines their behaviour
- When born our mind is ‘tabula rasa’ (a blank slate).
- There is little difference between the learning that takes place in humans and that in other animals. Therefore research can be carried out on animals as well as humans.
- Behaviour is the result of stimulus – response (*i.e.* all behaviour, no matter how complex, can be reduced to a simple stimulus – response association). Watson described the purpose of psychology as: “

To predict, given the stimulus, what reaction will take place; or, given the reaction, state what the situation or stimulus is that has caused the reaction”.

All behaviour is learnt from the environment. We learn new behaviour through classical or operant conditioning.

VARIETIES OF BEHAVIOURISM

Historically, the most significant distinction among versions of Behaviourism is that between Watson's original classical Behaviourism, and forms of Behaviourism later inspired by his work, known collectively as neo-behaviourism.

In his book, Psychology as the Behaviourist Views It Watson outlines the principles of all behaviourists:

Psychology as the behaviourist views it is a purely objective experimental branch of natural science. Its theoretical goal is the prediction and control of behaviour. Introspection forms no essential part of its methods, nor is the scientific value of its data dependent upon the readiness with which they lend themselves to interpretation in terms of consciousness. The Behaviourist, in his efforts to get a unitary scheme of animal response, recognizes no dividing line between man and brute. The behaviour of man, with all of its refinement and complexity, forms only a part of the Behaviourist's total scheme of investigation.

SOCIAL LEARNING

However, the theory strongly implies that there are types of learning wherein direct reinforcement is not the causal mechanism; rather, the so called social element can result to the development of new learning among individuals. Social Learning Theory has been useful in explaining how people can learn new things and develop new Behaviours by observing other people. It is to assume, therefore, that Social Learning Theory is concerned on observational learning process among people.

BASIC CONCEPTS

- **Observational Learning:** The Social Learning Theory says that people can learn by watching other people perform the behaviour. Observational learning explains the nature of children to learn Behaviours by watching the behaviour of the people around them, and eventually, imitating them. With the “Bobo Doll” experiment(s), Bandura included an adult who is tasked to act aggressively Towards a Bobo Doll while the children observe him. Later, Bandura let the children play inside a room with the Bobo Doll. He affirmed that these children imitated the aggressive behaviour Towards the doll, which they had observed earlier.
- *After his studies, Bandura was able to determine 3 basic models of observational learning, which include:*
 - A Live Model, which includes an actual person performing a behaviour.
 - A Verbal Instruction Model, which involves telling of details and descriptions of a behaviour.
 - A Symbolic Model, which includes either a real or fictional character demonstrating the behaviour via movies, books, television, radio, online media and other media sources.
- The state of mind (mental states) is crucial to learning. In this concept, Bandura stated that not only external reinforcement or factors can affect learning and behaviour. There is also what he called intrinsic reinforcement, which is in a

form of internal reward or a better feeling after performing the behaviour (e.g. sense of accomplishment, confidence, satisfaction, etc.)

- Learning does not mean that there will be a change in the behaviour of an individual.

MODELING PROCESS

The Modeling Process developed by Bandura helps us understand that not all observed Behaviours could be learned effectively, nor learning can necessarily result to Behavioural changes.

The modeling process includes the following steps in order for us to determine whether social learning is successful or not:

- *Step 1: Attention:* Social Cognitive Theory implies that you must pay attention for you to learn. If you want to learn from the behaviour of the model (the person that demonstrates the behaviour), then you should eliminate anything that catches your attention other than him. Also, the more interesting the model is, the more likely you are to pay full attention to him and learn.
- *Step 2: Retention:* Retention of the newly learned behaviour is necessary. Without it, learning of the behaviour would not be established, and you might need to get back to observing the model again since you were not able to store information about the behaviour.
- *Step 3: Reproduction:* When you are successful in paying attention and retaining relevant information, this step requires you to demonstrate the behaviour. In this phase, practice of the behaviour by repeatedly doing it is important for improvement.
- *Step 4: Motivation:* Feeling motivated to repeat the behaviour is what you need in order to keep on performing it. This is where reinforcement and punishment come in. You can be rewarded by demonstrating the behaviour properly, and punished by displaying it inappropriately.

SOCIAL CONSTRUCTIVISM

The level of potential development is the level at which learning takes place. It comprises cognitive structures that are still in the process of maturing, but which can only mature under the guidance of or in collaboration with others.

Social constructivism is a variety of cognitive constructivism that emphasizes the collaborative nature of much learning. Social constructivism was developed by post-revolutionary Soviet psychologist Lev Vygotsky. Vygotsky was a cognitivist, but rejected the assumption made by cognitivists such as Piaget and Perry that it was possible to separate learning from its social context. He argued that all cognitive functions originate in, and must therefore be explained as products of social interactions and that learning was not simply the assimilation and accommodation of new knowledge by learners; it

was the process by which learners were integrated into a knowledge community. According to Vygotsky, Every function in the child's cultural development appears twice: first, on the social level and, later on, on the individual level; first, between people (interpsychological) and then inside the child (intrapsychological). This applies equally to voluntary attention, to logical memory, and to the formation of concepts. All the higher functions originate as actual relationships between individuals. Vygotsky's theory of social learning has been expanded upon by numerous later theorists and researchers.

View of Knowledge

Cognitivists such as Piaget and Perry see knowledge as actively constructed by learners in response to interactions with environmental stimuli. Vygotsky emphasized the role of language and culture in cognitive development. According to Vygotsky, language and culture play essential roles both in human intellectual development and in how humans perceive the world. Humans' linguistic abilities enable them to overcome the natural limitations of their perceptual field by imposing culturally defined sense and meaning on the world. Language and culture are the frameworks through which humans experience, communicate, and understand reality. Vygotsky states, A special feature of human perception ... is the perception of real objects ... I do not see the world simply in colour and shape but also as a world with sense and meaning. I do not merely see something round and black with two hands; I see a clock ... Language and the conceptual schemes that are transmitted by means of language are essentially social phenomena. As a result, human cognitive structures are, Vygotsky believed, essentially socially constructed. Knowledge is not simply constructed, it is co-constructed.

View of Learning

Vygotsky accepted Piaget's claim that learners respond not to external stimuli but to their interpretation of those stimuli. However, he argued that cognitivists such as Piaget had overlooked the essentially social nature of language. As a result, he claimed they had failed to understand that learning is a collaborative process. Vygotsky distinguished between two developmental levels:

The level of actual development is the level of development that the learner has already reached, and is the level at which the learner is capable of solving problems independently. The level of potential development (the "zone of proximal development") is the level of development that the learner is capable of reaching under the guidance of teachers or in collaboration with peers. The learner is capable of solving problems and understanding material at this level that they are not capable of solving or understanding at their level of actual development; the level of potential development is the level at which learning takes place. It comprises cognitive structures that are still in the process of maturing, but which can only mature under the guidance of or in collaboration with others.

View of Motivation

Behavioural motivation is essentially extrinsic — a reaction to positive and negative reinforcements. Cognitive motivation is essentially intrinsic — based on the learner's internal drive. Social constructivists see motivation as both extrinsic and intrinsic. Because learning is essentially a social phenomenon, learners are partially motivated by rewards provided by the knowledge community. However, because knowledge is actively constructed by the learner, learning also depends to a significant extent on the learner's internal drive to understand and promote the learning process.

Implications for Teaching

Collaborative learning methods require learners to develop teamwork skills and to see individual learning as essentially related to the success of group learning. The optimal size for group learning is four or five people. Since the average section size is ten to fifteen people, collaborative learning methods often require GSIs to break students into smaller groups, although discussion sections are essentially collaborative learning environments. For instance, in group investigations, students may be split into groups that are then required to choose and research a topic from a limited area. They are then held responsible for researching the topic and presenting their findings to the class. More generally, collaborative learning should be seen as a process of peer interaction that is mediated and structured by the teacher. Discussion can be promoted by the presentation of specific concepts, problems, or scenarios; it is guided by means of effectively directed questions, the introduction and clarification of concepts and information, and references to previously learned material.

PSYCHOLOGY: MOTIVATION AND LEARNING

RESEARCH FUNDAMENTALS

Extrinsic motivation and fear of failure: When asked what makes them study and work hard in college, undergraduates typically say that they are trying to get the best grade possible. Grades, in fact, are the primary focus of most students (an extrinsic motivation). Only as secondary reasons do students list the desires to become competent, to prove themselves, and to avoid mistakes (intrinsic motivations). Learning about the content of the course for its own sake is the last of the reasons students give.

Conflicting expectations: As a result, students' goals tend to be mismatched with their instructors' objectives. First-year students, and many students beyond the freshmen level, tend to believe that it is their responsibility simply to follow directions and to absorb course content. They think that instructors should simply present and explain material, and that instructors are responsible for making that material interesting, relevant, and fun. Although there may be some truth to these perceptions, most instructors want to do far more — to encourage active engagement with the material, to foster

students' ability to self-monitor their learning, and to create a safe environment in which students can be inquisitive learners.

Leveraging intrinsic motivation: Several strategies can connect with students' interests so that they come closer to their instructors' goals for them:

- Organize a lesson plan around a problem for students to solve using the course material. The problem should relate to things students are already interested in — for example, in a pre-med microbiology course, having students put themselves in the place of a pediatrician figuring out which systems are at issue given a hypothetical patient's symptoms.
- If you are creating your own course, organize the entire course around a large question or problem that the course material will eventually enable them to accurately explain and solve.
- Connect course material with the non-academic world.
- Take advantage of events the students are interested in to explore how your field views the issues involved.
- Curiosity is great for priming intrinsic interest. Bring in anomalies or curiosities that students can use the course material to analyze and explain.

THEORIES OF TEACHING

Theories of learning, whether explicit or tacit, informed by study or intuition, well-considered or not, play a role in the choices instructors make concerning their teaching. The major trend in understanding how students learn has been a movement away from the behaviourist model to a cognitive view of learning.

IMPLICATIONS FOR TEACHING PRACTICE OF SOME KEY IDEAS FROM LEARNING THEORIES

Learning is a Process of Active Construction

Learning is the interaction between what students know, the new information they encounter, and the activities they engage in as they learn. Students construct their own understanding through experience, interactions with content and others, and reflection.

- *Teaching Implication:* Provide opportunities for students to connect with your content in a variety of meaningful ways by using cooperative learning, interactive lectures, engaging assignments, hands-on lab/field experiences, and other active learning strategies.

Students' Prior Knowledge is an Important Determinant of what they will Learn

Students do not come to your class as a blank slate. They use what they already know about a topic to interpret new information. When students cannot relate new material to

what they already know, they tend to memorize—learning for the test—rather than developing any real understanding of the content.

- *Teaching Implication:* Learn about your students' experiences, preconceptions, or misconceptions by using pre-tests, background knowledge probes, and written or oral activities designed to reveal students' thinking about the topic.

Organizing Information into a Conceptual Framework helps Students Remember and Use Knowledge

Students must learn factual information, understand these facts and ideas in the context of a conceptual framework, and organize knowledge in ways that facilitate retrieval and application in order to develop competence in a new topic.

- *Teaching Implication:* Support students by using concept maps, flowcharts, outlines, comparison tables, etc., to make the structure of the knowledge clear.

Learning is a Social Phenomenon

Students learn with greater understanding when they share ideas through conversation, debate, and negotiation. Explaining a concept to one's peers puts knowledge to a public test where it can be examined, reshaped, and clarified.

- *Teaching Implication:* Use Cooperative learning strategies, long-term group projects, class discussions, and group activities to support the social side of learning.

Learning is Context-specific

It is often difficult for students to use what they learn in class in new contexts (*i.e.*, other classes, the workplace, or their personal lives).

- *Teaching Implication:* Use problem-based learning, simulations or cases, and service learning to create learning environments similar to the real world.

Students' Metacognitive Skills (Thinking about Thinking) are Important to their Learning

Many students utilize few learning strategies and have a limited awareness of their thinking processes.

- *Teaching Implication:* Help students become more metacognitively aware by modeling your thinking as you solve a problem, develop an argument, or analyze written work in front of the class. Teach metacognitive strategies, such as setting goals, making predictions, and checking for consistency. Focus attention on metacognition by having students write in a learning journal or develop explanations of their problem-solving processes.

Modern Theories of Intelligence

Early intelligence theory emerged from an emphasis on a unitary concept of general ability, as can be seen in the definitions of Binet and Spearman. Spearman created a statistical technique called factor analysis to explore his approach. From his studies with this technique, he was able to report that about half of the variance in tests of mental ability was due to the general (*g*) factor. The remainder was due to the special ability (*e.g.*, numerical reasoning, vocabulary, mechanical skill) that was required of a person to enable performance of the specific tasks on the test. Later approaches tended to emphasize expanded abilities. For example, Cattell divided *g* into fluid (*gf*) and crystallized (*gc*) intelligence. *Fluid intelligence* encompasses abilities involved in thinking, reasoning, and in learning, while *crystallized intelligence* represents the knowledge and broader understanding that has developed through learning in the environmental setting.

Other theories further recognized the diversity of intelligent performance. In his 1967 book *Structure of Intellect*, Joy Paul Guilford (1897–1987), using factor analysis, devised a model of intelligence he termed the *Structure of Intellect* in which he proposed three aspects of intelligence: operations, products, and contents. Each of these is broken down further into specific kinds of intellectual activity which Guilford considered interrelated to produce intelligent functioning of specific tasks.

Gardner (1993) proposed a theory of multiple intelligences based on the differential cognitive processing required for demonstration of intelligent or creative performance in different areas. Gardner's theory references eight intelligences. Linguistic and logico-mathematical intelligences are most often associated with academic performance,

although others could be relevant depending on the task. Other intelligences identified by Gardner were musical, kinesthetic, spatial, naturalistic, and personal (intrapersonal and interpersonal).

Robert Sternberg also proposed a theory informed by research from cognitive psychology. Sternberg's model is named the *triarchic theory* of intelligence because it is composed of three kinds of components: memory-analytic, creative-synthetic, and practical-contextual. The first component is related to the academic view of intelligence and is most similar to what most intelligence tests measure. The second component is necessary for creative endeavours, including traditionally academic areas such as science and mathematics. Finally, the practical-contextual is necessary for success in an everyday environment like school or business. Daniel Golman (1995) has proposed an emotional intelligence that bears some similarity to the personal intelligences of Gardner and the practical intelligence of Sternberg, but goes further in tying emotion and personality to the capacity for intelligent behaviour. For example, fear, excitement, or anger may contribute to how one behaves regardless of one's knowledge or capacity to reason.

In the first half of the 20th century, research in intelligence was heavily influenced by factor analysis (first used by Spearman). Factor analysis is a statistical procedure that enables the systematic study of the relationships within a set of variables in order to find the common aspects. Research and theory emerging in the second half of the 20th into the 21st century has begun to have an impact on the methods of assessment of intelligence. For example, research in cognitive psychology, neuro-psychology, cognitive science, biopsychology and evolutionary processes, and cultural psychology has begun to affect theory and research in intelligence, its measurement and applications.

NATURE OF INTELLIGENCE

Intelligence is the *manipulation of motor mechanisms* to derive happiness for the organism – inclusive of surviving – and entails the gathering of information external to the organism so by nonverbal communication and wise *motor manipulation*, it can maximize happiness. And, at root, all living organisms are nothing more than a physical organization of their genotype: the initial genetic blueprint of an organism.

Ultimately, at its root, the genotype is actually nothing more than immaterial energy configuring atoms and molecules into genetic code. As it is immaterial – as a thought or idea is immaterial – the genotype is more like a “thought” or “idea” interacting with the other “thoughts” and “ideas” and set to interact in harmony with its environment – also a “thought.” Interaction is a “*thought being thought-out.*” This idea is not new nor original. Aristotle theorized that God was pure Thought and prior to the Greeks, the Judeo-Christian idea of the Logos, and Egyptian idea of their God, Amun. The physical is not the blueprint – the immaterial electromagnetic is.

The immaterial is idea-like – a living *thought*. As thinking is mediated by plus and minus ions (NaCl^- , KCl^- , Na^+ , K^+ , Cl^-) moving in and out of the nerve fibres of our brain and nervous system, *thought* and *thinking* is the sophisticated summation of the

flows and interplay of positive and negative energies. Both the medium for thinking (i.e. the brain) and the ability of thinking itself exist in “seed-like” form in the genotype and is “species-specific.”

The nature of the genotype is that it is a *hand-glove model* of its environment both in the sense that all features of a given animal’s environment are modelled in the genotype’s blueprint and in that the genotype creates the animal’s body and behaviour to fit *hand-glove* into its unique environment. In essence, in tiny cells, genotypes taken as a whole – at some snapshot of geological time – serve as a perfect multimedia picture-record written, imprinted in genetic code, of the heavens, the skies, the waters, and the living earth.

Paul Davies, writing of the ability of birds to use stars in their astro-navigation, comments that, “The necessary astro-navigational information is built in DNA... given a sufficient understanding of the nature of DNA, one could ‘decode’ this information and reconstruct a map of the stars.” DNA is built on an immaterial blueprint. In the case of you and I, we are, in essence, just the immaterial blueprint of the one-celled genotype that became us: eyes, brain, charming smile, wit, and great brilliance included.

Language ability is nothing more than the ability of the genotype to associate random sounds with contemporary objects of evolution (i.e. today’s world is different from the first human’s world). Reading and writing is nothing less than the ability of the genome to devise and associate random signs (otherwise known as written alphabets, words, and languages) with contemporary objects of evolution such as the words “ice cream” and the “Internet.” Rats evolve right along with us finding great comfort in living structures and types of food the prehistoric world had no imagination would ever exist. Insects have also kept pace with human evolution. Even viruses: computer viruses.

We are living witnesses of co-evolution. We can observe the ability of organisms worldwide to harmoniously co-evolve ever upwards: e.g. in the microscopic-level arm’s race, viral innovations against medicines force humans to develop better medicines. Harmonious tit for tat co-evolution finds origins in the organizational marvel of the immaterial dimension. One unfortunate effect of imbalance is that it creates new “harmonious” balances and the imbalances the human has produced have eliminated many life forms from the environment that man is destroying in 200 years that took 2 million years to create. Nevertheless, for living creatures, harmony and balance with nature are written and achieved in genotypes and together, genotypes make the whole living picture of life on earth – each specie of genotype one piece of the puzzle fitting perfectly with the rest. With the right equipment, the genotype could play an organism’s life like a video in a VCR. Through life – via the interaction of living things – Nature manifests on yet another level, the balance and harmony found throughout the universe. Balance is found, for instance, in every animal having a strength of defence against others, but a weakness making it available as fodder and the strong have the greatest likelihood of surviving, but death takes an old life away to make way for new. While from the perspective of the individual, intelligence serves the purpose of striving for,

realizing, and maintaining individual happiness, taken as a whole intelligence serves the greater purpose of bringing the whole of the living world and nature into harmonious oneness. Let us remember several things: living things have not the foggiest idea of the technologies that make their organs and forms and make them interact harmoniously within the organism and with the whole environment from starlight to raindrops.

Differences in Intelligence

In this module, we will consider the extremes of intelligence—from the very bright to the very dull. We will also look at a few of the many studies that have been done with IQ scores, correlating many things from sense of humour (Brumbaugh, 1939) to racial background. A few of these correlation studies and their results will be described in this module.

As you read the text, try to answer the following questions.

- When is a child considered gifted? Retarded?
- What are the degrees of mental retardation?
- What were Terman's findings in his study of gifted children?
- How does intelligence correlate with sex? Race? Socioeconomic level?

Theory of Intelligences: Single, Two Factor and Multifactor Theories

The theory of multiple intelligences was proposed by Howard Gardner in 1983, to more accurately define the concept of intelligence and address whether methods which claim to measure intelligence (or aspects thereof) are truly scientific.

Gardner's theory argues that intelligence, particularly as it is traditionally defined, does not sufficiently encompass the wide variety of abilities humans display. In his conception, a child who masters multiplication easily is not necessarily more intelligent *overall* than a child who struggles to do so.

The second child may be stronger in another *kind* of intelligence, and therefore may best learn the given material through a different approach, may excel in a field outside of mathematics, or may even be looking through the multiplication learning process at a fundamentally deeper level that hides a potentially higher mathematical intelligence than in the one who memorizes the concept easily.

Development of the Theory

"Howard Gardner, multiple intelligences and education" by given field (child prodigies, autistic savants);

- Neurological evidence for areas of the brain that are specialized for particular capacities (often including studies of people who have suffered brain damage affecting a specific capacity);
- The evolutionary relevance of the various capacities;
- Psychometric studies; and

- Susceptibility to encoding in a symbol system or notation (e.g. written language, musical notation, choreography).
- An identifiable core operation or set of operations.
- A distinctive development history, along with a definable set of 'end-state' performances.

Gardner originally identified seven core intelligences: linguistic, logical-mathematical, spatial, bodily-kinesthetic, musical, interpersonal and intrapersonal. In 1997 at the symposium "MIND 97" (Multiple Intelligences New Directions) he added an eighth, the "Naturalist" Intelligence. Investigation continues on whether there are Existentialist (existential) and Spiritualist (spiritual) Intelligences.

The theory has been widely criticized in the psychology and educational theory communities. The most common criticisms argue that Gardner's theory is based on his own intuition rather than empirical data and that the intelligences are just other names for talents or personality types.

Despite these criticisms, the theory has enjoyed a great deal of popularity amongst educators over the past twenty years. There are several schools which espouse MI as a pedagogy, and many individual teachers who incorporate some or all of the theory into their methodology. Many books and educational materials exist which explain the theory and how it may be applied to the classroom.

Creativity and Intelligence

There has been debate in the psychological literature about whether intelligence and creativity are part of the same process (the conjoint hypothesis) or represent distinct mental processes (the disjoint hypothesis). Evidence from attempts to look at correlations between intelligence and creativity from the 1950s onwards, by authors such as Barron, Guilford or Wallach and Kogan, regularly suggested that correlations between these concepts were low enough to justify treating them as distinct concepts.

Some researchers believe that creativity is the outcome of the same cognitive processes as intelligence, and is only judged as creativity in terms of its consequences, i.e. when the outcome of cognitive processes happens to produce something novel, a view which Perkins has termed the "nothing special" hypothesis.

A very popular model is what has come to be known as "the threshold hypothesis", proposed by Ellis Paul Torrance, which holds that a high degree of intelligence appears to be a necessary but not sufficient condition for high creativity. This means that, in a general sample, there will be a positive correlation between creativity and intelligence, but this correlation will not be found if only a sample of the most highly intelligent people are assessed. Research into the threshold hypothesis, however, has produced mixed results ranging from enthusiastic support to refutation and rejection.

An alternative perspective, Renzulli's three-rings hypothesis, sees giftedness as based on both intelligence and creativity. More on both the threshold hypothesis and Renzulli's work can be found in O'Hara and Sternberg.

Kinds of intelligent performance that matter in everyday life, such as social intelligence. Most of theories of multiple intelligences are relatively recent in origin, though Louis Thurstone proposed a theory of multiple “primary abilities” in the early 20th Century.

Triarchic Theory of Intelligence

Robert Sternberg’s triarchic theory of intelligence proposes three fundamental aspects of intelligence — analytic, creative, and practical — of which only the first is measured to any significant extent by mainstream tests. His investigations suggest the need for a balance between analytic intelligence, on the one hand, and creative and especially practical intelligence on the other.

Emotional Intelligence

Daniel Goleman and several other researchers have developed the concept of emotional intelligence and claim it is at least as “important” as more traditional sorts of intelligence. These theories grew from observations of human development and of brain injury victims who demonstrate an acute loss of a particular cognitive function — e.g. the ability to think numerically, or the ability to understand written language — without showing any loss in other cognitive areas.

Empirical Evidence

IQ proponents have pointed out that IQ’s predictive validity has been repeatedly demonstrated, for example in predicting important non-academic outcomes such as job performance, whereas the various multiple intelligence theories have little or no such support. Meanwhile, the relevance and even the existence of multiple intelligences have not been borne out when actually tested. A set of ability tests that do not correlate together would support the claim that multiple intelligences are independent of each other.

Factors Affecting Intelligence

Intelligence is an ill-defined, difficult to quantify concept. Accordingly, the IQ tests used to measure intelligence provide only approximations of the posited ‘real’ intelligence. In addition, a number of theoretically unrelated properties are known to correlate with IQ such as race, gender and height but since correlation does not imply causation the true relationship between these factors is uncertain. Factors affecting IQ may be divided into biological and environmental.

Biological

Evidence suggests that genetic variation has a significant impact on IQ, accounting for three fourths in adults. Despite the high heritability of IQ, few genes have been

found to have a substantial effect on IQ, suggesting that IQ is the product of interaction between multiple genes. Other biological factors correlating with IQ include ratio of brain weight to body weight and the volume and location of gray matter tissue in the brain. Because intelligence appears to be at least partly dependent on brain structure and the genes shaping brain development, it has been proposed that genetic engineering could be used to enhance the intelligence of animals, a process sometimes called biological uplift in science fiction.

Experiments on mice have demonstrated superior ability in learning and memory in various behavioural tasks.

Environmental

Evidence suggests that family environmental factors may have an effect upon childhood IQ, accounting for up to a quarter of the variance. On the other hand, by late adolescence this correlation disappears, such that adoptive siblings are no more similar in IQ than strangers.

Moreover, adoption studies indicate that, by adulthood, adoptive siblings are no more similar in IQ than strangers, while twins and full siblings show an IQ correlation. Consequently, in the context of the nature versus nurture debate, the “nature” component appears to be much more important than the “nurture” component in explaining IQ variance in the general population.

There are indications that, in middle age, intelligence is influenced by life style choices (e.g., long working hours). Cultural factors also play a role in intelligence. For example, on a sorting task to measure intelligence, Westerners tend to take a taxonomic approach while the Kpelle people take a more functional approach.

For example, instead of grouping food and tools into separate categories, a Kpelle participant stated “the knife goes with the orange because it cuts it”

Ethical Issues

Since intelligence is susceptible to modification through the manipulation of environment, the ability to influence intelligence raises ethical issues. Transhumanist theorists study the possibilities and consequences of developing and using techniques to enhance human abilities and aptitudes, and ameliorate what it regards as undesirable and unnecessary aspects of the human condition; eugenics is a social philosophy which advocates the improvement of human hereditary traits through various forms of intervention. The perception of eugenics has varied throughout history, from a social responsibility required of society, to an immoral, racist stance. Neuroethics considers the ethical, legal and social implications of neuroscience, and deals with issues such as difference between treating a human neurological disease and enhancing the human brain, and how wealth impacts access to neurotechnology. Neuroethical issues interact with the ethics of human genetic engineering.

Creativity: Views of Torrance, Getzels, Guilford

Creativity is a phenomenon whereby something new and somehow valuable is formed. The created item may be intangible (such as an idea, a scientific theory, a musical composition or a joke) or an original physical object (such as an invention, a literary work or a painting).

Scholarly interest in creativity involves many definitions and concepts pertaining to a number of disciplines: psychology, cognitive science, education, philosophy (particularly philosophy of science), technology, theology, sociology, linguistics, business studies, songwriting, and economics, covering the relations between creativity and general intelligence, mental and neurological processes, personality type and creative ability, creativity and mental health; the potential for fostering creativity through education and training, especially as augmented by technology; and the application of creative resources to improve the effectiveness of teaching and learning.

Definition

In a summary of scientific research into creativity, Michael Mumford suggested: “Over the course of the last decade, however, we seem to have reached a general agreement that creativity involves the production of novel, useful products”, or, in Robert Sternberg’s words, the production of “something original and worthwhile”. Authors have diverged dramatically in their precise definitions beyond these general commonalities: Peter Meusburger reckons that over a hundred different analyses can be found in the literature. As an illustration, one definition given by Dr. E. Paul Torrance described it as “a process of becoming sensitive to problems, deficiencies, gaps in knowledge, missing elements, disharmonies, and so on; identifying the difficulty; searching for solutions, making guesses, or formulating hypotheses about the deficiencies; testing and retesting these hypotheses and possibly modifying and retesting them; and finally communicating the results.”

Aspects

Theories of creativity (particularly investigation of why some people are more creative than others) have focused on a variety of aspects. The dominant factors are usually identified as “the four Ps” — process, product, person and place. A focus on *process* is shown in cognitive approaches that try to describe thought mechanisms and techniques for creative thinking. Theories invoking divergent rather than convergent thinking (such as Guilford), or those describing the staging of the creative process (such as Wallas) are primarily theories of creative process. A focus on creative *product* usually appears in attempts to measure creativity and in creative ideas framed as successful memes. The psychometric approach to creativity reveals that it also involves the ability to produce more. A focus on the nature of the creative *person* considers more general intellectual habits, such as openness, levels of ideation, autonomy, expertise, exploratory behaviour

and so on. A focus on *place* considers the circumstances in which creativity flourishes, such as degrees of autonomy, access to resources and the nature of gatekeepers. Creative lifestyles are characterized by non-conforming attitudes and behaviours as well as flexibility.

Etymology

The lexeme in the English word *creativity* comes from the Latin term *creô* “to create, make”: its derivational suffixes also come from Latin. The word “create” appeared in English as early as the 14th century, notably in Chaucer, to indicate divine creation. However, its modern meaning as an act of human creation did not emerge until after the Enlightenment.

HISTORY OF THE CONCEPT

Ancient Views

Most ancient cultures, including thinkers of Ancient Greece, Ancient China, and Ancient India, lacked the concept of creativity, seeing art as a form of discovery and not creation. The ancient Greeks had no terms corresponding to “to create” or “creator” except for the expression “*poiein*” (“to make”), which only applied to *poiesis* (poetry) and to the *poietes* (poet, or “maker”) who made it. Plato did not believe in art as a form of creation. Asked in *The Republic*, “Will we say, of a painter, that he makes something?”, he answers, “Certainly not, he merely imitates.”

It is commonly argued that the notion of “creativity” originated in Western culture through Christianity, as a matter of divine inspiration. According to the historian Daniel J. Boorstin, “the early Western conception of creativity was the Biblical story of creation given in the *Genesis*.” However, this is not creativity in the modern sense, which did not arise until the Renaissance. In the Judaeo-Christian tradition, creativity was the sole province of God; humans were not considered to have the ability to create something new except as an expression of God’s work. A concept similar to that of Christianity existed in Greek culture, for instance, Muses were seen as mediating inspiration from the Gods. Romans and Greeks invoked the concept of an external creative “daemon” (Greek) or “genius” (Latin), linked to the sacred or the divine. However, none of these views are similar to the modern concept of creativity, and the individual was not seen as the cause of creation until the Renaissance. It was during the Renaissance that creativity was first seen, not as a conduit for the divine, but from the abilities of “great men”.

The Enlightenment and After

The rejection of creativity in favour of discovery and the belief that individual creation was a conduit of the divine would dominate the West probably until the Renaissance

and even later. The development of the modern concept of creativity begins in the Renaissance, when creation began to be perceived as having originated from the abilities of the individual, and not God.

However, this shift was gradual and would not become immediately apparent until the Enlightenment. By the 18th century and the Age of Enlightenment, mention of creativity (notably in art theory), linked with the concept of imagination, became more frequent.

In the writing of Thomas Hobbes, imagination became a key element of human cognition; William Duff was one of the first to identify imagination as a quality of genius, typifying the separation being made between talent (productive, but breaking no new ground) and genius. As a direct and independent topic of study, creativity effectively received no attention until the 19th century. Runco and Albert argue that creativity as the subject of proper study began seriously to emerge in the late 19th century with the increased interest in individual differences inspired by the arrival of Darwinism. In particular they refer to the work of Francis Galton, who through his eugenicist outlook took a keen interest in the heritability of intelligence, with creativity taken as an aspect of genius.

In the late 19th and early 20th centuries, leading mathematicians and scientists such as Hermann von Helmholtz (1896) and Henri Poincaré (1908) began to reflect on and publicly discuss their creative processes.

Twentieth Century to the Present Day

The insights of Poincaré and von Helmholtz were built on in early accounts of the creative process by pioneering theorists such as Graham Wallas and Max Wertheimer. In his work *Art of Thought*, published in 1926, Wallas presented one of the first models of the creative process. In the Wallas stage model, creative insights and illuminations may be explained by a process consisting of 5 stages:

- (i) *preparation* (preparatory work on a problem that focuses the individual's mind on the problem and explores the problem's dimensions),
- (ii) *incubation* (where the problem is internalized into the unconscious mind and nothing appears externally to be happening),
- (iii) *intimation* (the creative person gets a "feeling" that a solution is on its way),
- (iv) *illumination* or insight (where the creative idea bursts forth from its preconscious processing into conscious awareness);
- (v) *verification* (where the idea is consciously verified, elaborated, and then applied).

Wallas' model is often treated as four stages, with "intimation" seen as a sub-stage. Wallas considered creativity to be a legacy of the evolutionary process, which allowed humans to quickly adapt to rapidly changing environments. Simonton provides an updated perspective on this view in his book, *Origins of genius: Darwinian perspectives on creativity*.

In 1927, Alfred North Whitehead gave the Gifford Lectures at the University of Edinburgh, later published as *Process and Reality*. He is credited with having coined the term “creativity” to serve as the ultimate category of his metaphysical scheme: “Whitehead actually coined the term – our term, still the preferred currency of exchange among literature, science, and the arts... a term that quickly became so popular, so omnipresent, that its invention within living memory, and by Alfred North Whitehead of all people, quickly became occluded”.

The formal psychometric measurement of creativity, from the standpoint of orthodox psychological literature, is usually considered to have begun with J. P. Guilford’s 1950 address to the American Psychological Association, which helped popularize the topic and focus attention on a scientific approach to conceptualizing creativity.

(It should be noted that the London School of Psychology had instigated psychometric studies of creativity as early as 1927 with the work of H. L. Hargreaves into the Faculty of Imagination, but it did not have the same impact.) Statistical analysis led to the recognition of creativity (as measured) as a separate aspect of human cognition to IQ-type intelligence, into which it had previously been subsumed. Guilford’s work suggested that above a threshold level of IQ, the relationship between creativity and classically measured intelligence broke down.

“Four C” Model

James C. Kaufman and Beghetto introduced a “four C” model of creativity; *mini-c* (“transformative learning” involving “personally meaningful interpretations of experiences, actions and insights”), *little-c* (everyday problem solving and creative expression), *Pro-C* (exhibited by people who are professionally or vocationally creative though not necessarily eminent) and *Big-C* (creativity considered great in the given field).

This model was intended to help accommodate models and theories of creativity that stressed competence as an essential component and the historical transformation of a creative domain as the highest mark of creativity. It also, the authors argued, made a useful framework for analyzing creative processes in individuals.

The contrast of terms “Big C” and “Little c” has been widely used. Kozbelt, Beghetto and Runco use a little-c/Big-C model to review major theories of creativity. Margaret Boden distinguishes between h-creativity (historical) and p-creativity (personal).

Robinson and Anna Craft have focused on creativity in a general population, particularly with respect to education. Craft makes a similar distinction between “high” and “little c” creativity, and cites Ken Robinson as referring to “high” and “democratic” creativity. Mihály Csíkszentmihályi has defined creativity in terms of those individuals judged to have made significant creative, perhaps domain-changing contributions. Simonton has analysed the career trajectories of eminent creative people in order to map patterns and predictors of creative productivity.

Theories of Creative Processes

There has been much empirical study in psychology and cognitive science of the processes through which creativity occurs. Interpretation of the results of these studies has led to several possible explanations of the sources and methods of creativity.

Incubation

Incubation is a temporary break from creative problem solving that can result in insight. There has been some empirical research looking at whether, as the concept of “incubation” in Wallas’ model implies, a period of interruption or rest from a problem may aid creative problem-solving. Ward lists various hypotheses that have been advanced to explain why incubation may aid creative problem-solving, and notes how some empirical evidence is consistent with the hypothesis that incubation aids creative problem-solving in that it enables “forgetting” of misleading clues.

Absence of incubation may lead the problem solver to become fixated on inappropriate strategies of solving the problem. This work disputes the earlier hypothesis that creative solutions to problems arise mysteriously from the unconscious mind while the conscious mind is occupied on other tasks.

Convergent and Divergent Thinking

J. P. Guilford drew a distinction between convergent and divergent production (commonly renamed convergent and divergent thinking). Convergent thinking involves aiming for a single, correct solution to a problem, whereas divergent thinking involves creative generation of multiple answers to a set problem. Divergent thinking is sometimes used as a synonym for creativity in psychology literature. Other researchers have occasionally used the terms *flexible* thinking or fluid intelligence, which are roughly similar to (but not synonymous with) creativity.

Creative Cognition Approach

In 1992, Finke et al. proposed the “Geneplore” model, in which creativity takes place in two phases: a generative phase, where an individual constructs mental representations called preinventive structures, and an exploratory phase where those structures are used to come up with creative ideas. Some evidence shows that when people use their imagination to develop new ideas, those ideas are heavily structured in predictable ways by the properties of existing categories and concepts. Weisberg argued, by contrast, that creativity only involves ordinary cognitive processes yielding extraordinary results.

The Explicit–Implicit Interaction (EII) Theory

Helie and Sun recently proposed a unified framework for understanding creativity in problem solving, namely the Explicit–Implicit Interaction (EII) theory of creativity.

This new theory constitutes an attempt at providing a more unified explanation of relevant phenomena (in part by reinterpreting/integrating various fragmentary existing theories of incubation and insight). The EII theory relies mainly on five basic principles, namely 1) The co-existence of and the difference between explicit and implicit knowledge; 2) The simultaneous involvement of implicit and explicit processes in most tasks; 3) The redundant representation of explicit and implicit knowledge; 4) The integration of the results of explicit and implicit processing; and 5) The iterative (and possibly bidirectional) processing. A computational implementation of the theory was developed based on the CLARION cognitive architecture and used to simulate relevant human data. This work represents an initial step in the development of process-based theories of creativity encompassing incubation, insight, and various other related phenomena.

Conceptual Blending

In *The Act of Creation*, Arthur Koestler introduced the concept of *bisociation*—that creativity arises as a result of the intersection of two quite different frames of reference. This idea was later developed into conceptual blending. In the '90s, various approaches in cognitive science that dealt with metaphor, analogy and structure mapping have been converging, and a new integrative approach to the study of creativity in science, art and humour has emerged under the label conceptual blending.

Honing Theory

Honing theory posits that creativity arises due to the self-organizing, self-mending nature of a worldview, and that it is by way of the creative process the individual hones (and re-hones) an integrated worldview. Honing theory places equal emphasis on the externally visible creative outcome and the internal cognitive restructuring brought about by the creative process. Indeed one factor that distinguishes it from other theories of creativity is that it focuses on not just restructuring as it pertains to the conception of the task, but as it pertains to the worldview as a whole. When faced with a creatively demanding task, there is an interaction between the conception of the task and the worldview. The conception of the task changes through interaction with the worldview, and the worldview changes through interaction with the task. This interaction is reiterated until the task is complete, at which point not only is the task conceived of differently, but the worldview is subtly or drastically transformed. Thus another distinguishing feature of honing theory is that the creative process reflects the natural tendency of a worldview to attempt to resolve dissonance and seek internal consistency amongst its components, whether they be ideas, attitudes, or bits of knowledge; it mends itself as does a body when it has been injured.

Yet another central, distinguishing feature of honing theory is the notion of a potentiality state. Honing theory posits that creative thought proceeds not by searching through and randomly 'mutating' predefined possibilities, but by drawing upon

associations that exist due to overlap in the distributed neural cell assemblies that participate in the encoding of experiences in memory. Midway through the creative process one may have made associations between the current task and previous experiences, but not yet disambiguated which aspects of those previous experiences are relevant to the current task. Thus the creative idea may feel ‘half-baked’. It is at that point that it can be said to be in a potentiality state, because how it will actualize depends on the different internally or externally generated contexts it interacts with.

Honing theory can account for many phenomena that are not readily explained by other theories of creativity. For example, creativity was commonly thought to be fostered by a supportive, nurturing, trustworthy environment conducive to self-actualization. However, research shows that creativity is actually associated with childhood adversity, which would stimulate honing. Honing theory also makes several predictions that differ from what would be predicted by other theories. For example, empirical support has been obtained using analogy problem solving experiments for the proposal that midway through the creative process one’s mind is in a potentiality state. Other experiments show that different works by the same creator exhibit a recognizable style or ‘voice’, and that this same recognizable quality even comes through in different creative outlets. This is not predicted by theories of creativity that emphasize chance processes or the accumulation of expertise, but it is predicted by honing theory, according to which personal style reflects the creator’s uniquely structured worldview. This theory has been developed by Liane Gabora.

Everyday Imaginative Thought

In everyday thought, people often spontaneously imagine alternatives to reality when they think “if only...”. Their counterfactual thinking is viewed as an example of everyday creative processes. It has been proposed that the creation of counterfactual alternatives to reality depends on similar cognitive processes to rational thought.

Measurement of Intelligence

Schools and placement agencies have been interested in the subject of intelligence testing because of its implications for placement. Intelligence tests attempt to predict the individual’s capacity to function generally in tasks that require scholastic aptitude. They measure the individual’s ability to cope with situations requiring the exercise of mental processes and try to measure the ability to comprehend the situation, apply past experiences, and solve the problem presented.

Tests are designed on the premise that intelligence can not be measured directly but is inferred by the observation or evaluation of behaviour. Behaviour has been grouped at various chronological age levels, and decisions have been made as to what is intelligent behaviour for a given age level. The measurements are converted into norms for representative groups of a population, so that eventually individuals can be classified

as having intelligence like a typical five year old or eight year old. Some tests, like the Kuhlmann-Anderson, used a median score in an attempt to avoid the averaging of scores in determining the norms. Most tests of intelligence place a premium on the ability to do abstract thinking through the use of symbols. Vocabulary also is of great importance in measuring intelligence. Many psychologists believe that as a child matures his vocabulary is the best single indicator of intelligence as related to scholastic aptitude. This measurement of vocabulary can be made in an individual test, such as the Stanford-Binet or the Wechsler Intelligence Scale for Children, or in group testing. For a long time, test makers have been concerned with the problem of avoiding measuring only differences in cultural opportunities or education. An effective test item combines a criterion of novelty and universality. It is novel in the sense that it has not been encountered before by individuals taking the test; it is universal to the extent that all, at a certain level, may have had some experience related to it. The group intelligence test has frequently been shown to have a high relationship to the extent, level, and quality of the child's educational experience. When we speak of measuring intellectual potential or "native capacity," therefore, we need to be aware of the overlapping of hereditary and environmental factors.

The teacher and school administrator must be aware of some of the problems involved in the measurement of intelligence. We must not assume that tests with similar titles measure similar behaviours. We must always determine whether the group on which the norms for this test were based is comparable to the children we are testing. It is also pertinent to recognise that tests with dissimilar titles may actually measure the same thing. It is quite applicable in testing to ask, "What's in a name" Intelligence testing with children must hold to a definite concept about what the test items are seeking to predict.

Psychologists engaged in test construction also need to recognise the cyclic nature of mental development. At certain times mental development proceeds more rapidly than at other times. Those who use intelligence tests in the schools should recognise that children with identical IQ's seldom function identically. One reason for this could very well be that they had acquired their mental-age points on different kinds of abilities and tasks.

Inhelder and Piaget also found differences in the modal age of children at a given stage on two different tasks. The age level for attainment of a given stage appears to be specific to a task rather than general, so that the child can operate at stage two on one task and at stage one on another.

The first individual intelligence test was the Stanford-Binet. Originally devised for use in dealing with a specific problem in France, it was later revised by Terman and then Terman and Merrill, and has been used extensively in the United States. The test items in this scale have been arranged in the order of empirically determined difficulty from the age of two upward. Items are always scored on a pass or fail basis. There are six test items at each half-year level from age two to five and six items at each yearly

age level thereafter until the age of fourteen. Thereafter, the test provides for average and superior adult levels. The items have been arranged so that the normal child of a given age is expected to perform at his chronological-age level. Retarded children perform below the level, gifted children above.

Another frequently used individual intelligence test is the Wechsler Intelligence Scale for children (WISC), which the psychologist uses diagnostically to determine areas of strength and weakness. The Wechsler test, developed and originally copyrighted by David Wechsler in 1949, abandoned the concept of mental age originally introduced by Binet in 1908. The Wechsler test has instead worked on the basis of a deviation intelligence quotient. The IQ is obtained by comparing the child's test performance with the scores earned by individuals in a single age group, not with the composite age group.

This was done in an attempt to keep the standard deviation of IQ's identical from year to year, so that a child's obtained IQ does not vary unless his actual test performance as compared with his peers varies. The Wechsler test also permits development of a verbal, performance, and total or full-scale intelligence quotient. The verbal factors include information, comprehension, arithmetic, similarities, vocabulary, and digit span. The performance factors include picture completion, picture arrangement, block design, object assembly, coding, and mazes. Group intelligence tests are frequently used in the schools to make educational decisions. Close analysis reveals that many mental-ability tests being widely used are really strictly measures of classroom achievement, while others are quite independent of classroom instruction. The person who selects or uses tests needs to know what the test actually measures. Individual deviations due to motivation, fatigue, distraction and the like may not be noted in a group-testing situation.

The following list includes some of the tests now being used in the public schools:

1. The California Test of Mental Maturity. Authors, E. T. Sullivan, W. W. Clark, and E. W. Tiegs. Publisher, California Test Bureau. Grades four to twelve.
2. Cattell Culture-fair Intelligence Test. Author, R. B. Cattell. Publisher, Bobbs-Merrill Co., Inc. School-learned skills not required. Grades three to twelve.
3. Differential Aptitude Tests. Authors, C. Bennett, H. Seashore, and A. Wesman. Publisher, Psychological Corporation. Items generally well-constructed and frequently used in grades eight to twelve to make educational and vocational decisions.
4. Lorge-Thorndike Intelligence Test. Authors, I. Lorge, R. L. Thorndike. Publisher, Houghton Mifflin Company. A promising instrument for predicting achievement in grade one. Series considered to be a well-constructed set of growth scales.
5. Science Research Associates Tests of General Ability (TOGA). Author, J. C. Flanagan. Publisher, Science Research Associates. Measures cultural understandings and reasoning with geometric forms.

6. Henmon-Nelson Tests of Mental Ability. Authors, T. A. Lanken, M. J. Nelson. Publisher, Houghton Mifflin Company. Places a great emphasis on reading; scores tend to correlate well with teacher grades. Grades three to twelve.
7. Otis Quick Scoring Mental Ability Tests. Author, A. S. Otis. Publisher, Harcourt, Brace & World, Inc. Grades four to twelve. Places a great emphasis on school skills.
8. School and College Ability Tests (SCAT). Author, Educational Testing Service Staff. Publisher, Educational Testing Service. Grades four to twelve. A new test, which provides both a verbal and a quantitative score. Considered to be a good measure of school-learned abilities.
9. Kuhlman-Anderson Intelligence Tests. Authors, F. Kuhlman and R. G. Anderson. Publisher, Personnel Press. Grades three to twelve. Well-constructed and considered to be one of the better measures of growth available. Emphasises both reading and achievement skills.
10. S.R.A. Primary Ability Test. Grades four to twelve. Authors, L. L. Thurstone and T. C. Thurstone. Publisher, Science Research Associates. Grades four to twelve. Stresses reading, arithmetic and reasoning abilities.
Because the testing held is frequently under attack from critics of education, it is important that the teacher and the parent understand more adequately the school testing programme and some of the significant concepts related to it. Robert Bauernfeind's *Building a School Testing Programme* (Boston: Houghton Mifflin Company, 1963) is recommended to those interested in investigating testing problems further.
11. what are the crucial questions to ask in setting up a school testing programme to measure intelligence and achievement? Plan an elementary school testing programme to cover just these two areas. Defend your programme and the instruments selected in terms of your school's educational objectives.

MEASUREMENT OF INTELLIGENCE IN EDUCATIONAL SETTINGS

Two kinds of intelligence tests will be presented here. The first, group tests, are used to identify the range of IQ scores in a group, usually for research or administrative purposes. Group tests tend to have more specific content and question formats, and are designed for more specific purposes than individually administered tests, which aim for a relatively more comprehensive clinical picture of the individual's cognitive functioning. The individual tests are designed to provide much more information about the individual. They are only used by professionals who are trained and licensed to administer and interpret these assessment instruments as part of a comprehensive clinical assessment that contributes information useful for planning educational or therapeutic interventions. These tests provide a variety of scores and clinical information that licensed

professional psychologists may use to plan interventions in schools, family, or other settings. In other words, the usefulness of individual intelligence tests goes well beyond the scores on the test. For this reason, group tests are not suitable substitutes for individual assessments when planning clinical or educational interventions.

Group Tests of Intelligence. Two examples of group intelligence test. The Cognitive Abilities Test, and the Multidimensional Aptitude Battery, Second Edition (MAB-II) were chosen because they may be used in school or educational settings.

The Multidimensional Aptitude Battery-II (MAB-II) is a multiple-choice assessment of aptitude and intelligence that can be administered to groups or individuals above the age of 16. The instrument was designed to obtain scores similar to that of the individually administered Wechsler Adult Intelligence Scale-Revised (WAIS-R) in a group, as opposed to individual administration. The MAB-II produces composite scores: Verbal Scale (Information, Comprehension, Arithmetic, Similarities, Vocabulary), Performance Scale (Digit Symbol, Picture Completion, Spatial, Picture Arrangement, and Object Assembly) and Full Scale. The test can be pencil and paper or computer-administered.

The MAB-II is considered to be a useful tool for assessing cognitive abilities when large numbers of students must be screened. It is a well-developed and empirically sound instrument that provides correlations between subtest scores and occupational strengths. The test can be administered by a proctor, rather than by a post-master's-level professional. As a result, the MAB-II should not be used for making clinical diagnoses about intelligence. Additionally, MAB-II should not be administered to students with a learning disability related to reading comprehension or whose reading level is below ninth grade because the test relies on the test-takers' reading ability.

The Cognitive Abilities Test—Multilevel Edition, Form 6 (CogAT-6) is one of the more widely used group ability tests for students in kindergarten through 12th grades. The test is intended to guide instruction to match the cognitive abilities and needs of each student, to provide an "alternative" measure of cognitive development, and to identify achievement-ability discrepancies. The test has a multitheoretical foundation as it is based on Vernon's model of hierarchical abilities, Cattell's model of crystallized and fluid abilities, and Carroll's work specifically on general abstract reasoning. The CogAT-6 is composed of two editions: the Primary and Multilevel Editions, both of which are intended to assess reasoning and problem-solving abilities and can be broken down into Verbal, Non-verbal, and Quantitative Batteries. The Primary Edition is designed for students in K–2nd grade and consists of six subtests. The Multilevel Edition is a nine-subtest instrument that is based on the Lorge-Thorndike Intelligence Tests and is appropriate for students in 3rd to 12th grade. The CogAT-6 has a mean of 100 and standard deviation of 16.

DiPerna's 2005 review of the CogAT-6 suggested that the strengths of the test include a large, representative standardization sample, co-norming with the Iowa Tests, and a theoretical basis. In spite of these positive attributes, significant weaknesses were cited relative to the CogAT's purposes as described earlier. Criticisms included insufficient

empirical evidence to support basing instructional recommendations on test results, a lack of reliability and predictive validity to measure cognitive ability and to predict cognitive ability-achievement discrepancies.

Individual Intelligence Tests. The following individual intelligence tests will be reviewed:

- Wechsler Intelligence Scale for Children-IV (WISC-IV)
- Woodcock-Johnson Tests of Cognitive Abilities-III (WJ COG III)
- Stanford-Binet Intelligence Scales, Fifth Edition (SB5)
- Das-Naglieri Cognitive Assessment System (CAS)
- Kaufman Assessment Battery for Children (K-ABC)

The Wechsler scales continue to be the most widely utilized individually administered tests of intelligence. The Wechsler tests are based on the *g* factor or the “overall capacity of the individual to act purposefully, to think rationally, and to deal effectively with the environment”. The Wechsler Intelligence Scale for Children, Fourth Edition (WISC-IV) is the current revision of the Wechsler scales for children 6.0 to 16.11 years of age. The WISC-IV comprises 15 sub-tests (5 of which are supplemental).

Scores on these tasks contribute to the four composite indices (Verbal Comprehension, Perceptual Reasoning, Working Memory, and Processing Speed) in addition to a Full Scale IQ (FSIQ) score that can range from 40 (very low) to 160 (very high), with a mean of 100 representing the average score, and a standard deviation of 15. A profile of the test taker’s learning strengths and weaknesses are derived from the test performance. The FSIQ is derived from a combination of all subtest scores and is considered the most representative estimate of global intellectual functioning. However, when large discrepancies between Index scores exist, the FSIQ can be invalid and misleading. In this case, it is most helpful to describe the strengths and weaknesses in the profile and de-emphasize the FSIQ.

The WISC-IV continues to be a reliable and valid instrument. WISC-IV scores can be interpreted in combination with Wechsler Individual Achievement Test, Second Edition (WIAT-II) scores for comparisons between ability and achievement. Flanagan and Kaufman (2004) provide an extensive description of the strengths and limitations of this assessment tool.

They cite the WISC-IV’s most significant strengths as being “a robust four-factor structure across the age range of the test, increased developmental appropriateness, de-emphasis on time, improved psychometric properties, and an exemplary standardization sample.” Flanagan and Kaufman (2004) indicate that none of the WISC-IV’s limitations are very serious, although they suggest ways in which its validity could be improved. Other Wechsler scales have been developed to assess IQ for young children and adults. The Wechsler Preschool and Primary Scale of Intelligence, Third Edition (WPPSI-III) (Harcourt Assessment, 2002) can be administered to children 2.6–7.3 years old and the Wechsler and the Wechsler Adult Intelligence Scale, Third Edition (WAIS-III) (Harcourt Assessment, 1997) is given to adults between 16 and 89 years of age.

The Woodcock-Johnson Tests of Cognitive Abilities-III (WJ COG III) is also widely used instrument in school settings. The WJ COG III is often administered as the examiner's second choice, if the age-appropriate Wechsler scale has been administered less than three years from the current assessment date. The WJ COG III is based on Catell, Horn, and Carroll's (CHC) concept of intelligence. The WJ COG III has been normed on individuals age 2 through 90+. The standard battery includes 10 tests, while the extended battery consists of 20. Based on these scores, the examinee earns three Cluster Scores (Verbal Ability, Thinking Ability, and Cognitive Efficiency) and a General Intellectual Ability (GIA) score. Test scores in the standard and extended batteries are not weighted equally when the GIA score is computed.

The WJ COG III has high technical quality and is based on a well-respected theory of cognitive abilities. It assesses cognitive abilities for a wide age range and uses sophisticated scoring procedures to calculate scores and discrepancies (Sattler, 2001). Additionally, ability and achievement discrepancy norms are provided as the WJ COG III was co-normed with the Woodcock Johnson Tests of Achievement, Third Edition (WJ ACH III). Greater evidence is necessary to understand the utility of the clinical clusters. The WJ COG III may also overestimate abilities if interpreted incorrectly. For example, the Written Language test score emphasizes one's ability to write brief sentences rather than to develop and organize paragraphs.

The Stanford-Binet Intelligence Scales, Fifth Edition (SB5), a direct descendent of Terman's adaptation of the Binet test developed more than 100 years ago, is used occasionally in the educational setting. The SB5 is based on the CHC theory of intelligence. It is designed for assessing intelligence and cognitive abilities among individuals between the ages of 2 and 85+. The SB5 consists of ten subtests and these scores are used to calculate four composite scores: factor, domain, abbreviated, and full scale (score range 40–160, mean = 100, SD = 15). The five factors measured include Fluid Reasoning, Knowledge, Quantitative Reasoning, Visual-Spatial Reasoning, and Working Memory, each with verbal and Non-verbal components. The SB5 contains two domain scales: Non-verbal IQ (NVIQ) and Verbal IQ (VIQ). An Abbreviated Battery IQ (ABIQ) can be determined with two routing subtest scores and the Full Scale IQ (FSIQ) is calculated using all 10 subtests.

The SB5 is advantageous as it emphasizes both verbal and Non-verbal abilities. The instrument is technically sound, according to Johnson and D'Amato (2005), although Kush's 2005 study cites technical limitations (*e.g.*, lower stability for young children and individuals with low cognitive abilities, problematically high correlations with achievement, uncertain factor structure). In spite of these weaknesses, the SB5 is referred to as an outstanding measurement instrument for the assessment of cognitive abilities of children, adolescents, and adults.

The Das-Naglieri Cognitive Abilities System (CAS) is a tool based on Luria's cognitive processing model that is intermittently used to evaluate cognitive abilities in schools. The CAS is useful for assessing Planning, Attention, Simultaneous, and

Successive (PASS) abilities among students between the ages of 5.0 and 17.11. The instrument's basic battery comprises 8 subtests and the standard battery consists of 12 subtests.

The CAS is an innovative instrument and its development meets high standards of technical adequacy. When compared to other individually administered general ability tests, it takes less time to administer. Additional empirical research must be completed to support the PASS construct. Factor analyses of subtests support both a 4-factor PASS model as well as a 3-factor model, which suggests that Planning and Attention may or may not be separate factors. The Kaufman Assessment Battery for Children, Second Edition (KABC-II) is occasionally utilized in schools as a culture-fair assessment of cognitive abilities for students between the ages of 3 and 18.

The KABC-II is based on a dual theoretical framework, Luria's neuropsychological model and the CHC approach. The authors suggest that its multitheoretical base allows the examiner to select which model is most appropriate for interpretation of results depending on the culture and/or verbal skills of the examinee. This assessment instrument contains 20 subtests that contribute to 4 scale scores (Sequential Processing, Simultaneous Processing, Planning, Learning, and Knowledge). The KABC-II has a mean of 100 and standard deviation of 15.

The KABC-II is an acceptable option for measuring cognitive abilities as it provides a reasonable, well-normed, clinically appealing, and technically sound approach to measuring cognitive abilities and generating diagnoses. Other strengths of the KABC-II include smaller score discrepancies between ethnic groups and the ability to compare ability and achievement differences with the Kaufman Test of Educational Achievement, Second Edition, Comprehensive Form. However, it has been criticized for the suggestion that examiners can select the model (Luria or CHC) by which to interpret results. Braden (2005) and Thorndike (2005) suggest that the interchangeability of theoretical models is inappropriate and illogical.

Intelligence testing should not be conducted in a vacuum. In addition to using previously mentioned norm-referenced tests, an assessment should include a variety of other data from a multitude of informants.

Sattler (2001) suggests that norm-referenced testing should be accompanied by interviews with a parent, teacher, and student; observations of the student during both the formal testing and natural environment (*e.g.*, classroom, lunchroom, playground); and informal assessment procedures (*e.g.*, district-wide criterion-referenced tests, school records). Such an assessment will provide the most accurate information by which educators can most effectively serve the student. Group versus Individual Intelligence Tests. Whereas group-administered IQ tests can be very well constructed (Aiken 2003) and can provide valid and reliable information suitable for certain non-clinical applications, they do not provide the same kind of information as individual tests and should not be used for the same purposes. One reason is that group tests primarily emphasize multiple-choice question format, whereas individually administered tests

provide a variety of response formats across the test. This allows clinicians to gather considerable clinically useful information about the person being tested, such as the approaches used in problem solving, quality of verbal expression, and other observational data.

Group tests were not designed to be used for clinical purposes and therefore should not be utilized in clinical settings or to substitute for individually administered intelligence tests for designing clinical interventions or individual educational programmes (IEPs) in school settings.

Individual intelligence tests are designed to be administered by highly trained professionals who interpret data through the lens of learning, cognition, emotion, language, culture, health, and development. The one-on-one setting provides advantages to the test-taker and the examiner. First, it allows the examiner to develop rapport with the test-taker, which can benefit the shy or anxious test-taker. Second, the examiner has the opportunity to observe important test-taking behaviour such as impulsiveness, compulsiveness, confidence, anxiety, and wandering attention. Which may vary according to the task and contribute to the interpretation of the performance on the test.

Third, the examiner can observe specific problem-solving approaches, which may also vary with the task. The integration of these observations with test scores allows the examiner to take a holistic approach in interpreting scores and developing interventions.

Critical Educational Psychology

Educational psychology remains one of the prominent ways through which psychological theory and research enter the public sphere. In schools and other educational settings, educational psychologists occupy a potentially powerful position. Their work to support children's learning in educational institutions engages them in close professional relationships with educational authorities, teachers, headteachers, parents and other professionals from health and social care.

Moreover, educational psychologists are expected to respond to an ever changing educational and social policy landscape in which childhood, learning, teaching and assessment are regularly revised. This leads to huge impacts upon the meaning of education and the experience of learning. Furthermore, educational psychologists are tasked with assessing the specific learning and educational needs of children which inevitably ensures that they are party to the inclusion and exclusion of children from educational settings. Just as educational psychology has developed so has psychology. Increasingly psychology has fostered a number of relationships with political, social and cultural discourses. The contemporary emphasis on the importance of community; understanding social exclusion, and working towards social inclusion; stakeholder interests in social change and 'bottom up' or participatory ways of working; the development of social capital and community participation; urban regeneration, and citizenship, for example, all coincide with the particular emphases within community and critical psychologies of understanding marginalised human behaviour and experience within different social contexts and of working in partnership with those affected by both psychological work and social changes.

Furthermore, developments in relation to what has been termed critical psychology has had a number of impacts upon the ways in which educational psychology theories, conceptualises and practices alongside children, childhood, difference, education and disability.

TRADITIONAL EDUCATION PSYCHOLOGY

Traditionally, educational psychology has sometimes been a bewildered traveller at the crossroads of Psychology and Education. Educational psychology can be defined as any area of education which is informed by psychological theories or techniques. At its broadest outlines, this field encompasses a number of areas related to student learning and individual difference considered to be crucial knowledge for teachers in training requiring more in-depth knowledge for managing difficult students or for their own professional development. There is a focus on the normative learner, with a tendency to position students with 'special educational needs' as exceptions to the norm. Professional educational psychologists (or school psychologists, in some countries) tend to be employed by education authorities and to work with students who have been identified as having learning or behavioural difficulties in the school system. Such psychologists receive postgraduate training in educational psychology with more emphasis on dealing with disabilities. Professional educational psychology includes applications of research in educational psychology but also overlaps with clinical fields of child and adolescent psychology. Teachers who undergo further postgraduate training in special education would also receive grounding in these issues, but with specific emphasis on applying knowledge to the classroom. The influence of the parent discipline can be seen throughout educational psychology, particularly in the emphasis on psychology as a 'science'. Many contemporary textbooks lean heavily upon 'scientific method', especially experimentation and objectivity (*e.g.*, Child, 1993; McCormick and Pressley, 1997). Thomas, writing critically of British practice, argued that professional educational psychologists 'are steeped in the view of themselves as applied scientists'. The field may be marked by some fears of being 'not scientific enough' to count as a real branch of psychology, creating tensions regarding the status of the field.

Within education, educational psychology has sometimes been seen as an atavism, a reminder of earlier times and values. During the 1960s and early 1970s the debates between 'behaviourism' and more humanistic views of learning and emotion left their scars on the visage of the field. Today some tertiary education programmes in New Zealand and Australia have no courses titled 'Educational Psychology'; instead they offer similar material under more general titles such as 'social contexts of learning'.

CRITICAL PSYCHOLOGY

Critical psychology is a perspective on psychology that draws extensively on critical theory. Critical psychology challenges main stream psychology and attempts to apply

psychological understandings in more progressive ways, often looking towards social change as a means of preventing and treating psychopathology. One of critical psychology's main criticisms of conventional psychology is that it fails to consider or deliberately ignores the way power differences between social classes and groups can impact the mental and physical well-being of individuals or groups of people. It does this, in part, because it tends to explain behaviour at the level of the individual.

ORIGINS

Criticisms of mainstream psychology consistent with current critical psychology usage have existed since psychology's modern development in the late 19th century. Use of the term "critical psychology" started in the 1970s in Berlin at Freie Universität Berlin. The German branch of critical psychology predates and has developed largely separately from the rest of the field.

As of May 2007, only a few works have been translated into English. The German Critical Psychology movement is rooted in the post-war babyboomers' student revolt of the late '60s; see German student movement. Marx's Critique of Political Economy played an important role in the German branch of the student revolt, which was centered in Berlin. Then Berlin was a capitalist city surrounded by communist-ruled East Germany, represented a "hot spot" of political and ideological controversy for the revolting German students. The sociological foundations of critical psychology are decidedly Marxist.

Klaus Holzkamp

One of the most important and sophisticated books in the field is the *Grundlegung der Psychologie* (Foundations of Psychology) by Klaus Holzkamp, who might be considered the theoretical founder of critical psychology. Holzkamp, who had written two books on theory of science and one on sensory perception before publishing the *Grundlegung der Psychologie* in 1983, thought this major work provided a solid paradigm for psychological research, as he viewed psychology as a pre-paradigmatic scientific discipline. Holzkamp mostly based his sophisticated attempt to provide a comprehensive and integrated set of categories defining the field of psychological research on Aleksey Leontyev's approach to cultural-historical psychology and activity theory. Leontyev had seen human action as a result of biological as well as cultural evolution and, drawing on Marx's materialist conception of culture, stressed that individual cognition is always part of social action which in turn is mediated by man-made tools (cultural artifacts), language and other man-made systems of symbols, which he viewed as a major distinguishing feature of human culture and, thus, human cognition.

Another important source was Lucien Séve's theory of personality, which provided the concept of "social activity matrices" as mediating structure between individual and social reproduction. At the same time, the *Grundlegung* systematically integrated

previous specialized work done at Free University of Berlin in the '70s by critical psychologists who also had been influenced by Marx, Leontyev and Seve. This included books on animal behaviour/ethology, sensory perception, motivation and cognition. He also incorporated ideas from Freud's psychoanalysis and Merleau-Ponty's phenomenology into his approach.

One core result of Holzkamp's historical and comparative analysis of human reproductive action, perception and cognition is a very specific concept of meaning that identifies symbolic meaning as historically and culturally constructed, purposeful conceptual structures that humans create in close relationship to material culture and within the context of historically specific formations of social reproduction.

Coming from this phenomenological perspective on culturally mediated and socially situated action, Holzkamp launched a devastating and original methodological attack on behaviourism (which he termed S–R (stimulus–response) psychology) based on linguistic analysis, showing in minute detail the rhetorical patterns by which this approach to psychology creates the illusion of “scientific objectivity” while at the same time losing relevance for understanding culturally situated, intentional human actions. Against this approach, he developed his own approach to generalization and objectivity, drawing on ideas from Kurt Lewin in Chapter 9 of *Grundlegung der Psychologie*.

His last major publication before his death in 1995 was about learning. It appeared in 1993 and contained a phenomenological theory of learning from the standpoint of the subject. One important concept Holzkamp developed was “reinterpretation” of theories developed by conventional psychology.

This meant to look at these concepts from the standpoint of the paradigm of critical psychology, thereby integrating their useful insights into critical psychology while at the same time identifying and criticizing their limiting implications while (which in the case of S–R psychology were the rhetorical elimination of the subject and intentional action, and in the case of cognitive psychology which did take into account subjective motives and intentional actions, methodological individualism). The first part of the book thus contains an extensive look at the history of psychological theories of learning and a minute re-interpretation of those concepts from the perspective of the paradigm of critical psychology, which focuses on intentional action situated in specific socio-historical/cultural contexts.

The conceptions of learning he found most useful in his own detailed analysis of “classroom learning” came from cognitive anthropologists Jean Lave (situated learning) and Edwin Hutchins (distributed cognition). The book's second part contained an extensive analysis on the modern state's institutionalized forms of “classroom learning” as the cultural–historical context that shapes much of modern learning and socialization. In this analysis, he heavily drew upon Michel Foucault's *Discipline and Punish*. Holzkamp felt that classroom learning as the historically specific form of learning does not make full use of student's potentials, but rather limits her or his learning potentials by a number of “teaching strategies.” Part of his motivation for the book was to look for

alternative forms of learning that made use of the enormous potential of the human psyche in more fruitful ways. Consequently, in the last section of the book, Holzkamp discusses forms of “expansive learning” that seem to avoid the limitations of classroom learning, such as apprenticeship and learning in contexts other than classrooms.

This search culminated in plans to write a major work on life leadership in the specific historical context of modern (capitalist) society. Due to his death in 1995, this work never got past the stage of early (and premature) conceptualizations, some of which were published in the journals *Forum Kritische Psychologie* and *Argument*.

1960 through 1970

In the 1960s and 1970s the term radical psychology was used by psychologists to denote a branch of the field which rejected conventional psychology’s focus on the individual as the basic unit of analysis and sole source of psychopathology. Instead, radical psychologists examined the role of society in causing and treating problems and looked towards social change as an alternative to therapy to treat mental illness and as a means of preventing psychopathology.

Within psychiatry the term anti-psychiatry was often used and now British activists prefer the term critical psychiatry. Critical psychology is currently the preferred term for the discipline of psychology keen to find alternatives to the way the discipline of psychology reduces human experience to the level of the individual and thereby strips away possibilities for radical social change.

In the 1990s

Starting in the 1990s a new wave of books started to appear on critical psychology, the most influential being the edited book *Critical Psychology* by Dennis Fox and Isaac Prilleltensky. Various introductory texts to critical psychology written in the United Kingdom have tended to focus on discourse, but this has been seen by some proponents of critical psychology as a reduction of human experience to language which is as politically dangerous as the way mainstream psychology reduces experience to the individual mind. Attention to language and ideological processes, others would argue, is essential to effective critical psychology - it is not simply a matter of applying mainstream psychological concepts to issues of social change.

Ian Parker

In 1999 Ian Parker published an influential manifesto in both the online journal *Radical Psychology* and the *Annual Review of Critical Psychology*. This manifesto argues that critical psychology should include the following four components:

- Systematic examination of how some varieties of psychological action and experience are privileged over others, how dominant accounts of “psychology” operate ideologically and in the service of power;

- Study of the ways in which all varieties of psychology are culturally historically constructed, and how alternative varieties of psychology may confirm or resist ideological assumptions in mainstream models;
- Study of forms of surveillance and self-regulation in everyday life and the ways in which psychological culture operates beyond the boundaries of academic and professional practice; and
- Exploration of the way everyday “ordinary psychology” structures academic and professional work in psychology and how everyday activities might provide the basis for resistance to contemporary disciplinary practices.

ABILITIES AND INTELLIGENCE

Earlier this century there was more focus on describing individual differences amongst students, in order to help teachers to work with different students. Cyril Burt was influential not only for his studies of the heritability of intelligence, but for his typology of ‘backward children’ (1946) which included the ‘lazy’, and the left-handed as well as the ‘dull’ student. There has been a long history of critique of intelligence testing and its tendencies towards labelling of individuals. Gould (1981) has written a sobering account of the nefarious activities of researchers determined to prove the heritability of intelligence.

Recent work on the ‘architecture’ of intelligence has expanded from the focus of Spearman on a single general factor of intelligence (thought to be largely hereditary) towards more complex models. One model that has become influential with educational psychologists in New Zealand and the US is Howard Gardner’s theory of ‘multiple intelligences’. Gardner has expanded the idea of intelligence beyond the narrow versions favoured within the traditional psychometric approaches to include linguistic, musical, logical-mathematical, spatial, bodily-kinaesthetic, interpersonal and intrapersonal intelligences.

Though Gardner recognised the importance of cultural input in determining the value of one domain of competence, his seven intelligences are tied to an unproblematic ‘biological’ basis.

MI [Multiple Intelligence] theory is framed in light of the biological origins of each problem-solving skill. Only those skills that are universal to the human species are treated. Even so, the biological proclivity to participate in a particular form of problem solving must also be coupled with the cultural nurturing of that domain.’.

The focus on culture seems undercut here by the search for a ‘universal’ underpinned by the ‘biological’. The dichotomy of the social and biological has been subject to critique for some time. A critical educational psychology could be strengthened with a focus on multiple competencies if this were extended by a critique of nature/nurture as a socially constructed dichotomy. There are also strong normative concerns about the unfolding of each intelligence in Gardner’s approach, since the evidence for these is

based on ‘knowledge about normal development’ and ‘data about the evolution of cognition over the millenia’.

What are not considered are the cultural processes that make some domains of endeavour (*e.g.*, the chess players favoured by studies of gifted children) higher in status or more indicative of a worthwhile talent than others. For example, much feminist work has considered the ways that mathematics is circumscribed as a domain of male rationality, even when activities of girls and boys may be similar.

Adding ‘bodily-kinesthetic’ intelligence to a list alongside ‘logical-mathematical’ intelligence does not necessarily make these two equal intelligences for determining future educational or career possibilities. While it is helpful that this model goes beyond narrow cognitive definitions of ability to consider a plurality of talents, many assumptions used by more conventional psychometric theories of intelligence have not been challenged in this work. The concept of multiple intelligences may unintentionally reinforce popular beliefs that intelligence(s) is inherited and immutable.

Given the moves towards widening views of abilities and intelligence in the past 15 years, it is worrying that many contemporary textbooks still make assumptions based on the idea that intelligence is a more or less fixed capacity in individuals, bounded by some biological or genetic capacity.

In the fifth edition of a popular UK text, a chapter on ‘Intelligence’ discussed ways to help teachers to be efficient:

- The detection and measurement of differences are important for the teacher. It would be disastrous for children if we did not quickly recognize their cognitive strengths and weaknesses, because the intellectually dull cannot, in general, cope with the same cognitive tasks as the intellectually bright of the same age...’.

In this view, there are dull bodies and bright bodies, far from the multiplicities of the postmodern body. A more far-reaching analysis of human ‘ability’ needs to take the deconstruction of the individual further, for example, in making moves towards Deleuze and Guattari’s (1988) idea of ‘bodies without organs’.

The competence of a body in one setting or another does not need to be conceptualised in terms of an internal capacity rooted in biology (as intelligence, personality and other human characteristics are conventionally described). The multiplicities of movement, of skills, can be viewed as belonging to an order beyond the division of individual body and external environment. Instead, a competence such as ‘emotional stability’ could be viewed as an event which occurs in a milieu which includes parts of various kinds, including hands, sound waves, electric lights, papers and pens, grey hair, pink cheeks, wooden surfaces.

It would be this collection of various items together which seem to create a recognised performance of ‘competence’. It is difficult to imagine competence in this way, as something not connected intimately with an individual body. Part of the difficulty in such imagining is that our language is already formatted into subjects and objects, by

individuals who take action and meet outcomes. Attempting to broaden language beyond the fixities of individual bodies with set quantities of ability could disrupt stereotypical interactions between students, teachers, parents and psychologists.

INTELLIGENCE AND INFORMATION PROCESSING

According to Robert Sternberg's theory, there are three basic mental processes that underlie all intelligent behaviour. Although what is viewed as intelligent in one culture might not be viewed as intelligent in another, the basic mental processes are the same across different cultures.

The three basic mental processes are:

- *Metacomponents*
- *Performance components*
- *Knowledge-acquisition components*

Metacomponents are the executive processes that we use to solve problems, plan what to do, make decisions, and evaluate outcomes. Performance components carry out the directions of the metacomponents. It is performance components that allow us to store information in short-term memory, compare two concepts, compare solutions to the task, etc. Knowledge-acquisition components are what we use to learn and store new information. In other words, metacomponents tell us what to do, performance components actually do it, and knowledge-acquisition components make sure we learn things along the way. For example, you may plan to read a book (metacomponents), grab a book off the shelf and actually read it (performance components), and learn some new vocabulary words while reading (knowledge-acquisition components).

THREE PARTS OF INTELLIGENCE

So what makes a person intelligent? According to Sternberg, intelligence cannot be defined by intelligence tests such as the Stanford-Binet scales. Rather, intelligence should be defined in terms of how you perform in your everyday world. Sternberg refers to what he calls *successful intelligence*. People who are successfully intelligent are able to define and achieve their own idea of success within their culture. People who are successfully intelligent are skilled at adapting to and modifying their environment to fit their needs. Because your intelligence is highly dependent upon the culture that you live in, an individual that is considered intelligent in one culture might not be considered intelligent in another. There are three components of successful intelligence: analytical, creative, and practical intelligence. It is not enough to possess the three components. One must also know when and how to use these components in order to be effective.

Analytical Intelligence

Have you ever been called 'book smart?' If so, you were being complimented on your analytical intelligence. Analytical intelligence, also called componential

intelligence, refers to your ability to problem-solve, process information effectively, and complete academic tasks. People with high analytical intelligence perform well on traditional IQ tests (*i.e.*, Stanford-Binet scales), college admission exams (*i.e.*, Scholastic Assessment Test), and school exams (*i.e.* math quizzes). They are skilled at critical thinking and analytical thinking. People with high analytical intelligence can examine problems from multiple points of view.

Creative Intelligence

Creative intelligence is our ability to call upon existing knowledge and skills to effectively handle new and unusual situations. Because creative intelligence involves using past experiences, it is also called experiential intelligence. People with high creative intelligence have great insight, imagination, and are able to formulate new ideas. For example, painters are often able to use their imagination to paint original artwork. Unlike analytical intelligence, creative intelligence is often overlooked by intelligence tests.

Intelligence is one of the most complex, talked about concepts within the field of psychology. In the past century, several theories about what constitutes intelligence have been created. One of the most well known theories of intelligence was created by Robert Sternberg, an American psychologist. Unlike most theories of intelligence, Sternberg's theory calls for the integration of intelligence and creativity.

EDUCATIONAL PSYCHOLOGY: THE CONCEPT OF INTELLIGENCE IS AN ARTIFICIAL CONSTRUCT

The concept of IQ as a singular phenomena has been called into question. The absent minded professor is nearly archetypal in popular culture. There may well be some truth behind a concept so prevalent and pervasive in western culture. If intelligence is a composite of many factors then the image of an intelligent person behaving as a fool would not be incongruent.

Spearman's concept of "g" was a theory of intelligence wherein "g" or general mental ability is a set of which many factors of intelligence are elements.

There are differences in the way people deal with problems of daily life There is a positive correlation between tests that observe mental processing, retrieval or manipulation of information and tests that measure cognitive ability. In other words, someone who does well on one kind of cognitive test is likely to do well on other different kinds of cognitive tests. Despite the fact that there is no agreed upon definition of intelligence, there is a continuum as such whereupon people differ. Spearman, due to the consistency of scores across different tests, proposed a single entity concept of intelligence as a factor of general mental ability or "g". Some theorists equate general mental ability as information processing. Intelligence is an abstract concept. It is not

even a metaphysical thing. Intelligence is, at best, a set of Behaviours. General mental ability is systematic individual differences in the performance of tasks that involve the manipulation, retrieval, evaluation, or processing of information. General mental ability is related to basic cognitive and physiological processes, but it is not certain whether these underlying processes can be described. Behaviours can be observed, so intelligence can be described in terms of Behaviours..

Scores on any cognitive tests are influenced by general mental ability or “g”. Correlations among test scores are due to each test presenting a partial measure of g. Each test measures some factor of g and is also influenced by measurement error. For example, the score on a test measuring intuitive geometry is affected by measurement error and the test takers level of intuitive geometry and the individual test taker’s level of g. This theory asks us to make intelligence tests highly g loaded, that reducing the influence of measurement error and the influence of any specific factor of intelligence will give better results. The upshot of this is that an intelligence test could be measuring some aspect of intelligence and not general mental ability *per se*. There is considerable evidence that intelligence is generalizable. There are broad and general differences between individuals on a broad range of cognitive tasks and a general factor arises out on nearly every analysis, One’s standing or level on this general factor is the best single predictor of performance on any cognitive task. Once g has been factored, there are no specific abilities that add any predictive reliability or validity.

Gardner’s theory of multiple intelligences is that the concept of general intelligence is not useful as people can be intelligent in many different ways.

Global comparisons between people or groups emphasizes only a small part of intelligence. The single score might be a composite score made up of many disparate factors. Other theorists posit other types of intelligence from the existence of other abilities such as a ballet dancer or basketball player said to possess bodily-Kinesthetic intelligence. While there may be some justification for considering intrapersonal intelligence, the capacity for empathy and compassion are more than just a bridge between the two. Furthermore the linguistic intelligence from the mathematico-logical intelligence and musical intelligence could well be spurious, as it is often said that mathematics is also another language. Concepts are concatenated in language. Subtraction is merely the addition of negatives. Sentences are diagrammed into formulae noun plus adjective plus verb plus adverb = statement in logic concepts are manipulated like integers in math. AND the later career of Ludwig Von Beethoven demonstrates the congruence of music and mathematics. Are these multiple intelligences of Gardner merely different facets of the same jewel? As is said in the Rig Veda, “the truth is the same, the sages know it by many names.”

STERNBERG’S TRIARCHIC THEORY OF INTELLIGENCE

Sternberg however divides intelligence horizontally instead of vertically into context, experience and process. I would argue that context is no more than the predeterminants

of intelligence and process is after all a process. Once again we are not looking at different jewels here, but looking separately at the cut, clarity and quality of a single gem. The why, what, and how can be lost if one concentrates too much on g, but multiple intelligences lack empirical support. Mental activities have components and different components are used to acquire different sorts of information. Adaptation to the environment is the critical measure of intelligence. Intelligence is coping and coping is self efficacy. Intelligence is more than information processing.

CRITICAL FOUNDATIONS FOR EDUCATIONAL PSYCHOLOGY

In general terms, the field of educational psychology is in many ways one of the furthest from the critical project in psychology mooted by overviews such as provided by Henriques, Hollway, Urwin, Venn and Walkerdine (1984) or Parker and Shotter (1990). There has been a longstanding focus on the individual as the source of problems in the classroom, since the under-achieving child was put under scrutiny by Cyril Burt, a founder of psychological services for education in England.

In my view, a more critical educational psychology would provide more possibilities for innovative practice by deconstructing notions such as normality, competence and coping. Important foundational texts would include Foucault's (1977) work on discourses involved in the history of western schooling, and Walkerdine's (1984) use of Foucauldian analysis to critique 'child-centred' discourses in modern pedagogy.

The critique of individualism in psychology can be applied par excellence to educational psychology, which has so often spotlighted the struggling individual in the classroom, whether an unmotivated learner or a child with 'brain damage'.

Earlier this century such a child might be labelled as deviant, while in more recent times there have been attempts to ensure that some children get extra help. This individualist bias led in the past to an unbalanced emphasis on individuals as causal agents of their problems.

Contemporary critical work has begun to move away from the intense focus on the individual, acknowledging that such a focus misses not only the interactions amongst people in a group, but the complexities of lived culture and language.

A critical educational psychology informed by a more collective focus should be reflexive about its progressivist and normalising tendencies. Critiques of western views of development by indigenous scholars who question the focus on the individual are emerging in the Pacific. My own institution has begun to consider seriously the indigenous Maori perspectives on human learning and development. These are oppositional to views of the individual based on Cartesian dualisms of mind and body. A Maori view of human existence encompasses corporeal and spiritual aspects of the whole family or tribal group, including both the living and the dead. This work could go further, in interrogating the facile dichotomy often assumed between hereditary and

environmental factors ('nature/nurture') which contribute to individual differences. For example, there are deconstructive possibilities for considering competence as something beyond the internalised abilities of individuals. Contributing forces appear different when the lens is widened to take in a larger group of people, and considering those influences that are not so readily visible, such as influences of parents (even imagined/ previously voiced expectations of long gone caregivers) and the impact of particular government policies about delivery of the curriculum or of welfare benefits to sole parents.

A critical educational psychology informed by other poststructural questioning about the individual self in western culture must also put at the forefront a consideration of language and social practices that create multiple possibilities for subject-positionings in the classroom and beyond.

Given the diversity of topics in the field of educational psychology, it is worth considering each separately, to give at least a Cook's tour of some critical possibilities. There are several broad areas covered by educational psychology: learning, intelligence/abilities, education for special needs, motivation (with some mention of 'self esteem' and personality issues), educational assessment and empirical research methods. There are developmental aspects of many of the above fields, though developmental psychology is another complex sub-discipline of psychology.

LEARNING

There are interesting moves within the field of 'learning' that offer possibilities beyond the focus on individual learners. Learning has become the biggest area in educational psychology, since it encompasses many paradigms. Approaches to research in learning have been described as a move from the 'behaviourist/empiricist' to the 'cognitive/rationalist', with the recent addition of an approach more timely for the critical project, the 'situative/pragmatist-sociohistoric' perspective.

Research on metacognition, which refers to reflection on or strategic use of cognition, emerged later in the cognitive/rationalist era. When I began to question my 'training' as a cognitive psychologist, metacognition was a breath of fresh air in the highly technical world of cognition. Instead of the view that some children were just born with bright or dull minds, metacognitive work opened up the new possibility that some students had more sophisticated strategies, techniques and management skills for organising and using their knowledge than other students. This has led to an entire field of work with children who have 'learning problems' (perhaps better called 'teaching problems', to change the emphasis from the deficit of an individual learner). Such students have been found to benefit from specific instruction or reflection on particular learning strategies and techniques.

Much recent work within a developmental framework has also moved away from the Piagetian focus of earlier decades towards a now more fashionable Vygotskian purview. Vygotsky differed from western developmental theorists in his attention to the

social and cultural milieu in which all people, including the newborn, are immersed. Neo-Vygotskian work which takes the emphasis on collective, community understandings of information is Moll and Greenberg's (1990) study of the transmission of knowledge in a Latin American community in Arizona. A literacy project began in a bilingual Spanish/English classroom, and unfolded into an exchange of information between school staff, researchers, and extended family members throughout the community. Students and their families began to set the agenda for the learning they wanted to document, using the languages of the community.

A more recent shift in learning studies, that of situative knowledge, offers further possibilities for considering learning as milieu, rather than as an internal change in a body. The situative approach has been described as viewing 'knowledge as distributed among people and their environments, including the objects, artefacts, tools, books, and the communities of which they are a part'. An interesting example of situative knowledge is found in work on 'distributed cognitions' among groups of students and teachers. In work also influenced by Vygotsky, Ann Brown and colleagues considered ways that groups of students share information in working together on tasks in human biology, tracking this through on-line conversations with the tutor and amongst students.

A wider field of research on cooperative learning also centres on the functioning of learning outcomes for a group rather than dividing the learning by individuals. This approach has exciting possibilities for moving away from a focus on the individual student as responsible for particular actions, based on his/her capacities. There is more scope for considering learning as a group outcome, in which bodies and other objects play their part. Though much of this work centres on the specific bodies, technologies and curriculum content of a particular learning setting, this could be broadened to consider a wider milieu framed by historically-shaped language and cultural practice.

Motivation

The field of motivation in educational psychology has a well-documented historical narrative. Graham and Weiner (1996) described the change from the 'Mechanistic Period: 1930-1960', which emphasised biological drives and processes, to a cognitive approach (1960-1970), in which the personality variable of achievement motivation became dominant, and finally to 'Contemporary Motivation Research: 1970-1990', centring on attributional approaches to motivation. At present, much research is broadly concerned with the reasoning processes involved in students' attributions about successful or poor performances. The field has branched into concerns beyond controlled testing towards more naturalistic, classroom-based contexts for tapping attributions and expectations. There has been work on widening the cultural context of motivational concerns beyond those of unmarked white, middle class students, in, for example, Sandra Graham's research with African-American students and teachers.

There are a number of problems with the focus on attributions as the dominant approach to motivation at present, given recent critiques of attribution research. Much

of the research of the 1980s was based on written vignettes, which have been criticised for their lack of acknowledgement of textual effects. The above-mentioned review by Greeno et al. suggested that 'Engaged Participation' might characterise motivation from a situative or sociohistorical position rather than the cognitive/rational approach of attribution theorists. This seems an intriguing possibility for a critical psychology of motivation. Unfortunately the studies listed in that review did not appear to be studies of motivation, but rather studies of learning influenced by such wider issues as identity and reflection in communities of learners. A major problem that has not been addressed is that motivation needs to be reconceptualised within a critical educational psychology.

Recent work on motivation is still fixed firmly on the individual, even if the context for the individual is considered more interactively than in the past. Part of the problem is that traditional psychological research has defined motivation as an inner property of the individual. This reinstates an uninterrupted dualism of inner self/outer context, which has been the subject of so much philosophical critique in recent times. This is also a problematic issue for the study of 'self esteem'.

'Self Esteem'

The issue of self esteem is usually raised briefly in many educational psychology textbooks, within a section on motivation. There has been a fortunate move away from the psychometric tradition of 'self esteem' scales. Contemporary work on self esteem has moved from uni-dimensional omnibus scales towards work on multiple aspects of self esteem such as academic, sporting and social components. The scaling paradigm has been examined by critical psychologists who have considered other aspects of testing.

The use of scales to examine self esteem is subject to the same limitations as other simple measurements of this type, such as the reliance on verbal expression to identify an 'inner' personality dimension. Given the complexities of theoretical work on the 'self' this century, from psychoanalysis through contemporary poststructural theorising, the attempt to map out a particular individual's 'self esteem' on the basis of a paper and pencil scale seems extraordinary. Perhaps it is an indication that there is still institutional support to be found for researchers who have not ventured outside the psychometric certainties of the 1960s.

There is a newer branch of work on self esteem related to attribution approaches. Weiner's work on educational attributions made clear links between expectations of future success or failure and attributions to stable causes. (In this theory, attributions to stable factors such as ability should lead to expectations to perform similarly in future, while attributions to unstable factors such as effort should lead to more variable expectations of future performance.) This idea was taken further in the 'self-worth' theory of Covington (1992) which proposes that students' self-concepts are shaped by their ideas of their 'ability'. What is needed in a more critical study of 'self esteem' is an analysis of the cultural vocabularies available to construct the kind of self that is characterised by terms such as 'ability'.

The related issue of children's marginalisation at school was the focus of the Disaffected Pupils project. This English study focused on student 'alienation' in schools, relating this to school management rather than interactions solely in the classroom. The work involved teachers, psychologists and researchers. There was a focus on a number of categories of marginalisation for students labelled by disability as well as gender and/or race. I like the word 'disaffection', with its non-punitive allusions to the idea that some children just don't like school (perhaps for good reason). However, I am less sanguine about the normative assumptions of future happiness for the disaffected pupils. In the Preface to the book, Jones provided an overview of the goals of the work on disaffection as inclusion in the normative: 'The aim should be that all pupils feel normal, valued, and achieving.' Foucault's work has surely made norms less attractive as a given standard for acceptability. Perhaps Jones was arguing for a wider definition of normality than conventionally offered, but the concept is fraught with difficult assumptions. Again, the cultural vocabulary relating to norms, happiness and the value of particular people needs more critical scrutiny.

A more critical approach to the whole issue of esteem was taken by the various projects mentioned in Kenway and Willis's (1990) provocative book. They included whole-school programmes for 'self esteem' which go beyond intervention for individual children identified as 'low self esteem'. In the project reported by Jonas (1990), an entire secondary school was involved in creating a climate supportive of democratic student processes, which led to formal questioning of the university entrance procedures that disadvantaged their students.

This supportive climate was created by policies within the school, no doubt the product of a concerted effort by many teachers, students and senior staff at the school. Instead of focusing on coping of individual students, there could be a wider concern with the practices and resources that are linked with energised, creative action in people working against racism, sexism and 'ableism' in schools.

EDUCATIONAL ASSESSMENT

Educational assessment is an interesting interdisciplinary field, having its traditions in the psychometric testing field including IQ testing, but also being well grounded at present in issues of comparative educational policy. One reason for this is that assessment, particularly national assessment in which countries compete for the best test scores in science, maths and other fields, can lead to huge investments of money in particular educational regimes. Researchers and practitioners can become overly grounded in the political fashions and government edicts of the day (perhaps for sound survival reasons). A number of British writers have become influential writers on the purposes and contexts of assessment.

The critique of testing this century has led to great awareness of the problems involved in norm-referenced assessment. While such standardised tests are used for determining university entrance in the US, they have few enthusiasts in the UK, Australia or New

Zealand. There are subtle uses of norm-referencing which can be quite influential, however, such as in the use of such tests to ‘moderate’ results across classes or schools.

Caroline Gipps referred to the ‘paradigm shift’ in assessment over the last 25 years or so from a focus on testing towards greater use of more fluid, situation-specific assessment. This has meant a turn away from norm-referenced tests towards criterion-referenced, formative and performance-based assessment. She also argued that contemporary assumptions about the learner have changed from the psychometrists’ assumption that individual differences are based largely on innate factors. Now there is more recognition of the importance of context, curriculum and pedagogies in learning.

The influence of Foucault’s analyses on schooling and disciplinary regimes is found in Patricia Broadfoot’s work on the system by which particular areas of knowledge come to be regarded as higher in status, with assessment functioning as a legitimating authority for particular practices and hierarchical social structures. She has also been an advocate for more use of self-assessment by students. Students who are involved in reflecting on their own learning and performance provide a more open-ended, fluid and ongoing process of assessment closer to a critical educational psychology than assessment centred solidly on technicist concerns about ‘objective’ measurement.

It is ironic that the work of British researchers mentioned in this section has been used in New Zealand to argue against the national testing of all children at particular ages (*e.g.*, as done with SATs in the UK) in favour of testing samples of children from different schools. This helps to avoid creating records of individual student achievement based on norm-referenced tests. Unfortunately the New Zealand government is at present planning to follow where Britain leads in introducing national testing of children, under the guise of accountability to parents.

Research Methods

Like traditional research in psychology, educational psychology has been defined by its positivist methodologies, and was earlier aligned with experimental research and quantitative statistical analyses. Fortunately there is now more attention given to introducing both qualitative and quantitative methods in educational psychology textbooks. The clinical case study has also been a key method throughout this century, especially as related to the practising educational psychologist in the field. Though the research summarised by contemporary educational psychology textbooks would not usually refer to interdisciplinary work, there is more openness to post-positivist approaches and a consideration of complexities beyond the old quantitative/qualitative divide.

There are encouraging hints that methodologies in educational psychology are changing as they are elsewhere in the ‘social sciences’. A critical educational psychology should surely include the diversity of issues and paradigms in qualitative methodologies. Of particular interest would be a contemporary overview of grounded theory as well as an introduction to contemporary tensions regarding crises of legitimation and multiple

perspectives. These approaches emphasise reflexive practices in contemporary research informed by postmodern questioning.

The use of reflexivity incorporated as part of a research methodology is more than a contemporary fashion. It is a way to link the conceptual and practical, to make research fit some of the dimensions of 'action' research so that educational psychologists, students, teachers and researchers work together on issues of relevance. A good example of this kind of research is Patti Lather's (1990) study of Women's Studies students' 'resistance' to the liberatory curriculum. During the research, Lather began to question her own practice, and the study changed towards a more collaborative project with students and other staff, questioning the hidden agenda that students would become liberated along a predetermined path of enlightenment. Such reflexive and open-ended exploration of issues such as student power, teachers' agendas, discourses about liberation, and educational goals towards personal development holds much promise for critical educational psychology research.

'Special' Educational Needs

The field of education concerned with students who have disabilities or other 'special' requirements as learners different to the generic 'normal child' is an area of central concern for educational psychologists. The creators of the first formal test of intelligence, Binet and Simon, were concerned with differentiating children thought to be capable of benefiting from mass education from those thought to need special educational facilities. At about the same time, the field of child development was being established with strong progressivist views explicitly tied to Darwin's evolutionary ideas. The 'normal child' was created as a subject-position in opposition to its 'other', the abnormal child, who did not fit the developmental stages and timetables being canonised by writers such as Gesell. Even today, most textbooks of human development focus on the 'normal' child, with little attention to the 'outliers' on the IQ distribution.

The field of 'special education' is contested ground between parents, teachers, educational psychologists and government policies about funding. Political moves were underway in the 1960s to bring some children formerly labelled as 'mildly retarded' into regular classrooms. Special facilities for segregationist residential care were put under scrutiny in many countries. The IQ test had been criticised for its racist biases in the 1960s. Suddenly the dominant medical discourses which defined a divide between the able-bodied and the 'handicapped' did not seem so convincing.

The political movement for disabled people's rights provided an alternative to medical discourses: that of social oppression. In the 1970s and 1980s, humanist calls for equal rights began to add disability to the inequities of race, gender, and sexuality. In education, 'mainstreaming', in which children with disabilities were placed in ordinary classrooms, became a rights issue for students and their parents. Unfortunately, some of these ordinary classrooms did not have facilities to be able to assist students with disabilities with their education.

In the 1980s there was a more postmodern move towards the idea of 'inclusion' in education. There was a shift in perspective away from the idea that some students with 'mild' or 'moderate' disabilities should be moved into ordinary, business-as-usual classrooms. Instead, inclusion began to consider the processes by which a student becomes a member of the classroom group, in a two-way process in which both 'regular' and 'disabled' students learn to get along and work with each other.

Recently the work on inclusion in education has become more critical about the dualism of able/disabled as another socially constructed divide. Stone (1993) and others argued that inclusive approaches may hide an assumption that there is still a dominant group which makes the decision about which children will be 'included'. Inclusion, in this view, is a more sophisticated strategy of assimilation. The 'disabled' must still 'fit in' to a reconstituted 'diverse' classroom or be excluded from school life. One future project in a critical educational psychology could be to consider the similarities between political aspirations of groups who wish to claim an identity based on disability with peoples from minority ethnic groups. Deaf parents have lobbied in several countries against forcing non-hearing children to assimilate to the norm of spoken language, arguing that it is a suppression of Deaf culture.

This has some similarities to the kura kaupapa movement in New Zealand, which is a parallel schooling system for indigenous students who are taught in the Maori language. In future there could be more attention paid to the preferences of students with disabilities regarding the kinds of classrooms and facilities they would prefer. There might also be further critical deconstruction of the able/disabled dichotomy that could lead to more recognition that each person has multiple selves that may be abled or disabled at different times and places in life.

The local and national context for considering issues of special need is directly affected by government policy changes. As Dessent noted regarding the situation in the UK, educational psychologists now do not just focus on defining which children are special and deciding how to work with such children; they must also work towards 'defining resource-worthiness'. New Zealand has undergone similar policy changes. Resourcing of special needs has become the province of individual advocacy, making it harder for communities to work together to create schools that make room for all kinds of difference amongst students. These policy moves point to the urgent need for critical work in educational psychology which challenges these individualising assumptions. There are already strong links between people working together towards more responsive, inclusive education across various countries.

FORWARD MOVES

Some emerging examples in practice are already around to show what educational psychology might look with a stronger critical direction. Rather than focus on the specific areas identified earlier as crucial parts of educational psychology (*e.g.*, learning, motivation, etc.), a challenging possibility for the future is to create more fractures

across areas. A more radical deconstruction of the individual would have huge implications for reconceptualising areas such as learning, abilities, motivation and self esteem.

There could also be new kinds of relationships between research and practice, between writing, reading, deciding, acting, discussing. Research in educational psychology could become part of the ongoing revisioning of the field and its place in the lives of children and adults as part of lives reflective about their governing circumstances. Bringing together collaborative research groups with some of the new methodological techniques provides other possibilities. There could be a widening of the collaborative research process to include children and young people as well as researchers, teachers and educational psychologists, all providing different perspectives on problems of mutual concern.

What I have focused on in this chapter are some of the innovative directions taken in the here-and-now by concerned people struggling within the broad boundaries of 'educational psychology'. The people to whom I have referred in this chapter have been mostly English-speaking writers whose work can be linked in some way to the British-American origins of the field earlier this century. This is to some extent inevitable in writing about a particular 'discipline' which, as Foucault suggested, contains the production of its own future.

I have also mentioned the work of some indigenous writers and people further from the discipline's barbed-wire boundaries. To expand critical educational psychology with further energies will require more voices of innovation from around the world, and from a diversity of educational and cultural communities. Part of this work will involve western teachers, researchers and academics who have been entrusted by the governments of developing countries with the training of their future elites. A greater focus on critical work in educational psychology will provide an interruption to the cycle of academic cloning whereby traditional ideas about IQ testing, normed assessment and the culture of positivism are sent to another generation in southeast Asia or Africa. Such a shift - which would lead to more equitable dialogue between parts of the world about issues such as special needs or assessment - would be much more troubling for the discipline. And surely unsettling the discipline so that it becomes more reflexive about its positionings and unforeseen consequences is what critical psychology is all about.

Educational Psychology: A Foundation for Teaching

The teacher's tasks are: motivating students, managing the classroom, assessing prior knowledge, communicating ideas effectively, taking into account the characteristics of the learners, assessing learning outcomes, and reviewing information-must be attended to at all levels of education, in or out of schools.

Can good teaching be taught? The answer is definitely yes. Good teaching has to be observed and practiced, but there are principles of good teaching that teachers need to know, which can then be applied in the classroom.

The major components of effective instruction are:

- Knowledge of subject and teaching resources
- Critical thinking and problem-solving skills
- Knowledge of students and their learning
- Teaching and communication skills

One attribute seems to be characteristic of outstanding teachers: intentionality. Intentionality means doing things for a reason, on purpose. Intentional teachers are those who are constantly thinking about the outcomes they want for their students and about how each decision they make moves children Towards those outcomes. Intentional teachers know that maximum learning does not happen by chance. Yes, children do learn in unplanned ways all the time, and many will learn from even the most chaotic lesson. But to really challenge students, to get their best efforts, to help them make

conceptual leaps and organize and retain new knowledge, teachers need to be purposeful, thoughtful, and flexible, without ever losing sight of their goals for every child. In a word, they need to be intentional. Intentional teachers are constantly asking themselves what goals they and their students are trying to accomplish; whether each portion of their lesson is appropriate to students' background knowledge, skills, and needs; whether each activity or assignment is clearly related to a valued outcome; whether each instructional minute is used wisely and well. An intentional teacher trying to build students' synonym skills during follow-up time might have them work in pairs to master a set of synonyms in preparation for individual quizzes.

Research finds that one of the most powerful predictors of a teacher's impact on students is the belief that what he or she does makes a difference. This belief, called teacher efficacy, is at the heart of what it means to be an intentional teacher. Teachers who believe that success in school is almost entirely due to children's inborn intelligence, home environment, or other factors that teachers cannot influence, are unlikely to teach in the same way as those who believe that their own efforts are the key to children's learning. An intentional teacher, one who has a strong belief in his or her efficacy, is more likely to put forth consistent effort, to persist in the face of obstacles, and to keep trying relentlessly until every student succeeds.

An intentional teacher achieves a sense of efficacy by constantly assessing the results of his or her instruction, constantly trying new strategies if their initial instruction didn't work, and constantly seeking ideas from colleagues, books, magazines, workshops, and other sources to enrich and solidify their teaching skills.

The goal of research in educational psychology is to carefully examine obvious as well as less than obvious questions, using objective methods to test ideas about the factors that contribute to learning. The products of this research are principles, laws, and theories. A principle explains the relationship between factors, such as the effects of alternative grading systems on student motivation. Laws are simply principles that have been thoroughly tested and found to apply in a wide variety of situations. A theory is a set of related principles and laws that explains a broad aspect of learning, behaviour, or another area of interest. Without theories the facts and principles that are discovered would be like disorganized specks on a canvas.

Theories tie together these facts and principles to give us the big picture. However, different theorists may interpret the same facts and principles in different ways. As in any science, progress in educational psychology is slow and uneven. A single study is rarely a breakthrough, but over time evidence accumulates on a subject and allows theorists to refine and extend their theories.

It is probably true that the most important things teachers learn, they learn on the job—in internships, while student teaching, or during their first years in the classroom. However, teachers make hundreds of decisions every day and each decision has a theory behind it, whether or not the teacher is aware of it. The quality, accuracy, and usefulness of those theories are what ultimately determine the teacher's success.

Developmental research indicates that as students enter adolescence, the peer group becomes all-important to them, and they try to establish their independence from adult control, often by flouting or ignoring rules. Basic research on behavioural learning theories shows that when a behaviour is repeated many times, some reward must be encouraging the behaviour, and that if the behaviour is to be eliminated, the reward must first be identified and removed. Teachers can use this information to plan effective teaching strategies and avoid being sidetracked by class management issues.

Research in educational psychology not only provides evidence for principles of effective practice, but it also provides evidence about the effectiveness of particular programmes or practices. For example, in the vignette at the beginning of this chapter, Leah Washington was using a specific approach to creative writing instruction that has been extensively evaluated as a whole. In other words, there is evidence that, on average, children whose teachers are using such methods learn to write better than those whose teachers use more traditional approaches.

There is evidence on the effectiveness of dozens of widely used programmes, from methods in particular subjects to strategies for reforming entire schools. An intentional teacher should be aware of research on programmes for his or her subject and grade level, and should seek out professional development opportunities to learn methods known to make a difference for children.

Many researchers and educators have bemoaned the limited impact of research in educational psychology on teachers' practices. Indeed, research in education has nowhere near as great an impact on practice as research in medicine or agriculture or engineering. Yet research in education does have a profound indirect impact on educational practice, even if teachers are not aware of it. It affects educational policies, professional development programmes, and teaching materials. It is important for educators to become intelligent consumers of research, not to take every finding or every expert's pronouncement as truth from Mount Olympus.

How do we know what we know in educational psychology? As in any scientific field, knowledge comes from many sources. Sometimes researchers study schools, teachers, or students as they are, and sometimes they create special programmes, or treatments, and study their effects on one or more variables (anything that can have more than one value, such as age, sex, achievement level, or attitudes). There is no one best or most useful approach to research; any method can be useful when applied to the right set of questions. The principal methods educational researchers use to learn about schools, teachers, students, and instruction are experiments, correlational studies, and descriptive research. In an experiment, researchers can create special treatments and analyze their effects. In one classic study, Lepper, Greene, and Nisbett (1973) set up an experimental situation in which children used felt-tipped markers to draw pictures. Children in the experimental group (the group that receives a treatment) were given a prize (a "good player award") for drawing pictures. Children in a control group received no prizes. At the end of the experiment, all students were allowed to choose among

various activities, including drawing with felt-tipped markers. The children who had received the prizes chose to continue drawing with felt-tipped markers about half as frequently as did those who had not received prizes. This result was interpreted as showing that rewarding individuals for doing a task they already liked could reduce their interest in doing the task when they were no longer rewarded.

The Lepper study illustrates several important aspects of experiments. First, the children were randomly assigned to receive prizes or not. For example, the children's names might have been put on slips of paper that were dropped into a hat and then drawn at random for assignment to a "prize" or "no-prize" group. Random assignment ensured that the two groups were essentially equivalent before the experiment began. This equivalence is critical, because if we were not sure that the two groups were equal before the experiment, we would not be able to tell whether it was the prizes that made the difference in their subsequent behaviour.

A second feature of the above study that is characteristic of experiments is that everything other than the treatment itself (the prizes) was kept the same for the prize, and no-prize groups. The children played in the same rooms with the same materials and with the same adults present. The researcher who gave the prize spent the same amount of time watching the no-prize children draw. Only the prize itself was different for the two groups. The goal was to be sure that it was the treatment, not some other factor, which explained the difference between the two groups.

The Lepper study is an example of a laboratory experiment. Even though the experiment took place in a school building, the researchers created a highly artificial, structured setting that existed for a very brief period of time. The advantage of laboratory experiments is that they permit researchers to exert a very high degree of control over all the factors involved in the study. Such studies are high in internal validity, which is to say that we can confidently attribute any differences they find to the treatments themselves (rather than to other factors).

The primary limitation of laboratory experiments is that they are typically so artificial and so brief that their results may have little relevance to real-life situations. For example, the Lepper study, which was later repeated several times, was used to support a theory that rewards can diminish individuals' interest in an activity when the rewards are withdrawn. This theory served as the basis for attacks on the use of classroom rewards, such as grades and stars. However, later research in real classrooms using real rewards has generally failed to find such effects. This finding does not discredit the Lepper and colleagues study; it does show that theories based on artificial laboratory experiments cannot be assumed to apply to all situations in real life but must be tested in the real settings.

The randomized field experiment, in which instructional programmes or other practical treatments are evaluated over relatively long periods in realistic conditions, is often used in educational research. Randomized field experiments are very difficult to do in education, as it is rare that teachers are willing to be assigned by chance to one

group or another. For this reason, field experiments more often use matching, in which teachers or schools using one method would be matched with those using a different method, or a control group. Matching is much more practical than random assignment, but its results must be carefully interpreted, since there may be reasons that one group of educators took on one method while another group did not.

In single-case experiments, a single student's behaviour may be observed for several days. Then a special programme is begun, and the student's behaviour under the new programme is observed. Finally, the new programme is withdrawn. If the student's behaviour improves under the special programme but the improvement disappears when the programme is withdrawn, the implication is that the programme has affected the student's behaviour. Sometimes the "single case" can be several students, or a whole class. Perhaps the most frequently used research method in education is the correlational study. In contrast to an experiment, in which the researcher deliberately changes one variable to see how this change will affect the other variables, in correlational research the researcher studies variables as they are to see whether they are related. Variables can be positively correlated, negatively correlated, or uncorrelated. An example of a positive correlation is the relationship between reading achievement and mathematics achievement. In general, someone better than average in reading will also be better than average in math. When one variable is high, the other tends also to be high. An example of a negative correlation is days absent and grades. The more days a student is absent, the lower his or her grades will tend to be.

The principal disadvantage of correlational methods is that while they may tell us that two variables are related, they do not tell us what causes what. Indeed, correlation does not imply causation-this is a frequent pitfall for novice researchers. Experimental and correlational research looks for relationships between variables. However, some research in educational psychology simply seeks to describe something of interest. One type of descriptive research is a survey or interview. Another, called ethnography, involves observation of a social setting (such as a classroom or school) over an extended period. Descriptive research usually does not have the scientific objectivity of correlational or experimental research, but it makes up for this lack in richness of detail and interpretation.

Developmental psychologists use descriptive research extensively to identify characteristics of children at different ages. The Swiss psychologist Jean Piaget, who began by carefully observing his own children, did the most important research in developmental psychology. As a result of his observations, he developed a theory that describes the cognitive development of children from infancy through adolescence.

Action research is a particular form of descriptive research that is carried out by educators in their own classrooms or schools. In action research, a teacher or principal might try out a new teaching method or school organization strategy, collect information about how it worked, and communicate this information to others. Because the people involved in the experiment are the educators themselves, action research lacks the objectivity sought in other forms of research, but it can provide deeper insight from

front-line teachers or administrators than would be possible in research done by outsiders. To teach in the public school system, it is necessary to obtain a license or certificate. The state requirements vary but all states require graduation from a four-year college with some specialized classes in teacher preparation. Although requirements are sometimes similar, there are many unique requirements also, especially in the area of licensure tests and alternative certification requirements. Continuing professional development is one of the basic requirements considered to be an ongoing one. Experience teachers find that teaching is a professional service that requires continuous updating. In this updating and sharing of information among their colleagues and/or the educational professional community, teachers become “educational researchers” as they critically examine their own practice, testing, and various strategies to help students learn.

Basic Principles of Teaching

In his classic handbook, *Teaching Tips*, Wilbert McKeachie notes that the unpredictability of teaching both frustrates and fascinates college instructors. You cannot anticipate every eventuality or problem that may pop up during a semester. Understanding and applying a few basic principles of teaching, however, not only will enhance student learning but also make your life as a teacher more rewarding and fun.

Rhythms of the Semester

Much like the rise and fall of the plot in a novel, semester courses unfold in an arc of development. After the flush of excitement during the first week of class, students and instructors begin to settle into more of a routine. When you think of your own undergraduate courses, you’ll recall a definite rhythm to the semester—a brief “honeymoon” period at the beginning when everything seems fresh and exciting; a longer span when students and instructors get to know each other and start establishing patterns of interaction; a stretch about three-quarters of the way into the semester when everyone seems concerned about deadlines, stressed, and, perhaps, ready for a break; and a few weeks before the semester ends, the final push, accompanied, we hope, by a sense of accomplishment. If you recognize these rhythms, you can design lecture materials, readings, and assignments to fit well with student learning needs at particular points along the continuum. Scheduling all the “fun” learning activities during the first few weeks of the course probably isn’t a good idea. On the other hand, waiting until the end of the semester to incorporate “fun” learning might send the message that you’ve “given up” on helping students work through the final, challenging push of the semester.

Keeping Students Engaged

One of the tenets of American education is that students learn and retain information and skills better if they are actively involved in the learning process. Explain to your students that you expect them to prepare for class, think carefully about course content,

take intellectual risks, and participate in class discussion. It is their responsibility to wrestle with the issues and concepts explored in the course; it is your responsibility to support their active involvement with the subject matter. You do this not so much by providing answers as by posing excellent questions.

A substantial literature exists on how students learn and how best to engage them in the learning process. Here are a few tips; many more are available in the Center for Teaching library and online resources.

Lecturing

- If you lecture, remember that research has shown most people actively listen for about 20 minutes. Plan your session accordingly with a break for small-group work, Q&A, or other active learning before resuming the lecture.
- Try to leave the lectern now and then. Walking “into the crowd” will help you make eye contact with students and keep them alert and curious about what you might do next.
- Vary the tone and rhythm of your voice. Use body language effectively.
- Define new vocabulary several times. Avoid jargon.
- Refer to and expand upon material presented in the textbook—don’t repeat it.
- Use technology—PowerPoint, film clips, student response systems (“clickers”), tablet PCs, etc.—effectively.
- Speak clearly and at a moderate pace.
- Summarize major points.

Active Learning

- usually involves students in more than listening;
- entails less emphasis on transmitting information and more on developing students’ learning and discipline-based skills;
- engages students in higher-order thinking skills such as analysis, synthesis, and evaluation;
- encourages students to explore their own attitudes and values; and
- does not mean you must abandon the lecture format, which is one of several effective ways to convey information.

Active Learning Techniques

- Small-group activities encourage many more students to speak in class. It’s harder to be a passive learner in a group of three than in a group of thirty. Small-group activities can be done even in classes of 500 students. All it takes is about five minutes and a well thought-out question or task for groups to work on.
- Effective use of teaching technology can present visual representations via PowerPoint, overhead projectors, videotapes, and tablet PCs. The Center for Teaching and Campus Technology Services can help you incorporate technology in ways that enhance learning and recall of course content and skills.

- Minute papers are easy, fun, and adaptable to many purposes. The goal of this exercise—which can take no more than a minute—is to gather feedback. Minute papers can help you discover what did or did not work well, as well as provide ideas about how to teach in new ways. It can be used in any size class.
- If you are trying to gather general information about what interests or confuses the students or what they think of your teaching, the feedback should be anonymous. Occasionally, you may want to just use the minute paper as a quiz; in that case, of course, names are necessary. Students write for a brief time in response to a focused question from the instructor.

Here are a few examples:

- What is the most important point you've learned in today's class (or this week's readings, this unit, etc.)?
- Did anything confuse you during today's discussion? If so, write it as a question or two.
- What has been the most effective teaching technique used during this unit?

You can do this exercise at any point during the class session, although most instructors do it at the end of a unit or the period. After you've looked through the responses, let students know one or two things you learned and how that information will affect the course. Students are glad to have an opportunity to express themselves in a way that has an impact on their learning in the course.

- Surveys can be conducted several ways, including electronic student response systems ("clickers") and minute papers, anonymous or not. Another effective way to survey students about their opinions or responses to a question is to have them line up along a continuum, discuss their choices, and then ask them to realign themselves along the continuum. If anyone has changed his or her place, ask them to explain why.
- A combination of small-group activities and surveys works well when each group holds up a card to indicate its choice of several answer options. Groups then defend their choices and try to convince others to "join" them.
- Allowing students to jot down a few thoughts before discussing in class can improve the depth of discussion and help those who feel shy about speaking off-the-cuff.
- Board work, role-playing, panel discussions, case studies, posters, and projects also actively engage students by enlisting the fully panoply of learning styles (visual, auditory, reading/writing, and kinesthetic).

Writing

Writing demands critical thinking, organizational skills, patience, and the ability to critique others and to listen to criticism. It's a truism—but also true—that the act of writing forces us to construct our understanding of a topic—in other words, to create as

well as to convey meaning. Writing is one of the essential skills that defines an educated individual, yet teaching writing seems to be the juggernaut of many an otherwise gifted teacher. The following ideas about teaching writing might smooth your way:

- Make writing a regular part of the classroom experience. One-minute papers work well. They can be anonymous or signed. You also can invite students to read aloud what they've written.
- Think carefully about the purpose of the writing assignment:
 - to demonstrate learned knowledge (a quiz or test);
 - to show research abilities—analysis, synthesis, evaluation (term paper);
 - to wrestle with an idea (short essay);
 - to help students speak aloud or focus class discussion (one-minute paper);
 - to help students understand that writing improves thinking (expository writing);
 - to hone communication skills (series of drafts); or
 - to express feelings and opinions, and to reflect on what has been learned, how, and why (personal essay, reflection).
- Design the writing assignment with explicit questions and provide clear learning objectives.
- Invite students to reflect—in writing—on their writing process for any particular assignment and also across the semester.

Even veteran faculty members wonder how to assess written work. Instructors sometimes find assessing personal writing, such as essays or reflections, especially challenging. But simply because a written piece is personal or subjective does not mean it cannot be assessed.

A few suggestions to help make assessment of writing effective for your students and efficient for you:

- Help students avoid the temptation to plagiarize by assigning a series of short exercises that build to the final paper or essay.
- Provide written comments in the margins. Marginalia need not be extensive, however, and if you have many students or many short assignments, you can rely on the “check, check-minus, check-plus” technique.
- Writing many comments such as “Good,” “Weak,” or “Confusing” probably helps the student less than fewer comments that are more specific as to why you find a sentence or paragraph good, weak, or confusing.
- One technique works especially well to start a conversation about written work. Rather than giving students grades for their first draft, ask them to read your comments and then respond. This assignment assures that they actually read and think about your comments. You also can have them rewrite certain sentences, paragraphs, or sections.
- Resist rewriting your students' work—you don't have time and they will not learn if you do the work for them.

- Grammatical errors should not be ignored. On the other hand, you should not serve as a copy editor for your students. Undoubtedly, they have been told about run-on sentences, “their” vs. “there,” and misspelled words many times before. What to do?
 - Put check marks in the margins and tell students to find and correct the errors on the marked lines.
 - Mark and correct one paragraph and tell them to find identical errors and fix them in the rest of the piece.
 - Tell students they will be exchanging papers for peer review of grammar and usage.
- Devote some time throughout the semester to teaching a few of the most frequent—and egregious—examples of bad grammar and usage. These exercises can be amusing, and students almost always appreciate learning or brushing up on a few basic writing skills.
- Develop and use a rubric for grading writing. Be sure to hand it out to students at the same time as you give them the assignment so they know what is expected of them. Writing assessment rubrics generally cover nuts-and bolts such as grammar and usage, but also include content, organization, critical thinking skills, and stylistic considerations.

Service Learning

Today, many students come to college having already performed impressive volunteer work in their communities. They are eager to participate in volunteer and service-learning opportunities in college. Service-learning courses provide active learning experiences by integrating community service with academic course work. Service learning requires students to apply what they learn in class in their community service efforts and then bring their volunteer experiences to bear on their classroom learning. As a teaching assistant in a service-learning course, you may be helping the faculty instructor and students establish community partnerships. You also may be monitoring and helping to assess the service aspect of the course, and possibly even engaging in community service yourself.

Teaching in Laboratory or Studio Settings

- Students who take lab or studio courses necessarily engage in active learning. In these types of courses, teaching assistants work with students on a more individual basis. To enhance the lab or studio experience:
- Introduce the conceptual background for their activities in each session of the lab or studio.
- Inform them of what they are going to do and the learning objectives for the activities.
- Make certain you have already conducted the experiment, or, in a studio, are familiar with the materials and media being used. Point out where you had difficulties and how you resolved them.

- Circulate around the room as students work. Ask them what they're doing and why and to interpret their results.
- Ask them to link what they are doing at the moment to what has been learned in the classroom or through the textbook.
- Invite questions.
- Remind students about safety issues and where to find devices and individuals to help in an emergency.

Effective use of Technology

At one time, slate blackboards were considered “cutting-edge” technology. But an instructor who wrote on the blackboard with his back to students was not using this technology in a way to optimize student learning. Incorporating technology into a course should not be an end unto itself, but a way to broaden and enhance learning. Although there are excellent pedagogical reasons to use technology to teach, many students—including graduate student TAs—have experienced “death by PowerPoint” or comparable technology faux pas. Any technology—even blackboards—risk becoming the end rather than a means for better teaching and learning.

Nevertheless, many excellent reasons exist for incorporating technology such as PowerPoint, student response systems, tablet PCs, ICON (course development and management systems), and videotapes in the teaching process. The Center for Teaching and Campus Technology Services help TAs and faculty members effectively incorporate technology into their teaching. The center assists with pedagogy, while Campus Technology Services can help with any technical issues.

Assessment of Learning

The most obvious way to assess student learning is by graded assignments or exams. However, many other assessment techniques exist to help you and your students understand their mastery of knowledge and skills. For instance, one-minute papers can reveal much about what students learned during one class session and what might confuse them. Discussion, small-group work, class surveys, written reflections, periodic questionnaires, and self-evaluation of learning also help both instructor and students determine where they stand in terms of learning content and developing a facility with required skills.

Grading

Assigning grades can be an angst-filled time for TAs. Finding out the answers to these questions may help you feel more at ease in the grading process:

- How much responsibility do I have for assigning particular or overall grades?
- Am I following my supervising faculty member's instructions on grading, including adhering to the grading schedule?

- Does my department have a grading curve that I am supposed to use in the course?
- Does my department or the course faculty instructor employ a rubric for grading particular assignments or exams? If so, when will I provide it to students?
- Will other TAs and the course faculty instructor discuss standards and expectations for grading? Will TAs calibrate their grades via a practice grading exercise?
- How will I handle the grading of students for whom English is a second language?
- What is the departmental, course supervisor's, or my policy on makeup exams and late assignments?

Your course supervisor or department may dictate grading guidelines, although you may have discretion in assigning at least some percentage of grades in a course.

While some departments require that all student work be graded, others encourage a variety of approaches to learning assessment, including observations and comments by instructors throughout the semester with grades only assigned at the end of the term. Instructors are not required to employ pluses or minuses on grades.

The clearer you are about how you will assign grades prior to the actual grading process, and the better records you keep, the more likely you can avoid grade disputes later. TAs who serve as graders for a professor should receive clear instructions about how to evaluate subjective responses. If more than one TA works with one class, the assistants and professor should meet to hammer out consistent grading standards. TAs also can rehearse and calibrate their grading. During this process, TAs review identical exam answers or papers, compare the responses and grades they would have assigned, and discuss differing reactions member then develop a rubric that specifies the criteria everyone will use to evaluate student work.

Grades, of course, can be a source of friction between students and instructors. Students want to know that instructors have been consistent and fair in their grading. The following points will help you achieve consistency and fairness and demonstrate your commitment to those qualities should a student question his or her grade:

- Keep accurate and duplicate copies of your grading records.
- Maintain records that can be easily interpreted by your supervising faculty member.
- Inform students fairly often throughout the semester about their progress in the course. This does not need to be a grade, but it should be evaluative in some way.
- Create a rubric (or a scoring guide that assigns specific points to each answer) when grading papers or exams.
- After you've developed a rubric or scoring guide, "test" it by applying it to five or ten exams or papers. If your grading guidelines don't seem reasonable, alter them.

- After you have graded all the exams or papers, re-examine the first few that you graded. Are the standards you applied at the beginning of the grading process the same as the standards you used at the end?
- When evaluating written work, try to find at least one thing to praise and encourage further work. And even students who produce “A” writing need to be given new, higher goals to shoot for.
- Make clear that you have considered what a student is trying to say in written work as well as how they have managed to communicate it.

Attendance

The University’s Operations Manual defines excused and reported absences. You must allow work and exams to be made up if a student misses class because of an excused absence. Excused absences include absences due to participation in a school-sanctioned athletic or other event for which the student has provided prior notice of event dates.

Large-scale tragedies or unexpected events can affect many students and their instructors.

Research has shown that students appreciate instructors who acknowledge the event and provide an opportunity to discuss it and its impact during class time. Such consideration by instructors reinforces the shared sense of community among students.

Whether during the current class session or the next session, turning back to the course work at hand after thoughtful discussion can provide a welcome return to normal routine. Communicating with your students:

- You will communicate with your students through a variety of technological means and in a variety of venues. Planning ahead about how to communicate effectively can help avoid potential pitfalls in communication later.
- The syllabus should include the TA office or at least the departmental telephone number. It is up to you whether you provide your home telephone number to students.
- ICON offers many opportunities for communication, such as chat rooms, threaded discussions, feedback surveys, and more.
- TAs can make creative use of their time during office hours. To become familiar with your students and help ease them into the course, you might schedule individual conferences with them during the first month of the course. As the semester continues, you might also use office hours to host small-group workshops or reviews.
- E-mail is a popular way for students to communicate with instructors, and instructors sometimes incorporate online discussion via e-mail as a course requirement.
 - Most students are familiar with e-mail before arriving on campus and all students may sign up for a free e-mail account through the University.

The process is included in the ISIS sign-on. Nevertheless, many students prefer to use another account, so if you want a list of your students' e-mail accounts, you should ask them to e-mail you and save their addresses in your e-mail address book.

- The course planning and management system ICON also has e-mail capabilities, including a method to send to the entire class.
- Include your e-mail address and hours in your syllabus, as well as your expectations for e-mail communication. Inform students that you are not available via e-mail (or otherwise) except during certain hours, and that they cannot expect you to respond immediately.
- The Family Educational Rights and Privacy Act of 1974 (FERPA) generally prohibits disclosing educational records (including grades) to anyone other than the student or a university official with a legitimate educational need to view the record. Instructors communicating via e-mail about grades need to be certain they are, indeed, communicating with the students. In general, sending an e-mail to a student's University address or via ICON provides adequate assurance that the recipient is the student. E-mailing to another address such as gmail or AOL does not provide adequate assurance.
- Some instructors accept assignments as attachments, and some do not due to potential viruses or system incompatibility. Indicate in your syllabus whether you will accept assignments via e-mail attachment.
- Be alert to the possible risks of communicating via e-mail, in particular, the challenges of interpreting intent and tone online. Irony and light sarcasm can be devastating to a student. Maintain a friendly, but professional tone and focus, and insist your students do the same. Write your messages with the assumption that your supervisor or department chair will read them.
- If you encounter the rare electronic message that is inappropriate, hostile, or bullying, forward it immediately to your faculty supervisor. If your supervisor feels you should respond, maintain professional integrity, copy your supervisor, and let the student see that you have done so.
- Save all messages and make paper copies for your course file.

Psychology Applied to Teacher Training

It is not surprising that developmental psychologists should express an interest in teaching and education, nor that they should find a ready demand for their ideas among educational practitioners. There is a broad common interest between those who study, and those who seek to train and develop, psychological processes. Their mutual interest lies in the problems associated with the acquisition, organisation, retention, elaboration and use of knowledge, skills, attitudes and values. Those who study developmental

processes in young children seem to have a special passion for generating educational messages. Intimate contact with the workings of young minds appears to coincide with a belief in children's potential. When this is compared with a knowledge of, and concern about, the products of schooling, the contrast frequently provokes the publication of appeals for desperate remedies.

In parallel with psychologists' interest in education, educators have consistently afforded developmental psychology a place in educational theory. This is not the place to consider the general nature of educational theory, except to comment that at least some aspects of it must be value-laden, since the essential process of education is intended to be directed towards the attainment of valued ends. The definition of these ends and the specification of the means by which they might be attained are informed by a wide range of consideration including conceptions of the nature of man and society, conceptions of the forces which shaped educational organisations, and theories of how institutions and individuals work. Developmental studies are strongly implicated in this list.

Fundamental Issues

Whilst in this light the place of developmental psychology seems assured in the student teacher's curriculum, there are many problems in translating from the essentially descriptive findings of psychology to the prescriptive business of education. These issues have long been recognised but have rarely been given the exhaustive treatment they merit before being used in teacher education (Desforges 1981).

The first thing that must be established is the quality of the informing psychological theory. If educational practices are genuinely to be founded on psychology, then the theory must be sound. Munsterberg (1912:94) observed: 'There is nothing more reckless than to take fragments of an unsafe new doctrine and turn them into practical remedies.'

Piaget emphasised the importance of resting educational principles, said to emanate from child study, on scientifically valid foundations. Whilst recognising some consonance between his educational principles and those of Rousseau, Piaget none the less emphasised that Rousseau's view was '...no more than a sociological belie for a polemical weapon...' (1971 a: 140) and that 'what Rousseau lacked in order to constitute a science of education was a psychology of mental development' (p. 141). It is the claim to scientific validity which underpins the developmentalist's right to a hearing on educational matters.

Unfortunately, there is no body of developmental theory which is not open to fundamental reservation. For example, extensive criticisms of Piaget's theory are available. The flaws have been considered to be of such magnitude that it has been thought premature and misleading to apply Piaget's theory to education. Piaget's theory dominates contemporary developmental studies and, since 'new doctrines' have yet to emerge from his critics, this must provoke considerable caution in applications of his work.

Since it is in the nature of science to contest theories, this could lead to a complete embargo on the educational applications of much developmental theory. As a way of avoiding this conclusion, it has been suggested (Hilgard 1964) that educators are practical people who must deal with facts and that conflicting theories are rarely about facts, but about interpretations of facts. So long as theoreticians agree on the facts, then the practitioner can work on the basis of secure knowledge. 'Whilst eventually the correct interpretation might make some difference, it often makes little difference at the present stage of technology.' (Hilgard 1964:403) Thus navigators do not suspend their calculations whilst they await the resolution of theoretical debates about the origins of the universe.

Although the above example is an impeccable instance of Hilgard's view on the practical utility of contested theory, it might not be applicable to conflicts in contemporary social science. For one thing, educational prescriptions rarely arise from the notional facts of a body of knowledge.

Rather, they arise from interpretations. The educational prescriptions emanating from Piaget's work do not arise from the largely uncontested facts that, at particular ages, most children fail specific archetypal tasks. On the contrary, the prescriptions arise precisely from Piaget's contested interpretations of his observations. For example, Piaget's critics confirm his observations that children under the age of seven typically respond differently from adults on the whole range of tasks known in Piagetian terms as 'tests of conservation'. Piaget's interpretation is that these children lack the internalised operation of conservation and this is taken to be just one symptom of their being 'pre-operational'. From this it has been argued that for such children the curriculum should be dominated by direct sensory experience. However, Piaget's critics—whilst endorsing his observations—reject the interpretation. Children's performance limitations on these tasks have been variously attributed to memory limitations (Case 1974), the level of experience with specific task content (Engelman 1971) or inexperience in interpreting the social demands of the task. Some of these interpretations have been associated with educational prescriptions at odds with the Piagetian approach. These authors do not assume that young children lack logical operations and do not limit pedagogies to direct sensory experience as *the* essential educational medium for young children.

Rather than searching for a solution to the problems of contested or condemned theory, as a basis for educational innovation, both psychologists and educationalists have tended to ignore the issues. For example, whilst Skinner's theory of language development was criticised by Chomsky (1959) and, whilst behaviourism generally has been criticised as naive in cognitive terms (Broadbent 1958) and untenable epistemologically (Piaget 1971b), both find extensive use in contemporary educational practice and teacher education syllabuses.

Whatever the quality of particular theories, there remains the overriding and general problem of the nature of the link between schooling and developmental studies. Piaget (1971:145) asserted: 'That schooling should be adapted to the child is something that

everyone has constantly urged.' This is simply not true. It is safer to say that no one has argued that schooling should be deliberately maladapted to the child. The practicalities of avoiding such maladaptation depend, in general terms, on what the nature of child development is taken to be.

For those theorists who believe that development is merely the associative products of learning, there are no special assumptions in applying developmental psychology to schooling. Special assumptions become necessary only when the claim is made that there is a natural course to human development or that there are natural limitations to human learning at certain stages. This claim is often advanced and, since it takes a variety of forms, it is necessary to examine its implications.

Taken at its most general, the view of the superiority of natural propensities is well expressed by Smith (1975:2) who asserts that children know how to learn and that 'A fundamental problem for any instructor is to avoid interfering with natural processes of comprehension and learning.' The educational implication of this is that 'There is only one reliable way to improve instruction, and that is to assist the instructor in understanding children.^{1/4}' (p. 245) A similar view is often deployed by popularisers of Piaget's work. Occasionally, a debilitating natural decline is posited. Bereiter and Scardamalia (1977), for example, have taken the view that there is a natural decline in spontaneous curiosity. In establishing the educational relevance of such views, it is necessary to ask whether the postulated 'natural processes' are indeed natural or whether they are themselves the products of schooling, or the knock-on effects of schooling mediated by parents.

Even if a natural order were established, its application to instruction depends on whether Nature were to be considered ideal, less than ideal, or simply immutable. It is interesting that Smith (1975) views children's natural ways of knowing as self-evident and good, whilst Bereiter and Scardamalia (1977) view children's natural decline in curiosity as simply an immutable fact.

It is clear that divining educational implications from developmental studies involves value judgements on whether natural development (when established) should set the limits to which schooling might aspire. There is a third stance on the form of development and its limitations which has its origins neither in an appeal to the superiority of Nature nor, initially at least, in the analysis of psychological processes.

This is the stance initiated by Piaget. Arising from his rejection, on epistemological grounds, of empiricism and rationalism as accounts of the growth of understanding and his adoption of a constructivist stance, Piaget's view of the impact of experience on the processes of equilibration was that it necessarily yielded cognitive structures of particular and temporarily limiting forms. Piaget's view of development is thus not based on a naive view of the wisdom or otherwise of Nature. He observed...let there be no misunderstanding. Memory, passive obedience, imitation of the adult, and the receptive factors in general are all as natural to the child as spontaneous activity.' (1971 a: 137-8) Piaget's theory has had enormous prominence in teacher education. This is not the

place to describe or evaluate the work. The concern here is to raise, in principle, the kinds of problems peculiar to the application of Piaget's work to schooling, and hence to evaluate its place in teacher education.

Instructional implications have been taken to arise from the functional aspect of the theory (that is the description of the dynamics of development) and from the formal aspect of the theory (that is the description of the sequential stages of underlying competence). Piaget's view of the dynamics of development (most notably his conception of equilibration) arises out of his position as a constructivist. Before applying this notion to education, it would be necessary to be convinced on epistemological grounds of the necessity of adopting this model. Piaget's own view was that equilibration was an unnecessary concept when contemplating teaching school subjects...whose contents have been invented or even developed by adults' (1971 a: 26).

However, there are some school subjects whose contents depend more on a process of research and discovery during the course of which the human intelligence affirms its own existence (1971a: 26). In these latter subjects, and Piaget quoted physics and mathematics as examples, the operation of the learner's own processes of equilibration were considered crucial.

In applying the stage model to education, different problems are met. The validity of the model would have to be established. This is extremely difficult since the theory is one of underlying competence. Even if the theory were considered valid, there are massive technical problems in translating descriptions of competence into taxonomies for analysing children's available structures (as revealed by their task-specific performances) and related taxonomies for sequencing curriculum tasks. Considerable effort has been put into such work (Shayer and Adey 1981) but the fundamental issues set out above seem to have been totally ignored (Desforges 1981).

Teacher Roles in a Collaborative Classroom

Across this nation, teachers are defining their roles in terms of mediating learning through dialogue and collaboration. While mediation has been defined in different ways by Reuven Feuerstein, Lev Vygotsky and others, we define mediation here as facilitating, modeling, and coaching. Most teachers engage in these practices from time to time. What is important here is that these behaviors (1) drive instruction in collaborative classrooms, and (2) have specific purposes in collaborative contexts.

Facilitator Facilitating involves creating rich environments and activities for linking new information to prior knowledge, providing opportunities for collaborative work and problem solving, and offering students a multiplicity of authentic learning tasks. This may first involve attention to the physical environment. For example, teachers move desks so that all students can see each other, thus establishing a setting that promotes true discussion. Teacher may also wish to move their desks from the front of the room to a less prominent space. Additionally, teachers may structure the resources in the classroom to provide a diversity of genres and perspectives, to use and build

upon cultural artifacts from the students' homes and communities, and to organize various learning activities. Thus, a collaborative classroom often has a multiplicity of projects or activity centers using everyday objects for representing numerical information in meaningful ways and for conducting experiments that solve real problems. These classrooms also boast a rich variety of magazines, journals, newspapers, audiotapes, and videos which allow students to experience and use diverse media for communicating ideas. In Video Conference 1, for example, students were shown investigating science concepts using everyday materials, such as paper and straw, found in their neighbourhoods. Facilitating in collaborative classrooms also involves people. Inside the classroom, students are organized into heterogeneous groups with roles such as Team Leader, Encourager, Reteller, Recorder, and Spokesperson. Additionally, collaborative teachers work to involve parents and community members.

Examples are: A workshop center in New York invites parents to come and experience the thinking processes involved in conducting experiments using everyday objects so that they can provide such learning experiences at home ; teachers in Tucson involve parents and the community in academic tasks their students engage in, and rural students in Colorado perform community services such as producing a local newspaper.

Another way that teachers facilitate collaborative learning is to establish classrooms with diverse and flexible social structures that promote the sort of classroom behaviour they deem appropriate for communication and collaboration among students. These structures are rules and standards of behaviors, fulfilling several functions in group interaction, and influencing group attitudes. Particular rules depend, of course, on the classroom context. Thus, teachers often develop them collaboratively with students and review or change them as needed. Examples of rules are giving all members a chance to participate, valuing others' comments, and arguing against (or for) ideas rather than people. Examples of group functions are: asking for information, clarifying, summarizing, encouraging, and relieving tension. To facilitate high quality group interaction, teachers may need to teach, and students may need to practice, rules and functions for group interaction.

Finally, teachers facilitate collaborative learning by creating learning tasks that encourage diversity, but which aim at high standards of performance for all students. These tasks involve students in high-level thought processes such as decision making and problem solving that are best accomplished in collaboration. These tasks enable students to make connections to real-world objects, events, and situations in their own and an expanded world, and tap their diverse perspectives and experiences. Learning tasks foster students' confidence and at the same time, are appropriately challenging.

Model Modeling has been emphasized by many local and state guidelines as sharing one's thinking and demonstrating or explaining something. However, in collaborative classrooms, modeling serves to share with students not only what one is thinking about the content to be learned, but also the process of communication and collaborative learning. Modeling may involve thinking aloud (sharing thoughts about something) or

demonstrating (showing students how to do something in a step-by-step fashion). In terms of content, teachers might verbalize the thinking processes they use to make a prediction about a scientific experiment, to summarize ideas in a passage, to figure out the meaning of an unfamiliar word, to represent and solve a problem, to organize complicated information, and so on. Just as important, they would also think aloud about their doubts and uncertainties. This type of metacognitive thinking and thinking aloud when things do not go smoothly is invaluable in helping students understand that learning requires effort and is often difficult for people. With respect to group process, teachers may share their thinking about the various roles, rules, and relationships in collaborative classrooms. Consider leadership, for example. A teacher might model what he or she thinks about such questions as how to manage the group's time or how to achieve consensus. Similarly, showing students how to think through tough group situations and problems of communication is as invaluable as modeling how to plan an approach to an academic problem, monitoring its progress, and assessing what was learned. A major challenge in mediating learning is to determine when it is appropriate to model by thinking aloud and when it is useful to model by demonstrating. If a teacher is certain that students have little experience with, say, a mathematical procedure, then it may be appropriate to demonstrate it before students engage in a learning task. (This is not to say that the teacher assumes or states that there is only one way to perform the procedure. It is also important to allow for individual variations in application.) If, on the other hand, the teacher believes students can come up with the procedure themselves, then he or she might elect to ask the students to model how they solved the problem; alternatively the teacher could give students hints or cues. Coach Coaching involves giving hints or cues, providing feedback, redirecting students' efforts, and helping them use a strategy. A major principle of coaching is to provide the right amount of help when students need it—neither too much nor too little so that students retain as much responsibility as possible for their own learning. For example, a collaborative group of junior high students worked on the economic development of several nations. They accumulated a lot of information about the countries and decided that the best way to present it was to compare the countries. But they were stymied as to how to organize the information so they could write about it in a paper, the product they chose to produce. Their teacher hinted that they use a matrix—a graphic organizer they had learned—to organize their information. When the group finished the matrix, the teacher gave them feedback. In so doing, he did not tell them it was right or wrong, but asked questions that helped them verbalize their reasons for completing the matrix as they did. The principle the teacher followed was to coach enough so that students could continue to learn by drawing on the ideas of other group members.

Student Roles in a Collaborative Classroom

Students also assume new roles in the collaborative classroom. Their major roles are collaborator and active participant. It is useful to think how these new roles influence

the processes and activities students conduct before, during, and after learning. For example, before learning, students set goals and plan learning tasks; during learning, they work together to accomplish tasks and monitor their progress; and after learning, they assess their performance and plan for future learning. As mediator, the teacher helps students fulfill their new roles.

Goal setting Students prepare for learning in many ways. Especially important is goal setting, a critical process that helps guide many other before-, during-, and after-learning activities. Although teachers still set goals for students, they often provide students with choices. When students collaborate, they should talk about their goals.

For example, one teacher asked students to set goals for a unit on garbage. In one group, a student wanted to find out if garbage is a problem, another wanted to know what happens to garbage, a third wanted to know what is being done to solve the problem of garbage. The fourth member could not think of a goal, but agreed that the first three were important and adopted them. These students became more actively involved in the unit after their discussion about goals, and at the end of the unit, could better evaluate whether they had attained them.

Designing Learning Tasks and Monitoring While teachers plan general learning tasks, for example, to produce a product to illustrate a concept, historical sequence, personal experience, and so on, students assume much more responsibility in a collaborative classroom for planning their own learning activities.

Ideally, these plans derive in part from goals students set for themselves. Thoughtful planning by the teacher ensures that students can work together to attain their own goals and capitalize on their own abilities, knowledge, and strategies within the parameters set by the teacher. Students are more likely to engage in these tasks with more purpose and interest than in traditional classrooms.

Self-regulated learning is important in collaborative classrooms. Students learn to take responsibility for monitoring, adjusting, self-questioning, and questioning each other. Such self-regulating activities are critical for students to learn today, and they are much better learned within a group that shares responsibility for learning. Monitoring is checking one's progress toward goals. Adjusting refers to changes students make, based on monitoring, in what they are doing to reach their goals. For example, a group of students decided that the sources of information on the Civil War they selected initially were not as useful as they had hoped, so they selected new materials. Another group judged that the paper they had planned to write would not accomplish what they thought it would the way they had organized it, so they planned a new paper.

Students can further develop their self-regulating abilities when each group shares its ideas with other groups and gets feedback from them. For example, in the first video conference, elementary students were shown collaborating in small groups to define and represent math problems. Working in small groups, the children determined what was being asked in story problems and thought of ways to solve the problems. Then each group shared its ideas with the whole class. Members of the class commented on

the ideas. As students developed problem-solving skills with feedback from other groups, they learned more about regulating their own learning which they could use in the future. **Assessment** While teachers have assumed the primary responsibility for assessing students' performance in the past, collaborative classrooms view assessment much more broadly. That is, a major goal is to guide students from the earliest school years to evaluate their own learning. Thus, a new responsibility is self-assessment, a capability that is fostered as students assess group work.

Self-assessment is intimately related to ongoing monitoring of one's progress toward achievement of learning goals. In a collaborative classroom, assessment means more than just assigning a grade. It means evaluating whether one has learned what one intended to learn, the effectiveness of learning strategies, the quality of products and decisions about which products reflect one's best work, the usefulness of the materials used in a task, and whether future learning is needed and how that learning might be realized.

Collaborative classrooms are natural places in which to learn self-assessment. And because decisions about materials and group performance are shared, students feel more free to express doubts, feelings of success, remaining questions, and uncertainties than when they are evaluated only by a teacher. Furthermore, the sense of cooperation (as opposed to competition) that is fostered in collaborative work makes assessment less threatening than in a more traditional assessment situation. Ideally, students learn to evaluate their own learning from their experiences with group evaluation.

Interactions in a Collaborative Classroom

The critical role of dialogue in collaborative classrooms has been stressed throughout this Guidebook. The collaborative classroom is alive with two-way communication. A major mode of communication is dialogue, which in a collaborative classroom is thinking made public. A major goal for teachers is to maintain this dialogue among students.

Consider examples of interactions in collaborative groups. Members discuss their approaches to solving a math problem, explain their reasoning, and defend their work. Hearing one student's logic prompts the other students to consider an alternative interpretation. Students are thus challenged to re-examine their own reasoning. When three students in a group ask a fourth student to explain and support her ideas, that is, to make her thinking public, she frequently examines and develops her concepts for herself as she talks. When one student has an insight about how to solve a difficult problem, the others in the group learn how to use a new thinking strategy sooner than if they had worked on their own. Thus, students engaged in interaction often exceed what they can accomplish by working independently.

Collaborative teachers maintain the same sort of high-level talk and interaction when a whole class engages in discussion. They avoid recitation, which consists primarily of reviewing, drilling, and quizzing; i.e., asking questions to which the answer is known by the teacher and there is only one right answer. In true discussion, students talk to

each other as well as to the teacher, entertain a variety of points of view, and grapple with questions that have no right or wrong answers. Sometimes both students and the teacher change their minds about an idea. In sum, interactions in whole group discussion mirror what goes on in small groups.

Still a third way interactions differ in collaborative classrooms has been suggested above. Teachers, in their new roles as mediators, spend more time in true interactions with students. They guide students' search for information and help them share their own knowledge. They move from group to group, modeling a learning strategy for one group, engaging in discussion with another, giving feedback to still another.

Challenges and Conflicts

When teachers and schools move from traditional to collaborative instruction, several important issues are likely to arise. They are important concerns for teachers, administrators, and parents. Classroom Control Collaborative classrooms tend to be noisier than traditional classrooms. This is a legitimate issue for a number of people. Some teachers believe that noisy classrooms indicate lack of discipline or teacher control. In such situations, they argue, students cannot learn.

Earlier in this essay we stressed that collaborative classrooms do not lack structure. Indeed, structure becomes critical. Students need opportunities to move about, talk, ask questions, and so on. Thus, we argue that the noise in a smoothly running collaborative classroom indicates that active learning is going on. However, students must be taught the parameters within which they make their choices. Rules and standards must be stressed from the beginning, probably before any collaboration is initiated, and reviewed throughout a school year.

Preparation Time for Collaborative Learning Teachers and administrators may believe that new lesson plans must be formed for these classrooms. To a certain extent, they are correct. But many teachers already have created engaging units and activities that are easily implemented in a collaborative classroom. Furthermore, teachers can begin slowly, making changes in one subject area or unit within a subject area, probably one they are already very comfortable teaching, and then add other subjects and units. Teachers can also share their plans with each other. Indeed, if we expect students to collaborate, we should encourage teachers to do the same! Principals and curriculum specialists can also collaborate with teachers to plan effective segments of instruction. Moreover, there is a tradeoff between the extra planning time needed and benefits such as less time correcting lessons, increased student motivation, and fewer attendance and discipline problems.

Individual Differences Among Students We have touched on this concern in the section on heterogeneous grouping. Nevertheless, many people will still doubt that individual differences can be better addressed in collaborative classrooms than in traditional classrooms with homogeneous grouping. A major question people have concerns the advantage collaboration affords gifted or high-achieving students. There

are two tough issues here. First, many teachers do not believe that low-achieving students have much to contribute to the learning situation; in effect, that they have no prior experiences or knowledge of value. Second, teachers worry that high-achieving students will be held back.

In response to the first issue, many collaborative teachers have expressed surprise when seemingly less-able students had insights and ideas that went way beyond what teachers expected. Further, if each student contributes something, the pool of collective knowledge will indeed be rich. In answer to the second concern, data suggest that high-achieving students gain much from their exposure to diverse experiences and also from peer tutoring. Also, students who may be high achieving in one area may need help in other areas. Teachers and others also wonder whether shy students can fully participate in a classroom that depends so much on dialogue. We suggest that these students might feel more comfortable talking in small groups that share responsibility for learning. Furthermore, interaction between learners can happen in ways other than oral dialogue, for example, writing and art.

A related concern is that many schools are structured homogeneously so that an individual teacher cannot form heterogeneous groups without involving changes in the entire school. A whole class of “low” readers are taught by one teacher, “average” by another. High school tracks are even more systematically entrenched. Clearly, these practices are not conducive to collaborative learning and require system-wide restructuring. Individual teachers or groups of teachers can initiate dialogue on the problem, however.

Individual Responsibility for Learning This concern is a difficult one to solve unless major changes in other areas of schooling are also undertaken. Students are used to being graded for individual work; parents expect to know how their students fare in school. School staff and state departments depend on traditional assessments. In collaborative classrooms, it is often difficult to assign individual grades. Some teachers give group grades, but many students and parents are uncomfortable with these.

Ideally, assessment practices should be changed so that they are consistent with collaboration, with a new view of learning and with a thinking curriculum. Video Conference 4 addresses recent research and practice on assessment. In the meantime, effective ways have been developed whereby individual students can be evaluated in collaborative classrooms. For example, David Johnson and Roger Johnson, as well as Robert Slavin, advise making individuals responsible for subtasks in group work and then determining both group and individual grades.

Present Scenario of Teacher Education

In a society that is undergoing a great deal of transformation consequent upon the abandonment of antiquated knowledge and skill, Teacher-training institutions cannot just perform the role of preparing teachers of the traditional type. Except for a smattering of Educational Psychology, Philosophy, School administration and Methods of Teaching

a trainee hardly gains functional knowledge, flexible teaching strategy and innovative teaching in most Training Colleges. A trained teacher is in no way superior to an untrained, raw graduate in terms of practising innovative methods of teaching. Since the Department of Education in every State stipulates Teacher Training as an essential pre-requisite there is a gate-crashing rush in all Teacher-training Colleges to seek entry and complete the course rather ritualistically. The sole motive of the vast majority of teachers is mainly to tag on a B.Ed. degree rather than acquiring a new insight into teaching. It would not be an exaggeration to say that under these circumstances, Teacher-training degenerates into a kind of ritual, devoid of any excitement. In view of the changes in high-school curriculum initiated by the National Council of Educational Research and Training (NCERT) and the Central Board of Secondary Education (CBSE), one has to take a second look at the Teacher-preparation level.

Stephen Corey and Action-Research

In the early Seventies an American Educationist by name *Stephen M. Corey* visited India and disseminated the concept of *Action-Research*. Action-research represents the implementation of *John Dewey's* idea of harnessing classroom teaching in solving particular problems.

Stephen Corey attempted to integrate Research and Classroom teaching. Usually researches are confined to Post-graduate Department of Education in Universities and seldom percolate to the level of high-school teachers.

Being fully aware of the limitations of an average teacher in understanding sophisticated Research-methodology, he introduced a new system called *Action-Research*.

Many of the problems facing a teacher require immediate attention. He cannot look for remedy or solution in Research journals because these may not be easily accessible or he might lack the necessary knowledge in interpreting the findings. Such on-the-spot research aimed at the solution of an immediate problem is known as *Action-Research*.

Conventional Research and Action-Research

How does an Action-research differ from Conventional research? Usually, the term 'Research' connotes a rigorous study of a clearly defined problem under carefully controlled conditions on a representative sample of the population with due care bestowed upon Research-design and statistical analysis of data. The findings of a Research-study extend far beyond the confines of the sample and help the investigator to arrive at broad generalisations. A good deal of time, resources and expertise are needed to carry out such a research-project.

An action-research, on the other hand, is less rigorous and more adapted to the immediate needs of a classroom teacher. Quite often a teacher gets dissatisfied with his own teaching techniques and wants to do better. A teacher habituated to a stereotyped

way of doing things cannot improve. Constant revision and modification are indispensable.

Significance of Educational Psychology to the Teacher

Over the years, educational psychology has been a part of teacher preparation, moving from a centerpiece in many programs, through periods when it was deemed irrelevant by some, to current concerns about its role in the reforming of teacher education and teaching. Today, psychological knowledge is used to ground reforms in teaching and schooling, particularly the call for teaching for understanding. Current standards for teacher certifications and licensure and suggestions for reform in teacher education assume that teachers will have a deep and generative understanding of learning, development, motivation, and individual differences. This article explores several themes that recur in the writing on reforms and teacher standards: the need to place learning at the centre of teaching, the call for integrated studies, and the value of collaboration with the public schools. These themes have both positive and negative implications for the role of educational psychology in teacher education.

For years, I have been interested in the role of educational psychology in teacher education. Discussions with colleagues across the country suggest that many share this concern. The overarching goal of this article is to examine the past and present status of educational psychology in the preparation of teachers. I focus on three themes that recur in the writing on teacher standards and reforms—themes that have implications for *what* about educational psychology teachers should learn and *how* that learning should take place. The themes are: (a) the centrality of student learning in teaching, (b) the call for integrated studies, and (c) the value of collaboration with the public schools.

This examination begins with a brief glance at the history of educational psychology in teacher education, then moves to consider how the content and focus of educational psychology have changed over the years as both teacher education faculty and students demanded greater relevance, application, and utility from the study of educational psychology.

Next, I look to the future by analysing selected perspectives on standards and reforms in teaching and teacher preparation—most written by educators and policymakers outside educational psychology. What messages do these views of the future hold for our field? In many of these reform agendas are affirmations of the importance for teaching and teacher education of topics historically studied by educational psychologists, such as learning, development, motivation, and individual differences. Although few teacher educators or policymakers over the years have questioned the value of psychological knowledge, there is less agreement about what exactly should be learned or how the learning should take place.

The third part of this article focuses on the three reform themes listed earlier. The first theme, student learning as the centre of teaching, has implications for what educational psychology should be taught. The last two themes, integrated studies and

collaboration with the schools, are related to how the learning of educational psychology might take place. I suggest a few opportunities and dangers in embracing these reforms and then conclude by describing challenges for educational psychology in teacher education.

Educational Psychology for Teachers: Reforming Our Courses, Rethinking Our Roles

The Educational Psychology Division (Division 15) of the American Psychological Association created a committee to examine the teaching of educational psychology in the broader context of teacher education reform. The committee recommends that educational psychologists who participate in teacher education should help prospective teachers develop contemporary psychological perspectives.

To select content and pedagogy that support this goal, educational psychology teachers should consider both recent conceptualizations of teaching and teachers' knowledge as complex phenomena and current assumptions about complex learning. Educational psychologists should also think about relationships with other teacher educators, the preparation and support of educational psychology teachers, and research and development about teaching educational psychology.

Becoming a K-12 teacher usually involves enrolment in a college or university program that culminates in teacher certification. Within the majority of these programs, there is a formal requirement for coursework about ideas that educational psychologists claim as their domain, such as learning and development, individual differences, instruction, motivation, and measurement. Such courses historically have been offered early in students' programs, and sometimes are considered (and even named) "foundations" for later study of teaching.

Educational psychology foundations courses were developed in the early part of this century to help upgrade the status of teacher education—to improve the scholarship and rigor of professional programs by posing a scientific basis for further study (L. Shulman, 1990). However, criticisms of the Requests for reprints should be sent to Linda Anderson, 116G Erickson Hall, Michigan State University, East Lansing, MI 48824, foundational metaphor and courses based upon it are now being raised by educational psychologists and other teacher educators (Doyle, 1990 a). Similar criticisms of foundational approaches to professional education are also heard in other fields.

Online Experience

While the vast majority of our survey respondents had been active in posting course resources, syllabi, or personal information on the Web, not all had previous experience in Web-based instruction. Nearly a quarter of the respondents had never taught even a portion of a course online. On the other hand, nearly 4 in 10 respondents had taught courses partially online; among this group, the average number of partially online courses

taught was about four. Another 18 per cent had experience teaching fully online courses, with an average of five such courses. In addition, 19 per cent had done both—partial and completely online courses—with an average of 10 such online course experiences. Calculations across these responses indicated that nearly 4 in 10 early Web adopters had taught completely online courses, while nearly 6 in 10 had taught at least part of a course online. Given these data, the respondents in this study certainly had extensive online teaching experiences on which to base their survey answers.

Respondent's Web-Related Skills

An instructor's degree of comfort in using different Web technologies has a direct bearing on classroom practices as well as the decision to teach even part of a course online. When instructors are hesitant or lacking in confidence, there is less likelihood for innovation and risk taking.

Therefore, we asked these early Web adopters about their degree of comfort with the following Web skills:

- Creating HTML pages,
- Hosting an online chat,
- Sending and receiving file attachments,
- Using Web-based course ware systems, and
- Moderating a Web-based asynchronous discussion.

The responses were interesting. For instance, over 90 per cent of these faculty members felt a high degree of comfort sending and receiving file attachments in e-mail. Fewer than one per cent of respondents were uncomfortable with this skill. Somewhat surprisingly, 62 per cent were highly comfortable with creating HTML pages and another 20 per cent had a medium level of comfort.

However, this acknowledged degree of comfort likely includes a range of skills from using standard software options such as “save as HTML” to actually being facile with HTML and other programming code. The degree of expertise with HTML remains a question for future surveys. These early adopting faculty were somewhat less comfortable moderating a Web-based asynchronous discussion forum or bulletin board. Still, nearly 50 per cent rated their degree of comfort as high, while another quarter of them reported a medium level of comfort. Similarly, 44 per cent were highly comfortable with Web-based course ware systems and another 34 per cent felt moderately comfortable.

On the low end was comfort with hosting an online chat session. Perceptions of online chat tools were roughly split across low, medium, and high comfort categories. These results indicate that these faculty members possessed at least some basic technology skills. Perhaps, as the NEA survey of traditional and distance learning higher education members revealed, workshops and training sessions on teaching via distance learning are now readily available. While such a skill base and comfort level may be expected of these early Web adopters, many of these faculty members are either taking

advantage of university training and support or are engaged in a heavy amount of self-teaching in regard to Web-based teaching tools. Or perhaps they are overstating their skills. In fact, latter parts of this report reveal a somewhat different picture.

Time Commitments

In terms of overall time investment, these college instructors almost unilaterally agreed that teaching online is more time-consuming than traditional classroom-based instruction. More than 4 in 5 faculty agreed that teaching online courses requires more time than traditional courses. Fewer than 10 per cent disagreed with that statement. Once again, this is consistent with the NEA report finding that more than half of college faculty teaching via distance learning spent more time on their online courses than their traditional ones regardless of the number of students or times they had previously taught the course. Such findings point to a need for greater course support and incentives that could ease time pressures felt by instructors involved in online teaching.

Attrition

Some reports and media releases contend that students are more likely to drop online courses than traditional ones. Those utilizing a mixed mode or blended approach—traditional and online in the same course—were less likely to experience significant student attrition than those teaching completely online courses. In fact, only 29 per cent of those utilizing a blended approach experienced more than 10 per cent drop the their courses, whereas 44 per cent of those teaching completely online courses had more than 10 per cent drop their course.

Perhaps more strikingly, only 2 per cent of blended courses experienced more than a 50 per cent attrition rate compared to 10 per cent of the completely online courses with such huge attrition rates.

Internet Access

Computer access does not appear to be a problem for these early adopters of Web technologies. Seventy-eight per cent of these college instructors had Internet access in their current or most recent classroom. Computer lab accessibility was even higher with 93 per cent indicating that they had access to an Internet-connected computer lab for class use.

Even more, 97 per cent, had Web access from home. This is more than double the 47 per cent of Americans who are users of the Internet at home as reported in a recent UCLA study. Such high level of technology access is not too surprising given that the majority of the respondents were early Web adopters who had a high level of education. In effect, these findings indicate that access to computers and Internet resources is no longer an obstacle for many college faculty.

Experiential Approach to Teaching Psychology

A distinction is sometimes made between academic and experiential approaches to teaching and learning. Whereas the former emphasises intellectual skills, the latter, emphasises learning by doing. A psychological education approach to teaching psychology at the secondary school or undergraduate level would incorporate experiential teaching of life skills and/or the skills of applied psychological practice as well as intellectual teaching of academic skills. In fact, Sprinthall argues that including an experiential emphasis may enhance academic results.

A Range of Outreach Activities

Sometimes psychological education is not only a form of outreach activity, but an overall term for the range of outreach activities.

Such activities include life skills training and the training of paraprofessionals in counselling skills in settings, such as living and working environments, outside the counsellor's office. Additional outreach activities may include consultancy to an institution or community, or to groups within it, and also providing individuals with psychological self-help information through a 'phone-in' service.

Educating the Public in Psychological Matters

Halmos observed that the counselling literature, ever since the time of Sigmund Freud, has had a far-reaching influence both on the cultural and moral leadership in society and on its rank and file. Sometimes this influence has been indirect.

On other occasions, for example in relation to the women's movement and to a greater understanding of the homosexual and bisexual experience, counsellors have joined in direct attempts at public education, including, where necessary, lobbying to alter discriminatory legislation.

In conclusion, though there is no single definition of psychological education, there are some recurrent themes which go some way to distinguishing it.

Such themes include: providing psychological services for all; no assumption of severe disturbance among the receivers of psychological assistance; a strong developmental and preventive rather than just a remedial emphasis; providing services outside as well as inside traditional counselling settings; training in skills as a preferred mode of treatment; the extension of the provision of psychological services to include the use of paraprofessionals and peer self-help; and the use of both small-group and large-group methods of training.

Humanistic Education

Humanistic education focuses on the whole pupil or student and sees the objective of both counselling and education as the production of self-actualising people. The role

of teachers is that of facilitating learning. As well as having subject matter knowledge, they need to possess an empathic attitude and be able to offer their pupils or students the person-centred 'core' interpersonal conditions.

Training Paraprofessionals in Counselling Skills

Counsellors can extend their effectiveness by training paraprofessionals and lay people in counselling skills. There are numerous instances in which paraprofessionals have been so trained. For instance, Anthony and Carkhuff have trained health care personnel in human relations skills, and Thorne and Da Costa have extended their Associates Programme at the University of East Anglia to include the training of selected Associates in the skills of personal counselling. A variant of paraprofessional training is that sometimes peers are trained to counsel each other, whether they are students in educational settings, women in the women's movement, or members of racial minorities. All agencies involved in training and using voluntary counsellors might be viewed as engaging in psychological education as defined in this section.

Some Examples of Psychological Education

Psychological education programmes vary in several ways. For instance, programmes may use a behavioural skills acquisition model, or may be based on humanistic assumptions, or may be a mixture of the two. The focus may be on one or more of a number of life skills or on bringing about broader changes, such as in community racial attitudes. The targets or clientele may be school children, young adults, parents, women, coloured people, the homosexual, bisexual and heterosexual, an institution or community, or the public at large. The personnel conducting the psychological education may be psychologists, counsellors, teachers, or peers who either have been or are in the same situation as the learners. Its various settings include education, industry, medicine, the social services, marriage guidance councils and other community settings. We now turn to some examples of psychological education, but the categories under which the programmes are discussed are not discrete. For instance, we discuss separately programmes in the schools and programmes targeted at women, yet the schools are one of the settings for women's programmes.

RELEVANCE OF EDUCATIONAL PSYCHOLOGY FOR SECONDARY TEACHERS

Educational psychology has contributed considerably to the creation of the modern system of education.

The knowledge of educational psychology helps the teacher in the following ways:

- *To understand the Stages of Development:* Psychology has clearly shown that human life passes through different stages of development before it reaches

adulthood. They are infancy, childhood, adolescence and adulthood. Psychologists have also thoroughly studied the characteristic behaviour patterns in these different periods of life. Identification of these periods with different sets of characteristics and attributes as regards physical, mental and emotional development greatly help educationists to design curriculum and determine appropriate methods of teaching for students at different stages.

- *To Know the Learner:* The child or the learner is the key factor in the teaching-learning process. Educational psychology helps the teacher to know his interests, attitudes, aptitudes and the other acquired or innate capacities and abilities; to know the stage of development linked with his social, emotional, intellectual, physical and aesthetic needs; to know his level of aspiration, his conscious and unconscious behaviour; his motivational and group behaviour; his conflicts, desires and other aspects of his mental health. So that perfect guidance and help can be provided and positive attitude towards the learner can be formed.
- *To Understand the Nature of Classroom Learning:* Educational Psychology helps the teacher to adapt and adjust his teaching according to the level of the learners. A teacher is teaching in a class but a large number of students do not understand the subject-matter which is being taught. To deal with the students effectively in the class the teacher must have the knowledge of the various approaches to the learning process, principles, laws and factors affecting it then only he/she can apply remedial measures in the learning situation.
- *To Understand the Individual Differences:* No two persons are exactly alike. Pupils differ in their level of intelligence, aptitudes, likes and dislikes and in other propensities and potentialities. There are gifted, backward, physically and mentally challenged children. Thus, psychology tells the teacher about the individual differences among the students in the class and the procedure, methodology and techniques to be adopted for them.
- *To Solve Classroom Problems:* There are innumerable problems like truancy, bullying, peer pressure, ethnic tensions, cheating in tests etc. Educational Psychology helps to equip the teacher by studying the characteristics of the problem children, the dynamics of the group, behavioural characteristics and adjustments.
- *To develop Necessary Skills and Interest in Teaching:* Educational psychology helps the teacher to acquire and develop necessary qualities and skills to deal with the problems created by the pupils, maintain a healthy atmosphere in the classroom and show concern regarding the progress of the child.
- *To Understand Effective Methods of Teaching:* Educational Psychology has discovered several new approaches, principles, methods and techniques of teaching which are very helpful in today's teaching-learning process.

Educational psychology tells us how significant play and recreation are for the children and how play-way methods turn learning into an interesting task.

- *To Understand the Influence of Heredity and Environment on the Child:* Educational psychology helps the teacher to know that the child is the product of heredity and environment. They are the two sides of a coin. Both play a prominent part in the all-round development of the child. While the child is born with a number of hereditary qualities, environment helps them to be modified according to the requirements of the society.
- *To Understand the Mental Health of the Child:* Educational Psychology helps the teacher to know what are the factors responsible for the mental ill-health and maladjustment of a student and to suggest improvement thereof. Besides this, it also provides the teacher with necessary insight to improve his own mental status to cope up with the situation.
- *To Understand the Procedure of Curriculum Construction:* Curriculum is an integral part of the teaching-learning process. Curriculum should be child-centred and fulfil the motives and psychological needs of the individual because child capacities differ from stage to stage. Educational psychology helps the teacher to suggest ways and means to curriculum framers to prepare sound and balanced curriculum for the children.
- *To Provide Guidance and Counselling:* Today guidance to a child at every stage of life is needed because psychological abilities, interests and learning styles differ from person to person. Similarly, what courses of study the child should undertake in future is also a vital question. All these can be answered well if the teacher knows the psychology of children.
- *To Understand Principles of Evaluation and Assessment:* Evaluation is an integral part of the teaching-learning process. How to test the potentialities of the child depends upon the evaluation techniques. The development of the different types of psychological tests for the evaluation of the individual is a distinct contribution of educational psychology.
- *To inculcate Positive and Creative Discipline:* The slogan of the traditional teachers was “spare the rod and spoil the child.” Flogging the child was the chief instrument. Educational Psychology has replaced the repressive system with the preventive system. Now teachers adopt a cooperative and scientific approach to modify the behaviour of the students. Emphasis is laid on self-discipline through creative and constructive activities.
- *Educational Psychology and Research:* Educational psychologists conduct research to improve the behaviour of human beings in the educational situation. For this purpose it helps in developing tools and devices to measure the performance and suggest remedial measures thereof.
- *To Know Himself/Herself:* Educational Psychology helps the teacher to know about himself/herself. His/her own behaviour pattern, personality

characteristics, likes and dislikes, motivation, anxiety, conflicts, adjustment etc. All this knowledge helps him in growing as a successful teacher.

- *Educational Psychology Helps in Professional Growth, Changing Attitude and Innovative Thinking:* Inside the classroom, educational psychology has enabled the teacher to achieve proper conditioning of pupils by achieving and directing classroom programmes on human lives. Not only this, educational psychologists are busy in finding out innovations in the field of education. These innovations will bring about professional growth of the teacher.

THE PRACTICAL VALUE OF PSYCHOLOGY TO THE TEACHER

It would be an interesting task to trace the history of psychology from its earliest crude and haphazard beginnings to its present state of advancement with its wide range of enquiries and interests, its struggles to attain exact scientific results, its efforts to employ experimental methods, its laboratories, its failures, its achievements. I have not to speak, however, of what psychology has been or is, but assuming that to be sufficiently familiar to you, I must attempt briefly to point out some of its applications to the great and noble art of teaching. We may consider the importance of psychology to the teacher in (1) the discovery of the inter-relations of different lines of study, (2) in organizing and systematizing his own mental life, (3) in guiding the process of bringing together the subject of study and the subject who studies, *i.e.*, in helping the teacher as (*a*), director; (*b*), student; (*c*), educator.

THE TEACHER AS DIRECTOR

The teacher must know something about the inter-relations of different studies. He has to arrange the time-table, and frequently to teach several of the subjects. Even where, he is restricted to the teaching of some speciality he should know how his special subject is related to the others pursued by the pupils he is teaching. Does psychology occupy such a place as to make it specially valuable in seeing the inter-relations of various studies? Let us examine. Wundt divides studies into three great classes, (*a*), the natural sciences; (*b*), the mental sciences; (*c*), the philosophical enquiries. He claims that psychology is complementary to (*a*), the natural sciences, assisting in the treatment of problems otherwise inadequately solved; is the foundation of (*b*), the mental sciences, as dealing with the simple data and underlying principles of all mental sciences, and lastly it is the natural preparation for and introduction to (*c*), the philosophical enquiries.

That psychology is complementary to the natural sciences may be illustrated by a number of commonplace and well-known instances as the case of the “personal equation” in astronomy, where it becomes necessary to account for the apperception and reaction times of the observer, who is using the transit instrument to prevent mistakes. Familiar

examples illustrate that the abstracted, mathematical and physical properties of the observed phenomenon do not alone explain the appearances, *e.g.*, the larger apparent size of the moon when near the horizon; the apparent motion of the sun. Other simple illustrations might be taken from the optical illusions arising when what is termed “pencils” of lines are drawn from a point between two parallel lines, cutting the parallel lines in various directions, make the parallel lines seem to curve outward; while lines drawn from points outside the parallel lines and terminating in an imaginary line midway between the parallel lines, make the parallel lines appear to curve inward, etc.

Cases of colour contrast afford other illustrations. A continuous strip of gray on contiguous surfaces of black and white appear darker on the white and lighter on the black background; the same gray placed on backgrounds of red and of green appear greenish on the red and reddish on the green background.

The British Scientific Association places psychology among the natural sciences in its meetings by making it a sub-section of physiology. The American Scientific Association places it under the second group of mental sciences by making it a subsection of anthropology. It belongs to both places.

Only a slight examination is required to see that for the mental sciences psychology is just as fundamental and underlying as mathematics is for the natural sciences. Note any recent advance in these and you will find it resting on insight into and appreciation of some psychological principle. Look at the new methods of teaching grammar, not *before*, but *through* the language to which it belongs. Look at the complete revolution in method in the manner of teaching and using *rules*, once first, now last in the process, once announced and memorized, now discovered and constructed by the pupil himself.

Look at the improvement in history in such works as Green’s *Short History of the English People*; going beneath the events to the life of the people, their aims and passions, and the analysis of the character and motives of the chief actors. Look at the improvement in political economy by the introduction of psychological and ethical considerations. What may we expect in law when some of the time spent on procedure in criminal law is applied to the study of the *criminal* himself?

As to the value of psychology as an introduction to the philosophical enquiries, an objection might be raised that all of them, philosophy, aesthetics and theology, claiming to deal with the true, the beautiful and the good as ideals, are ultimately based on metaphysics, and the less we have to do with metaphysics the better.

Modern philosophy, however, should not be confounded with the much-misunderstood and much-maligned medieval disputations any more than modern chemistry with alchemy, or modern biology and medical science with the views of Theophrastus Bombastus Paracelsus. And even the superseded past should be remembered with some gratitude and respect as the progenitor of the present. “Honour thy father and thy mother.” Those who cry out most loudly against metaphysics, past or present, are in almost every case the unconscious victims of the shallowest and most erroneous forms of metaphysical speculation.

It is philosophical speculation carefully conducted which has done most to expose false principles and to amend crude and erroneous standpoints. If we mean by philosophy, reflection on the meaning of experience, reconsideration of the significance of the results gained in scientific investigations, then, instead of saying no one should have anything to do with philosophy, we should rather say everyone should have something to do with philosophy.

Everyone who reflects on the meaning of life and its experiences, who desires to pass beyond the mere appearances and discover their worth and importance for life, conduct and destiny is to that extent a philosopher.

It is necessary to specialize in science to gain results. But every scientist in every field has not only the privilege but also the duty to give more than mere details connected with his speciality. He should endeavour to give hints concerning their ultimate meaning as this is revealed to him. At any rate, the teacher cannot be a mere pedant. He must be a man as well as a scholar, and he will give a respectful hearing to such investigations and cultivate an intelligent interest in them. For this, psychology is a useful introduction and preparation. May we not conclude that psychology stands in such a central position and in such intimate connection with every branch of enquiry that it is peculiarly fitted to assist in their co-ordination ?

THE TEACHER AS STUDENT

It is scarcely necessary to say anything about the importance of continual study to the teacher. He must keep alive his interest in what he is teaching by continually enriching his mind by new enquiries and acquisitions.

Our studies should be organized. Each new discovery should be made to throw light upon everything we already know.

By reflectively, actively organizing in this way the mind gains strength and insight, keeps alive its old interests and creates new ones. Thus study is made delightful and fruitful, thought is trained to become consecutive and successful. The teacher should himself be a thinker of this type and he should have psychological insight to enable him to guide his pupils to attain such an intellectual culture.

THE TEACHER AS EDUCATOR

What the teacher acquires and gains in his own self-culture is, as teacher, a means; the end sought by him is the training of pupils. He must stimulate and awaken interest. He desires to make the subject of study a means to transform the whole character of the subject who studies. In order to accomplish this, the teacher must keep in mind the logical order of a correct presentation of the subject of study; the stage of development and powers of his pupil and the laws of his mental growth; that he may gain the result, the developed pupil. In order of presentation, he must proceed from the simpler to the more complex; and the simpler is not the most abstract but the most concrete, for he

must also proceed from the known to the less known. He must arrange the presentation so that a puzzle or problem is proposed and suggested to the pupil, add his curiosity aroused to endeavour to solve it.

The teacher must sympathetically place himself at the pupil's standpoint, if he desires the pupil, to advance to his point of view. In order to do this, he should endeavour to recall the stages and processes whereby he as pupil proceeded, when he was at the stage now occupied by his pupil. The ability to do this, probably accounts for the fact that, in many cases an English-speaking teacher will be more successful in teaching pupils the rudiments of a foreign language than a native. It may also account for the fact that so large a proportion of young and inexperienced teachers succeed as well as they do.

The most important service of psychology to the teacher, is that it leads him to consciously and systematically study his pupils and thus awakens or intensifies his interest in them. Surely, if a doctor becomes interested in the discovery of new diseases and new remedies for them, a teacher should be interested in each new pupil and in each experiment for his improvement.

An individualized interest makes a teacher as careful of his pupils as a fond mother is of her children. He is on the alert to see that the physical well-being of the child is not neglected. Has the child had habits of sitting, or studying, or walking, or breathing? He discovers the cause and endeavours to correct kindly, wisely, and at once. Proper physical habits conduct to health and morality.

Is the child untidy or unmannerly? The teacher leads him by example and considerate advice. The child is respected and is taught to respect himself., Is the child dull and stupid? The teacher endeavours to find out if ill-health, or poor food, or ill-usage at home, is the cause; he encourages the child to play, and soon it will turn out that the teacher is found visiting the home and endeavouring to arouse parental solicitude and gain parental co-operation. This teacher will not neglect lighting, heating, or ventilation; he will be careful not to unduly fatigue his pupils, and will be found supervising their plays without officious interference. He will even be found guarding the out-houses and walls from the desecration of perverted vandalism. He will be the guide, counsellor and confidential friend of the adolescent pupils; guarding them with solicitude and watchfulness in this critical period of unstable equilibrium, when the nature is plastic and responsive to the promptings of the highest ideals; and when, on the other hand, the danger is so great of the beginnings of perverted habits and criminal tendencies arising, if the pupils are neglected, and allowed simply to "grow up" like Topsy or Ruth Bonnython.

Let us now recall some examples of assistance from psychology, in arrangement of time table and presentation of the subject of study.

The thoughtful teacher will distinguish between the more severely logical and mathematical subjects, and the historical and literary. For the former, more concentrated attention is required, and therefore, these should be placed in the early part of the programme.

When it comes to reviewing, it will turn out that the second class of studies requires more repetition and reviewing. Pupils should, however, be taught to recall directly what they have previously read and studied, without using the book to assist them. The memory should be trained in self-reliance. Perhaps it is in connection with memory that most people would think of the assistance of psychology to the student.

Kant says memory may be mechanical, ingenious, or judicious. I think it must be confessed that the earliest attempts to apply psychology in assisting and directing memory training, were chiefly of the “ingenious” kind. discovering curious and arbitrary connections in accordance with the law of the association of ideas, through similarity, contrast and contiguity.

Many text-books seem to be constructed with the view of employing the “mechanical” memory. It is supposed that the briefer the summary, the easier it will be to learn and remember. The student is supposed to con[sic] over the tables and learn them by sheer repetition. A deeper insight will indicate more “judicious” methods. The great rule for memory is “take care of knowing and recollecting will take care of itself.” Let the subject be taught and studied logically, systematically, thoroughly, and woven as widely as possible into the warp and woof of the mental interests and thoughts of the pupil. In this way the time spent in one subject is not taken from all others, but is contributing to all others. It is a popular fallacy to suppose that all the time spent in one subject is subtracted away from every other.

The trained and experienced teacher educates all the powers of his pupils, and utilizes every subject for this purpose. He keeps clearly before his view the result to be attained, carefully selects the most efficient means, and with solicitude and interest observes and directs the process. He desires the full and harmonious development of *all* the powers and capabilities of the pupil, physical, mental, social, moral and religious. He is aware that he is co-operating with the pupil in the formation of character. Is there anything of higher value? This thought makes the teacher reverent, it impresses him with a sense of his responsibility; it also enables him to respect his profession and see in it one of the noblest efforts of human endeavour. Although our Public Schools are sometimes accused of giving merely intellectual drill, no teacher worthy of the name is limiting his efforts to this. He is bending every energy to attain discipline and training of character, by means of the intellectual and the disciplinary; he strives to inculcate ideals and form habits of faithfulness, honesty, uprightness, industry, truthfulness, obedience, reverence/Mark, he is not teaching *definitions* of these, that would be a “merely intellectual drill.” He is moulding the character to these moral habits. It is just because the Public Schools are so efficient that Sunday School and home continually desire to relegate more and more to the Public Schools. The careful and reverent study of the child is destined to react upon home, Sunday School and Church. If child-nature had been studied should we find the text “Except ye become as little children ye cannot enter the Kingdom of Heaven,” so continually misinterpreted to mean that there should be passive admission of truth without questioning or enquiry? Is that the way the child learns or acts? Should

not our religious life exhibit the same fearless confidence in asking questions and the same readiness in putting into practice the answers that the active child displays? It would be a wide field to follow the pernicious effects of un-psychological methods of parents and teachers in the suppressing of questions, and stifling the religious cravings of children. We have too often “offended these little ones.”

Sooner or later truer psychological methods, as exemplified in the Kindergarten, will permeate the whole school system and overflow into the Sunday School, the Church and the home. Let me add to the teacher interested in the study of psychology and its applications to his profession:— Remember that the Science of Psychology, with all its intrinsic importance and immediate usefulness, is simply the portal and propaedeutic to the higher reflective problems of the ultimate significance of life, and, art, moral conduct, and religious aspiration. As in your teaching you desire the intellectual to be the means to lift up the pupil to higher ground, prepare him for the reception of the highest truths, so let these lofty themes be in your own life constant topics of interest, perennial sources of new insight, continual fountains of noblest inspiration.

Teacher Motivation and Performance

There are several factors that significantly impact on teacher performance and motivation. These include the actual amount of time the teacher spends in class teaching, the proportion of time spent in maintaining order and doing other administrative tasks, the pedagogic techniques used by them in the classroom and disciplinary policies adopted by the teacher and the school.

A number of official policies and programme documents over the years have reiterated the significance of teachers in the education of children, including the Education Commission Report, the National Policy on Education 1986, and the National Policy on Education: Programme of Action 1992.

A school without adequate numbers of teachers is non-functional, but a school which has an adequate number of teachers who are neither empowered nor professionally qualified nor committed to the basic ideals of their profession, with low morale and self-esteem, are likely to do more damage than good to students. Sub-optimal use of teachers is another serious concern in most states. Generally, teacher absenteeism is high in India. Even if teachers attend schools, their presence in classrooms is often low. This happens at least partially because of their engagement in non-teaching activities. A recent study on teacher absence in 20 states, nearly one quarter of teachers were absent from government primary schools on a typical school day. Absence rates ranged from around 15% in Maharashtra to 38% in Bihar and 42% in Jharkhand.

The absence rate was also more than 30% in Chhattisgarh, Uttarakhand, Assam, Punjab, Bihar and Jharkhand. One factor determining the motivation and performance of teachers is the status they enjoy in society. There is a general perception that in, recent years, the teaching profession has experienced a lowering of its status. Batra suggests that teaching has declined to a least favoured profession status. A recent study

reveals that teaching is considered as more of a 'safe fall back option' than a formal career choice. Further, the social distance between teachers and students is one reason why many teachers have limited commitment to the educational advancement of their pupils, as well as have limited understanding of their problems. Research indicates that the attitudes of teachers can be particularly discriminatory in the case of working children. Ramachandran et al points out in a study of working children that both children and parents see teachers treating poor children differently; they do not appreciate the predicament of children who have to work before and after school hours.

This not only has a significant negative impact on the participation of those children in schooling, but also creates negative attitudes towards schooling for those who continue with their education. There is, in fact, a discernible hierarchical division of schools in India with selected urban schools with good infrastructural facilities and teaching staff catering to rich students, while schools with poor facilities and inadequate numbers of teachers tend to cater to the poor and deprived sections of society. Educational provision has in recent years become much more diverse owing to the rapid growth of private sector provision. There is now a greater range of choice for parents who can afford to pay for their children's education, while government schools are usually the only option for those from lower economic strata who cannot exercise such choice.

As a result, many teachers working in state-run schools perceive themselves to be in a 'deficit' situation. A typical profile of a government school teacher tends to be of someone who is poorly educated, with little or no teacher education and training, limited or almost no contact with books, technology and science, and increasingly recruited from among low income backgrounds. The majority of untrained or under-trained teachers work in rural areas, where the poorest and most deprived schools are situated.

There is also a growing criticism of public schools which coincides with a decreased confidence in teachers, when in reality teachers are often not given the conditions to ensure teaching and even less learning. Teachers, who are themselves products of the poor education system, are often seen as obstacles to educational change rather than as key human resources. Consequently, teachers tend to respond defensively by overemphasizing problems outside their direct influence.

Pedagogy and Teaching

In education, teachers facilitate student learning, often in a school or academy or perhaps in another environment such as outdoors. A teacher who teaches on an individual basis may be described as a tutor. The objective is typically accomplished through either an informal or formal approach to learning, including a course of study and lesson plan that teaches skills, knowledge and/or thinking skills. Different ways to teach are often referred to as pedagogy. When deciding what teaching method to use teachers consider students' background knowledge, environment, and their learning goals as well as standardized curricula as determined by the relevant authority. Many times, teachers assist in learning outside of the classroom by accompanying students on

field trips. The increasing use of technology, specifically the rise of the internet over the past decade, has begun to shape the way teachers approach their roles in the classroom. The objective is typically a course of study, lesson plan, or a practical skill. A teacher may follow standardized curricula as determined by the relevant authority.

The teacher may interact with students of different ages, from infants to adults, students with different abilities and students with learning disabilities. Teaching using pedagogy also involve assessing the educational levels of the students on particular skills. Understanding the pedagogy of the students in a classroom involves using differentiated instruction as well as supervision to meet the needs of all students in the classroom. Pedagogy can be thought of in two manners. First, teaching itself can be taught in many different ways, hence, using a pedagogy of teaching styles. Second, the pedagogy of the learners comes into play when a teacher assesses the pedagogic diversity of his/her students and differentiates for the individual students accordingly. Perhaps the most significant difference between primary school and secondary school teaching is the relationship between teachers and children.

In primary schools each class has a teacher who stays with them for most of the week and will teach them the whole curriculum. In secondary schools they will be taught by different subject specialists each session during the week and may have 10 or more different teachers. The relationship between children and their teachers tends to be closer in the primary school where they act as form tutor, specialist teacher and surrogate parent during the course of the day. This is true throughout most of the United States as well. However, alternative approaches for primary education do exist. One of these, sometimes referred to as a “platoon” system, involves placing a group of students together in one class that moves from one specialist to another for every subject. The advantage here is that students learn from teachers who specialize in one subject and who tend to be more knowledgeable in that one area than a teacher who teaches many subjects. Students still derive a strong sense of security by staying with the same group of peers for all classes. Co-teaching has also become a new trend amongst educational institutions. Co-teaching is defined as two or more teachers working harmoniously to fulfill the needs of every student in the classroom. Co-teaching focuses the student on learning by providing a social networking support that allows them to reach their full cognitive potential. Co-teachers work in sync with one another to create a climate of learning.

Rights to Enforce School Discipline

Throughout the history of education the most common form of school discipline was corporal punishment. While a child was in school, a teacher was expected to act as a substitute parent, with all the normal forms of parental discipline open to them. In past times, corporal punishment was one of the most common forms of school discipline throughout much of the world. Most Western countries, and some others, have now banned it, but it remains lawful in the United States following a US Supreme Court decision in 1977 which held that paddling did not violate the US Constitution. 30 US

states have banned corporal punishment, the others have not. It is still used to a significant degree in some public schools in Alabama, Arkansas, Georgia, Louisiana, Mississippi, Oklahoma, Tennessee and Texas.

Private schools in these and most other states may also use it. Corporal punishment in American schools is administered to the seat of the student's trousers or skirt with a specially made wooden paddle. This often used to take place in the classroom or hallway, but nowadays the punishment is usually given privately in the principal's office. Official corporal punishment, often by caning, remains commonplace in schools in some Asian, African and Caribbean countries. Currently detention is one of the most common punishments in schools in the United States, the UK, Ireland, Singapore and other countries. It requires the pupil to remain in school at a given time in the school day; or even to attend school on a non-school day, *e.g.*, "Saturday detention" held at some US schools. During detention, students normally have to sit in a classroom and do work, write lines or a punishment essay, or sit quietly.

A modern example of school discipline in North America and Western Europe relies upon the idea of an assertive teacher who is prepared to impose their will upon a class. Positive reinforcement is balanced with immediate and fair punishment for misbehaviour and firm, clear boundaries define what is appropriate and inappropriate behaviour. Teachers are expected to respect their students, and sarcasm and attempts to humiliate pupils are seen as falling outside of what constitutes reasonable discipline. Whilst this is the consensus viewpoint amongst the majority of academics, some teachers and parents advocate a more assertive and confrontational style of discipline. Such individuals claim that many problems with modern schooling stem from the weakness in school discipline and if teachers exercised firm control over the classroom they would be able to teach more efficiently. This viewpoint is supported by the educational attainment of countries—in East Asia for instance—that combine strict discipline with high standards of education.

It's not clear, however that this stereotypical view reflects the reality of East Asian classrooms or that the educational goals in these countries are commensurable with those in Western countries. In Japan, for example, although average attainment on standardized tests may exceed those in Western countries, classroom discipline and behaviour is highly problematic. Although, officially, schools have extremely rigid codes of behaviour, in practice many teachers find the students unmanageable and do not enforce discipline at all. Where school class sizes are typically 40 to 50 students, maintaining order in the classroom can divert the teacher from instruction, leaving little opportunity for concentration and focus on what is being taught. In response, teachers may concentrate their attention on motivated students, ignoring attention-seeking and disruptive students. The result of this is that motivated students, facing demanding university entrance examinations, receive disproportionate resources, while the rest of the students are allowed, perhaps expected to, fail. Given the emphasis on attainment of university places, administrators and governors may regard this policy as appropriate.

Triangular Basis of Teacher Education

Construction of the relevant knowledge base for each stage of education requires a high degree of academic and intellectual understanding of matter related to teacher education at each stage. This involves selection of theoretical knowledge from disciplines cognate to education, namely, psychology, sociology and philosophy, and converting it into forms suitable for teacher education. Teacher education derives its content from the disciplines of Philosophy, Sociology and Psychology. These disciplines provide the base for better understanding and application of Teacher education.

The Philosophical basis provides insights to the student teachers about the implications of- the various schools of philosophy, ancient and modern philosophical thoughts, educational thoughts of philosophical thinkers on education and its various aspects such as curriculum construction and discipline. The Sociological basis helps the student teachers to understand the role of society and its dynamics in the educational system of a nation and the world at large. It encompasses the ideals that influence national and international scenes. The Psychological basis helps the student teachers develop insights into students' psychological make-up. This enables the student teachers to understand their self, their students and the learning situations such that they are able to provide meaningful and relevant learning experiences to their students.

Aspects of Teacher Education

Teacher education is concerned with the aspects such as:

- Who (Teacher Educator),
- Whom (Student teacher),
- What (Content) and
- How (Teaching Strategy).

Teacher education is dependent upon the quality of teacher educators. The quality of pedagogical inputs in teacher education programmes and their effective utilization for the purpose of preparing prospective teachers depend largely on the professional competence of teacher educators and the ways in which it is utilized for strengthening the teacher education programme.

Teacher education, thus, first deals with the preparation of effective teacher educators. Teacher education reaches out to the student teachers by providing the relevant knowledge, attitude and skills to function effectively in their teaching profession. It serves to equip the student teachers with the conceptual and theoretical framework within which they can understand the intricacies of the profession. It aims at creating the necessary attitude in student teachers towards the stakeholders of the profession, so that they approach the challenges posed by the environment in a very positive manner. It empowers the student teachers with the skills (teaching and soft skills) that would enable them to carry on the functions in the most efficient and effective manner. Teacher education therefore pays attention to its content matter.

THE INTERNATIONAL LEVEL

UNESCO

At no time in human history was the welfare of nations so closely linked to the quality and outreach of their higher education systems and institutions.. As the only United Nations agency with a mandate in higher education, UNESCO facilitates the development of evidence-based policies in response to new trends and developments in this field emphasizing its role in achieving the Millennium Development Goals and particularly poverty eradication.

The Organization fosters innovation to meet education and workforce needs and examines ways of increasing higher education opportunities for young people from vulnerable and disadvantaged groups. It deals with cross-border higher education and quality assurance, with a special focus on mobility and recognition of qualifications, and provides tools to protect students and other stakeholders from low-quality provision of higher education.

UNESCO promotes policy dialogue and contributes to enhancing quality education, strengthening research capacities in higher education institutions, and knowledge sharing across borders.

Teacher Education

- Global leadership on teachers,
- Their status,
- Their professional training,
- Their management and administration and key policy issues.
- The UNESCO/ILO Recommendations concerning the Status of Teachers and provide the framework for the same.
- The Teacher Training Initiative for Sub-Saharan Africa (TTISSA) is a core initiative addressing key issues in the African context.

What UNESCO is doing for Teacher Training- (Role and Functions)

UNESCO promotes the development of a professionally-trained corps of teachers who provided the human contact, understanding and judgment necessary to prepare our children for the world of tomorrow.

UNESCO and Teachers

Good teachers are the cornerstone of quality education. On a daily basis, teachers contribute to sustainable development by building its human foundation – nurturing each child’s capacity and desire to learn. Without teachers, Education for All (EFA) by 2015 would be an unobtainable dream.

Teachers: Creating hope for Tomorrow

- Teachers are at the very heart of UNESCO's work. Each day, over 60 million teachers care for 1 billion children, cultivating their souls and minds. Any process that attempts to improve the quality of education promote peace and harmony and eliminate discrimination requires teachers. Teachers work with children who will be the leaders of tomorrow.
- But for teachers to be effective, they must be well-trained, motivated, have a decent work environment, good pay and an attractive career path. UNESCO enables the world's teachers by building on the standards for the professional, social, ethical and material concerns of teachers set in the 1966 and 1997 recommendation concerning the status of teachers and education personnel.
- There is currently a severe shortage of teachers worldwide. UNESCO helps adjust national policies to reverse teacher flight, teacher drop-out and assists countries with the professionalization of "volunteer" teachers recruited by hard-pressed governments to fill crisis-level gaps.

UNESCO and Teacher Education

- Emphasizing the essential role teacher training and education policy play in national development goals.
- Producing and disseminating policy guidelines on open and distance learning, e-learning, and use of ICTs in teacher education.
- Advocacy to improve the training and status of teachers worldwide.
- Integrating international standards regarding HIV/AIDS and life skills into national teacher education policies.
- Promoting exchange of good national practices and lessons learnt within groups of countries with common teacher-related agendas through networking and exchange.
- UNESCO promotes the development of a professionally-trained corps of teachers who provided the human contact, understanding and judgment necessary to prepare our children for the world of tomorrow.

Understanding Life Skills a Teacher's Prospective

Life skills have been defined as "*the abilities for adaptive and positive behaviour that enable individuals to deal effectively with the demands and challenges of everyday life*" (WHO). 'Adaptive' means that a person is flexible in approach and is able to adjust in different circumstances.

Positive behaviour implies that a person is forward looking and even in difficult situations, can find a ray of hope and opportunities to find solutions. The terms '*Livelihood skills*' or occupational/vocational skills refer to capabilities, resources and opportunities to pursue individual and household economic goals and relate to income generation. Thus, Life skills are distinct from livelihood skills.

Key Life Skills

Life skills include psychosocial competencies and interpersonal skills that help people make informed decisions, solve problems, think critically and creatively, communicate effectively, build healthy relationships, empathize with others, and cope with managing their lives in a healthy and productive manner. Essentially, there are two kinds of skills - those related to thinking termed as “*thinking skills*”; and skills related to dealing with others termed as “*social skills*”. While thinking skills relate to reflection at a personal level, social skills include interpersonal skills and do not necessarily depend on logical thinking.

It is the combination of these two types of skills that are needed for achieving assertive behaviour and negotiating effectively. “*Emotional*” can be perceived as a skill not only in making rational decisions but also in being able to make others agree to one’s point of view. To do that, coming to terms first with oneself is important. Thus, selfmanagement is an important skill including managing/coping with feelings, emotions, stress and resisting peer and family pressure. Young people as advocates need both thinking and social skills for consensus building and advocacy on issues of concern.

The Ten core Life Skills as laid down by WHO are:

- Self-awareness includes recognition of ‘self’, our character, our strengths and weaknesses, desires and dislikes. Developing self-awareness can help us to recognize when we are stressed or feel under pressure. It is often a prerequisite to effective communication and interpersonal relations, as well as for developing empathy with others.
- Empathy - To have a successful relationship with our loved ones and society at large, we need to understand and care about other peoples’ needs, desires and feelings. Empathy is the ability to imagine what life is like for another person. Without empathy, our communication with others will amount to one-way traffic. Worst, we will be acting and behaving according to our self-interest and are bound to run into problems. No man is an island, no woman either! We grow up in relationships with many people – parents, brothers and sisters, cousins, uncles and aunts, classmates, friends and neighbours. When we understand ourselves as well as others, we are better prepared to communicate our needs and desires. We will be more equipped to say what we want people to know, present our thoughts and ideas and tackle delicate issues without offending other people. At the same time, we will be able to elicit support from others, and win their understanding. Empathy can help us to accept others, who may be very different from ourselves. This can improve social interactions, especially, in situations of ethnic or cultural diversity. Empathy can also help to encourage nurturing behaviour towards people in need of care and assistance, or tolerance, as is the case with AIDS sufferers, or people with mental disorders, who may be stigmatized and ostracized by the very people they depend upon for support.

- Critical thinking is an ability to analyze information and experiences in an objective manner. Critical thinking can contribute to health by helping us to recognize and assess the factors that influence attitudes and behaviour, such as values, peer pressure and the media.
- Creative thinking is a novel way of seeing or doing things that is characteristic of four components:
 - Fluency (generating new ideas),
 - Flexibility (shifting perspective easily),
 - Originality (conceiving of something new), and
 - Elaboration (building on other ideas).
- Decision making helps us to deal constructively with decisions about our lives. This can have consequences for health. It can teach people how to actively make decisions about their actions in relation to healthy assessment of different options and, what effects these different decisions are likely to have.
- Problem solving helps us to deal constructively with problems in our lives. Significant problems that are left unresolved can cause mental stress and give rise to accompanying physical strain.
- Interpersonal relationship skills help us to relate in positive ways with the people we interact with. This may mean being able to make and keep friendly relationships, which can be of great importance to our mental and social well-being. It may mean keeping, good relations with family members, which are an important source of social support. It may also mean being able to end relationships constructively.
- Effective communication means that we are able to express ourselves, both verbally and non-verbally, in ways that are appropriate to our cultures and situations. This means being able to express opinions and desires, and also needs and fears. And it may mean being able to ask for advice and help in a time of need.
- Coping with stress means recognizing the sources of stress in our lives, recognizing how this affects us, and acting in ways that help us control our levels of stress, by changing our environment or lifestyle and learning how to relax.
- Coping with emotions means involving recognizing emotions within us and others, being aware of how emotions influence behaviour and being able to respond to emotions appropriately. Intense emotions like anger or sadness can have negative effects on our health if we do not respond appropriately.

Measurement and Evaluation in Psychology and Education

Measurement is the assignment of a number to a characteristic of an object or event, which can be compared with other objects or events. The scope and application of a measurement is dependent on the context and discipline. In the natural sciences and engineering, measurements do not apply to nominal properties of objects or events, which is consistent with the guidelines of the *International vocabulary of metrology* published by the International Bureau of Weights and Measures. However, in other fields such as statistics as well as the social and behavioral sciences, measurements can have multiple levels, which would include nominal, ordinal, interval, and ratio scales. Measurement is a cornerstone of trade, science, technology, and quantitative research in many disciplines. Historically, many measurement systems existed for the varied fields of human existence to facilitate comparisons in these fields. Often these were achieved by local agreements between trading partners or collaborators. Since the 18th century, developments progressed towards unifying, widely accepted standards that resulted in the modern International System of Units (SI). This system reduces all physical measurements to a mathematical combination of seven base units. The science of measurement is pursued in the field of metrology.

METHODOLOGY

The measurement of a property may be categorized by the following criteria: type, magnitude, unit, and uncertainty.

They enable unambiguous comparisons between measurements.

- The *type* or *level* of measurement is a taxonomy for the methodological character of a comparison. For example, two states of a property may be compared by ratio, difference, or ordinal preference. The type is commonly not explicitly expressed, but implicit in the definition of a measurement procedure.
- The *magnitude* is the numerical value of the characterization, usually obtained with a suitably chosen measuring instrument.
- A *unit* assigns a mathematical weighting factor to the magnitude that is derived as a ratio to the property of an artifact used as standard or a natural physical quantity.
- An *uncertainty* represents the random and systemic errors of the measurement procedure; it indicates a confidence level in the measurement. Errors are evaluated by methodically repeating measurements and considering the accuracy and precision of the measuring instrument.

STANDARDIZATION OF MEASUREMENT UNITS

Measurements most commonly use the International System of Units (SI) as a comparison framework. The system defines seven fundamental units: kilogram, metre, candela, second, ampere, kelvin, and mole. Six of these units are defined without reference to a particular physical object which serves as a standard (artifact-free), while the kilogram is still embodied in an artifact which rests at the headquarters of the International Bureau of Weights and Measures in Sèvres near Paris. Artifact-free definitions fix measurements at an exact value related to a physical constant or other invariable phenomena in nature, in contrast to standard artifacts which are subject to deterioration or destruction. Instead, the measurement unit can only ever change through increased accuracy in determining the value of the constant it is tied to.

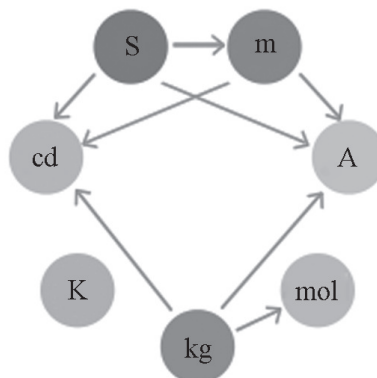


Fig. The seven base units in the SI system. Arrows point from units to those that depend on them.

The first proposal to tie an SI base unit to an experimental standard independent of fiat was by Charles Sanders Peirce (1839–1914), who proposed to define the metre in terms of the wavelength of a spectral line. This directly influenced the Michelson–Morley experiment; Michelson and Morley cite Peirce, and improve on his method.

STANDARDS

With the exception of a few fundamental quantum constants, units of measurement are derived from historical agreements. Nothing inherent in nature dictates that an inch has to be a certain length, nor that a mile is a better measure of distance than a kilometre. Over the course of human history, however, first for convenience and then for necessity, standards of measurement evolved so that communities would have certain common benchmarks. Laws regulating measurement were originally developed to prevent fraud in commerce.

Units of measurement are generally defined on a scientific basis, overseen by governmental or independent agencies, and established in international treaties, pre-eminent of which is the General Conference on Weights and Measures (CGPM), established in 1875 by the Metre Convention, overseeing the International System of Units (SI) and having custody of the International Prototype Kilogram. The metre, for example, was redefined in 1983 by the CGPM in terms of light speed, while in 1960 the international yard was defined by the governments of the United States, United Kingdom, Australia and South Africa as being *exactly* 0.9144 metres.

In the United States, the National Institute of Standards and Technology (NIST), a division of the United States Department of Commerce, regulates commercial measurements. In the United Kingdom, the role is performed by the National Physical Laboratory (NPL), in Australia by the National Measurement Institute, in South Africa by the Council for Scientific and Industrial Research and in India the National Physical Laboratory of India.

UNITS AND SYSTEMS

UNITS OF MEASUREMENT

A unit of measurement is a definite magnitude of a quantity, defined and adopted by convention or by law, that is used as a standard for measurement of the same quantity. Any other value of that quantity can be expressed as a simple multiple of the unit of measurement. For example, length is a physical quantity. The metre is a unit of length that represents a definite predetermined length. When we say 10 metres (or 10 m), we actually mean 10 times the definite predetermined length called “metre”. Measurement is a process of determining how large or small a physical quantity is as compared to a basic reference quantity.

The definition, agreement, and practical use of units of measurement have played a crucial role in human endeavour from early ages up to this day. Different systems of units used to be very common. Now there is a global standard, the International System of Units (SI), the modern form of the metric system. In trade, weights and measures is often a subject of governmental regulation, to ensure fairness and transparency. The International Bureau of Weights and Measures (BIPM) is tasked with ensuring worldwide uniformity of measurements and their traceability to the International System of Units (SI). Metrology is the science for developing nationally and internationally accepted units of weights and measures. In physics and metrology, units are standards for measurement of physical quantities that need clear definitions to be useful. Reproducibility of experimental results is central to the scientific method. A standard system of units facilitates this. Scientific systems of units are a refinement of the concept of weights and measures developed long ago for commercial purposes.

Science, medicine, and engineering often use larger and smaller units of measurement than those used in everyday life and indicate them more precisely. The judicious selection of the units of measurement can aid researchers in problem solving.

In the social sciences, there are no standard units of measurement and the theory and practice of measurement is studied in psychometrics and the theory of conjoint measurement. A unit of measurement is a standardised quantity of a physical property, used as a factor to express occurring quantities of that property. Units of measurement were among the earliest tools invented by humans. Primitive societies needed rudimentary measures for many tasks: constructing dwellings of an appropriate size and shape, fashioning clothing, or bartering food or raw materials.

The earliest known uniform systems of weights and measures seem to have all been created sometime in the 4th and 3rd millennia BC among the ancient peoples of Mesopotamia, Egypt and the Indus Valley, and perhaps also Elam in Persia as well.

Weights and measures are mentioned in the Bible (Leviticus 19:35–36). It is a commandment to be honest and have fair measures. In the *Magna Carta* of 1215 (The Great Charter) with the seal of King John, put before him by the Barons of England, King John agreed in Clause 35 “There shall be one measure of wine throughout our whole realm, and one measure of ale and one measure of corn—namely, the London quart;—and one width of dyed and russet and hauberk cloths—namely, two ells below the seluage...”

As of the 21st Century, multiple unit systems are used all over the world such as the United States Customary System, the British Customary System, and the International System. However, the United States is the only industrialized country that has not yet completely converted to the Metric System.

The systematic effort to develop a universally acceptable system of units dates back to 1790 when the French National Assembly charged the French Academy of Sciences to come up such a unit system. This system was the precursor to the metric system which was quickly developed in France but did not take on universal acceptance until

1875 when. The Metric Convention Treaty was signed by 17 nations. After this treaty was signed, a General Conference of Weights and Measures (CGPM) was established. The CGPM produced the current SI system which was adopted in 1954 at the 10th conference of weights and measures. Currently, the United States is a dual-system society which uses both the SI system and the US Customary system.

SYSTEMS OF UNITS

Traditional Systems

Historically many of the systems of measurement which had been in use were to some extent based on the dimensions of the human body. As a result, units of measure could vary not only from location to location, but from person to person.

Metric Systems

A number of metric systems of units have evolved since the adoption of the original metric system in France in 1791. The current international standard metric system is the International System of Units (abbreviated to SI). An important feature of modern systems is standardization. Each unit has a universally recognized size. Both the imperial units and US customary units derive from earlier English units. Imperial units were mostly used in the British Commonwealth and the former British Empire. US customary units are still the main system of measurement used in the United States despite Congress having legally authorized metric measure on 28 July 1866. Some steps towards US metrication have been made, particularly the redefinition of basic US and imperial units to derive exactly from SI units. Since the international yard and pound agreement of 1959 the US and imperial inch is now defined as exactly 0.0254 m, and the US and imperial avoirdupois pound is now defined as exactly 453.59237 g.

Natural Systems

While the above systems of units are based on arbitrary unit values, formalised as standards, some unit values occur naturally in science. Systems of units based on these are called natural units. Similar to natural units, atomic units (au) are a convenient system of units of measurement used in atomic physics. Also a great number of unusual and non-standard units may be encountered. These may include the solar mass (2×10^30 kg) and the megaton (the energy released by detonating one million tons of trinitrotoluene, TNT).

Legal Control of Weights and Measures

To reduce the incidence of retail fraud, many national statutes have standard definitions of weights and measures that may be used (hence “statute measure”), and these are verified by legal officers.

Base and Derived Units

Different systems of units are based on different choices of a set of base units. The most widely used system of units is the International System of Units, or SI. There are seven SI base units. All other SI units can be derived from these base units. For most quantities a unit is absolutely necessary to communicate values of that physical quantity. For example, conveying to someone a particular length without using some sort of unit is impossible, because a length cannot be described without a reference used to make sense of the value given.

But not all quantities require a unit of their own. Using physical laws, units of quantities can be expressed as combinations of units of other quantities. Thus only a small set of units is required. These units are taken as the *base units*. Other units are *derived units*. Derived units are a matter of convenience, as they can be expressed in terms of basic units. Which units are considered base units is a matter of choice.

The base units of SI are actually not the smallest set possible. Smaller sets have been defined. For example, there are unit sets in which the electric and magnetic field have the same unit. This is based on physical laws that show that electric and magnetic field are actually different manifestations of the same phenomenon.

Real-world Implications

One example of the importance of agreed units is the failure of the NASA Mars Climate Orbiter, which was accidentally destroyed on a mission to Mars in September 1999 instead of entering orbit due to miscommunications about the value of forces: different computer programmes used different units of measurement (newton versus pound force). Considerable amounts of effort, time, and money were wasted.

On 15 April 1999, Korean Air cargo flight 6316 from Shanghai to Seoul was lost due to the crew confusing tower instructions (in metres) and altimeter readings (in feet). Three crew and five people on the ground were killed. Thirty-seven were injured.

In 1983, a Boeing 767 (which came to be known as the Gimli Glider) ran out of fuel in mid-flight because of two mistakes in figuring the fuel supply of Air Canada's first aircraft to use metric measurements. This accident was the result of both confusion due to the simultaneous use of metric and Imperial measures and confusion of mass and volume measures.

SYSTEM OF MEASUREMENT

A system of measurement is a collection of units of measurement and rules relating them to each other. Systems of measurement have historically been important, regulated and defined for the purposes of science and commerce. Systems of measurement in modern use include the metric system, the imperial system, and United States customary units. The French Revolution gave rise to the metric system, and this has spread around the world, replacing most customary units of measure. In most systems, length (distance),

mass, and time are *base quantities*. Later science developments showed that either electric charge or electric current could be added to extend the set of base quantities by which many other metrological units could be easily defined. (However, electrical units are not necessary for such a set. Gaussian units, for example, have only length, mass, and time as base quantities, and the ampere is defined in terms of other units.) Other quantities, such as power and speed, are derived from the base set: for example, speed is distance per unit time. Historically a wide range of units was used for the same type of quantity: in different contexts, length was measured in inches, feet, yards, fathoms, rods, chains, furlongs, miles, nautical miles, stadia, leagues, with conversion factors which were not powers of ten. Such arrangements were satisfactory in their own contexts.

The preference for a more universal and consistent system (based on more rational base units) only gradually spread with the growth of science. Changing a measurement system has substantial financial and cultural costs which must be offset against the advantages to be obtained from using a more rational system. However pressure built up, including from scientists and engineers for conversion to a more rational, and also internationally consistent, basis of measurement.

In antiquity, *systems of measurement* were defined locally: the different units might be defined independently according to the length of a king's thumb or the size of his foot, the length of stride, the length of arm, or maybe the weight of water in a keg of specific size, perhaps itself defined in *hands* and *knuckles*. The unifying characteristic is that there was some definition based on some standard. Eventually *cubits* and *strides* gave way to "customary units" to met the needs of merchants and scientists.

In the metric system and other recent systems, a single basic unit is used for each base quantity. Often secondary units (multiples and submultiples) are derived from the basic units by multiplying by powers of ten, *i.e.* by simply moving the decimal point. Thus the basic metric unit of length is the metre; a distance of 1.234 m is 1, 234 millimetres, or 0.001234 kilometres.

Metrication is complete or nearly complete in almost all countries. US customary units are heavily used in the United States and to some degree in Liberia. Traditional Burmese units of measurement are used in Burma. U.S. units are used in limited contexts in Canada due to the large volume of trade; there is also considerable use of Imperial weights and measures, despite *de jure* Canadian conversion to metric.

A number of other jurisdictions have laws mandating or permitting other systems of measurement in some or all contexts, such as the United Kingdom – whose road signage legislation, for instance, only allows distance signs displaying imperial units (miles or yards) – or Hong Kong. In the United States, metric units are used almost universally in science, widely in the military, and partially in industry, but customary units predominate in household use. At retail stores, the liter is a commonly used unit for volume, especially on bottles of beverages, and milligrams, rather than grains, are used for medications. Some other standard non-SI units are still in international use, such as nautical miles and knots in aviation and shipping.

Metric System

Metric systems of units have evolved since the adoption of the first well-defined system in France in 1795. During this evolution the use of these systems has spread throughout the world, first to non-English-speaking countries, and then to English speaking countries. Multiples and submultiples of metric units are related by powers of ten and their names are formed with prefixes. This relationship is compatible with the decimal system of numbers and it contributes greatly to the convenience of metric units. In the early metric system there were two base units, the metre for length and the gram for mass. The other units of length and mass, and all units of area, volume, and derived units such as density were derived from these two base units.

Mesures usuelles (French for *customary measurements*) were a system of measurement introduced as a compromise between the metric system and traditional measurements. It was used in France from 1812 to 1839.

A number of variations on the metric system have been in use. These include gravitational systems, the centimetre–gram–second systems (cgs) useful in science, the metre–tonne–second system (mts) once used in the USSR and the metre–kilogram–second system (mks).

The current international standard metric system is the International System of Units (*Système international d'unités* or SI). It is an mks system based on the metre, kilogram and second as well as the kelvin, ampere, candela, and mole. The SI includes two classes of units which are defined and agreed internationally. The first of these classes includes the seven SI base units for length, mass, time, temperature, electric current, luminous intensity and amount of substance.

The second class consists of the SI derived units. These derived units are defined in terms of the seven base units. All other quantities (*e.g.* work, force, power) are expressed in terms of SI derived units.

Imperial and US Customary Units

Both imperial units and US customary units derive from earlier English units. Imperial units were mostly used in the former British Empire and the British Commonwealth, but in all these countries they have been largely supplanted by the metric system. They are still used for some applications in the United Kingdom but have been mostly replaced by the metric system in commercial, scientific, and industrial applications. US customary units, however, are still the main system of measurement in the United States. While some steps towards metrication have been made (mainly in the late 1960s and early 1970s), the customary units have a strong hold due to the vast industrial infrastructure and commercial development.

While imperial and US customary systems are closely related, there are a number of differences between them. Units of length and area (the inch, foot, yard, mile etc.) are identical except for surveying purposes. The Avoirdupois units of mass and weight

differ for units larger than a pound (lb). The imperial system uses a stone of 14 lbs, a long hundredweight of 112 lbs and a long ton of 2240 lb. The stone is not used in the US and the hundredweights and tons are short: 100 lbs and 2000 lbs respectively. Where these systems most notably differ is in their units of volume. A US fluid ounce (fl oz), about 29.6 millilitres (ml), is slightly larger than the imperial fluid ounce (about 28.4 ml). However, as there are 16 US fl oz to a US pint and 20 imp fl oz per imperial pint, the imperial pint is about 20% larger. The same is true of quarts, gallons, etc. Six US gallons are a little less than five imperial gallons. The Avoirdupois system served as the general system of mass and weight. In addition to this there are the Troy and the Apothecaries' systems. Troy weight was customarily used for precious metals, black powder and gemstones. The troy ounce is the only unit of the system in current use; it is used for precious metals. Although the troy ounce is larger than its Avoirdupois equivalent, the pound is smaller. The obsolete troy pound was divided into 12 ounces, rather than the 16 ounces per pound of the Avoirdupois system. The Apothecaries' system was traditionally used in pharmacology, but has now been replaced by the metric system; it shared the same pound and ounce as the troy system but with different further subdivisions.

Natural Units

Natural units are physical units of measurement defined in terms of universal physical constants in such a manner that selected physical constants take on the numerical value of one when expressed in terms of those units. Natural units are so named because their definition relies on only properties of nature and not on any human construct. Various systems of natural units are possible.

Some other examples are as follows:

- Geometric unit systems are useful in relativistic physics. In these systems the base physical units are chosen so that the speed of light and the gravitational constant are set equal to unity.
- Planck units are a form of geometric units obtained by also setting Boltzmann's constant, the Coulomb force constant and the reduced Planck constant to unity. They are based only on properties of free space rather than any object or particle.
- Stoney units are similar to Planck units but set the elementary charge to unity and allow Planck's constant to float (*i.e.* it is a number that has to be determined by experiment).
- "Schrödinger" units are also similar to Planck units and also set the elementary charge to unity, but allow the speed of light to float.
- Atomic units are a system of units used in atomic physics, particularly for describing the properties of electrons. The atomic units have been chosen such that several the constants relating to the electron are all equal to one. They are similar to "Schrödinger" units but set the electron mass to unity and

allow the gravitational constant to float. The unit of energy in this system is the total energy of the electron in the Bohr atom and called the Hartree energy. The unit of length is the Bohr radius.

- Electronic units are similar to Stoney units but set the electron mass to unity and allow the gravitational constant to float. They are also similar to atomic units but set the speed of light to unity and allow Planck's constant to float.
- Quantum electrodynamical units are similar to the electronic system of units except that the proton mass is normalised rather than the electron mass.

Non-Standard Units

Non-standard measurement units, sometimes found in books, newspapers etc., include:

Area

- The American football field, which has a playing area 100 yards (91.4 m) long by 160 feet (48.8 m) wide. This is often used by the American public media for the sizes of large buildings or parks. It is used both as a unit of length (100 yd or 91.4 m, the length of the playing field excluding goal areas) and as a unit of area (57,600 sq ft or 5,350 m), about 1.32 acres (0.53 ha).
- British media also frequently uses the football pitch for equivalent purposes, although soccer pitches are not of a fixed size, but instead can vary within defined limits (100–130 yd or 91.4–118.9 m long, and 50–100 yd or 45.7–91.4 m wide, giving an area of 5,000 to 13,000 sq yd or 4,181 to 10,870 m). However the UEFA Champions League field must be exactly 105 by 68 m (114.83 by 74.37 yd) giving an area of 7,140 m (0.714 ha) or 8,539 sq yd (1.764 acres). Example: HSS vessels are aluminium catamarans about the size of a football pitch... - Belfast Telegraph 23 June 2007
- Larger areas are sometimes expressed as a multiple of the areas of certain American states, or subdivisions of the U.K. etc.

Energy

- A ton of TNT equivalent, and its multiples the kiloton, the megaton, and the gigaton. Often used in stating the power of very energetic events such as explosions and volcanic events and earthquakes and asteroid impacts. A gram of TNT as a unit of energy has been defined as 1000 thermochemical calories (1,000 cal or 4,184 J).
- The atom bomb dropped on Hiroshima. Its force is often used in the public media and popular books as a unit of energy. (Its yield was roughly 13 kilotons, or 60 TJ.)
- One stick of dynamite

Units of Currency

A unit of measurement that applies to money is called a unit of account in economics and unit of measure in accounting. This is normally a currency issued by a country or a fraction thereof; for instance, the US dollar and US cent (D_{100} of a dollar), or the euro and euro cent. ISO 4217 is the international standard describing three letter codes (also known as the currency code) to define the names of currencies established by the International Organization for Standardization (ISO).

IMPERIAL AND US CUSTOMARY MEASUREMENT SYSTEMS

The imperial and US customary systems of measurement are two closely inter-related systems of measurement both derived from earlier English system of measurement units which can be traced back to Ancient Roman units of measurement, and Carolingian and Saxon units of measure.

US Customary units, developed and used in the United States after the American Revolution, are based on a subset of the English units used in the Thirteen Colonies, while the Imperial system of units was developed and used after 1824 in the United Kingdom and subsequently used in the rest of the Commonwealth. US Customary units are the predominant system of units in the United States, but in all Commonwealth countries the metric system has, to varying degrees, replaced the imperial system.

Most of the units of measure have been adopted in one way or another since the Norman Conquest (1066). The units of linear measure have changed the least – the yard (which replaced the ell) and the chain were measures derived in England. The foot used by craftsman supplanted the foot used in agriculture which was reduced in size by a factor of D_{11} to bring it into alignment. This resulted in there being $16 D_2$ feet in a rod, pole or perch instead of 15 agricultural feet. The furlong and the acre, once it became a unit of the size of a piece of land rather than its value, remained relatively unchanged. In the last thousand years, three principal pound were used in England. The troy pound (5760 grains) was used for precious metals, the apothecaries' pound, (also 5760 grains) was used by pharmacists and the avoirdupois pound (7000 grains) was used for general purposes. The apothecaries and troy pounds are divided into 12 ounces (of 480 grains) while the avoirdupois pound has 16 ounces (of 437.5 grains). The unit of volume, the gallon, has different values in the United States and in the United Kingdom – the US fluid gallon being about 0.83 imperial gallons and the US dry gallon being about 0.97 imperial gallons.

Both systems of measure were widely used in mechanical engineering, though not in electrical engineering. Some units of measure such as the horsepower or the British thermal unit (BTU) have special names but by and large unit names are generated from their constituent components – for example, pounds per square inch. In contrast, the metric system has a special name for pressure—the pascal. After the United States Declaration of Independence the units of measurement in the United States developed

into what is now known as customary units. The United Kingdom overhauled its system of measurement in 1824, when it introduced the imperial system of units. This resulted in both countries having different gallons. Later in the century, efforts were made to align the definition of the pound and the yard in both countries by using copies of the standards adopted by the British Parliament in 1855. However, these standards were of poor quality compared with those produced for the Convention of the Metre. In 1960 both countries agreed to common definitions of the yard and the pound based on definitions of the metre and the kilogram. This change, which amounted to a few parts per million, had little effect in the United Kingdom, but resulted in the United States having two slightly different systems of linear measure – the international system, and the surveyors system.

English Units of Measure

English units of measure, were derived from a combination of Roman, Carolignian and Saxon units of measure. They were a precursor to both the Imperial system of units (first defined in 1824) and United States customary units which evolved from English Units from 1776 onwards. The earliest records of English units of measure relate to Saxon coinage – at that time coins were both weights and currency. The penny introduced by Offa was about 20 grains (1.296 g). Edward the Elder increased the weight of the English penny to 26 grains (1.685 g), thereby aligning it with the penny of Charlemagne, but by the time of the conquest (1066), it has decreased to 24 grains (1.555 g). This value was subsequently called the *pennyweight* and formed the basis of the Troy units of weight—the troy ounce being used to this day for weighting precious metals. In 965 AD, King Edgar decreed “that only one weight and one measure should pass throughout the King’s dominion”. In his report to Congress in 1817, John Quincy Adams wrote that the link between coinage and weight (or mass) was broken during the reign of Edward I (1272–1307) when he debased the English coinage by introducing a groat (four pence) which weighed of 89 grains rather than the expected 96 grains. In the 1350s the groat was further devalued by reducing its weight to 72 grains.

One of the enduring conflicts during medieval times was who had the right to define weights and measures. In 1197 Richard I decreed that the measures of corn and pulse, and of wine and ale should be the same throughout all England. The Magna Carta, signed by King John in 1215 extended this to include cloth.

During Saxon times land was described in both terms of its economic value and in terms of an absolute size. In the Domesday Book, the hide, an economic unit of measure was used, while there are references elsewhere relating to the furlong and to the rood that appear to be related to ploughing procedures. Of particular interest was the rood which was 15 [North German or Saxon] feet in length, the North German foot being equivalent to 335 mm (13.2 inches). Craftsmen, on the other hand used a shorter Roman foot. Some time between 1266 and 1303 the weights and measures of England were radically revised by a law known as the *Composition of Yards and Perches* (*Compositio*

ulnarum et perticarum) often known as the *Compositio* for short. This law, attributed to either Henry III or his successor Edward I, instituted a new foot that was exactly D_{11} the length of the old foot, with corresponding reductions in the size of the yard, ell, inch, and barleycorn. Furlongs and rods, however, remained the same, but the rod changed from 15 old feet to $16D_2$ new feet.

Units of length were systematized by Edward II who in 1324 defined the inch as being three barleycorns, the foot as twelve inches, the yard as three feet, the perch five and half yards and the acre as being an area four perches in width by forty perches in length. Apart from the ell (45 inches or 114.3 cm) which continued to be used in the cloth trade and the chain which was introduced by Edmund Gunter in 1620, these units formed the basis of the units of length of the English system of measurement. The units were however redefined many times – during Henry VIII's time standard yards and ells made of brass were manufactured, during Elizabeth I's time these were replaced with standards made of bronze and in 1742, after scientific comparisons showed a variation of up to 0.2% from the mean, a definitive standard yard was proposed (but not manufactured).

During the medieval era agricultural products, apart from wool, were mostly sold by volume, with various bushels and gallons being introduced over the years for different commodities. In the early fourteenth century the wool trade traditionally used the avoirdupois system of weights, a process that was formalized by Edward III in 1340. At the same time, the stone, when used to weigh wool, was formalized as being 14 pounds.

During the Tudor period, numerous reforms were made to English weights and measures. In 1496 Henry VII ordered that reference copies of the yard, pound and gallon should be made of brass and distributed to specified towns and cities throughout the kingdom. Many weights and measures that had crept into use were banned: in 1527 Henry VIII banned the Tower pound (5400 grains against the 5760 grains of the apothecaries and troy pounds) and the mercantile pound (6750 grains against the 7000 grains of the pound avoirdupois) and in 1592 Elizabeth I ordered the use of the “statute mile” (5280 feet against the 5000 feet of the London or Old English mile).

Under the Act of Union of 1707, Scotland, which had developed its own system of weights and measures independently of England, abandoned them in favour of English weights and measures. The Acts of Union 1800 which united Ireland with Great Britain had less of an effect on weights and measures—Irish weights and measures having been based on the English foot and pound avoirdupois since 1351, though the Irish acre and mile were based on a perch of 7 yards, not $5 D_2$ yards as in England.

By the early nineteenth century many commodities had their own set of units, the units of measure for the wool and cloth industries had units of measure specific to those commodities, albeit derived on the pound avoirdupois or the foot while wine and beer used units with the same names but different sizes – the wine gallon being 231 cubic inches and the beer or ale gallon being 282 cubic inches. Agricultural produce was sold by the bushel which was based on yet another gallon – the dry gallon of 268.8 cubic

inches. Even though not explicitly permitted by statute, many markets used bushels based on weight rather than volume when selling wheat and barley.

Imperial Units

The British Weights and Measures Act of 1824 repealed all existing British weights and measures legislation, some dating back to the 1300s, and redefined existing units of measure. In particular, a new standard yard and troy pound were manufactured as the standards for length and weight respectively. A new measure, the imperial gallon, which replaced the many gallons in use, was defined as being the volume of 10 pounds of water at 62 °F which, after the authorized experiments, was found to be 277.274 cubic inches. The bushel, which like the gallon, had definitions reflecting the various gallons, was defined as 8 imperial gallons.

The 1824 Act also introduced some changes to the administration of the standards of weights and measures: previously Parliament had been given the custody of the standards but the act passed this responsibility on to the Exchequer. The act also set up an inspectorate for weights and measures.

The standard yard and pound were lost in 1834 when a fire partially destroyed the Palace of Westminster. Following a report published in 1841 by a commission new standard yard and pound were manufactured using the best available secondary sources. Unlike the previous standard, the new pound standard was a pound avoirdupois. They were accepted by an Act of Parliament as the standards for length and weight in 1855. Following the debacle over the different gallons that had been adopted by the United States and the United Kingdom thirty years earlier, one of the copies of the standard yard was offered to and accepted by the United States Government.

The 1835 Weights and Measures Act tidied up a number of shortcomings in the 1825 Act. In response to representations from traders, the stone and the hundredweight were formally defined as being 14 pounds and 112 pounds respectively and the experiment of defining a “heaped” measure as outlined in the 1824 Act was abandoned. Not all trades followed the use of the 14 stone—Britten, in 1880 for example, catalogued a number of different values of the stone in various British towns and cities ranging from 4 lb to 26 lb. The 1835 Act also restricted the use of Troy measure to precious metals and required that coal be sold by weight and not by volume.

The Weights and Measures Act of 1878 overhauled the inspection regime of weights and measures used in trade. The act also reaffirmed the use of the brass standard yard and platinum standard pound as the standards for use in the United Kingdom, reaffirmed the use of apothecaries measures in the pharmaceutical industry, reaffirmed the 1824 definition of the gallon, removed the Troy pound from the list of legal units of measure, added the fathom to the list of legal units and fixed the ratio of metric to imperial units at one metre being equal to 39.3708 inches and one kilogram being equal to 15432.3487 grains (1 lb = 0.453592654 kg). Subsequent to the passing of the act, the volume of the gallon which had been defined as being the volume of 10 lb distilled water at 62 °F (17

°C) was remeasured and set at 277.42 cubic inches though HM Customs and Excise continued to use the 1824 definition for excise purposes.

The Weights and Measures Act of 1878 effectively prohibited the use of metric weights for trade, the United Kingdom having declined to sign the Convention of the Metre three years previously. The standard imperial yard was not stable – in 1947 its rate of shrinkage was quantified and found to be one part per million every 23 years. In April 1884 HJ Chaney, Warden of Standards in London unofficially contacted the BIPM (custodians of the standard metre) inquiring whether the BIPM would calibrate some metre standards that had been manufactured in the United Kingdom. Broch, director of the BIPM replied that he was not authorised to perform any such calibrations for non-member states. On 17 September 1884, the British Government signed the convention on behalf of the United Kingdom. The Weights and Measures Act of 1897 authorized the use of metric units for trade; a list of metric to imperial equivalents being published the following year.

Under the Weights and Measures Act of 1824 custody of the standard yard and pound and custody of the administration of weights and measures was entrusted to the Exchequer but verification was administered locally. The 1835 Act formally described the office and duties of Inspectors of Weights and Measures and required every borough to appoint such officers and the 1866 Act passed responsibility for weights and measures to the Board of Trade. In 1900 the Board of Trade established the National Physical Laboratory (NPL) to provide laboratory facilities for weights and measures.

After the passage of the 1897 Act, weights and measures in the United Kingdom remained relatively unchanged until after the Second World War. By the middle of the century the difference of 2 parts per million between the British and US standard yards was causing problems—in 1900 a tolerance of 10 parts per million was adequate for science, but by 1950 this tolerance had shrunk to 0.25 parts per million. In 1960 representatives from the NPL and other national laboratories from the United States and Commonwealth agreed to redefine the yard as being exactly 0.9144 metres, an action that was ratified by the British Government as part of the 1963 Weights and Measures Act.

Metrication in the United Kingdom began in the mid-1960s. Initially this metrication was voluntary and by 1985 many traditional and imperial units of measure had been voluntarily removed from use in the retail trade. The Weights and Measures Act of 1985 formalized their removal for use in trade, though imperial units were retained for use on road signs and the most common imperial weights such as the foot, inch, pound, ounce, gallon and pint continued to be used in the retail trade for the sale of loose goods or goods measured or weighed in front of the customer. Since 1 January 2000 it has been unlawful to use imperial units for weights and measures in retail trade in the United Kingdom except as supplementary units or for the sale of draught beer and cider by the pint or milk that is sold in returnable containers.

The British Empire

When colonies attained dominion status, they also attained the right to control their own systems of weights and measures. Many adopted the imperial system of units with local variations. India and Hong Kong supplemented the imperial system of units with their own indigenous units of measure, parts of Canada and South Africa included land survey units of measure from earlier colonial masters in their systems of measure while many territories used only a subset of the units used in the United Kingdom—in particular the stone, quarter and cental were not catalogued in, amongst others, Australian, Canadian and Indian legislation. Furthermore, Canada aligned her ton with US measures by cataloguing the ton of 2000 lb as being legal for trade, but kept the imperial gallon. The standardization of the yard in 1960 required not only agreement between the United States and the United Kingdom, but also of Canada, Australia, New Zealand and South Africa, all of whom had their own standards laboratories.

United States Customary Units

Prior to their declaration of independence in 1776, the thirteen colonies that were to become the United States used the English system of measurement. The Articles of Confederation which predated the Constitution gave the central government “the sole and exclusive right and power of...fixing the Standard of Weights and Measures throughout the United States.” Subsequent to the formation of the United States, the Constitution reaffirmed the right of Congress to “fix the Standard of Weights and Measures” but reserved the right to regulate commerce and weights and measures to the individual states. During the First Congress of the United States in 1789, Thomas Jefferson was detailed to draw up a plan for the currency and weights of measures that would be used in the new republic. In his response in 1790 he noted that the existing system of measure was sound but that control of the base artefact was not under the control of the United States. His report suggested a means of manufacturing a local standard and also left the way open for an adoption of a decimal-based system should this be appropriate. In the event, the existing standards were retained.

For many years no action was taken at the federal level to ensure harmony in units of measure – the units acquired by the early colonists appeared to serve their purpose. Congress did nothing, but Ferdinand Hassler, Superintendent of the East Coast survey, who in 1790 had met using contacts in his native Switzerland acquired a copy of the [French] *mètre des Archives*. In 1810 Ferdinand Hassler was dispatched to Europe by the Treasury to acquire measuring instruments and standards.

In 1827 Albert Gallatin, United States minister at London acquired an “exact copy” of the troy pound held by the British Government which in 1828 was adopted as the reference copy of weight in the United States.

In 1821 John Quincy Adams, then Secretary of State submitted a report based on research commissioned by the Senate in 1817 which recommended the adoption of the

metric system. Congress did nothing and in 1832 the Treasury adopted the yard of 36 inches as the unit of length for customs purposes, the avoirdupois pound of 7000 grains as the unit of weight and the gallon of 231 cubic inches (the “Queen Anne gallon”) and the bushel of 2150.42 cubic inches as the units of volume. Congress did little to promote standards across the United States other than fixing the size of the yard and the gallon.

Throughout the nineteenth century individual states developed their own standards and in particular a variety of bushels based on weight (mass) rather than volume emerged, dependent on both commodity and state. This lack of uniformity crippled inter-state trade and in 1905 the National Bureau of Standards called a meeting of the states to discuss the lack of uniform standards and in many cases, a means of regulatory oversight. A meeting was held the following year and subsequently became an annual gathering known as the National Conference on Weights and Measures (NCWM). In 1915 the conference published its first model standards. The bushel was not fully standardized and the Chicago Mercantile Exchange still (May 2013) uses different bushels for different commodities—a bushel of corn being 56 lb, a bushel of oats 38 lb and a bushel of soybeans 60 lb and a bushel of red winter wheat (both hard and soft) also 60 lb. Other commodities at the exchange are reckoned in pounds, in short tons or in metric tons.

One of the actions taken by Congress was to permit the use of the metric system in trade (1866), made at the height of the metrication process in Latin America. Other actions were to ratify the Metre Convention in 1875 and under the Mendenhall Order of 1897, to redefine the pound and the yard in terms of the International Prototype Kilogram and the International Prototype Metre respectively. In 1901 the administration of weights and measures was handed to a federal agency, the National Bureau of Standards, which in 1988 became the National Institute of Standards and Technology. Inactivity by Congress and the lack of uniformity of weights and measures which were crippling US economic growth in the nineteenth century led to the National Bureau of Standards to call a meeting of states in 1905 which resulted in the setting up of the National Conference on Weights and Measures (NCWM). This organisation is the *de facto* controlling body for weights and measures in the United States, though in respect of international relations such as membership of the General Conference on Weights and Measures (an intergovernmental organization) the US Government itself has to take the lead. During the twentieth century the principal change in the customary system of weights and measures was an agreement between NIST and the corresponding bodies in Australia, Canada, New Zealand, South Africa and the United Kingdom, signed in 1960, that redefined the yard and the pound in terms of the metre and the kilogram respectively. These new units became known as the international yard and pound. Congress has neither endorsed nor repudiated this action.

Energy, Power and Temperature

Imperial and US customary units have long been used in many branches of engineering. Two of the earliest such units of measure to come into use were the *horsepower* and the

degree Fahrenheit. The *horsepower* was defined by James Watt in 1782 as the power required to raise 33,000 pounds of water through a height of one foot in one minute and the degree Fahrenheit was first defined by Daniel Fahrenheit in about 1713 as being a temperature scale having its lower calibration point (0 °F) at temperature where a supersaturated salt/ice mixture froze and its upper calibration point at body temperature (96 °F). In 1777 the Royal Society, under the chairmanship of Henry Cavendish, proposed the definition of the Fahrenheit scale be modified such that the temperature corresponding to the melting point of ice be 32 °F and the boiling point of water under standard atmospheric conditions be 212 °F. The British thermal unit (Btu) is defined as the heat needed to raise the temperature of one pound of water by one degree Fahrenheit. It was in use before 1859 as a unit of heat based on imperial units rather than the metric units used by the French—Clément-Desormes having defined the calorie in terms of the kilogram and degrees centigrade in 1824. In 1873 a committee of the British Association for the Advancement of Science under the chairmanship of William Thomson (Lord Kelvin) introduced the concept of a coherence into units of measure and proposed the names dyne and erg as the units of force and work in the CGS system of units. Two years later James Thomson, older brother of William Thomson introduced the term poundal as a coherent unit of force in the Foot–pound–second system (FPS) of measurement. The equivalent FPS unit of work was the foot-poundal. Other systems for the measurement of dynamic quantities that used imperial and US customary units are the British Gravitational System (BG) proposed by Arthur Mason Worthington and the English Engineering System (EE). Both systems depend on the gravitational acceleration, and use the pound-force as the unit of force but use different approaches when applying Newton’s laws of motion. In the BG system, force, rather than mass is a base unit while the slug is a derived unit of inertia (rather than mass). On the other hand, the EE system uses a non-coherent approach and introduce the acceleration due to gravity (g) into its equations. Both these approaches led to slight variations in the meaning of the pound-force (and also of the kilogram force) in different parts of the world. Various countries published standard values that should be used for g and in 1901 the CGPM published a standard value for g that should be used in the “International Service of Weights and Measures”, namely 9.80665 m/s² (32.174 ft/s²) which is equal to the value of g at 45° latitude.

Newton’s second law in these systems becomes:

BG: Force (lbf) = inertia (slugs) $\times$ acceleration (ft/s²)

EE: Force (lbf) = mass (lb) $\times g \times$ acceleration (ft/s²)

FPS: Force (poundals) = mass (lb) $\times$ acceleration (ft/s²)

FPS is ignored in many engineering courses and textbooks while some, such as Darby only uses EE (alongside SI), having described the BG and EE systems as “archaic”.

Metric Equivalents

The standard yard and [Troy] pound were lost in 1834 when a fire partially destroyed the Palace of Westminster. Following a report published in 1841 by a commission new

standard yard and pound were manufactured using the best available secondary sources. Unlike the previous standard, the new pound standard, made of platinum, was a pound avoirdupois. The new yard, slightly longer than a yard to prevent wear as was experienced with the *mètre des Archives*, was made of brass and had two gold plugs close to its end. Scratch marks on the plugs denoted the length of the yard. They were accepted by an Act of Parliament as the standards for length and weight in 1855. Following the debacle over the different gallons that had been adopted by the United States and the United Kingdom thirty years earlier, one of the copies of the standard yard and avoirdupois pound (known in the United States as the “Mint pound”) was offered to and accepted by the United States Government.

In the years that followed the passing of the 1878 Act, the standard imperial yard was found to be shrinking at a rate, confirmed in 1950, to be nearly one part per million every 30 years. On the other hand, the international prototype metre, manufactured from a platinum-iridium alloy rather than brass by a British firm, which in 1889 replaced the *metre des archives* as the standard for the metre, was found to be more stable than the standard yard. Both the United States and the United Kingdom, as signatories of the Metre Convention, took delivery of copies of both the standard metre and the standard kilogram. The “Mint pound” was also found to be of poor workmanship.

In 1866 the United States Government legalised use of metric units in contract law, defining them in terms of the equivalent customary units to five significant figures which was sufficient for purposes of trade. In 1893, under the Mendenhall Order the United States abandoned the 1855 yard as a standard of length and the “Mint pound” as a standard of mass, redefining them in terms of the metre and kilogram using the values of the 1866 legislation. In the United Kingdom fresh comparisons of the imperial and metric standards of length and mass were made and were used in the Weights and Measures Act 1897 to redefine the yard and pound in terms of the metre and kilogram respectively.

In addition, the definitions of both the yard and the pound in terms of the artifacts held by the British Government was reaffirmed giving both the yard and the pound two different definitions. The differences between the British and the US yard and pound was of the order of a few parts per million.

Table. Metric Equivalents (1890s)

United States	United Kingdom	Difference
Mendenhall Order (1893)	Weights and Measures Act (1897)	parts per million
1 metre = 39.37 inches	1 metre = 1.0936143 yards	2.9
1 kilogram = 2.2046 pounds	1 kilogram = 2.2046223 pounds	10.1

By the end of the Second World War, the standards laboratories of Canada, Australia, New Zealand and South Africa also had their own copies of the pound and the yard. These legal and technical discrepancies, described by McGreevy (pg 290) as being “unsound” led to the Commonwealth Science Conference of 1946. proposing that the Commonwealth countries and the United States should all redefine the yard and the

pound in terms of an agreed fraction of the metre and kilogram respectively. Agreement was reached by the standards laboratories in 1960 to redefine the yard and the metre as

1 international yard = 0.9144 metres

1 international pound = 0.453592 37 kilograms

The final digit of the value chosen for the pound was chosen so as to make the number exactly divisible by 7, making the grain exactly 64.79891 milligrams. This agreement was ratified by the United Kingdom in 1963 while Canada pre-empted the decision by adopting these values in 1951, nine years ahead of the full international agreement. The United States Congress has neither ratified nor repudiated the agreement.

Comparison of Imperial and US Customary Systems

Prior to 1960 the imperial and customary yard and the pound were sufficiently close to each other that for most practical purposes the differences in the sizes of units of length, area, volume and mass could be disregarded, though there were differences in usage - for example, in the United States short road distances are specified in feet while in the United Kingdom they are specified in yards. The introduction of the international yard in 1960 caused small but noticeable effects in surveying in the United States which resulted in some states retaining the original definitions of the customary units of measure which are now known as the survey mile, foot, while other states adopted the international foot. The definition of units of weight above a pound differed between the customary and the imperial system - the imperial system employed the stone of 14 pounds, the hundredweight of 8 stone and the ton of 2240 pounds (20 hundredweight), while the customary system of units did not employ the stone but has a hundredweight of 100 pounds and a ton of 2000 pounds. In international trade, the ton of 2240 pounds was often referred to as the "long ton" and the ton of 2000 pounds as the "short ton". When using customary units, it is usual to express body weight in pounds, but when using imperial units, to use stones and pounds.

In his *Plan for Establishing Uniformity in the Coinage, Weights, and Measures of the United States*, Thomas Jefferson, then secretary of state, identified 14 different gallons in English statutes varying in size from 224 to 282 cubic inches (3.67 to 4.62 litres). In 1832, in the absence of any direction by Congress, the United States Treasury chose the second smallest gallon, the "Queen Anne gallon" of 231 cubic inches (3.785 litres) to be the official gallon in the United States for fiscal purposes. Sixteen US fluid ounces make a US pint (8 pints equals 1 gallon in both customary and imperial systems). During the reform of Weights and measures legislation in the United Kingdom in 1824, old gallons were replaced by the new imperial gallon which was defined to be the volume to 10 pounds of water at 62 °F (17 °C) and determined experimentally to be 277.42 cubic inches (4.546 litres). Twenty imperial fluid ounces make an imperial pint, the imperial fluid ounce being 0.96 US fluid ounces. The US Customary system of units makes use of set of dry units of capacity that have a similar set of names to those of liquid capacity, though different volumes - the dry pint having a volume of 33.6

cubic inches (550 ml) against the US fluid pint's volume of 28.875 cubic inches (473 ml) and the imperial pint of 34.68 cubic inches (568 ml). The imperial system of measure does not have an equivalent to the US customary system of "dry measure".

In the international commodities markets, the barrel (42USgallons, ~159 litres) is used in both London and New York/Chicago for trading in crude oil and the troy ounce (~31.10 grams) for trading in precious metals, otherwise the London markets use metric units and the Chicago Board of Trade uses customary units.

Units in Use

The tables below catalogue the imperial units of measure that were permitted for use in trade in the United Kingdom on the eve of metrication (1976) and the customary "units of measurement that have traditionally been used in the United States". In addition, named units of measure that are used in the engineering industry are also catalogued. Prior to metrication, the units of measure used in the Ireland were the same as those used in the United Kingdom while those used in the British Commonwealth and in South Africa were in most cases a subset of those used in the United Kingdom with, in certain cases, local differences. Unless otherwise specified, the units of measure quoted below were used in both the United States, the United Kingdom. The SI equivalents are quoted to four significant figures.

Current Status

Since its inception, the metric system has displaced the traditional system of units in many countries. In the 1960s a metrication programme was initiated in most English-speaking countries, resulting in either the partial or total displacement of the imperial system or the US customary system of measure in those countries. The current status of imperial and US customary units, as summarised by NIST, is that "the SI metric system is now the official system of units in the United Kingdom, while the customary units are still predominantly used in the United States". The situation is however not as clear-cut as this. In the United States, for example, the metric system is the predominant system of measure in certain fields such as automobile manufacture even though customary units are used in aircraft manufacture. In the United Kingdom, metric units are required for almost all regulated use of units of measure except for a few specifically exempt areas such as road signs, speedometers and draught beer. The NIST statement is also incomplete as it made no mention of metrication in the Commonwealth countries of Australia, India, New Zealand or South Africa where metrication is all but complete, nor did it mention the status of metrication in Canada where the imperial system has been displaced for many areas. As of June 2013 the imperial and US customary systems of measurement were dependent on the SI for their formal definitions, the yard being defined as 0.9144 metres exactly, the pound avoirdupois as 0.45359237 kilograms exactly while both systems of measure share the definition of the second.

INTELLIGENCE TESTS AS CLINICAL INSTRUMENTS

INTELLIGENCE

Intelligence has been defined in several different methods including logic, abstract thought, understanding, self-awareness, communication, learning, having emotional knowledge, retaining, planning, and problem solving.

Intelligence is most widely studied in humans, but has also been observed in other animals and in plants. Artificial intelligence is the simulation of intelligence in machines.

Within the discipline of psychology, several approaches to human intelligence have been adopted. The psychometric approach is especially familiar to the general public, as well as being the most researched and through distant the most widely used in practical settings.

Definitions

The definition of intelligence is controversial. Some groups of psychologists have suggested the following definitions:

From “Mainstream Science on Intelligence” (1994), an editorial statement through fifty-two researchers: A very general mental capability that, among other things, involves the skill to cause, plan, solve problems, think abstractly, comprehend intricate ideas, learn quickly and learn from experience. It is not merely book learning, a narrow academic ability, or test-taking smarts. Rather, it reflects a broader and deeper capability for comprehending our surroundings— “catching on,” “making sense” of things, or “figuring out” what to do.

From “Intelligence: Known’s and Unknowns” (1995), a report published through the Board of Scientific Affairs of the American Psychological Association: Individuals differ from one another in their skill to understand intricate ideas, to adapt effectively to the environment, to learn from experience, to engage in several forms of reasoning, to overcome obstacles through taking thought. Although these individual differences can be substantial, they are never entirely constant: a given person’s intellectual performance will vary on different occasions, in different domains, as judged through different criteria.

Concepts of “intelligence” are attempts to clarify and organize this intricate set of phenomena. Although considerable clarity has been achieved in some areas, no such conceptualization has yet answered all the significant questions, and none commands universal assent. Indeed, when two dozen prominent theorists were recently asked to describe intelligence, they gave two dozen, somewhat different, definitions.

Theories

Intelligence theorists fall into two categories. In one group are those who argue for a “general intelligence” that characterizes a person’s actions and thinking in all areas. Their critics consider that intelligence is composed of several separate types of aptitudes and abilities, and that a person who excels in one area will not necessarily excel in all areas.

Early Theories: Spearman and Thurstone

Spearman whispered that intelligence is general: People who are bright in one area are bright in other areas as well. Thurstone disagreed: He whispered that intelligence encompasses seven mental abilities that are relatively independent of one another. In contrast, Cattell divided mental abilities into two clusters. The first is crystallized intelligence, or abilities such as reasoning and the verbal and numerical skills that are stressed in school. The second is fluid intelligence, or skills such as spatial and visual imagery, the skill to notice visual details, and rote memory.

Contemporary Theories: Sternberg and Gardner

In the mid-1980s, Yale psychologist Robert Sternberg proposed a triarchic theory of intelligence that comprises a much broader range of skills and abilities. According to this theory, intelligence consists of three overarching characteristics: componential intelligence, the traditional mental processes or skills accentuated through earlier theories of intelligence, such as the skill to acquire new knowledge and perform tasks efficiently; experiential intelligence, characterized through insight and creative adaptability as well as efficient and quick processing of information without conscious thought; and contextual intelligence, marked through responsiveness to the environment. Intelligent people, according to Sternberg, are adept at making the most of their strengths and compensating for their weaknesses. Howard Gardner has proposed his theory of multiple intelligences, which asserts that what we refer to as intelligence actually consists of several separate abilities, each of which is relatively independent of the others.

Formal theories of intelligence serve as the foundation for the design and administration of intelligence tests. And because experts do not view intelligence in exactly the same method that non-experts do, it is understandable that most tests of intelligence do not contain items that several non-experts think of as part of intelligence.

Measures of Intelligence

Alfred Binet and Theodore Simon, commissioned through the French government to devise a test that would differentiate flunked by children who could not learn because of low skill and those who would not learn because of poor motivation, created the first intelligence test in 1905. Binet did not really have a theory of mental skill, but he

thought that intelligence could be best expressed in and assessed through intricate cognitive tasks. The first intelligence test consisted of items that assessed memory, knowledge, and reasoning skills arranged in a sequence of increasing difficulty and grouped into age levels, or when the “average” child could accomplish them. This first test has undergone numerous revisions and is now recognized as the Stanford-Binet Intelligence Scale, Fifth Edition.

Currently, the most widely-used intelligence tests are the Wechsler Scales. Intended through David Wechsler, a clinical psychologist who used the test to help identify cognitive strengths and weaknesses in his clients and assist in diagnoses, the tests were based on his definition of intelligence as “the aggregate or global capability of the individual to act purposefully, to think rationally, and to deal effectively with his environment”.

Because the test seemed to be a valid measure of general intelligence, its use expanded to other populations.

There are three separate Wechsler scales:

- Wechsler Preschool and Primary Scale of Intelligence, 3rd ed. (WPPSI-III)
- Wechsler Intelligence Scale for Children, 4th ed. (WISC-IV)
- Wechsler Adult Intelligence Scale, 3rd ed. (WAIS-III)
- The WISC-IV is intended for use with children and adolescents ages 6 to 16 years 11 months and will be the focus of this section. Sample items from many of the 10 core subtests are presented below:
 - *Similarities*: In what method are wool and cotton alike?
 - *Vocabulary*: What does corrupt mean?
 - *Comprehension*: Why do people buy fire insurance?
 - *Digit Span*: I am going to say some numbers. Listen cautiously and when I am through, say the numbers right after me: 734186
 - *Coding*: In the top row, each figure is shown with a number. Fill in the number that goes with each figure in the second row
 - *Matrix Reasoning*: For each item the child looks at a partial matrix and selects the missing portion from five response options.
 - *Picture Concepts*: The child identifies objects that share some common property.

Some of the items in this test require children and adolescents to arrange materials rather than talk with the test administrator. This procedure allows non-English-speaking children, as well as youth with speech or language disorders, to demonstrate their intellectual capabilities. Some professionals find the division flanked by verbal and performance items very useful for assessment purposes.

The items create up four index scores—Verbal Comprehension, Perceptual Organization, Freedom from Distractability, and Processing Speed—that are combined to provide an overall skill score. When the four index scores are added together, they produce a total skill score, or intelligence quotient (IQ). On the Wechsler scales, the

mean total IQ is set at 100 with a standard deviation of 15. Most other major intelligence tests have set the same mean so it is easy to compare the IQ score from one intelligence test to another.

DETERMINES

Heredity

Historically, research on the determinants of intelligence has focused on identical twins—some reared together; others reared separately in separate households. The correlation flanked by the IQs of all identical twins is usually very high, indicating that their identical genetic inheritance is a more powerful determinant of intelligence than their experiences. But critics of this research create many strong points: (1) It is hard to find identical twins who have been separated at birth, so that there are only a few such studies; (2) identical twins tend to be placed in households similar in socioeconomic background to those of their biological parents; and (3) even twins separated at birth have had almost identical prenatal experiences.

Environment

Research on rats as well as on humans strengthens the case for environment as a factor in the development of superior intellectual skill. Therefore, even though certain mental abilities are inherited, without the necessary stimulation a child's intelligence will not develop.

This finding is significant because lower-income families don't have access to the kinds of resources that other families do. Significantly, when they are placed in more stimulating environments, economically deprived children show an improvement in their level of intelligence. For instance, lower-income children raised in middle-class homes display important gains in IQ compared with their counterparts rising up in low-income households. Similarly, children who participate in intervention programmes such as Head Start regularly exhibit improvements in cognitive abilities, although the long-term effects of such programmes have yet to be confirmed.

The IQ Debate: A Continuing Controversy

Accounting for group differences in IQ poses a vexing problem in psychology. A milestone in this debate was the 1969 publication of an article through psychologist Arthur Jensen, claiming that overall differences in IQ scores flanked by the races are largely inherited. Jensen's article raised a storm of controversy, which swelled up again in 1994 with the publication of a book on this topic through Richard Herrnstein and Charles Murray. Significantly, most participants in this debate agree that both heredity and environment affect IQ scores.

TESTING FOR SPECIAL APTITUDES

Aptitude

An aptitude is a component of a competency to do a certain kind of work at a certain level, which can also be measured “talent”. Aptitudes may be physical or mental. Aptitude is not knowledge, understanding, learned or acquired abilities (skills) or attitude. The innate nature of aptitude is in contrast to achievement, which represents knowledge or skill that is gained.

Intelligence

Aptitude and intelligence quotient are related, and in some methods opposite views of human mental skill. Whereas intelligence quotient sees intelligence as being a single measurable characteristic affecting all mental skill, aptitude refers to one of several different characteristics which can be independent of each other, such as aptitude for military flight, air traffic control, or computer programming. This is more similar to the theory of multiple intelligences.

Concerning a single measurable characteristic affecting all mental skill, analysis of any group of intelligence test scores will almost always show them to be highly correlated. The U.S. Department of Labour’s General Learning Skill, for instance, is determined through combining Verbal, Numerical and Spatial aptitude subtests. In a given person some are low and others high.

In the context of an aptitude test the “high” and “low” scores are usually not distant separately, because all skill test scores tend to be correlated. Aptitude is better applied intra-individually to determine what tasks a given individual is more skilled at performing.

Inter-individual aptitude differences are typically not very important due to IQ differences. Of course this assumes individuals have not already been pre-screened for aptitude through some other process such as SAT scores, GRE scores, or finishing medical school.

Combined Aptitude and Knowledge Tests

Tests that assess learned skills or knowledge are regularly described achievement tests. Though, certain tests can assess both types of constructs. An instance that leans both methods is the Armed Services Vocational Aptitude Battery (ASVAB), which is given to recruits entering the armed forces of the United States.

Another is the SAT, which is intended as a test of aptitude for college in the United States, but has achievement elements. For instance, it tests mathematical reasoning, which depends both on innate mathematical skill and education received in mathematics.

Aptitude tests can typically be grouped according to the type of cognitive skill they measure:

- *Fluid intelligence:* the skill to think and cause abstractly, effectively solve problems and think strategically. It's more commonly recognized as 'street smarts' or the skill to 'quickly think on your feet'. Examples of what employers can learn from your fluid intelligence in relation to the your suitability for the role for which you are applying
- *Crystallized intelligence:* the skill to learn from past experiences and relevant learning, and to apply this learning to work-related situation. Work situations that require crystallized intelligence contain producing and analyzing written reports, comprehending work instructions, using numbers as a tool to create effective decisions, etc.

The Stanford-Binet Intelligence Scale

The Binet-Simon Scale, the first test of intelligence, was urbanized in France through Alfred Binet and Theodore Simon for testing children. Originally issued in 1908, it consisted of 30 tests arranged in order of increasing difficulty. From the average scores of children, Binet urbanized the concept of mental age.

The best-recognized Binet version, created through Stanford University's L. M. Terman in 1916, is the Stanford-Binet Intelligence Scale. Terman introduced the term intelligence quotient (IQ), which is a numerical value given to scores on an intelligence test (a score of 100 corresponds to average intelligence). The Stanford-Binet is intended to measure skills in four areas: verbal reasoning, abstract/visual reasoning, quantitative reasoning, and short-term memory.

The Wechsler Intelligence Scales

The Wechsler Adult Intelligence Scale-Third Edition (WAIS-III) was urbanized through David Wechsler especially for adults. The test measures both verbal and performance abilities. Wechsler also created the Wechsler Intelligence Scale for Children-Third Edition (WISC-III), which is meant to be used with school-aged children. It measures verbal and performance abilities separately, though it also yields an overall IQ score.

Group Tests

Group tests are administered through one examiner to several people at one time. Group tests are most commonly used through schools. The California Test of Mental Maturity (CTMM) and the SAT are group tests. Group tests aim to overcome the problems of time and expense associated with individual tests and to eliminate bias on the part of the examiner. Though, in a group setting the examiner is less likely to notice whether an individual test taker is tired, ill, or confused through the directions.

Emotionally disturbed children and people who have less experience taking tests usually do better on individual tests than on group tests.

Performance and Culture-Fair Tests

Some intelligence tests may discriminate against members of certain cultural or ethnic groups. Performance tests are intelligence tests that do not involve language, so they can be useful for testing people who lack a strong command of English. The Seguin Form Board, the Porteus Maze, and the Bayley Scale of Infant Development are performance tests. Culture-fair tests are intended to eliminate cultural bias through minimizing skills and values that vary from one culture to another. The Good enough-Harris Drawing Test and the Progressive Matrices are examples of culture-fair tests.

MOTIVATION

Motives as Inference

There is some reference (be it conscious or unconscious) underlying our behaviour as it is not random or purposeless. So, motivation plays a role of energizing behaviour from inside. A motivated individual initiates and accomplishes several responses in the environment, and achieves his/her goals faster, more vigorously, and more persistently than does one who is less motivated. Motivation is noted to be long lasting. It accounts for performed behaviors and explains the why of behaviour. We can so say that motivation has direct impact on behaviour as it creates behaviour much more goal and purpose oriented.

Through nature, most motives go cyclically. There is usually a start point, which ends to become itself after some time laps. (Here, note that there are few motives that do not go in cycles). In explaining the cycle of motives, let us first describe motivation. "Motivation refers to states within a person or animal that drive behaviour toward some goal." In this definition, there are three points that need emphasis;

Motivating state- (this state could also be described the 'drive.') It arouses the motivating behaviour. The arousal could start to function mainly through three factors:

- Internal bodily needs,
- External sensory stimulation, and
- Cognitive experiences

The behaviour that is aroused through the drive - this one is instrumental in satisfying our drive. This behaviour is a means to an end. It is going to settle the unsettled feelings in us. The goal (end) - depending on the drive that activates (incites) the present motivating state, this is the reduction of the drive. This one is an end through itself hence it is the satisfaction of the need we experienced at the first stage.

Depending on the motivating state that is moving us at a certain point in time, goals could be positive or negative. We approach positive goals and try to satisfy them. On

the other hand, we avoid or escape from negative goals that frustrate or endanger us. Both satisfy a certain motive though we approach the positive as pleasant and avoid the negative as unpleasant.

Theories of Motivation

- *Drive-reduction theory ('push' theory):* States that the reduction of biological drives, like hunger, thirst, oxygen, pain avoidance, ..., is pleasure producing. A drive is an internal states of tension that motivates (pushes) an organism to engage in activities that should reduce this tension. This theory is simple with several limitations since the source of pleasure is not always associated with biological drive reduction. But it has significance in explaining the fact that when we lack something that is significant for our survival, there is the 'drive' that motivates us to satisfy it.
- *Incentive ('pull') theory:* This is described 'pull' theory because the goal objects pull behaviour toward them. In contrast from the drive theory, this theory says that the response to what is motivating us at a certain point in time is valued more than anything in the environment. This is so because people expect pleasure from the attainment of what are described positive incentives, and from avoidance of negative incentives. So, the response is acting as a goal (incentive) to our motivating behaviour. Experiments on the proof of this theory show that organisms tend to like the goal they achieved when they really wanted it and when it is something they like more.
- *Arousal Theory:* This theory tells us in relation to the things arouse (excite) organisms to respond in a certain manner to their nearby. It says that both high and low arousal states do not create our performance effective. So, to become effective, it is good to be in an intermediate state of arousal.

There are four general sources of arousal;

- Physiological (internal) drives and incentives are arousing,
- Intensity of environmental stimuli affect our arousal,
- Surprising or new events arouse curiosity,
- Drugs arouse people to act in an excited manner.

Self-Actualization Theory

A. Maslow put what drive human beings from the bottom to the higher in order of hierarchy. The most significant ones are found at the base. Human beings necessity first satisfies their needs for survival – named *physiological needs*. After being able to eat, to drink, to sleep, ..., (after securing survival), people continue to care in relation to their.

Safety needs that are avoidance of accidents and painful sensations. They contain concern for long-term survival, job security, pension, buying insurance, saving money, and the like.

Love and belongingness (social) needs continue through having someone to love and through affiliating with others as a contributing member in the society. Here, an individual is able to identify his/her personal being.

Esteem (ego) needs are the fourth elements in the hierarchy that Maslow described human beings get motivated through. People want to report their success to get prestige. This is the feeling of self-worth through knowing that others are aware of one's competence and value.

At the top of Maslow's hierarchy comes the last of the motivating states for human beings, *i.e. self-actualization*. Through this we explore and understand our potentialities to the full and influence the world. This is the state of self-fulfillment. Getting motivated through the self-actualization needs ultimately goes in making us actualize who we really are. At this stage, people feel satisfied of their current state of affairs. This fifth need is highly influenced through the society we live in. This is because a urbanized society helps one get facilitated through the materials (the infrastructure) one can use in the process of actualizing his/her own self (inner potentialities), while less urbanized societies are not conducive places to satisfy this need. Of these five hierarchies, Maslow labels the last three as psychological states while the first two are basic physiological ones.

Biological Motives

Biological Motives to a large extent, rooted in the physique (body). Since they are innate and internal, their arousal is basically unlearned. Most of them necessity be met for survival.

They may show up in our behaviour starting from birth, or they may come in relation to the maturation. The most significant primary motives contain hunger, pain avoidance, a need for oxygen, sleep, elimination of wastes, and regulation of body temperature. Several of these motives are triggered through departures from balanced state of equilibrium described homeostasis in their internal physiological processes. Homeostasis, body equilibrium, works automatically to maintain balance among internal physiological circumstances.

Social Motives

These are motives that are learned and satisfied in the context of others. As they are the wellsprings of several of human actions, they are intricate motive states. These human motives can be looked upon as general states that lead to different kinds of scrupulous behaviors. Not only do they help to determine much of what a person does, they persist, never fully satisfied, over the years. No sooner is one goal reached than the motive is directed to another one. These motives are significant components of personality. As they are learned, they also have different strengths from one person to another.

Motives to Know and to be Effective

Motives to know and to be effective just like the physiological drives, these ones are also innate or unlearned. They are persistent and seem to exist to one degree or another in everyone. But their root is not the physique. Drives under this category highly contribute to the normal behavioral development of human beings. In some instances, there could be some interconnection flanked by the physiological needs and the ones in this classification.

These motives, though, are motives that drive a person to seek diversity in stimulation, to process information in relation to the nearby world, to explore, and to be effective in mastering challenges from the environment. So, they seek out sensory stimulation through continual interaction with the environment.

INTEGRITY INVENTORY

The *Integrity Inventory* (stylized as *I*), is a nationally normed entry-level personnel selection tool that incorporates employment integrity testing. It was developed by industrial organizational psychologist Fred Rafilson, Ph.D., CEO and Founder of I/O Solutions, Inc. or Industrial/Organizational Solutions Inc, referred to as IOS in the 2009 United States Supreme Court case, *Ricci v. DeStefano*. In the United States, vocations within the public safety sector, (*i.e.*, firefighter, sheriff and police officer, correctional officer, emergency medical services including emergency medical technician); and employment in for-profit retail and wholesale business, and non-profit sectors often require Industrial and Organizational Psychology employment testing, for initial employment and advancement throughout the ranks.

The Integrity Inventory, when used as a pre-employment screening tool, predicts individuals' likelihood of engaging in counterproductive workplace behaviors including assessing: ethics and moral character, work attitudes, theft attitudes, potential for substance abuse (*i.e.*, alcohol or drug use), emotional stability, turnover intentions, and/or behaviors that are hazardous and place the civilian population and coworkers at heightened risks. Qualified individuals with a reduced risk of such counterproductive workplace behaviors are selected for employment, leading to a more productive workforce. Counterproductive workplace behaviors lead to real financial losses for business; in the public safety and private sector, such behaviors are hazardous and place the civilian population and coworkers at heightened risk. As such, the prediction of counterproductive workplace behaviors constitutes a business necessity as outlined in the U.S. Equal Employment Opportunity Commission's Uniform Guidelines on Employee Selection Procedures. Psychological testing is a standard and growing practice for human resources departments throughout the United States who seek high integrity employees.

This exam has the sensitivity needed to accurately assess the individuals that will be charged with protecting our lives, families, property, and businesses. The Integrity

Inventory displays no adverse impact against protected classes and allows you to select and rank-order candidates most likely to succeed based on their integrity and honesty.

PROJECTIVE TEST

In psychology, a projective test is a personality test designed to let a person respond to ambiguous stimuli, presumably revealing hidden emotions and internal conflicts projected by the person into the test. This is sometimes contrasted with a so-called “objective test” or “self-report test” in which responses are analyzed according to a presumed universal standard (for example, a multiple choice exam), and are limited to the content of the test. The responses to projective tests are content analyzed for meaning rather than being based on presuppositions about meaning, as is the case with objective tests. Projective tests have their origins in psychoanalytic psychology, which argues that humans have conscious and unconscious attitudes and motivations that are beyond or hidden from conscious awareness.

THEORY

The general theoretical position behind projective tests is that whenever a specific question is asked, the response will be consciously-formulated and socially determined. These responses do not reflect the respondent’s unconscious or implicit attitudes or motivations. The respondent’s deep-seated motivations may not be consciously recognized by the respondent or the respondent may not be able to verbally express them in the form demanded by the questioner. Advocates of projective tests stress that the ambiguity of the stimuli presented within the tests allow subjects to express thoughts that originate on a deeper level than tapped by explicit questions, and provide content that may not be captured by responsive tools that may lack appropriate items. After some decrease in interest in the 1980s and 1990s, newer research suggesting that implicit motivation is best captured in this way has increased the research and use of these tools.

PROJECTIVE HYPOTHESIS

This holds that an individual puts structure on an ambiguous situation in a way that is consistent with their own conscious and unconscious needs. It is an indirect method—testee is talking about something that comes spontaneously from the self without conscious awareness or editing.

- Reduces temptation to fake
- Does not depend as much on verbal abilities
- Taps both conscious and unconscious traits
- Focus is clinical perspective - not normative - but has developed norms over the years.

COMMON VARIANTS

Rorschach Test

The Rorschach test is a psychological test in which subjects' perceptions of inkblots are recorded and then analyzed using psychological interpretation, complex algorithms, or both. Some psychologists use this test to examine a person's personality characteristics and emotional functioning. It has been employed to detect underlying thought disorder, especially in cases where patients are reluctant to describe their thinking processes openly. The test is named after its creator, Swiss psychologist Hermann Rorschach. In the 1960s, the Rorschach was the most widely used projective test.

Although the Exner Scoring System (developed since the 1960s) claims to have addressed and often refuted many criticisms of the original testing system with an extensive body of research, some researchers continue to raise questions. The areas of dispute include the objectivity of testers, inter-rater reliability, the verifiability and general validity of the test, bias of the test's pathology scales towards greater numbers of responses, the limited number of psychological conditions which it accurately diagnoses, the inability to replicate the test's norms, its use in court-ordered evaluations, and the proliferation of the ten inkblot images, potentially invalidating the test for those who have been exposed to them.

Using interpretation of "ambiguous designs" to assess an individual's personality is an idea that goes back to Leonardo da Vinci and Botticelli. Interpretation of inkblots was central to a game, *Gobolinks*, from the late 19th century. Rorschach's, however, was the first systematic approach of this kind. The ink blots were hand drawn by Rorschach. It has been suggested that Rorschach's use of inkblots may have been inspired by German doctor Justinus Kerner who, in 1857, had published a popular book of poems, each of which was inspired by an accidental inkblot. French psychologist Alfred Binet had also experimented with inkblots as a creativity test, and, after the turn of the century, psychological experiments where inkblots were utilized multiplied, with aims such as studying imagination and consciousness.

After studying 300 mental patients and 100 control subjects, in 1921 Rorschach wrote his book *Psychodiagnostik*, which was to form the basis of the inkblot test (after experimenting with several hundred inkblots, he selected a set of ten for their diagnostic value), but he died the following year. Although he had served as Vice President of the Swiss Psychoanalytic Society, Rorschach had difficulty in publishing the book and it attracted little attention when it first appeared.

In 1927, the newly founded Hans Huber publishing house purchased Rorschach's book *Psychodiagnostik* from the inventory of Ernst Bircher. Huber has remained the publisher of the test and related book, with Rorschach a registered trademark of Swiss publisher Verlag Hans Huber, Hogrefe AG. The work has been described as "a densely written piece couched in dry, scientific terminology".

After Rorschach's death, the original test scoring system was improved by Samuel Beck, Bruno Klopfer and others. John E. Exner summarized some of these later developments in the *comprehensive system*, at the same time trying to make the scoring more statistically rigorous. Some systems are based on the psychoanalytic concept of object relations. The Exner system remains very popular in the United States, while in Europe other methods sometimes dominate, such as that described in the textbook by Evald Bohm, which is closer to the original Rorschach system and rooted more deeply in the original psychoanalysis principles.

Rorschach never intended the inkblots to be used as a general personality test, but developed them as a tool for the diagnosis of schizophrenia. It was not until 1939 that the test was used as a projective test of personality, a use of which Rorschach had always been skeptical. Interviewed in 2012 for a BBC Radio 4 documentary, Rita Signer, curator of the Rorschach Archives in Bern, Switzerland, suggested that far from being random or chance designs, each of the blots selected by Rorschach for his test had been meticulously designed to be as ambiguous and "conflicted" as possible.

Method

The Rorschach test is appropriate for subjects from the age of five to adulthood. The administrator and subject typically sit next to each other at a table, with the administrator slightly behind the subject. Side-by-side seating of the examiner and the subject is used to reduce any effects of inadvertent cues from the examiner to the subject. In other words, side-by-side seating mitigates the possibility that the examiner will accidentally influence the subject's responses. This is to facilitate a "relaxed but controlled atmosphere". There are ten official inkblots, each printed on a separate white card, approximately 18 by 24 cm in size. Each of the blots has near perfect bilateral symmetry. Five inkblots are of black ink, two are of black and red ink and three are multicolored, on a white background. After the test subject has seen and responded to all of the inkblots (*free association* phase), the tester then presents them again one at a time in a set sequence for the subject to study: the subject is asked to note where he sees what he originally saw and what makes it look like that (*inquiry* phase). The subject is usually asked to hold the cards and may rotate them. Whether the cards are rotated, and other related factors such as whether permission to rotate them is asked, may expose personality traits and normally contributes to the assessment. As the subject is examining the inkblots, the psychologist writes down everything the subject says or does, no matter how trivial. Analysis of responses is recorded by the test administrator using a tabulation and scoring sheet and, if required, a separate location chart.

The general goal of the test is to provide data about cognition and personality variables such as motivations, response tendencies, cognitive operations, affectivity, and personal/interpersonal perceptions. The underlying assumption is that an individual will class external stimuli based on person-specific perceptual sets, and including needs, base motives, conflicts, and that this clustering process is representative of the process used in real-life

situations. Methods of interpretation differ. Rorschach scoring systems have been described as a system of pegs on which to hang one's knowledge of personality. The most widely used method in the United States is based on the work of Exner. Administration of the test to a group of subjects, by means of projected images, has also occasionally been performed, but mainly for research rather than diagnostic purposes.

Test administration is not to be confused with test interpretation:

- The interpretation of a Rorschach record is a complex process. It requires a wealth of knowledge concerning personality dynamics generally as well as considerable experience with the Rorschach method specifically. Proficiency as a Rorschach *administrator* can be gained within a few months. However, even those who are able and qualified to become Rorschach *interpreters* usually remain in a "learning stage" for a number of years.

Features or Categories

The interpretation of the Rorschach test is not based primarily on the contents of the response, *i.e.*, *what* the individual sees in the inkblot (the *content*). In fact, the contents of the response are only a comparatively small portion of a broader cluster of variables that are used to interpret the Rorschach data: for instance, information is provided by the time taken before providing a response for a card can be significant (taking a long time can indicate "shock" on the card), as well as by any comments the subject may make in addition to providing a direct response. In particular, information about *determinants* (the aspects of the inkblots that triggered the response, such as form and colour) and *location* (which details of the inkblots triggered the response) is often considered more important than content, although there is contrasting evidence. "Popularity" and "originality" of responses can also be considered as basic dimensions in the analysis.

- *Content:* The goal in coding content of the Rorschach is to categorize the objects that the subject describes in response to the inkblot. There are 27 established codes for identifying the name of the descriptive object. The codes are classified and include terms such as "human", "nature", "animal", "abstract", "clothing", "fire", and "x-ray", to name a few. Content described that does not have a code already established should be coded using the code "idiographic contents" with the shorthand code being "Idio." Items are also coded for statistical popularity (or, conversely, originality). More than any other feature in the test, content response can be controlled consciously by the subject, and may be elicited by very disparate factors, which makes it difficult to use content alone to draw any conclusions about the subject's personality; with certain individuals, content responses may potentially be interpreted directly, and some information can at times be obtained by analyzing thematic trends in the whole set of content responses (which is only feasible when several responses are available), but in general content cannot be analyzed outside of the context of the entire test record.

- *Location:* Identifying the location of the subject's response is another element scored in the Rorschach system. Location refers to how much of the inkblot was used to answer the question. Administrators score the response "W" if the whole inkblot was used to answer the question, "D" if a commonly described part of the blot was used, "Dd" if an uncommonly described or unusual detail was used, or "S" if the white space in the background was used. A score of W is typically associated with the subject's motivation to interact with his or her surrounding environment. D is interpreted as one having efficient or adequate functioning. A high frequency of responses coded Dd indicate some maladjustment within the individual. Responses coded S indicate an oppositional or uncooperative test subject.
- *Determinants:* Systems for Rorschach scoring generally include a concept of "determinants": These are the factors that contribute to establishing the similarity between the inkblot and the subject's content response about it. They can also represent certain basic experiential-perceptual attitudes, showing aspects of the way a subject perceives the world. Rorschach's original work used only *form*, *colour* and *movement* as determinants. However currently, another major determinant considered is *shading*, which was inadvertently introduced by poor printing quality of the inkblots. Rorschach initially disregarded shading, since the inkblots originally featured uniform saturation, but later recognized it as a significant factor. *Form* is the most common determinant, and is related to intellectual processes. *Colour* responses often provide direct insight into one's emotional life. *Movement* and *shading* have been considered more ambiguously, both in definition and interpretation. Rorschach considered *movement* only as the experiencing of actual motion, while others have widened the scope of this determinant, taking it to mean that the subject sees something "going on". More than one determinant can contribute to the formation of the subject's perception. Fusion of two determinants is taken into account, while also assessing which of the two constituted the primary contributor. For example, "*form-colour*" implies a more refined control of impulse than "*colour-form*". It is, indeed, from the relation and balance among determinants that personality can be most readily inferred.

Symmetry of the Test Items

A striking characteristic of the Rorschach inkblots is their symmetry. Many unquestionably accept this aspect of the nature of the images but Rorschach, as well as other researchers, certainly did not. Rorschach experimented with both asymmetric and symmetric images before finally opting for the latter.

He gives this explanation for the decision:

- Asymmetric figures are rejected by many subjects; symmetry supplied part of the necessary artistic composition. It has a disadvantage in that it tends to

make answers somewhat stereotyped. On the other hand, symmetry makes conditions the same for right and left handed subjects; furthermore, it facilitates interpretation for certain blocked subjects. Finally, symmetry makes possible the interpretation of whole scenes.

The impact of symmetry in the Rorschach inkblot's has also been investigated further by other researchers.

Exner Scoring System

The *Exner scoring system*, also known as the *Rorschach Comprehensive System* (RCS), is the standard method for interpreting the Rorschach test. It was developed in the 1960s by Dr. John E. Exner, as a more rigorous system of analysis. It has been extensively validated and shows high inter-rater reliability. In 1969, Exner published *The Rorschach Systems*, a concise description of what would be later called “the Exner system”. He later published a study in multiple volumes called *The Rorschach: A Comprehensive system*, the most accepted full description of his system.

Creation of the new system was prompted by the realization that at least five related, but ultimately different methods were in common use at the time, with a sizeable minority of examiners not employing any recognized method at all, basing instead their judgment on subjective assessment, or arbitrarily mixing characteristics of the various standardized systems.

The key components of the Exner system are the clusterization of Rorschach variables and a sequential search strategy to determine the order in which to analyze them, framed in the context of standardized administration, objective, reliable coding and a representative normative database. The system places a lot of emphasis on a cognitive triad of *information processing*, related to how the subject processes input data, *cognitive mediation*, referring to the way information is transformed and identified, and *ideation*. In the system, responses are scored with reference to their level of vagueness or synthesis of multiple images in the blot, the location of the response, which of a variety of determinants is used to produce the response (*i.e.*, what makes the inkblot look like what it is said to resemble), the form quality of the response (to what extent a response is faithful to how the actual inkblot looks), the contents of the response (what the respondent actually sees in the blot), the degree of mental organizing activity that is involved in producing the response, and any illogical, incongruous, or incoherent aspects of responses. It has been reported that popular responses on the first card include bat, badge and coat of arms.

Using the scores for these categories, the examiner then performs a series of calculations producing a structural summary of the test data. The results of the structural summary are interpreted using existing research data on personality characteristics that have been demonstrated to be associated with different kinds of responses.

With the Rorschach plates (the ten inkblots), the area of each blot which is distinguished by the client is noted and coded—typically as “commonly selected” or

“uncommonly selected”. There were many different methods for coding the areas of the blots. Exner settled upon the area coding system promoted by S. J. Beck (1944 and 1961). This system was in turn based upon Klopfer’s (1942) work.

As pertains to response form, a concept of “form quality” was present from the earliest of Rorschach’s works, as a subjective judgment of how well the form of the subject’s response matched the inkblots (Rorschach would give a higher form score to more “original” yet good form responses), and this concept was followed by other methods, especially in Europe; in contrast, the Exner system solely defines “good form” as a matter of word occurrence frequency, reducing it to a measure of the subject’s distance to the population average.

Rorschach Performance Assessment System

Rorschach performance assessment system (R-PAS) is a scoring method created by several members of the Rorschach Research Council. They believed that the Exner scoring system was in need of an update, but after Exner’s death, the Exner family forbade any changes to be made to the Comprehensive System. Therefore, they established a new system: the R-PAS. It is an attempt at creating a current, empirically based, and internationally focused scoring system that is easier to use than Exner’s Comprehensive System. The R-PAS manual is intended to be a comprehensive tool for administering, scoring, and interpreting the Rorschach. The manual consists of two chapters that are basics of scoring and interpretation, aimed for use for novice Rorschach users, followed by numerous chapters containing more detailed and technical information. In terms of updated scoring, the authors only selected variables that have been empirically supported in the literature. To note, the authors did not create new variables or indices to be coded, but systematically reviewed variables that had been used in past systems. While all of these codes have been used in the past, many have been renamed to be more face valid and readily understood. Scoring of the indices has been updated (*e.g.* utilizing percentiles and standard scores) to make the Rorschach more in line with other popular personality measures.

In addition to providing coding guidelines to score examinee responses, the R-PAS provides a system to code an examinee’s behaviour during Rorschach administration. These behavioral codes are included as it is believed that the behaviors exhibited during testing are a reflection of someone’s task performance and supplements the actual responses given.

This allows generalizations to be made between someone’s responses to the cards and their actual behaviour.

Cultural Differences

Comparing North American Exner normative data with data from European and South American subjects showed marked differences in some features, some of which

impact important variables, while others (such as the average number of responses) coincide. For instance, texture response is typically zero in European subjects (if interpreted as a need for closeness, in accordance with the system, a European would seem to express it only when it reaches the level of a *craving* for closeness), and there are fewer “good form” responses, to the point where schizophrenia may be suspected if data were correlated to the North American norms. Form is also often the only determinant expressed by European subjects; while colour is less frequent than in American subjects, colour-form responses are comparatively frequent in opposition to form-colour responses; since the latter tend to be interpreted as indicators of a defensive attitude in processing affect, this difference could stem from a higher value attributed to spontaneous expression of emotions.

The differences in form quality are attributable to purely cultural aspects: different cultures will exhibit different “common” objects (French subjects often identify a chameleon in card VIII, which is normally classed as an “unusual” response, as opposed to other animals like cats and dogs; in Scandinavia, “Christmas elves” (*nisser*) is a popular response for card II, and “musical instrument” on card VI is popular for Japanese people), and different languages will exhibit semantic differences in naming the same object (the figure of card IV is often called a *troll* by Scandinavians and an *ogre* by French people).

Many of Exner’s “popular” responses (those given by at least one third of the North American sample used) seem to be universally popular, as shown by samples in Europe, Japan and South America, while specifically card IX’s “human” response, the crab or spider in card X and one of either the butterfly or the bat in card I appear to be characteristic of North America. Form quality, popular content responses and locations are the only coded variables in the Exner systems that are based on frequency of occurrence, and thus immediately subject to cultural influences; therefore, cultural-dependent interpretation of test data may not necessarily need to extend beyond these components.

The cited language differences mean that it’s imperative for the test to be administered in the subject’s native language or a very well mastered second language, and, conversely, the examiner should master the language used in the test. Test responses should also not be translated into another language prior to analysis except possibly by a clinician mastering both languages.

For example, a bow tie is a frequent response for the center detail of card III, but since the equivalent term in French translates to “butterfly tie”, an examiner not appreciating this language nuance may code the response differently from what is expected.

Motivation in Educational Psychology

INTRODUCTION

Education must transfer from generation to generation the core of our culture's accumulated body of knowledge. In our day, many think that to believe in an accepted body of knowledge that prioritizes what is important to learn and what is not smacks of elitism and exclusivity. In part, this charge cannot be denied because discernment often demands that we play the role of intellectual hatchet-men; however, if you will reject the notion of a "canon" of knowledge, you are faced with the task of creating a rationale for your own curriculum that can give a convincing answer to that most awkward but ubiquitous question, "Why do we have to learn this?"

Having cut themselves free of the constraints that guide classical education, our large educational institutions have resorted to an ever increasingly frantic attempt to construct a convincing rationale for their methodology out of the vacuum of their own errant psyches. Yet, whether they resort to ethnicity, technology, gender or the deviancy of pop culture, their attempts to give meaning to their teaching end up being mere exercises in personal assertiveness.

Even though our secular school system abandoned any true foundation for absolutes by rejecting God's authority over all of life, academia has been able to keep a coherent

system together by merely coasting on the inertia built up by our culture's rich intellectual tradition. This has been especially true in Mathematics where a healthy regard for disciplined rigorous thinking and applied mental struggle were thought to guarantee mental fruit, but now, even this field seems vulnerable to the prevailing epistemological meltdown. The "new new math" curricula currently being promoted by the National Council of Teachers of Mathematics seems bent on destroying mathematical rigour in hopes of instilling students with such dubiously mathematical abilities as "group learning skills", "guess and check techniques" and "awareness of diverse cultural approaches to mathematics". By attempting to make math more "interesting" and "accessible", they have removed precisely what makes Mathematics a joy to learn. No longer will students be able to savor the hard won pleasure of successfully working complex mathematical procedures. Even in the field of Mathematics where you think it would be nearly impossible for the mind to be led astray, our schools are abandoning rigour and sense for mere gimmicks and social agendas. What is the philosophical lesson from all this? The closer and closer man comes to pure epistemological autonomy (that is, the ability to accept only those ideas he finds within himself) the closer he comes to resigning himself to complete nonsense. For students to be motivated in their studies, they need to know that what they are studying is indeed of real significance. They need to know that they are not being feed some new-sprung agenda or half-baked innovation that will simply go the way of the faddish educational chaff that, once having gleaned its profits, goes to the winds never to be seen or thought of again. Students need to know that they are being feed the best that our civilization has to offer- that they are studying something that is much, much larger than themselves. As I guide students through the study of the proofs in Euclid's Elements, it is always a pleasure to point out to them the fact that their geometry book is the same that was used by Thomas Jefferson, James Garfield, Lewis Carroll and a host of other intellectual witnesses going back before 200 BC. Even though Euclid makes absolutely no attempt to show you how his system is "practical", I do not find my students asking, "what will I ever us this for?" Because the Element's is truly a classic work, the students come to see why mathematicians have admire Euclid's beautifully constructed proofs through the ages.

When we climb out of the broad stream that comprises the wisdom of the ages, it is very easy to lose our educational bearings, being blown to and fro be the winds of opinion. Furthermore, without a good rational for our curriculum, as suggested, be unable to resist the student's desire to find the path of least exertion between now and breaktime. To be motivated to work, we need to instill in our children first godly character and then the conviction that their studies are indeed significant. Despite the mantras that are continually chanted around us, the motivation for pursuing an education does not come from looking at charts of the average salary levels of various degree recipients, or from following the educational atomism that reduces all educational accomplishment to a single GPA, or by explaining all labours as just steps in the great "ordu salutis" culminating in acceptance by that Ivy-league dream college. By putting before students

these poor reasons for getting an education we are drumming into them the idea that education is a means and not an end. Until they understand that education is an end in itself, that indeed, the creation in which we dwell and the historical saga in which we take part are truly worthy of our interest and concentrated study, as suggested, only see them labour with a slave's reluctance.

Kind of Motives

A MOTIVE may be defined as meaning that which acts as an inducement to preference or choice. 'In other words it is a very strong influence toward some object to be attained. Just what these influences are depends entirely upon the nature of the person and on the object to be attained. We often hear the expression, and doubtless use it as often ourselves, "What was your motive in doing this?"' meaning what caused you or what influenced you to act in such a manner. We may state with truth that every wilful act has its inducement or cause. Just as in mathematics there is a reason for every step, so also in every act there is a motive or cause. From the child on up the individual can give a reason for each and every one of his voluntary acts however small and mean the reason may be. Still it is a reason. It may only be a certain state of consciousness which happens to be uppermost in a man's mind as he acts. Very often Intention and Motive are confused; in fact they are used interchangeably in society. Notwithstanding this fact there is a vast difference in their meaning. Intention is what a man means to do, while Motive is the personal frame of mind which indicates why he means to do it. Therefore Intention has the stronger moral value, while Motive renders the consequences interesting and attractive. Motive is that which makes the difference between one act and another.

As we now know the difference between Intention and Motive, let us examine the different kinds of Motives. Motive may be divided into two classes, Egoistical and Altruistical. Both classes are absolutely necessary in our daily life. Let us first look at egoistic motives or the ones which concern the self only. Our original instincts are such that their objects are to look after the advantages of the self. However everything depends upon the sort of self maintained. What would become of a person if he would not struggle for food and strive against obstacles? What would become of society? Self-preserving instincts must be therefore socially conservative. No one has a right to neglect his own interests, hoping some one else will care for them. The man who takes exercise because he thinks of his health must be commanded, but the one who is thinking continually of his health and excludes other thoughts must be condemned.

THE HIERARCHAL MODEL OF ACHIEVEMENT MOTIVATION

Achievement motivation has been conceptualized in many different ways. Our understanding of achievement-relevant effects, cognition, and behaviour has improved.

Despite being similar in nature, many achievement motivation approaches have been developed separately, suggesting that most achievement motivation theories are in concordance with one another instead of competing. Motivational researchers have sought to promote a hierarchical model of approach and avoidance achievement motivation by incorporating the two prominent theories: the achievement motive approach and the achievement goal approach. Achievement motives include the need for achievement and the fear of failure. These are the more predominant motives that direct our behaviour toward positive and negative outcomes. Achievement goals are viewed as more solid cognitive representations pointing individuals toward a specific end. There are three types of these achievement goals: a performance-approach goal, a performance-avoidance goal, and a mastery goal. A performance-approach goal is focused on attaining competence relative to others, a performance-avoidance goal is focused on avoiding incompetence relative to others, and a mastery goal is focused on the development of competence itself and of task mastery. Achievement motives can be seen as direct predictors of achievement-relevant circumstances. Thus, achievement motives are said to have an indirect or distal influence, and achievement goals are said to have a direct or proximal influence on achievement-relevant outcomes.

These motives and goals are viewed as working together to regulate achievement behaviour. The hierarchical model presents achievement goals as predictors for performance outcomes. The model is being further conceptualized to include more approaches to achievement motivation. One weakness of the model is that it does not provide an account of the processes responsible for the link between achievement goals and performance. As this model is enhanced, it becomes more useful in predicting the outcomes of achievement-based behaviours.

IMPLICIT AND SELF-ATTRIBUTED MOTIVES

Motivational researchers share the view that achievement behaviour is an interaction between situational variables and the individual subject's motivation to achieve. Two motives are directly involved in the prediction of behaviour, implicit and explicit. Implicit motives are spontaneous impulses to act, also known as task performances, and are aroused through incentives inherent to the task. Explicit motives are expressed through deliberate choices and more often stimulated for extrinsic reasons. Also, individuals with strong implicit needs to achieve goals set higher internal standards, whereas others tend to adhere to the societal norms. These two motives often work together to determine the behaviour of the individual in direction and passion.

Explicit and implicit motivations have a compelling impact on behaviour. Task behaviours are accelerated in the face of a challenge through implicit motivation, making performing a task in the most effective manner the primary goal. A person with a strong implicit drive will feel pleasure from achieving a goal in the most efficient way. The increase in effort and overcoming the challenge by mastering the task satisfies the individual. However, the explicit motives are built around a person's self-image. This

type of motivation shapes a person's behaviour based on their own self-view and can influence their choices and responses from outside cues. The primary agent for this type of motivation is perception or perceived ability. Many theorists still can not agree whether achievement is based on mastering one's skills or striving to promote a better self-image. Most research is still unable to determine whether these different types of motivation would result in different behaviours in the same environment.

Concepts of Motivation

Intrinsic motivation refers to motivation that is driven by an interest or enjoyment in the task itself, and exists within the individual rather than relying on any external pressure. Intrinsic motivation has been studied by social and educational psychologists since the early 1970s. Research has found that it is usually associated with high educational achievement and enjoyment by students. Explanations of intrinsic motivation have been given in the context of Fritz Heider's attribution theory, Bandura's work on self-efficacy, and Deci and Ryan's cognitive evaluation theory. Students are likely to be intrinsically motivated if they:

- Attribute their educational results to internal factors that they can control (e.g. the amount of effort they put in),
- Believe they can be effective agents in reaching desired goals (i.e. the results are not determined by luck),
- Are interested in mastering a topic, rather than just rote-learning to achieve good grades.

Extrinsic motivation comes from outside of the individual. Common extrinsic motivations are rewards like money and grades, coercion and threat of punishment. Competition is in general extrinsic because it encourages the performer to win and beat others, not to enjoy the intrinsic rewards of the activity. A crowd cheering on the individual and trophies are also extrinsic incentives. Social psychological research has indicated that extrinsic rewards can lead to overjustification and a subsequent reduction in intrinsic motivation. In one study demonstrating this effect, children who expected to be (and were) rewarded with a ribbon and a gold star for drawing pictures spent less time playing with the drawing materials in subsequent observations than children who were assigned to an unexpected reward condition and to children who received no extrinsic reward. Self-determination theory proposes that extrinsic motivation can be internalised by the individual if the task fits with their values and beliefs and therefore helps to fulfil their basic psychological needs.

Self-control

The self-control of motivation is increasingly understood as a subset of emotional intelligence; a person may be highly intelligent according to a more conservative definition (as measured by many intelligence tests), yet unmotivated to dedicate this

intelligence to certain tasks. Yale School of Management professor Victor Vroom's "expectancy theory" provides an account of when people will decide whether to exert self control to pursue a particular goal.

Drives and desires can be described as *a deficiency or need that activates behaviour that is aimed at a goal or an incentive*. These are thought to originate within the individual and may not require external stimuli to encourage the behaviour. Basic drives could be sparked by deficiencies such as hunger, which motivates a person to seek food; whereas more subtle drives might be the desire for praise and approval, which motivates a person to behave in a manner pleasing to others.

By contrast, the role of extrinsic rewards and stimuli can be seen in the example of training animals by giving them treats when they perform a trick correctly. The treat motivates the animals to perform the trick consistently, even later when the treat is removed from the process.

Theories of Motivation

Ever wonder why some people seem to be very successful, highly motivated individuals? Where does the energy, the drive, or the direction come from? Motivation is an area of psychology that has gotten a great deal of attention, especially in the recent years. The reason is because we all want to be successful, we all want direction and drive, and we all want to be seen as motivated. There are several distinct theories of motivation we will discuss in this section. Some include basic biological forces, while others seem to transcend concrete explanation. Let's talk about the five major theories of motivation.

Instinct Theory

Instinct theory is derived from our biological make-up. We've all seen spider's webs and perhaps even witnessed a spider in the tedious job of creating its home and trap. We've all seen birds in their nests, feeding their young or painstakingly placing the twigs in place to form their new home. How do spiders know how to spin webs? How do birds now how to build nests?

The answer is biology. All creatures are born with specific innate knowledge about how to survive. Animals are born with the capacity and often times knowledge of how to survive by spinning webs, building nests, avoiding danger, and reproducing. These innate tendencies are preprogrammed at birth, they are in our genes, and even if the spider never saw a web before, never witnessed its creation, it would still know how to create one. Humans have the same types of innate tendencies. Babies are born with a unique ability that allows them to survive; they are born with the ability to cry. Without this, how would others know when to feed the baby, know when he needed changing, or when she wanted attention and affection? Crying allows a human infant to survive. We are also born with particular reflexes which promote survival. The most important of

these include sucking, swallowing, coughing, blinking. Newborns can perform physical movements to avoid pain; they will turn their head if touched on their cheek and search for a nipple (rooting reflex); and they will grasp an object that touches the palm of their hands.

Drive Reduction Theory

According to Clark Hull (1943, 1952), humans have internal biological needs which motivate us to perform a certain way. These needs, or drives, are defined by Hull as internal states of arousal or tension which must be reduced. A prime example would be the internal feelings of hunger or thirst, which motivates us to eat. According to this theory, we are driven to reduce these drives so that we may maintain a sense of internal calmness.

Arousal Theory

Similar to Hull's Drive Reduction Theory, Arousal theory states that we are driven to maintain a certain level of arousal in order to feel comfortable. Arousal refers to a state of emotional, intellectual, and physical activity. It is different from the above theory, however, because it doesn't rely on only a reduction of tension, but a balanced amount. It also does better to explain why people climb mountains, go to school, or watch sad movies.

Psychoanalytic Theory

Remember Sigmund Freud and his five part theory of personality. As part of this theory, he believed that humans have only two basic drives: Eros and Thanatos, or the Life and Death drives. According to Psychoanalytic theory, everything we do, every thought we have, and every emotion we experience has one of two goals: to help us survive or to prevent our destruction. This is similar to instinct theory, however, Freud believed that the vast majority of our knowledge about these drives is buried in the unconscious part of the mind. Psychoanalytic theory therefore argues that we go to school because it will help assure our survival in terms of improved finances, more money for healthcare, or even an improved ability to find a spouse. We move to better school districts to improve our children's ability to survive and continue our family tree. We demand safety in our cars, toys, and in our homes. We want criminal locked away, and we want to be protected against poisons, terrorists, and any thing else that could lead to our destruction. According to this theory, everything we do, everything we are can be traced back to the two basic drives

Humanistic Theory

Although discussed last, humanistic theory is perhaps the most well know theory of motivation. According to this theory, humans are driven to achieve their maximum

potential and will always do so unless obstacles are placed in their way. These obstacles include hunger, thirst, financial problems, safety issues, or anything else that takes our focus away from maximum psychological growth.

The best way to describe this theory is to utilize the famous pyramid developed by Abraham Maslow (1970) called the Hierarchy of Needs. Maslow believed that humans have specific needs that must be met and that if lower level needs go unmet, we can not possible strive for higher level needs. The Hierarchy of Needs shows that at the lower level, we must focus on basic issues such as food, sleep, and safety. Without food, without sleep, how could we possible focus on the higher level needs such as respect, education, and recognition?

Motivation and Learning

Basic principles of motivation exist that are applicable to learning in any situation.

1. The environment can be used to focus the student's attention on what needs to be learned.

Teachers who create warm and accepting yet businesslike atmospheres will promote persistent effort and favourable attitudes toward learning. This strategy will be successful in children and in adults. Interesting visual aids, such as booklets, posters, or practice equipment, motivate learners by capturing their attention and curiosity.

2. Incentives motivate learning.

Incentives include privileges and receiving praise from the instructor. The instructor determines an incentive that is likely to motivate an individual at a particular time. In a general learning situation, self-motivation without rewards will not succeed. Students must find satisfaction in learning based on the understanding that the goals are useful to them or, less commonly, based on the pure enjoyment of exploring new things.

3. Internal motivation is longer lasting and more self-directive than is external motivation, which must be repeatedly reinforced by praise or concrete rewards. Some individuals — particularly children of certain ages and some adults — have little capacity for internal motivation and must be guided and reinforced constantly. The use of incentives is based on the principle that learning occurs more effectively when the student experiences feelings of satisfaction. Caution should be exercised in using external rewards when they are not absolutely necessary. Their use may be followed by a decline in internal motivation.

4. Learning is most effective when an individual is ready to learn, that is, when one wants to know something.

Sometimes the student's readiness to learn comes with time, and the instructor's role is to encourage its development. If a desired change in behaviour is urgent, the instructor may need to supervised directly to ensure that the desired behaviour occurs. If a student is not *ready to learn*, he or she

may not be reliable in following instructions and therefore must be supervised and have the instructions repeated again and again.

5. Motivation is enhanced by the way in which the instructional material is organized.

In general, the best organized material makes the information meaningful to the individual. One method of organization includes relating new tasks to those already known. Other ways to relay meaning are to determine whether the persons being taught understand the final outcome desired and instruct them to compare and contrast ideas.

None of the techniques will produce sustained motivation unless the goals are realistic for the learner. The basic learning principle involved is that *success is more predictably motivating than is failure*. Ordinarily, people will choose activities of intermediate uncertainty rather than those that are difficult (little likelihood of success) or easy (high probability of success). For goals of high value there is less tendency to choose more difficult conditions. Having learners assist in defining goals increases the probability that they will understand them and want to reach them. However, students sometimes have unrealistic notions about what they can accomplish. Possibly they do not understand the precision with which a skill must be carried out or have the depth of knowledge to master some material. To identify realistic goals, instructors must be skilled in assessing a student's readiness or a student's progress toward goals.

1. Because learning requires changed in beliefs and behaviour, it normally produces a mild level of anxiety.

This is useful in motivating the individual. However, severe anxiety is incapacitating. A high degree of stress is inherent in some educational situations. If anxiety is severe, the individual's perception of what is going on around him or her is limited. Instructors must be able to identify anxiety and understand its effect on learning. They also have a responsibility to avoid causing severe anxiety in learners by setting ambiguous or unrealistically high goals for them.

2. It is important to help each student set goals and to provide informative feedback regarding progress toward the goals.

Setting a goal demonstrates an intention to achieve and activates learning from one day to the next. It also directs the student's activities toward the goal and offers an opportunity to experience success.

3. Both affiliation and approval are strong motivators.

People seek others with whom to compare their abilities, opinions, and emotions. Affiliation can also result in direct anxiety reduction by the social acceptance and the mere presence of others. However, these motivators can also lead to conformity, competition, and other behaviours that may seem as negative.

4. Many behaviours result from a combination of motives.

It is recognized that no grand theory of motivation exists. However, motivation is so necessary for learning that strategies should be planned to organize a continuous and interactive motivational dynamic for maximum effectiveness. The general principles of motivation are interrelated. A single teaching action can use many of them simultaneously.

Finally, it should be said that an enormous gap exists between knowing that learning must be motivated and identifying the specific motivational components of any particular act. Instructors must focus on learning patterns of motivation for an individual or group, with the realization that errors will be common.

Motivation Activates and Behaviour

Motivation is an internal state that activates, guides and sustains behaviour. Educational psychology research on motivation is concerned with the volition or will that students bring to a task, their level of interest and intrinsic motivation, the personally held goals that guide their behaviour, and their belief about the causes of their success or failure. As intrinsic motivation deals with activities that act as their own rewards, extrinsic motivation deals with motivations that are brought on by consequences or punishments. A form of attribution theory developed by Bernard Weiner describes how students' beliefs about the causes of academic success or failure affect their emotions and motivations. For example, when students attribute failure to lack of ability, and ability is perceived as uncontrollable, they experience the emotions of shame and embarrassment and consequently decrease effort and show poorer performance. In contrast, when students attribute failure to lack of effort, and effort is perceived as controllable, they experience the emotion of guilt and consequently increase effort and show improved performance.

Motivational theories also explain how learners' goals affect the way they engage with academic tasks. Those who have mastery goals strive to increase their ability and knowledge. Those who have performance approach goals strive for high grades and seek opportunities to demonstrate their abilities. Those who have performance avoidance goals are driven by fear of failure and avoid situations where their abilities are exposed. Research has found that mastery goals are associated with many positive outcomes such as persistence in the face of failure, preference for challenging tasks, creativity and intrinsic motivation. Performance avoidance goals are associated with negative outcomes such as poor concentration while studying, disorganized studying, less self-regulation, shallow information processing and test anxiety. Performance approach goals are associated with positive outcomes, and some negative outcomes such as an unwillingness to seek help and shallow information processing.

Locus of control is a salient factor in the successful academic performance of students. During the 1970s and '80s, Cassandra B. Whyte did significant educational research studying locus of control as related to the academic achievement of students pursuing higher education coursework. Much of her educational research and publications focused

upon the theories of Julian B. Rotter in regard to the importance of internal control and successful academic performance. Whyte reported that individuals who perceive and believe that their hard work may lead to more successful academic outcomes, instead of depending on luck or fate, persist and achieve academically at a higher level. Therefore, it is important to provide education and counseling in this regard.

Achievement Motivation on Behaviour

Motivation can be defined as the driving force behind all the actions of an individual. The influence of an individual's needs and desires both have a strong impact on the direction of their behaviour. Motivation is based on your emotions and achievement-related goals. There are different forms of motivation including extrinsic, intrinsic, physiological, and achievement motivation. There are also more negative forms of motivation. Achievement motivation can be defined as the need for success or the attainment of excellence. Individuals will satisfy their needs through different means, and are driven to succeed for varying reasons both internal and external.

Motivation is the basic drive for all of our actions. Motivation refers to the dynamics of our behaviour, which involves our needs, desires, and ambitions in life. Achievement motivation is based on reaching success and achieving all of our aspirations in life. Achievement goals can affect the way a person performs a task and represent a desire to show competence (Harackiewicz, Barron, Carter, Lehto, & Elliot, 1997). These basic physiological motivational drives affect our natural behaviour in different environments. Most of our goals are incentive-based and can vary from basic hunger to the need for love and the establishment of mature sexual relationships. Our motives for achievement can range from biological needs to satisfying creative desires or realizing success in competitive ventures. Motivation is important because it affects our lives everyday. All of our behaviors, actions, thoughts, and beliefs are influenced by our inner drive to succeed.

Implicit and Self-Attributed Motives

Motivational researchers share the view that achievement behaviour is an interaction between situational variables and the individual subject's motivation to achieve. Two motives are directly involved in the prediction of behaviour, implicit and explicit. Implicit motives are spontaneous impulses to act, also known as task performances, and are aroused through incentives inherent to the task. Explicit motives are expressed through deliberate choices and more often stimulated for extrinsic reasons. Also, individuals with strong implicit needs to achieve goals set higher internal standards, whereas others tend to adhere to the societal norms. These two motives often work together to determine the behaviour of the individual in direction and passion (Brunstein & Maier, 2005).

Explicit and implicit motivations have a compelling impact on behaviour. Task behaviors are accelerated in the face of a challenge through implicit motivation, making

performing a task in the most effective manner the primary goal. A person with a strong implicit drive will feel pleasure from achieving a goal in the most efficient way. The increase in effort and overcoming the challenge by mastering the task satisfies the individual. However, the explicit motives are built around a person's self-image. This type of motivation shapes a person's behaviour based on their own self-view and can influence their choices and responses from outside cues. The primary agent for this type of motivation is perception or perceived ability. Many theorists still can not agree whether achievement is based on mastering one's skills or striving to promote a better self-image (Brunstein & Maier, 2005). Most research is still unable to determine whether these different types of motivation would result in different behaviors in the same environment.

The Hierarchal Model of Achievement Motivation

Achievement motivation has been conceptualized in many different ways. Our understanding of achievement-relevant effects, cognition, and behaviour has improved. Despite being similar in nature, many achievement motivation approaches have been developed separately, suggesting that most achievement motivation theories are in concordance with one another instead of competing. Motivational researchers have sought to promote a hierarchal model of approach and avoidance achievement motivation by incorporating the two prominent theories: the achievement motive approach and the achievement goal approach.

Achievement motives include the need for achievement and the fear of failure. These are the more predominant motives that direct our behaviour toward positive and negative outcomes. Achievement goals are viewed as more solid cognitive representations pointing individuals toward a specific end.

There are three types of these achievement goals: a performance-approach goal, a performance-avoidance goal, and a mastery goal. A performance-approach goal is focused on attaining competence relative to others, a performance-avoidance goal is focused on avoiding incompetence relative to others, and a mastery goal is focused on the development of competence itself and of task mastery. Achievement motives can be seen as direct predictors of achievement-relevant circumstances. Thus, achievement motives are said to have an indirect or distal influence, and achievement goals are said to have a direct or proximal influence on achievement-relevant outcomes (Elliot & McGregor, 1999).

These motives and goals are viewed as working together to regulate achievement behaviour. The hierarchal model presents achievement goals as predictors for performance outcomes. The model is being further conceptualized to include more approaches to achievement motivation. One weakness of the model is that it does not provide an account of the processes responsible for the link between achievement goals and performance. As this model is enhanced, it becomes more useful in predicting the outcomes of achievement-based behaviors (Elliot & McGregor, 1999).

Intrinsic Motivation and Achievement Goals

Intrinsic motivation is defined as the enjoyment of and interest in an activity for its own sake. Fundamentally viewed as an approach form of motivation, intrinsic motivation is identified as an important component of achievement goal theory. Most achievement goal and intrinsic motivational theorists argue that mastery goals are facilitative of intrinsic motivation and related mental processes and performance goals create negative effects. Mastery goals are said to promote intrinsic motivation by fostering perceptions of challenge, encouraging task involvement, generating excitement, and supporting self-determination while performance goals are the opposite. Performance goals are portrayed as undermining intrinsic motivation by instilling perceptions of threat, disrupting task involvement, and creating anxiety and pressure (Elliot & Harackiewicz, 1996).

An alternative set of predictions may be derived from the approach-avoidance framework. Both performance-approach and mastery goals are focused on attaining competence and foster intrinsic motivation. More specifically, in performance-approach or mastery orientations, individuals perceive the achievement setting as a challenge, and this likely will create excitement, encourage cognitive functioning, increase concentration and task absorption, and direct the person toward success and mastery of information which facilitates intrinsic motivation. The performance-avoidance goal is focused on avoiding incompetence, where individuals see the achievement setting as a threat and seek to escape it (Elliot & Harackiewicz, 1996). This orientation is likely to elicit anxiety and withdrawal of effort and cognitive resources while disrupting concentration and motivation.

A Tripartite Model of Motivation for Achievement: Attitude/Drive/Strategy

Motivating students to achieve in school is a topic of great practical concern to teachers and parents, and of great theoretical concern to researchers. New books on the topic appear with increasing frequency and relevant research is proliferating at a rapid rate. Higher education institutions are beginning to provide assistance to students, especially new ones, in developing so-called study skills and self-regulatory skills such as time management. One of the greatest *challenges and opportunities of the 21st century* will be for schools at all levels to focus more on assisting students to become motivated in order that they can succeed in school.

The purpose of this paper is to present a proposed model of motivation for achievement, as applied particularly to the educational setting. This model focuses on three generic variables: (1) *attitude* or beliefs that people hold about themselves, their capabilities, and the factors that account for their outcomes; (2) *drive* or the desire to attain an outcome based on the value people place on it; (3) *strategy* or the techniques that people employ to gain the outcomes they desire. Each of these variables will be described in more detail, and evidence will be provided to support the contention that

each exerts an important influence on motivation to achieve in an academic environment. Achievement outcomes have been regarded as a function of two characteristics, “skill” and “will” (McCombs and Marzano, 1990), and these must be considered separately because possessing the will alone may not insure success if the skill is lacking. The focus in this model is on will, or the motivation to achieve the outcome, and it will be considered separately from level of skill. Where achievement measures such as scores on course examinations or grades are used as criteria of motivation to achieve, measures of skill have to be separated out or controlled for. To measure motivation for achievement directly, measures of *engagement* must be examined.

Cognitive engagement represents the amount of effort spent in either studying or completing assignments. It is the result of motivation, not its source. Pintrich and Schrauben (1992) review a large body of research that suggests that (1) the value of an outcome to the student affects that student’s motivation, and (2) motivation leads to cognitive engagement, such engagement manifesting itself in the use or application of various learning strategies.

Many of the studies Pintrich and Schrauben describe employed self-reports of learning strategy use as a measure of cognitive engagement. Such studies become dependent on what students claimed to be doing as a way of determining that they were indeed engaged in a task. To avoid this dependence on students’ self-reports, the studies I carried out (either alone or in collaboration with Tom Sexton) and report on in this paper operationalized cognitive engagement as a manifestation of effort expenditure or actual performance on the homework task of writing test items on text chapters in which students had the option to perform assignments for extra credit. In studies I carried out and report on using achievement test scores as a measure of motivation to achieve, relevant data on students’ skill level was used as a control.

Motivation and Motivating in the Foreign Language Classroom

In this paper, we will briefly examine a variety of techniques, strategies and macrostrategies which teachers can employ in order to motivate their students. As Dornyei (2001: 116) notes, ‘teacher skills in *motivating* learners should be seen as central to teaching effectiveness’. Even though there have been a lot of education-oriented publications providing taxonomies of classroom-specific motives, they fall short of offering an efficient guide to practitioners. Thus, our main goal is to familiarise any putative “practitioners” with a set of techniques and strategies (henceforward, “motivational strategies”) for motivating foreign language students.

Power in the Classroom

Prior to presenting some of these motivational strategies, it would be of relevance to say a few things about the teacher / learner relationship. Whichever way we look at it, this relationship is riddled with power and status. For many, power plays a large part in

the relationship (see “Language and Power in Education” for further details). The rights and duties of teachers and learners are related to power. For example, many teachers might assert that they have the right to punish those learners who misbehave. In any social encounter involving two or more people, there are certain power relationships ‘which are almost always asymmetrical’ (Wright, 1987: 17). Social psychologists distinguish between three different types of power—*coercive*, *reward-based*, and *referent* (ibid.). The basis of coercive power is punishment. Some individuals or institutions have the authority to punish others. The basis of the second type of power is reward. Some individuals or institutions have the power to reward what they deem appropriate behaviour. For example, business organisations reward employees with a salary, a bonus etc. The basis of the third type of power is motivation. In this case, individuals or institutions appeal to the commitment and interest of others. In view of this three-fold paradigm, it is of importance to concern ourselves with the fostering of learner motivation, as it is considered to be the most effective and proactive, so to speak, power relationship.

Group Processes and Motivation

A discussion of motivation and motivational strategies would not be complete without a consideration of group processes, inasmuch as there is usually a group of people that we as teachers are called on to motivate. Tuckman (1969, quoted in Argyle, 1969) established that a group went through four stages from its formation, which has important implications for the study of the classroom and the use of group activities during teaching.

- Stage 1 **Forming** : At first, there is some anxiety among the members of the group, as they are dependent on the leader (that is, the teacher) and they have to find out what behaviour is acceptable.
- Stage 2 **Storming** : There is conflict between sub-groups and rebellion against the leader. Members of the group resist their leader and the role relations attending the function of the group are questioned.
- Stage 3 **Norming** : The group begins to develop a sort of cohesion. Members of the group begin to support each other. At this stage, there is co-operation and open exchange of views and feelings about their roles and each other.
- Stage 4 **Performing** : Most problems are resolved and there is a great deal of interpersonal activity. Everyone is devoted to completing the tasks they have been assigned.

Experience shows that almost every group goes through these four (or even more) stages until it reaches equilibrium and, thus, taps into its potential. In reality, this process may go on forever, since student lethargy and underachievement norms in the classroom are considered to be basic hindrances to effective teaching and learning (Daniels, 1994). Against this background, we will try to design a framework for motivational strategies.

A Framework for Motivational Strategies

As we have already said, skill in motivating students to learn is of paramount importance.

Until recently, however, teachers were forced to rely on “bag-of-tricks” approaches in their attempt to manage their classroom and motivate their learners. Good and Brophy (1994: 212) hold that these approaches have been influenced by two contradictory views:

- That learning should be fun and that any motivation problems that may appear should be ascribed to the teacher’s attempt to convert an enjoyable activity to drudgery; and
- That school activities are inherently boring and unrewarding, so that we must rely on extrinsic rewards and punishment with a view to forcing students to engage in these unpleasant tasks.

Rewards and punishments may be a mainstay of the teaching-learning process, but they are not the only tools in teachers’ arsenal. Dornyei (2001: 119) believes that ‘the spectrum of other potentially more effective motivational strategies is so broad that it is hard to imagine that none of them would work’.

The central question in designing a framework of motivational strategies is to decide how to organise them into separate themes. The following taxonomy, around which our main discussion will revolve, is based on the process-oriented model by Dornyei and Otto (1998). The key units in this taxonomy are as follows:

- *Creating the basic motivational conditions*, which involves setting the scene for the use of motivational strategies;
- *Generating student motivation*, which roughly corresponds to the preactional phase in the model;
- *Maintaining and protecting motivation*, which corresponds to the actional phase;
- *Encouraging positive self-evaluation*, which corresponds to the postactional phase

Creating the Basic Motivational Conditions

Motivational strategies cannot work in a vacuum. There are certain preconditions to be met before any attempts to generate motivation can be effective. Some of these conditions are the following:

- Appropriate teacher behaviour and good teacher-student rapport;
- A pleasant and supportive classroom atmosphere;
- A cohesive learner group characterised by appropriate group norms

Appropriate Teacher Behaviour and Good Teacher-student Rapport

Whatever is done by a teacher has a motivational, formative, influence on students. In other words, teacher behaviour is a powerful ‘motivational tool’ (Dornyei, 2001:

120). Teacher influences are manifold, ranging from the rapport with the students to teacher behaviours which “prevail upon” and / or “attract” students to engage in tasks. For Alison (1993), a key element is to establish a relationship of mutual trust and respect with the learners, by means of talking with them on a personal level. This mutual trust could lead to enthusiasm. At any rate, enthusiastic teachers impart a sense of commitment to, and interest in, the subject matter, not only verbally but also non-verbally—cues that students take from them about how to behave.

A Pleasant and Supportive Classroom Atmosphere

It stands to reason that a tense classroom climate can undermine learning and demotivate learners. On the other hand, learner motivation will reach its peak in a safe classroom climate in which students can express their opinions and feel that they do not run the risk of being ridiculed.

To be motivated to learn, students need both ample opportunities to learn and steady encouragement and support of their learning efforts. Because such motivation is unlikely to develop in a chaotic classroom, it is important that the teacher organise and manage the classroom as an effective learning environment. Furthermore, because anxious or alienated students are unlikely to develop motivation to learn, it is important that learning occurs within a relaxed and supportive atmosphere (Good and Brophy, 1994: 215).

A Cohesive Learner Group Characterised by Appropriate Group Norms

As was hinted at above, fragmented groups, characterised by lack of cooperativeness, can easily become ineffective, thus reducing the individual members’ commitment to learn. There are several factors that promote group cohesiveness, such as the time spent together and shared group history, learning about each other, interaction, intergroup competition, common threat, active presence of the leader [...].

As for group norms, they should be discussed and adopted by members, in order to be constructive and long-lasting. If a norm mandated by a teacher fails to be accepted as proper by the majority of the class members, it will not become a group norm.

Generating Student Motivation

Ideally, all learners exhibit an inborn curiosity to explore the world, so they are likely to find the learning experience *per se* intrinsically pleasant. In reality, however, this “curiosity” is weakened by such inexorable factors as compulsory school attendance, curriculum content, and grades—most importantly, the premium placed on them.

Apparently, unless teachers increase their learners’ ‘goal-orientedness’, make curriculum relevant for them, and create realistic learner beliefs, they will come up against a classroom environment fraught with lack of cohesiveness and rebellion.

Increasing the Learners' 'Goal-Orientedness'

In some classes, many, if not most, students do not understand why they are involved in an activity. It may be the case that the goal set by outsiders (i.e., the teacher or the curriculum) is far from being accepted by the group members. Thus, it would seem beneficial to increase the group's goal-orientedness, that is, the extent to which the group tunes in to the pursuit of its official goal. This could be achieved by allowing students to define their own personal criteria for what should be a group goal.

Making the Curriculum Relevant for the Learners

Many students do their homework and engage in many types of learning activities, even when a subject is not very interesting. Obviously, these students share the belief of the curriculum makers that what they are being taught will come in handy. In order to inspire learners to concern themselves with most learning activities, we should find out their goals and the topics they want to learn, and try to incorporate them into the curriculum. According to Chambers (1999: 37), '[i]f the teacher is to motivate pupils to learn, then relevance has to be the red thread permeating activities'.

Creating Realistic Learner Beliefs

It is widely acknowledged that learner beliefs about how much progress to expect, and at what pace, can, and do, lead to disappointment. Therefore, it is important to help learners get rid of their preconceived notions that are likely to hinder their attainment. To this end, learners need to develop an understanding of the nature of second language learning, and should be cognisant of the fact that the mastery of L2 can be achieved in different ways, using a diversity of strategies, and a key factor is for learners to discover for themselves the optimal methods and techniques.

Maintaining and Protecting Motivation

Unless motivation is sustained and protected when action has commenced, the natural tendency to get tired or bored of the task and succumb to any attractive distractions will result in demotivation. Therefore, there should be a motivational repertoire including several motivation maintenance strategies. Let us have a look at two of them: a) increasing the learners' self-confidence; and b) creating learner autonomy.

Research Methodology

The research methods used in educational psychology tend to be drawn from psychology and other social sciences. There is also a history of significant methodological innovation by educational psychologists, and psychologists investigating educational problems. Research methods address problems in both research design and data analysis. Research design informs the planning of experiments and observational studies to ensure that their results have internal,

external and ecological validity. Data analysis encompasses methods for processing both quantitative (numerical) and qualitative (non-numerical) research data. Although, historically, the use of quantitative methods was often considered an essential mark of scholarship, modern educational psychology research uses both quantitative and qualitative methods.

Quantitative Methods

Perhaps first among the important methodological innovations of educational psychology was the development and application of factor analysis by Charles Spearman. Factor analysis is mentioned here as one example of the many multivariate statistical methods used by educational psychologists. Factor analysis is used to summarize relationships among a large set of variables or test questions, develop theories about mental constructs such as self-efficacy or anxiety, and assess the reliability and validity of test scores. Over 100 years after its introduction by Spearman, factor analysis has become a research staple figuring prominently in educational psychology journals.

Because educational assessment is fundamental to most quantitative research in the field, educational psychologists have made significant contributions to the field of psychometrics. For example, alpha, the widely used measure of test reliability was developed by educational psychologist Lee Cronbach. The reliability of assessments are routinely reported in quantitative educational research. Although, originally, educational measurement methods were built on classical test theory, item response theory and Rasch models are now used extensively in educational measurement worldwide. These models afford advantages over classical test theory, including the capacity to produce standard errors of measurement for each score or pattern of scores on assessments and the capacity to handle missing responses. Meta-analysis, the combination of individual research results to produce a quantitative literature review, is another methodological innovation with a close association to educational psychology. In a meta-analysis, effect sizes that represent, for example, the differences between treatment groups in a set of similar experiments, are averaged to obtain a single aggregate value representing the best estimate of the effect of treatment. Several decades after Pearson's work with early versions of meta-analysis, Glass published the first application of modern meta-analytic techniques and triggered their broad application across the social and biomedical sciences. Today, meta-analysis is among the most common types of literature review found in educational psychology research. Other quantitative research issues associated with educational psychology include the use of nested research designs (*e.g.*, a student nested within a classroom, which is nested within a school, which is nested within a district, etc.) and the use of longitudinal statistical models to measure change.

Qualitative Methods

Qualitative methods are used in educational studies whose purpose is to describe events, processes and situations of theoretical significance. The qualitative methods

used in educational psychology often derive from anthropology, sociology or sociolinguistics. For example, the anthropological method of ethnography has been used to describe teaching and learning in classrooms. In studies of this type, the researcher may gather detailed field notes as a participant observer or passive observer. Later, the notes and other data may be categorized and interpreted by methods such as grounded theory. Triangulation, the practice of cross-checking findings with multiple data sources, is highly valued in qualitative research. Case studies are forms of qualitative research focusing on a single person, organization, event, or other entity. In one case study, researchers conducted a 150-minute, semi-structured interview with a 20-year-old woman who had a history of suicidal thinking between the ages of 14 to 18. They analyzed an audio-recording of the interview to understand the roles of cognitive development, identity formation and social attachment in ending her suicidal thinking.

Qualitative analysis is most often applied to verbal data from sources such as conversations, interviews, focus groups, and personal journals. Qualitative methods are thus, typically, approaches to gathering, processing and reporting verbal data. One of the most commonly used methods for qualitative research in educational psychology is protocol analysis. In this method the research participant is asked to think aloud while performing a task, such as solving a math problem. In protocol analysis the verbal data is thought to indicate which information the subject is attending to, but is explicitly not interpreted as an explanation or justification for behaviour. In contrast, the method of verbal analysis does admit learners' explanations as a way to reveal their mental model or misconceptions (*e.g.*, of the laws of motion).

The most fundamental operations in both protocol and verbal analysis are segmenting (isolating) and categorizing sections of verbal data. Conversation analysis and discourse analysis, sociolinguistic methods that focus more specifically on the structure of conversational interchange (*e.g.*, between a teacher and student), have been used to assess the process of conceptual change in science learning. Qualitative methods are also used to analyze information in a variety of media, such as students' drawings and concept maps, video-recorded interactions, and computer log records.

Indigenous Education, Learning, and Motivation

For many indigenous students (such as Native American children), motivation may be derived from social organization; an important factor educators should account for in addition to variations in Sociolinguistics and Cognition. While poor academic performance among Native American students is often attributed to low levels of motivation, Top-down classroom organization is often found to be ineffective for children of many cultures, who depend on a sense of community purpose and competence to effectively engage in material.

Horizontally-structured, community-based learning strategies often provide a more structurally supportive environment for motivating indigenous children, who tend to be driven by "social/affective emphasis, harmony, holistic perspectives, expressive

creativity, and non-verbal communication." This drive is also traceable to a cultural tradition of community-wide expectations of participation in the activities and goals of the greater group, rather than individualized aspirations of success or triumph.

Structure for social learning in indigenous communities also often allows siblings to co-parent younger children in their acquisition of behaviours and traditions, which fosters the dynamic of community-motivated engagement from a young age. Furthermore, it is commonplace for children to assist and demonstrate for their younger counterparts without being prompted by authority figures. Observation techniques are demonstrated in such examples as weaving in Chiapas, Mexico, where it is commonplace for children to learn by "a more skilled other" within the community. The assumption of responsibility amongst children is also apparent within Mayan weaving apprenticeships; often, when the "more skilled other" is tasked with multiple obligations, an older child will step in and guide the learner. Sibling guidance is supported from early youth, where learning through play encourages horizontally-structured environments through alternative educational models such as "Intent Community Participation." Research also suggests that formal Westernized schooling can actually reshape the traditionally collaborative nature of social life in indigenous communities. This research is supported cross-culturally, with variations in motivation and learning often reported higher between indigenous groups and their national Westernized counterparts than between indigenous groups across international continental divides.

Sudbury Model Schools' Approach

Sudbury Model schools adduce that the cure to the problem of procrastination, of learning in general, and particularly of scientific illiteracy is to remove once and for all what they call the underlying disease: compulsion in schools. They contend that human nature in a free society recoils from every attempt to force it into a mold; that the more requirements we pile onto children at school, the surer we are to drive them away from the material we are trying to force down their throats; that after all the drive and motivation of infants to master the world around them is legendary. They assert that schools must keep that drive alive by doing what some of them do: nurturing it on the freedom it needs to thrive.

Sudbury Model schools do not perform and do not offer evaluations, assessments, transcripts, or recommendations, asserting that they do not rate people, and that school is not a judge; comparing students to each other, or to some standard that has been set is for them a violation of the student's right to privacy and to self-determination. Students decide for themselves how to measure their progress as self-starting learners as a process of self-evaluation: real lifelong learning and the proper educational evaluation for the 21st century, they adduce. According to Sudbury Model schools, this policy does not cause harm to their students as they move on to life outside the school. However, they admit it makes the process more difficult, but that such hardship is part of the students learning to make their own way, set their own standards and meet their own goals. The

no-grading and no-rating policy helps to create an atmosphere free of competition among students or battles for adult approval, and encourages a positive cooperative environment amongst the student body.

Business

At lower levels of Maslow's hierarchy of needs, such as physiological needs, money is a motivator, however it tends to have a motivating effect on staff that lasts only for a short period (in accordance with Herzberg's two-factor model of motivation). At higher levels of the hierarchy, praise, respect, recognition, empowerment and a sense of belonging are far more powerful motivators than money, as both Abraham Maslow's theory of motivation and Douglas McGregor's theory X and theory Y (pertaining to the theory of leadership) demonstrate.

According to Maslow, people are motivated by unsatisfied needs. The lower level needs such as Physiological and Safety needs will have to be satisfied before higher level needs are to be addressed. We can relate Maslow's Hierarchy of Needs theory with employee motivation. For example, if a manager is trying to motivate his employees by satisfying their needs; according to Maslow, he should try to satisfy the lower level needs before he tries to satisfy the upper level needs or the employees will not be motivated. Also he has to remember that not everyone will be satisfied by the same needs. A good manager will try to figure out which levels of needs are active for a certain individual or employee.

Maslow has money at the lowest level of the hierarchy and shows other needs are better motivators to staff. McGregor places money in his Theory X category and feels it is a poor motivator. Praise and recognition are placed in the Theory Y category and are considered stronger motivators than money.

- Motivated employees always look for better ways to do a job.
- Motivated employees are more quality oriented.
- Motivated workers are more productive.

The average workplace is about midway between the extremes of high threat and high opportunity. Motivation by threat is a dead-end strategy, and naturally staff are more attracted to the opportunity side of the motivation curve than the threat side. Motivation is a powerful tool in the work environment that can lead to employees working at their most efficient levels of production. Nonetheless, Steinmetz also discusses three common character types of subordinates: ascendant, indifferent, and ambivalent who all react and interact uniquely, and must be treated, managed, and motivated accordingly. An effective leader must understand how to manage all characters, and more importantly the manager must utilize avenues that allow room for employees to work, grow, and find answers independently.

The assumptions of Maslow and Herzberg were challenged by a classic study at Vauxhall Motors' UK manufacturing plant. This introduced the concept of orientation to work and distinguished three main orientations: instrumental (where work is a means

to an end), bureaucratic (where work is a source of status, security and immediate reward) and solidaristic (which prioritizes group loyalty). Other theories which expanded and extended those of Maslow and Herzberg included Kurt Lewin's Force Field Theory, Edwin Locke's Goal Theory and Victor Vroom's Expectancy theory. These tend to stress cultural differences and the fact that individuals tend to be motivated by different factors at different times.

According to the system of scientific management developed by Frederick Winslow Taylor, a worker's motivation is solely determined by pay, and therefore management need not consider psychological or social aspects of work. In essence, scientific management bases human motivation wholly on extrinsic rewards and discards the idea of intrinsic rewards.

In contrast, David McClelland believed that workers could not be motivated by the mere need for money—in fact, extrinsic motivation (*e.g.*, money) could extinguish intrinsic motivation such as achievement motivation, though money could be used as an indicator of success for various motives, *e.g.*, keeping score. In keeping with this view, his consulting firm, McBer and Company, had as its first motto "To make everyone productive, happy, and free." For McClelland, satisfaction lay in aligning a person's life with their fundamental motivations.

Elton Mayo found that the social contacts a worker has at the workplace are very important and that boredom and repetitiveness of tasks lead to reduced motivation. Mayo believed that workers could be motivated by acknowledging their social needs and making them feel important. As a result, employees were given freedom to make decisions on the job and greater attention was paid to informal work groups. Mayo named the model the Hawthorne effect. His model has been judged as placing undue reliance on social contacts at work situations for motivating employees.

William Ouchi introduced Theory Z, a hybrid management approach consisting of both Japanese and American philosophies and cultures. Its Japanese segment is much like the clan culture where organizations focus on a standardized structure with heavy emphasis on socialization of its members. All underlying goals are consistent across the organization. Its American segment retains formality and authority amongst members and the organization. Ultimately, Theory Z promotes common structure and commitment to the organization, as well as constant improvement of work efficacy. In *Essentials of Organizational Behaviour*, Robbins and Judge examine recognition programmes as motivators, and identify five principles that contribute to the success of an employee incentive programme:

- Recognition of employees' individual differences, and clear identification of behaviour deemed worthy of recognition
- Allowing employees to participate
- Linking rewards to performance
- Rewarding of nominators
- Visibility of the recognition process

MOTIVATIONAL THEORIES

Incentive Theory

A reward, tangible or intangible, is presented after the occurrence of an action (i.e. behaviour) with the intent to cause the behaviour to occur again. This is done by associating positive meaning to the behaviour. Studies show that if the person receives the reward immediately, the effect is greater, and decreases as delay lengthens. Repetitive action-reward combination can cause the action to become habit. Motivation comes from two sources: oneself, and other people. These two sources are called intrinsic motivation and extrinsic motivation, respectively.

Reinforcers and reinforcement principles of behaviour differ from the hypothetical construct of reward. A reinforcer is any stimulus change following a response that increases the future frequency or magnitude of that response, therefore the cognitive approach is certainly the way forward as in 1973 Maslow described it as being the golden pineapple. Positive reinforcement is demonstrated by an increase in the future frequency or magnitude of a response due to in the past being followed contingently by a reinforcing stimulus. Negative reinforcement involves stimulus change consisting of the removal of an aversive stimulus following a response. Positive reinforcement involves a stimulus change consisting of the presentation or magnification of a positive stimulus following a response. From this perspective, motivation is mediated by environmental events, and the concept of distinguishing between intrinsic and extrinsic forces is irrelevant.

Applying proper motivational techniques can be much harder than it seems. Steven Kerr notes that when creating a reward system, it can be easy to reward A, while hoping for B, and in the process, reap harmful effects that can jeopardize your goals. Incentive theory in psychology treats motivation and behaviour of the individual as they are influenced by beliefs, such as engaging in activities that are expected to be profitable. Incentive theory is promoted by behavioural psychologists, such as B.F. Skinner and literalized by behaviourists, especially by Skinner in his philosophy of Radical behaviourism, to mean that a person's actions always have social ramifications: and if actions are positively received people are more likely to act in this manner, or if negatively received people are less likely to act in this manner.

Incentive theory distinguishes itself from other motivation theories, such as drive theory, in the direction of the motivation. In incentive theory, stimuli "attract", to use the term above, a person towards them, as opposed to the body seeking to reestablish homeostasis pushing it towards the stimulus. In terms of behaviourism, incentive theory involves positive reinforcement: the stimulus has been conditioned to make the person happier. For instance, a person knows that eating food, drinking water, or gaining social capital will make them happier. As opposed to in drive theory, which involves negative reinforcement: a stimulus has been associated with the removal of the punishment-the

lack of homeostasis in the body. For example, a person has come to know that if they eat when hungry, it will eliminate that negative feeling of hunger, or if they drink when thirsty, it will eliminate that negative feeling of thirst.

Escape-seeking Dichotomy Model

Escapism and seeking are major factors influencing decision making. Escapism is a need to breakaway from a daily life routine, turning on the television and watching an adventure film, whereas seeking is described as the desire to learn, turning on the television to watch a documentary. Both motivations have some interpersonal and personal facets for example individuals would like to escape from family problems (personal) or from problems with work colleagues (interpersonal). This model can also be easily adapted with regard to different studies.

Drive-reduction Theory

There are a number of drive theories. The Drive Reduction Theory grows out of the concept that we have certain biological drives, such as hunger. As time passes the strength of the drive increases if it is not satisfied (in this case by eating). Upon satisfying a drive the drive's strength is reduced. The theory is based on diverse ideas from the theories of Freud to the ideas of feedback control systems, such as a thermostat.

Drive theory has some intuitive or folk validity. For instance when preparing food, the drive model appears to be compatible with sensations of rising hunger as the food is prepared, and, after the food has been consumed, a decrease in subjective hunger. There are several problems, however, that leave the validity of drive reduction open for debate. The first problem is that it does not explain how secondary reinforcers reduce drive. For example, money satisfies no biological or psychological needs, but a pay check appears to reduce drive through second-order conditioning. Secondly, a drive, such as hunger, is viewed as having a "desire" to eat, making the drive a homuncular being-a feature criticized as simply moving the fundamental problem behind this "small man" and his desires. In addition, it is clear that drive reduction theory cannot be a complete theory of behaviour, or a hungry human could not prepare a meal without eating the food before he finished cooking it. The ability of drive theory to cope with all kinds of behaviour, from not satisfying a drive (by adding on other traits such as restraint), or adding additional drives for "tasty" food, which combine with drives for "food" in order to explain cooking render it hard to test.

Cognitive Dissonance Theory

Suggested by Leon Festinger, cognitive dissonance occurs when an individual experiences some degree of discomfort resulting from an inconsistency between two cognitions: their views on the world around them, and their own personal feelings and actions.

For example, a consumer may seek to reassure himself regarding a purchase, feeling, in retrospect, that another decision may have been preferable. His feeling that another purchase would have been preferable is inconsistent with his action of purchasing the item. The difference between his feelings and beliefs causes dissonance, so he seeks to reassure himself.

While not a theory of motivation, *per se*, the theory of cognitive dissonance proposes that people have a motivational drive to reduce dissonance. The cognitive miser perspective makes people want to justify things in a simple way in order to reduce the effort they put into cognition. They do this by changing their attitudes, beliefs, or actions, rather than facing the inconsistencies, because dissonance is a mental strain. Dissonance is also reduced by justifying, blaming, and denying. It is one of the most influential and extensively studied theories in social psychology.

Need Theories

Motivation, as defined by Pritchard and Ashwood, is the process used to allocate energy to maximize the satisfaction of needs.

Need Hierarchy Theory

The content theory includes the hierarchy of needs from Abraham Maslow and the two-factor theory from Herzberg. Maslow's theory is one of the most widely discussed theories of motivation. The American motivation psychologist Abraham H. Maslow developed the Hierarchy of needs consistent of five hierarchic classes. It shows the complexity of human requirements. According to him, people are motivated by unsatisfied needs. The lower level needs such as Physiological and Safety needs will have to be satisfied before higher level needs are to be addressed. We can relate Maslow's Hierarchy of Needs theory with employee motivation. For example, if a manager is trying to motivate his employees by satisfying their needs; according to Maslow, he should try to satisfy the lower level needs before he tries to satisfy the upper level needs or the employees will not be motivated.

Also he has to remember that not everyone will be satisfied by the same needs. A good manager will try to figure out which levels of needs are active for a certain individual or employee. The basic requirements build the first step in his pyramid. If there is any deficit on this level, the whole behaviour of a human will be oriented to satisfy this deficit. Subsequently we do have the second level, which awake a need for security. Basically it is oriented on a future need for security. After securing those two levels, the motives shift in the social sphere, which form the third stage. Psychological requirements consist in the fourth level, while the top of the hierarchy comprise the self- realization.

So theory can be summarized as follows:

- Human beings have wants and desires which influence their behaviour. Only unsatisfied needs influence behaviour, satisfied needs do not.

- Since needs are many, they are arranged in order of importance, from the basic to the complex.
- The person advances to the next level of needs only after the lower level need is at least minimally satisfied.
- The further the progress up the hierarchy, the more individuality, humanness and psychological health a person will show.

The needs, listed from basic (lowest-earliest) to most complex (highest-latest) are as follows:

- Physiology (hunger, thirst, sleep, etc.)
- Safety/Security/Shelter/Health
- Belongingness/Love/Friendship
- Self-esteem/Recognition/Achievement
- Self actualization

Herzberg's Two-Factor Theory

Frederick Herzberg's two-factor theory, a.k.a. intrinsic/extrinsic motivation, concludes that certain factors in the workplace result in job satisfaction, but if absent, they don't lead to dissatisfaction but no satisfaction. The factors that motivate people can change over their lifetime, but "respect for me as a person" is one of the top motivating factors at any stage of life.

He distinguished between:

- Motivators; (*e.g.* challenging work, recognition, responsibility) which give positive satisfaction, and
- Hygiene factors; (*e.g.* status, job security, salary and fringe benefits) that do not motivate if present, but, if absent, result in demotivation.

The name Hygiene factors is used because, like hygiene, the presence will not make you healthier, but absence can cause health deterioration. The theory is sometimes called the "Motivator-Hygiene Theory" and/or "The Dual Structure Theory." Herzberg's theory has found application in such occupational fields as information systems and in studies of user satisfaction.

Alderfer's ERG Theory

Alderfer, expanding on Maslow's hierarchy of needs, created the ERG theory. This theory posits that there are three groups of core needs - existence, relatedness, and growth, hence the label: ERG theory. The existence group is concerned with providing our basic material existence requirements. They include the items that Maslow considered to be physiological and safety needs. The second group of needs are those of relatedness- the desire we have for maintaining important personal relationships. These social and status desires require interaction with others if they are to be satisfied, and they align with Maslow's social need and the external component of Maslow's

esteem classification. Finally, Alderfer isolates growth needs as an intrinsic desire for personal development. These include the intrinsic component from Maslow's esteem category and the characteristics included under self-actualization.

Self-Determination Theory

Self-determination theory (SDT), developed by Edward Deci and Richard Ryan, focuses on the importance of intrinsic motivation in driving human behaviour. Like Maslow's hierarchical theory and others that built on it, SDT posits a natural tendency towards growth and development. Unlike these other theories, however, SDT does not include any sort of "autopilot" for achievement, but instead requires active encouragement from the environment. The primary factors that encourage motivation and development are autonomy, competence feedback, and relatedness.

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Index

A

Accompanied 158
Administration 239, 240, 242, 252, 253, 260, 262, 263
Adolescence 181, 183, 210, 211, 212
Aggressively 132
Analytical 167, 168
Antiquity 232
Application 226
Applications 233

C

Cognitive 129, 133, 134, 135, 136, 138, 139, 154, 155, 156, 157, 158, 166, 168, 169, 183
Combined Aptitude 251
Comparison 227, 245
Concerned 131, 132
Congruence 169
Consciousness 132
Considered 125, 126, 136, 138, 155, 156, 166, 167, 184
Constitutes 168
Contemporary Theories 248
Controversy 250
Correlations 155, 157, 169

Creativity 258
Critique 165
Current Status 246

D

Discrepancies 155, 156, 157, 158
Dramatic 124, 125
Drive theory 254

E

Economics 236
Efficacy 170, 180
Encouraging 128, 129, 130, 181
Endeavoring 215

G

Group Tests 252, 253

H

Hereditary 165, 211
Heredity 250
Heritability 165

I

Impulsiveness 159
Integrity Inventory 256
Inter-rater Reliability 258, 262
Interchangeability 158

J

Judgment 262, 263

L

Legal Control 230

M

Measurement of Intelligence 152
Medieval Era 238
Metacomponents 167

Metaphysical 169, 213
Method 228, 229, 248, 249, 257, 259, 260, 262, 263
Methodology 226
Metric Equivalents 243, 244
Metric System 229, 230, 231, 232, 233, 234, 236, 242, 246
Motivation 248, 253, 254, 257, 261
Motives as Inference 253

N

Natural Systems 230
Natural Units 230, 234
New Variables 263

P

Personality 255, 257, 258, 259, 260, 261, 262, 263
Population 152
Predictions 137
Pronouncement 181
Propensities 210
Putting 129, 217

R

Rationally 156
Rorschach Performance 263
Rorschach Test 258, 259, 260, 262

S

Scoring 258, 259, 260, 261, 262, 263
Scoring Systems 260
Self-Actualization Theory 254
Sequencing 126
Shortcomings 239
Sidetracked 181
Significance 214, 217
Simultaneous 157, 158
Social Motives 255

T

Technology 195
Tendencies 165, 215

Terminology 258
Theories 248, 254
Traditional Systems 230
Transmitted 134

U

Units of Measurement 228, 229, 230, 231, 232, 234, 236, 246

V

Various Directions 213